

THE ULTIMATE DATA STREAMING GUIDE

Concepts, Use Cases,
Industry Stories

SECOND EDITION

**Guest chapter
by Evi Schneider:**
*How to build a data
streaming community
in your own
organization*

Kai Waehner
with a foreword by
Jay Kreps



CONFLUENT





THE ULTIMATE DATA STREAMING GUIDE

Concepts, Use Cases,
Industry Stories

SECOND EDITION

Kai Waehner
with a foreword by
Jay Kreps

Guest chapter
by **Evi Schneider**:
*How to build a data
streaming community
in your own
organization*

Contents

Foreword by Jay Kreps	12
Preface	14
Content Matrix	18

01 Event-Driven Architecture and the Demand for Real-Time Data 20

1.1 The Business Impact of a Data Mess	22
1.2 The Need for an Event-Driven Architecture	24
1.3 Data Streaming Platform	26
1.4 Business Value of Data Streaming	31
1.5 Analytic Data and AI Platforms	36
1.6 The Shift-Left Architecture	39
1.7 Data Streaming vs Lakehouse	40

02 Data Streaming Use Cases 42

2.1 Data Products	43
• Relation of Data Products to Data Streaming	44
• Examples for Data Products and Data Streaming	44
• Data Products Build With Data Streaming for Data Consistency and Continuous Processing	45
2.2 IT Modernization	46
• Relation of IT Modernization to Data Streaming	46
• Bidirectional Integration With Legacy and Cloud Systems	46
• Gradual Rip and Replace of Legacy Systems	47
• Examples for IT Modernization With Data Streaming	48
• IT Modernization With Data Streaming for Seamless Integration to Ensure Data Accuracy	49
2.3 Observability	50
• Relation of Observability to Data Streaming	51
• Examples for Observability With Data Streaming	52
• Reliability, Security, and User Satisfaction With Data Streaming	52
2.4 Hybrid and Multicloud	54
• Relation of Hybrid and Multicloud to Data Streaming	54
• Examples for Hybrid and Multicloud With Data Streaming	58
• Hybrid and Multicloud With Data Streaming for Global Availability and Scalability	59
2.5 Event-Driven Microservices	60
• Relation of Event-Driven Microservices and Data Streaming	61
• Examples for Event-Driven Microservices and Data Streaming	61

- Event-Driven Microservices Using Data Streaming for an Evolving Architecture and Technology-Independent Integration 62

2.6 Data Sharing 63

- Relation of Data Sharing and Data Streaming 63
- Examples for Data Sharing 64
- Data Sharing Through Data Streaming for Consistent Data Accessibility in Real Time 65

2.7 Artificial Intelligence 66

- Challenges of Agentic AI Development 66
- Relation of AI and Data Streaming 67
- Examples for AI Using Data Streaming 68
- Data Streaming for More Powerful and Accurate Results in AI Use Cases 69

2.8 Internet of Things 72

- Relation of IoT and Data Streaming 73
- Examples for Data Streaming in the IoT World 74
- IoT with Data Streaming to Optimize the OEE and Provide Innovative Consumer Experiences 74

2.9 Data Governance 76

- Relation of Data Governance and Data Streaming 76
- Examples for Data Governance in Streaming Data Architectures 78
- Business Value of Data Governance with Data Streaming 78

2.10 Environmental, Social, and Governance 80

- Relation of ESG and Data Streaming 80
- Examples for ESG Use Cases With Data Streaming 81
- Business Value of ESG With Data Streaming 82

03 Industry Success Stories for Data Streaming 84

3.1 Financial Services 86

- Scenario 1: Legacy Offloading 88
- Scenario 2: Core Banking 89
- Scenario 3: Real-Time Payments 90
- Scenario 4: Fraud Detection 91
- Scenario 5: Regulatory Reporting 92
- Scenario 6: Embedded Finance 93

- Scenario 7: FinOps 94
- Scenario 8: Digital Currencies 95
- Scenario 9: Stablecoins 96
- Real-Life Examples (DNB ASA, Erste Group Bank, KOR) 97
- Conclusion 106

3.2 Manufacturing and Automotive 108

- Scenario 1: Condition Monitoring and Predictive Maintenance 110
- Scenario 2: Supply Chain 111
- Scenario 3: Track-and-Trace 112
- Scenario 4: Engineering Tool Integration for Product and Equipment Design 113
- Scenario 5: Connected Vehicles and Real-Time Car APIs 114
- Scenario 6: Real-Time Quality Intelligence and Plant Standardization 115
- Real-Life Examples (PAUL Tech AG, Knauf, Siemens, BMW Group) 116
- Conclusion 129

3.3 Retail and eCommerce 130

- Scenario 1: Inventory Management 132
- Scenario 2: Loyalty Platforms 133
- Scenario 3: Personalization 134
- Scenario 4: Live Commerce 135
- Scenario 5: Retail Media 136
- Real-Life Examples (Migros) 137
- Conclusion 141

3.4 Telecom and Media 142

- Scenario 1: Network Monitoring and Alerting 144
- Scenario 2: OTT Services 145
- Scenario 3: Advertisements 146
- Scenario 4: Network Security and Threat Detection 157
- Scenario 5: Fraud Detection and Prevention 148
- Scenario 6: Data Sharing With MVNOs and Partners 149
- Scenario 7: Open RAN Operations 150
- Scenario 8: Monetizing Network APIs 151
- Scenario 9: Live Event Analytics and Dynamic Ad Placement 152
- Scenario 10: Intelligent Media Production Pipelines 153
- Scenario 11: Digital Rights and Asset Management 154
- Scenario 12: Autonomous Network Operations 155
- Conclusion 156

3.5 Gaming 158

- Scenario 1: Infrastructure Operations 160
- Scenario 2: Game Telemetry 161

• Scenario 3: Monetization Network	162
• Scenario 4: Esports and Social Live Experiences	163
• Scenario 5: Anti-Cheat and Fair Play	164
• Scenario 6: Dynamic Content and Live Ops	165
• Conclusion	166
3.6 Government and Public Sector	168
• Scenario 1: Citizen Services	170
• Scenario 2: Smart Cities	171
• Scenario 3: National Security	172
• Scenario 4: Emergency Response and Disaster Management	173
• Scenario 5: Public Health and Pandemic Monitoring	174
• Scenario 6: Transportation and Public Infrastructure Operations	175
• Real-Life Examples (PNCC)	176
• Conclusion	180
3.7 Energy and Utilities	182
• Scenario 1: Smart Grid Monitoring	184
• Scenario 2: Industrial Control Systems	185
• Scenario 3: Energy Trading	186
• Scenario 4: Decarbonization	187
• Scenario 5: Virtual Power Plants and Distributed Energy Networks	188
• Scenario 6: Electric Vehicle Charging Networks	189
• Real-Life Examples (AMPEERS ENERGY, aedifion, Uniper)	190
• Conclusion	199
3.8 Transportation and Travel	200
• Scenario 1: Transport Management	202
• Scenario 2: On-Demand Mobility and Delivery Services	203
• Scenario 3: Travel Booking	204
• Scenario 4: Baggage and Cargo Tracking Services	205
• Scenario 5: Autonomous Fleet Operations	206
• Scenario 6: Border Control, Customs, and Tariff Management	207
• Real-Life Examples (Austrian Post, Deutsche Bahn, Flix, LKW Walter, Lufthansa, Swiss Post, Schiphol Group)	208
• Conclusion	225
3.9 Insurance	226
• Scenario 1: Workflow Orchestration	228
• Scenario 2: Full-Stack Contract Handling	229
• Scenario 3: Claims and Policy Fraud Prevention	230
• Scenario 4: Catastrophe Modeling and Risk Management	231
• Scenario 5: Embedded Insurance at Scale	232

• Scenario 6: Micro Insurance	233
• Real-Life Examples (Generali Switzerland)	234
• Conclusion	239
3.10 Healthcare and Life Sciences	240
• Scenario 1: Medical Diagnostics	242
• Scenario 2: Pharmaceutical R&D	243
• Scenario 3: Caregiving	244
• Scenario 4: Value-Based Care	245
• Scenario 5: Clinical Trials Monitoring	246
• Scenario 6: Hospital Operations and Patient Flow Optimization	247
• Real-Life Examples (City of Hope, Siemens Healthineers)	248
• Conclusion	255
3.11 Technology	256
• Scenario 1: Platforms and Mobile Apps	258
• Scenario 2: Infrastructure Monitoring	260
• Scenario 3: Digital Marketplaces	261
• Scenario 4: Community and Social Platforms	262
• Scenario 5: Subscription and Digital Experience Platforms	263
• Real-Life Examples (Character.AI, Wix, Grafana Labs)	264
• Conclusion	275

04 Creation and Positioning of a Data Streaming Organization and Community 276

4.1 The Journey to a Data Streaming Organization	278
4.2 Key Drivers for a Community Within a Data Streaming Organization	281
4.3 Fundamental Tactics for Developing a Data Streaming Community	284
4.4 Strategies for Long-Term Engagement in a Data Streaming Community	287

Guest
chapter by
Evi Schneider

05 Data Streaming Is a Journey: A Roadmap to Success 292

A Testament to the Value of Data Streaming

Foreword by Jay Kreps

As the original co-creator of Apache Kafka® and co-founder and CEO of Confluent, I've had the incredible opportunity to witness the evolution of data streaming—from its early days, when many viewed it as just a kind of scalable message broker, to its position today as a cornerstone of modern data architecture. Throughout this journey, one of my favorite parts of my job is getting to meet with users and customers who are actually putting the technology into practice. For me, it has been fascinating to watch how this technology found applications across virtually every industry, well beyond anything we originally imagined.

This book, *The Ultimate Data Streaming Guide 2nd Edition—Concepts, Use Cases, Industry Stories*, is a fantastic resource for anyone looking to dive deep into the world of data streaming. It captures not only the technology but also the stories behind its adoption and the value it delivers.

Apache Kafka, which started as a low-level infrastructure layer for data, has evolved into the basis for a rich and powerful ecosystem. Today, [more than 150,000 organizations](#) rely on Kafka, a testament to its robustness and versatility.

It's exciting to see the growing number of industry analysts who recognize data streaming as a distinct and vital software category in evaluations such as [Forrester Wave](#) and [IDC Marketscape](#). The recognition of data streaming as a new software category by research analysts underscores its significance. Traditional software players and emerging startups alike are joining the data streaming game, further validating its importance. For example, Confluent has reached a billion-dollar revenue run rate within just a decade—clear evidence that something is happening here.

"Apache Kafka, which started as a low-level infrastructure layer for data, has evolved into the basis for a rich and powerful ecosystem."

While Kafka remains central to enterprise architectures that demand decoupling, scalability, real-time capabilities, or

flexibility, we're seeing similar growth trajectories with Apache Flink® in the realm of stream processing. Flink is rapidly becoming the go-to standard for processing streaming data, often paired with Kafka to power cutting-edge applications that harness the latest in artificial intelligence (AI) and analytics to power real-time use cases. Together, these technologies are redefining what's possible in the data ecosystem.

Learning how to apply streaming can be a challenge. Although the technology has become far simpler to program to and cloud services have removed much of the operational burden, there is still a mindset shift in moving from batch to streaming architectures. That's where this book comes in. It dives into real-world use cases across industries, offering practical insights into what makes data streaming successful in practices in real applications. These applications are often harnessing data in new and creative ways and powering some of the most essential applications at the heart of modern businesses. By bringing together these examples, Kai has created a fantastic resource to help inspire and inform the rest of us how this technology can be applied in the real world. Whether you're an experienced technologist or just beginning your journey into data streaming, you'll find valuable insights and examples to guide you.

Jay Kreps

Co-founder and CEO, Confluent





The Real-Time Foundation for AI

Preface by Kai Waehner

In an AI-first world, the difference between success and irrelevance often comes down to one thing: real-time data.

No matter the industry, whether it's banking, telecom, healthcare, manufacturing, logistics, retail, or gaming, decisions are only as good as the data behind them. That data must be fresh, contextual, and available the moment it's needed. This is especially true for generative and agentic AI, which rely on continuous streams of high-quality information to deliver accurate and trustworthy outcomes.

Batch-driven architectures cannot support this. Outdated information leads to poor decisions, bad experiences, and increased risk. Agentic systems built on stale data, whether from yesterday's ETL job or last week's data lake refresh, simply fail. To work as intended, AI needs live data from operational systems—governed, consistent, and enriched in motion.

That's where the data streaming platform comes in. More than just Kafka or Flink, Confluent provides real-time infrastructure that powers event-driven architectures. It moves data continuously

and reliably across clouds, platforms, and applications. This bridges operational and analytical workloads in ways traditional systems cannot. Businesses no longer have to choose between fast data for operations and slow data for insights. With streaming, both are unified.

AI use cases benefit directly. Retrieval-augmented generation (RAG) pipelines, real-time analytics, autonomous decisioning, and personalization all depend on data in motion. Streaming agents can detect anomalies, react autonomously, and improve with every event—but only with a real-time data backbone in place.

Kafka provides the event log. Flink adds processing for transformation and enrichment. Apache Iceberg™ enables governed access to historical data. Together, they power systems that learn, adapt, and scale.

Not every AI agent will be built on Confluent Intelligence. Many organizations will use third-party tools or software-as-a-service (SaaS) platforms. But they all need accurate, up-to-date data from core systems. The data streaming platform acts as a context store—feeding the right data into any AI system at the right time, regardless of where it runs.

This book shows how organizations across industries are leaving behind batch-centric systems. It shares use cases from finance, manufacturing, retail, telecom, healthcare, energy,

and more. These stories highlight how data streaming supports AI, modernizes legacy systems, improves operations, and enables real-time responsiveness.

From fraud detection and instant payments to connected factories, personalized commerce, and smart grids, each example shows the value of streaming. Whether optimizing supply chains or supporting digital health, data streaming enables scalable, future-ready architectures.

"Data streaming is the central nervous system of the modern enterprise."

Confluent's fully managed platform makes open source standards available in enterprise-grade services. This lets teams focus on delivering business outcomes, not managing infrastructure. Real-time data streaming is no longer a niche capability. It's the central nervous system of the modern enterprise. For AI to scale, for insights to be trusted, and for systems to stay responsive, streaming must be at the core.

What matters most is not whether your organization uses streaming. It is how strategically you use it.

This book will show you how.

KAI WAEHNER

This book is structured to guide you through the world of data streaming:

- **Chapter 1:**
Event-Driven Architecture and the Demand for Real-Time Data
- **Chapter 2:**
Technical Data Streaming Use Cases
- **Chapter 3:**
Industry Success Stories for Data Streaming
- **Chapter 4:**
Creation of a Data Streaming Organization and Community
- **Chapter 5:**
Data Streaming Is a Journey: A Roadmap to Success

The structure of "The Ultimate Data Streaming Guide" is designed for easy access, whether you are responsible for data streaming platforms and services within your company or need to present specific examples to upper management. The book includes a matrix of technical data streaming use cases and vertical industry success stories, completed with customer examples. IT executives can leverage the book's insights to highlight business value findings.

For those reading the digital version, links are provided to seamlessly navigate from a case study in chapter 3 to the related technical use cases in chapter 2, ensuring a comprehensive understanding of how data streaming can drive success in your organization.

Data streaming is a journey, and by reading this book, you will learn about its transformative potential and how it can boost your business into the future while driving industry innovation. Through a maturity model, the book guides you from initial use cases and adoption across various lines of business, until data streaming, powered by an event-driven architecture, becomes the central nervous system of your organization.

Kai Waehner
Field CTO, Confluent

TAKE CONTROL OF YOUR DATA MONSTER

With  CONFLUENT

**Feedback? Questions?
Get in Touch!**

General Feedback or Questions:

✉ use-case-book@confluent.io

Kai Waehner

Data Streaming Trends,
Use Cases, Industry Stories:

 LinkedIn,  X (Twitter),
 Blog

Evi Schneider

Internal Community Creation,
Positioning of Data Streaming
Organization:  LinkedIn



Content Matrix

Industry Success Stories for Data Streaming	Financial Services	Manufacturing and Automotive	Retail and eCommerce	Telecom and Media	Gaming	Government and Public Sector	Energy and Utilities	Transportation and Travel	Insurance	Healthcare and Life Sciences	Technology
Data Streaming Use Cases											
Data Products	Erste Group Bank DNB ASA	Knauf Siemens	Migros			PNCC		Austrian Post Deutsche Bahn Flix		City of Hope	Character.AI
IT Modernization	Erste Group Bank KOR DNB ASA					PNCC		Lufthansa Schiphol Group Swiss Post Deutsche Bahn	Generali Switzerland		Character.AI Grafana Labs
Observability	KOR		Migros				aedifion	Austrian Post LKW Walter		City of Hope	Grafana Labs Wix
Hybrid and Multicloud	DNB ASA	BMW Group PAUL Tech AG Knauf Siemens				PNCC	AMPEERS ENERGY	Flix	Generali Switzerland	Siemens Healthineers	Grafana Labs Wix
Event-Driven Microservices	DNB ASA	Knauf Siemens	Migros				Uniper	Austrian Post LKW Walter Swiss Post Flix Deutsche Bahn		City of Hope	Wix
Data Sharing	Erste Group Bank	Knauf Siemens					Uniper	LKW Walter Lufthansa Schiphol Group Swiss Post			Character.AI Grafana Labs
Artificial Intelligence		BMW Group Siemens						Lufthansa		City of Hope Siemens Healthineers	Character.AI Wix
Internet of Things		BMW Group PAUL Tech AG Siemens	Migros			PNCC	AMPEERS ENERGY aedifion	Schiphol Group		City of Hope Siemens Healthineers	
Data Governance	Erste Group Bank KOR					PNCC			Generali Switzerland	Siemens Healthineers	Character.AI
Environmental, Social, and Governance		PAUL Tech AG					aedifion	Schiphol Group			

To be extended!
—
If you want to become
a part of it, don't forget
to reach out to
use-case-book@confluent.io



Event-Driven Architecture and the Demand for Real-Time Data

01

Frequent online shoppers are all too familiar with the following scenario:

You find a product, place your order, and complete the payment. With every click, the anticipation of receiving your new pair of pants, fancy espresso machine, or rowing ergometer grows—until you discover that the item is, unfortunately, no longer in stock because another fashion-lover or sportive buddy was quicker. It's frustrating.

The same disappointment happens with AI-driven experiences: The recommendation engine shows you the "perfect" shoes that cannot be purchased, an agent confirms delivery options that aren't available, or a digital shopping assistant reserves an item that turns out to be out of stock.

But is this just something we, as buyers, have to get used to because it's unavoidable? Are there simply "natural limits" to the availability of information that we must accept? Or are we—as private consumers and, even more so, as business users handling data every day—on the threshold of a paradigm shift in how we use data?

Let's be clear right from the start:

We absolutely don't have to accept delayed data as our fate. In fact, it could almost be stated as a universal rule:

Real-time data is better than slow or delayed data in nearly every application.

"Real-time data beats slow data or delayed data in nearly every application."

Accurate and up-to-date information empowers organizations to make informed decisions, enhance customer experiences, and drive innovation. However, unlocking data and analytics is much harder than it should be. In the current digital landscape, most enterprises across all industries are grappling with the limitations of fractured and siloed data architectures. These traditional systems, characterized by their reliance on traditional databases, periodic data processing, and point-to-point APIs, often fall short in meeting the demands of modern business operations. The inherent latency and complex integration across systems means that data is updated only at scheduled intervals, leading to significant delays in decision-making and operational responsiveness.

This lag can be particularly harmful in use cases where real-time data is crucial, such as in the financial sector, where the inability to access real-time data can hinder risk management and trading strategies, or in healthcare, where delayed data can impact patient care and treatment outcomes.

1.1 The Business Impact of a Data Mess

One of the primary challenges of a siloed data world is the inconsistency of information. As data is collected and processed in batches or through APIs, there is always a time gap between the occurrence of an event and its reflection in the system. This delay can result in outdated or stale data being used for critical business decisions and lead to suboptimal outcomes.

Most enterprise architectures store data across two distinct and different estates:

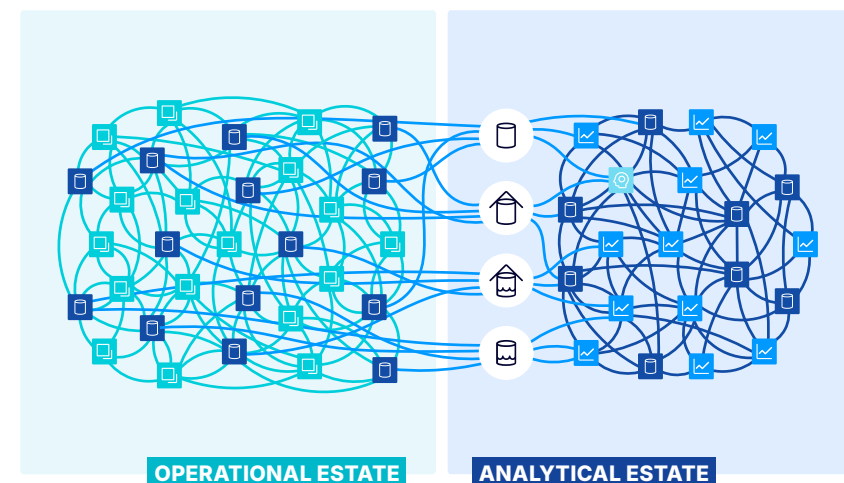
1 The **operational estate**, which serves the transactional and real-time systems that run the business

2 The **analytical estate** for after-the-fact analysis and business decision-making

Organizations typically employ a mish-mash of different approaches for data integration to connect the operational and analytical estates. Moreover, batch-driven architectures are often built on monolithic legacy systems that are inflexible and difficult to scale. These systems are typically tightly coupled. Any change in one component can have cascading effects on the entire system. This rigidity makes it challenging to adapt to new business requirements or integrate with modern technologies. For example, in the highly competitive world of ecommerce, the inability to quickly adapt to changing customer preferences or market trends can result in lost sales and decreased customer satisfaction.

The "data mess" is made worse by the increasing volume and velocity of data generated by modern enterprises. Traditional systems struggle to keep up with the sheer scale of data, leading to performance bottlenecks and inefficiencies. In industries like manufacturing, the Internet of Things (IoT) generates massive volumes of data from sensors and equipment. When this data can't be processed in real time, it disrupts critical use cases such as predictive maintenance and reduces overall operational efficiency.

This complexity leads to a so-called **"spaghetti architecture."**



Spaghetti architecture: Data complexity grows over time, creating tangled integrations and an unmanageable data landscape

Data streaming offers a transformative solution to these challenges. By enabling continuous data processing and immediate insights, data streaming ensures that businesses can respond quickly to emerging technical or business events. This capability is invaluable across all industries and provides a competitive edge in a data-driven world.

IT executives can ensure that their organizations remain agile, competitive, and positioned for future growth.

The shift from batch systems, databases, and APIs to data streaming is not just a technological upgrade but a strategic imperative to solve the giant data mess. Embracing data streaming can significantly enhance operational efficiency, reduce costs, and unlock new revenue streams. By investing in modern data architectures,

"By investing in modern data architectures, IT executives can ensure that their organizations remain agile, competitive, and positioned for future growth."

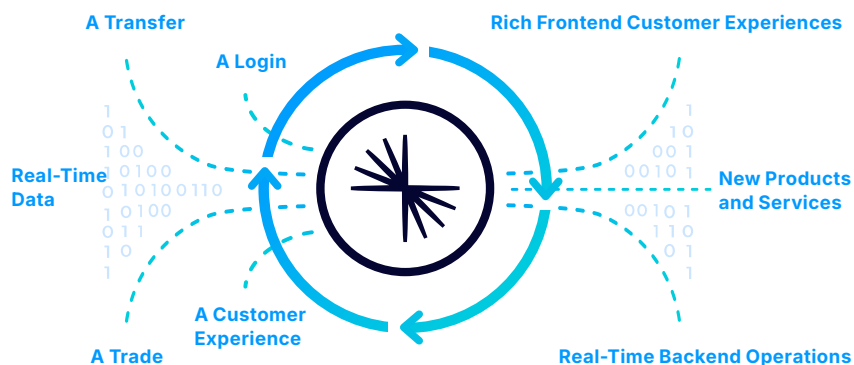
1.2 The Need for an Event-Driven Architecture

IT architectures are currently undergoing a significant evolution: the shift from tightly coupled point-to-point communication using databases and APIs to event-driven architectures. This marks a major change in how modern organizations design and manage their systems. While traditional point-to-point integrations may serve their purpose, they often lead to a web of dependencies that is difficult to manage and scale. In such systems, each application is directly connected to the others, creating a tightly coupled structure where changes to one application can affect the entire enterprise architecture. This inflexibility hinders innovation and makes it challenging to adapt quickly to new business needs and achieve rapid time to market.

In contrast, an event-driven architecture offers true decoupling of applications, enabling a more flexible and scalable system design. In an event-driven architecture, applications communicate through events, which are discrete pieces of information that signify a change in state. These events are published to a central event bus where they can be consumed by any interested consuming application.

This decoupling allows each application to:

- Operate independently
- Reduce dependencies
- Choose the right technology per use case
- Introduce new features or services faster and easier



The need for an event-driven architecture to set data in motion

One of the key benefits of an event-driven architecture is its alignment with domain-driven design. Domain-driven design is a software development approach that focuses on modeling software to match the business domain. By using events to represent business processes and changes, an event-driven architecture naturally supports the principles of domain-driven design, enabling a more intuitive and maintainable system design. This alignment ensures that the technical architecture closely mirrors the business processes, facilitating better communication and collaboration between technical and business stakeholders.

Furthermore, an event-driven architecture is a key component of a data mesh, a modern approach to data management that treats data as a product. A data product is a curated asset designed to deliver specific value or solve particular problems for stakeholders. Unlike raw data, a data product is structured, refined, and packaged in a way that it can be easily consumed, interpreted, and utilized by its intended audience. In a data mesh, data ownership is decentralized, with each domain responsible for its own data products.

Events play a crucial role in this model, as they serve as triggers of data sharing between the decoupled systems. Events are pre-defined changes in state and

"Events play a crucial role, as they serve as triggers of data sharing between the decoupled systems."

can be anything from a product being ordered to a user address being edited to other stimuli (like an alert issued by security systems).

In an event-driven architecture, events allow data products to be immediately available in the connected systems, making data sharing and integration across domains more seamless. This approach not only improves data quality and accessibility but also fosters a culture of data-driven decision-making in the organization.

The next step for event-driven systems is adopting the Kappa architecture. Instead of maintaining separate pipelines for batch and stream in a Lambda architecture, Kappa provides a single, event-based backbone where operational and analytical data flows converge. With the rise of generative AI, streaming agents, and autonomous decisioning, Kappa has become the de facto foundation to ensure consistent, high-quality, and low-latency data for both business processes and intelligent applications.

1.3 Data Streaming Platform

Data streaming is a fundamentally different paradigm, elevating data to a first-class citizen. Unlike traditional batch processing or point-to-point communication via APIs, data is not static or passive. It's constantly evolving and continuously being shared in real time. The fundamental notion is that data is moving, not sitting at rest waiting to be queried or extracted. Data streaming enables the continuous flow of data throughout a business.

This capability is particularly valuable in today's fast-paced business environment, where timely information can be a critical differentiator.

How does data streaming help solve the giant data mess and get to a central nervous system? It shifts your thinking from tightly coupled point-to-point integrations to a multi-subscriber, decoupled architecture leveraging data products.

Instead of repeatedly wrangling and cleansing data downstream, data quality and curation are shifted upstream.

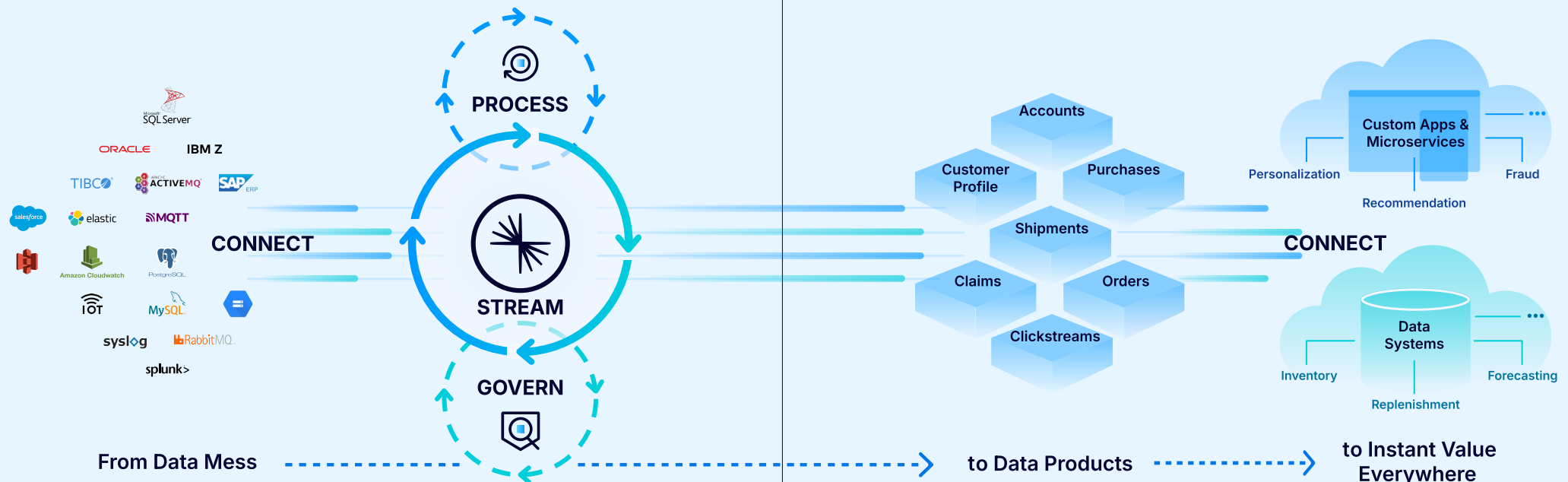
This allows domain experts to present ready-to-consume data that is governed, trustworthy, and reusable, enabling consumers to instantly apply those datasets to their use cases.

Instead of batch-based integration models like ETL or ELT, business units focus on a continuous flow of integrated

data that is always up to date. And finally, instead of querying data that's fixed in time, processing, analyzing, and acting on data happens continuously.

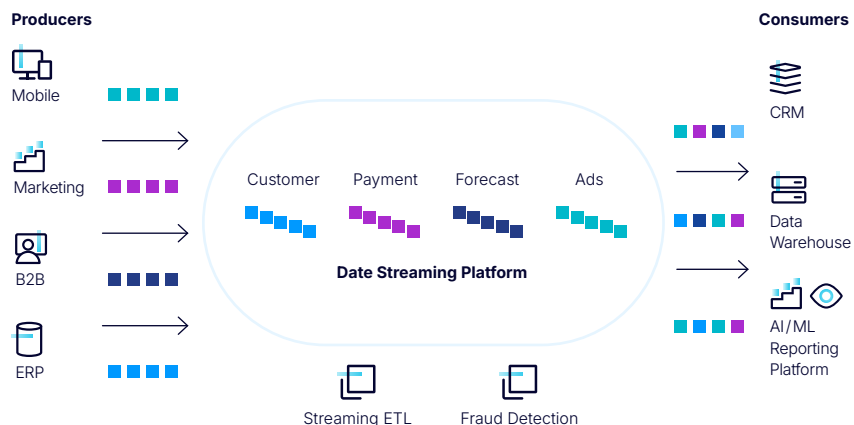
A data streaming platform powered by an event-driven architecture enables reusable, real-time, and trust worthy universal data products.

Data streaming enables teams to package and serve their data as if it were a ready-to-consume product—one that can be put to immediate use.



A data streaming platform powered by an event-driven architecture to enable reusable, real-time, and trustworthy universal data products

Event-Driven Architecture With Data Streaming



Event-driven data flow connecting producers and consumers in real time

This approach enables teams to package and serve their data as if it were a ready-to-consume product—one that can be put to immediate use.

Hence, companies rely on data streaming as the foundation for their data product strategies.

Data products should be reusable assets that represent key business entities—such as customers, accounts, claims, inventory, and shipments. And when done right, they are constantly up to date (real time) and continuously governed as well as reusable across both operational and analytical estates. Most data streaming infrastructures are powered by the open source technologies Apache Kafka and Apache Flink®. Apache Kafka is a distributed data streaming platform that enables the

real-time collection, storage, and processing of data. It's widely adopted across industries.

More than 100,000 organizations leverage Kafka's capabilities to build scalable and resilient data pipelines. The robust architecture and high throughput make it an ideal choice for both handling large volumes of data and critical transactional workloads with low latency. The persistence layer enables data consistency across real-time and non-real-time systems such as databases, data lakes, file systems, and request-response APIs.

The Kafka protocol has become the de facto standard for event streaming, powering a vast ecosystem of compatible tools, connectors, and services.

Apache Flink®, is a stream processing framework that complements Kafka by providing powerful capabilities for continuous processing of operational and analytical workloads. Flink's advanced features, such as stateful processing and complex event processing (CEP) enable real-time analytics at extreme scale. Together, Kafka and Flink form a powerful combination for building end-to-end data streaming solutions. While Kafka and Flink provide the foundational technologies for data streaming, Confluent takes it a step further by offering a complete data streaming platform. Confluent's platform extends the capabilities of Kafka and Flink with additional features and tools that simplify the development, deployment, management, and operations of data streaming pipelines and applications.

Connectors provide connectivity to legacy systems like a mainframe and also provide the latest technologies and cloud services like a lakehouse or an agentic AI SaaS service. If no connector exists, several options are available: integrating via APIs, clients for all major programming languages, implementing a custom connector, or leveraging a connector developed by a third-party vendor. Data governance is a critical add-on in a data streaming platform to ensure data quality, compliance, and security. It provides real-time data lineage and traceability and a metadata catalog for self-service consumption of existing data products to enable immediate actionability and consistent policy enforcement across distributed environments. By standardizing data

formats and protocols, data governance facilitates seamless integration and supports strategic business initiatives, fostering trust and collaboration among stakeholders that use the same data product in different business units or projects.

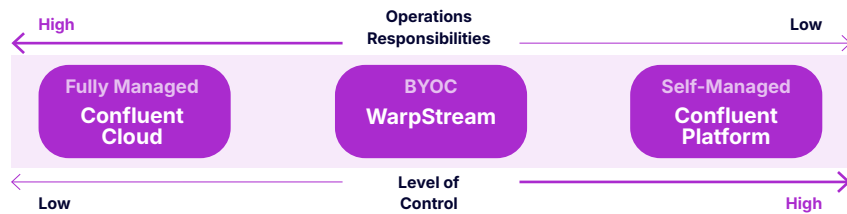
Data Streaming Everywhere: Deployments Options

Confluent Cloud, a fully managed service, provides a cloud-native platform that runs on all major cloud providers. This flexibility allows organizations to leverage the power of data streaming without the operational overhead of managing infrastructure.

"Confluent Cloud delivers a managed data streaming service that operates natively on every major cloud."

For organizations with the requirement to self-manage (usually because of cost, security, or latency reasons), Confluent Platform offers a comprehensive solution that can be deployed in data centers and public cloud customer virtual private clouds (VPCs), or even at the edge outside of a data center, such as in factories or retail stores. This versatility ensures that businesses can harness the benefits of data streaming regardless of their deployment environments.

Deployment Options: Fully Managed Cloud, BYOC, Self-Managed



Flexible deployment alternatives with the right trade-off between control, responsibility, and operational efficiency

As a third option in the Confluent product portfolio, WarpStream offers a cloud service using the Kafka protocol, which provides organizations with the flexibility of a Bring Your Own Cloud (BYOC) model. This approach allows businesses to maintain control over their data by deploying WarpStream in their own cloud environments or data centers while benefiting from the managed service's operational support. WarpStream's BYOC model empowers organizations to optimize their data streaming strategies according to their unique security and compliance requirements: It strikes a balance between the advantages of a managed service with some of the data isolation controls that come from using one's own infrastructure. In this zero access model, Confluent operates the service without ever having direct access to the customer's data, ensuring maximum privacy and control. WarpStream also uses a diskless Kafka architecture, storing all event data directly in object storage instead of local broker disks. This eliminates the cost and complexity of managing broker storage, improves scalability, and reduces cross-zone networking fees while maintaining full Kafka protocol compatibility. With built-in guarantees for zero data loss

and optional synchronous replication across regions, WarpStream ensures robust disaster recovery and business continuity. These capabilities make it suitable not only for high volume analytical workloads but also for mission-critical use cases where absolute reliability is essential.

The adoption of a data streaming platform is a strategic investment in the future of an organization's data architecture. By enabling real-time data products, companies are equipped to increase operational efficiency, enhance customer experiences, and unlock new revenue streams. The ability to deploy data streaming solutions across multicloud, on-premises, and edge environments ensures that they can remain agile and responsive to changing business needs and compliance requirements. Streaming and processing events is the core of a data streaming platform. Additionally, connectors and API integration play a crucial role, serving as the bridge between various data sources and sinks. They facilitate the seamless integration of disparate systems, enabling the flow of data across different environments.

1.4 Business Value of Data Streaming

The business value of data streaming goes far beyond its technical capabilities. At its core, data streaming transforms how organizations operate and compete. Instead of waiting for delayed insights from batch processes, businesses can react to events the moment they happen. This shift—from static data at rest to data in motion—creates new opportunities to generate revenue, reduce costs, and manage risks more effectively.

Driving Revenue Growth

Revenue impact is often the most visible outcome of streaming. By turning every customer interaction or market signal into actionable insight in real time, organizations can tailor experiences, optimize pricing, and capture demand that would otherwise be missed. For example, retailers use streaming to deliver personalized offers the moment a customer browses online or walks into a store. Financial institutions rely on streaming to detect and execute trading opportunities in milliseconds. The common thread is speed: Revenue is maximized, not delayed, when insights are immediate.

Reducing Costs and Improving Efficiency

Data streaming also unlocks significant cost savings by improving operational efficiency. Continuous monitoring of assets, processes, and services allows businesses to prevent problems rather than react to them. In manufacturing, predictive maintenance powered by streaming prevents unplanned downtime, cutting both repair costs and lost production. In healthcare, streaming data flows reduce manual administrative tasks and accelerate clinical decision-making, lowering costs while improving patient outcomes. Efficiency gains result from removing latency from processes across the organization.

Mitigating Risks and Strengthening Trust

Perhaps most critical, streaming reduces risk in ways batch systems cannot. Threats in cybersecurity, fraud, or compliance don't wait for batch windows. They require instant recognition and response. Real-time fraud detection prevents financial losses before they occur. Streaming-enabled cybersecurity tools can stop attacks before they escalate. In industries like banking, insurance, and healthcare, this is not just about protection; it's about building trust with customers, regulators, and partners.

Beyond revenue, cost, and risk, data streaming provides a foundation for agility. Organizations that can continuously integrate and act on data across systems are better positioned to innovate and respond to change. Whether an organization is launching a new product, scaling a digital service, or embedding AI into operations, streaming makes it possible to adapt at the speed of business. In today's environment, this agility is itself a source of competitive advantage.

Ultimately, the business value of data streaming lies in its ability to align technology with outcomes that executives care about: growth, efficiency, resilience, and agility. By clearly linking real-time data capabilities to these strategic priorities, project teams can secure executive buy-in and demonstrate measurable impact. In this sense, data streaming is not just an IT initiative; it's a catalyst for business transformation.

Stage	Yes	No but expecting	Not expecting
STAGE 5	63%	33%	4%
STAGE 4	62%	31%	7%
STAGE 3	51%	37%	10%
STAGE 2	49%	38%	11%
STAGE 1	43%	39%	15%

Legend: ● Yes ● No but expecting ● Not expecting

Insights from the Confluent Data Streaming Report 2025

Business Value

- Strategic Driver**
 - Increase Revenue (make money)**
 - Business Use Case**
 - Core Business Platform
 - Services
 - Decrease Costs (save money)**
 - Business Use Case**
 - Operational Efficiency
 - IT Modernization
 - Mitigate Risk (protect money)**
 - Business Use Case**
 - Fraud Prevention
 - Compliance
- Example Scenario**
 - Accelerate Time to Market**
 - Core Business Processes
 - Customer 360
 - Push Notifications
 - Future-Proof Architecture**
 - Legacy Offloading
 - Hybrid Cloud Integration
 - Middleware Replacement
 - Improve Customer Experience (CX)**
 - Cybersecurity
 - Fraud Detection
 - Regulatory
- Technical Use Case**
 - Data Pipelines
 - Data Products
 - Data Sharing
 - Event-Driven Microservices
 - Observability
 - AI/ML
 - IoT

Strategic drivers for data streaming initiatives and related use cases and scenarios

2025 Data Streaming Report

The Data Streaming Report 2025 shows how IT leaders confirm the business value of data streaming in their organizations.



LEVEL 2

74%

of those in Level 2 say they're achieving or anticipating 2x to 5x ROI compared to 67% in 2024—proving even companies at earlier stages in the maturity curve see excellent ROI results with data streaming.

Percentage of IT leaders expecting investment growth in these areas over the next two years:



DATA STREAMING TECHNOLOGIES

68%

GENERATIVE AI

74%

Executive Summary

In the age of AI, real-time data reigns supreme.

In essence, it's the lifeblood that fuels AI. And enabling easy, continuous access to high-quality data—at the right time and in the right format—is pivotal to driving success with AI.

Findings from our 2025 Data Streaming Report: *Moving the Needle on AI Adoption, Speed to Market, and ROI*—where we tap into insights from 4,175 IT leaders—underpin how [data streaming platforms](#) (DSPs) are foundational to this effort.

89%

of IT leaders see DSPs easing AI adoption by helping them directly address hurdles in the areas of data access, quality assurance, and governance.

4,175

IT leaders providing insights.

86%

IT leaders prioritizing investments in data streaming.

84% of IT leaders say data products enable more confident data sharing across business units.

83% say data products enable more robust risk management.

76% say they help foster a culture where data is viewed as a shared enterprise asset.

EXECUTIVE SUMMARY

DSPs drive significant advantages by helping businesses:

91%

break down data silos

84%

achieve faster time to market

84%

increase revenue

82%

enhance customer satisfaction

Overall, DSPs have emerged as essential catalysts for innovation, helping businesses unlock the full value of data.

These kinds of benefits underpin a continued commitment to DSPs. While 86% of IT leaders continue to prioritize investments in data streaming, 64% cite higher investments in DSPs in particular for 2025.

report.confluent.io

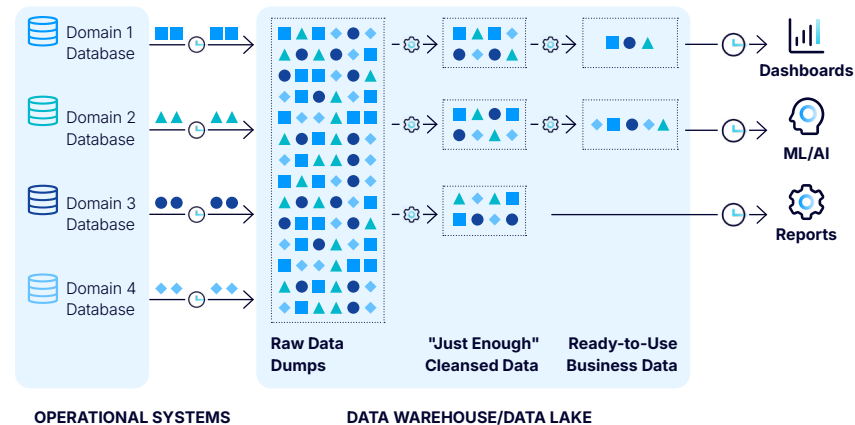
5

1.5 Analytic Data and AI Platforms

The landscape of data platforms has evolved significantly over the years, with each type of platform offering unique capabilities and benefits. Traditional data platforms primarily focus on building data pipelines that store and process data at rest. Data pipelines are the series of processes that move data from one system to another. The data is moved from sources (such as databases, IoT devices, or APIs), through an ETL process where the data is extracted, cleaned, transformed, and enriched before delivering it to data sinks (such as critical applications or analytical data

platforms). Alternatively, ELT first extracts and loads data into a target system before it's transformed within that system. Among those traditional data platforms are data warehouses, data lakes, and AI platforms. Data warehouses are designed for structured data and are optimized for complex queries and reporting. They are commonly used for business intelligence and analytics, providing historical insights based on large volumes of data. Data lakes, on the other hand, are designed to handle both structured and unstructured data.

ELT With Batch Workloads, Inconsistent Data, and High Compute Cost



A conventional ELT architecture using a lakehouse

They offer greater flexibility in terms of data storage and processing, making them suitable for big data analytics and machine learning (ML) applications. However, data lakes can become unwieldy and difficult to manage if not properly governed. When metadata, lineage, and data quality controls are lacking, a data lake can degrade into a data swamp: a repository filled with disorganized, incomplete, or outdated data that's hard to find, trust, or use effectively.

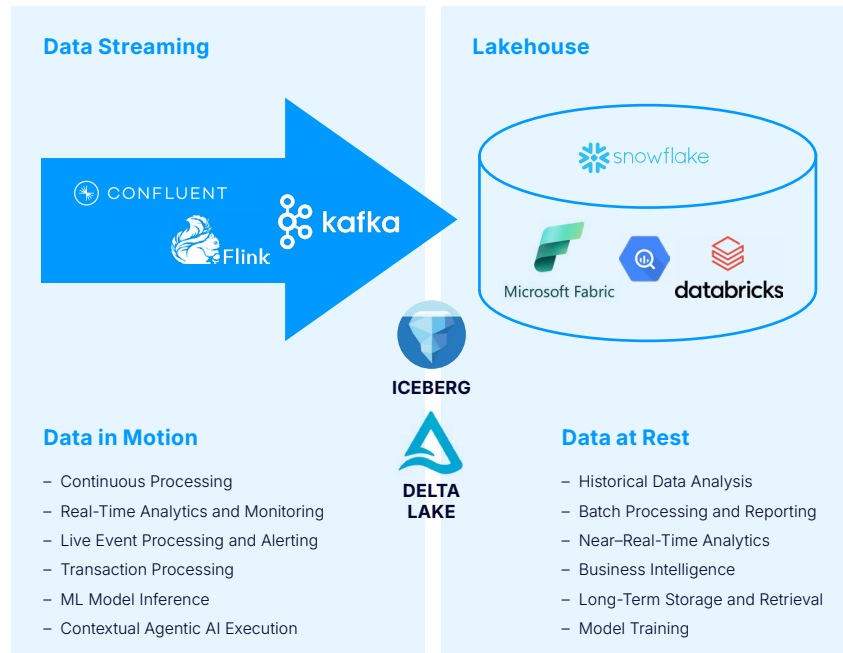
With the lakehouse, a hybrid concept has emerged that aims to combine the best of both data warehouses and data lakes. A lakehouse provides a unified platform to build data pipelines for all types of data to enable both analytical and transactional workloads. This approach simplifies data management and improves data accessibility, making it easier for organizations to derive insights from their data. The conventional data processing of data lakes and warehouses often leads to batch workloads, inconsistent data, and high compute costs because of repeated data processing. While these traditional data platforms focus on data at rest for analytical use cases, data streaming introduces the capability to process data in motion. The widely adopted approach of feeding data from the lakehouse back into operational systems using reverse ETL is considered an anti-pattern.

It complicates data architecture by moving data back from analytical systems to operational systems, leading to increased complexity, potential data inconsistencies, and higher maintenance costs.

In recent years, AI platforms have emerged as a critical addition to the data ecosystem. They provide the infrastructure to build, train, and deploy predictive, generative, and agentic AI models at scale. They often include tools for data preparation, feature engineering, experiment tracking, model management, and deployment pipelines. AI platforms can exist as stand-alone frameworks or SaaS offerings, or they can be embedded as part of broader analytical platforms such as a lakehouses.

The boundaries between AI platforms, data warehouses, and data lakes are increasingly blurred. For example, a lakehouse may include integrated machine learning capabilities, while an AI platform may provide its own storage and processing engine to support end-to-end workflows. This overlap offers flexibility but also adds complexity, as organizations must decide whether to adopt a dedicated AI platform or leverage embedded capabilities within their existing data infrastructures.

Data Streaming vs Lakehouse



Data streaming and lakehouse are complementary, not competing, in the enterprise architecture

A data streaming platform enables continuous data pipelines to provide immediate insights and enable timely decision-making for operational and analytical applications and platforms. This capability is particularly valuable in scenarios where the speed of data processing is critical, such as real-time fraud detection, predictive maintenance, and personalized marketing. Additionally, a data streaming platform provides the required service level agreements (SLAs) for uptime, availability, and transactional guarantees to support both. Understanding the interplay between different data platforms is crucial for

designing a robust, scalable, and cost-efficient data architecture. By combining the capabilities of data warehouses, data lakes, lakehouses, and data streaming, enterprises can create a unified data ecosystem that supports both analytical and operational workloads. This holistic approach ensures that organizations can derive maximum value from their data, driving innovation and competitive advantage. In this ecosystem, data streaming becomes the foundation for AI, particularly for operational AI workloads such as streaming agents, by ensuring that intelligent systems act on the most timely and reliable information.

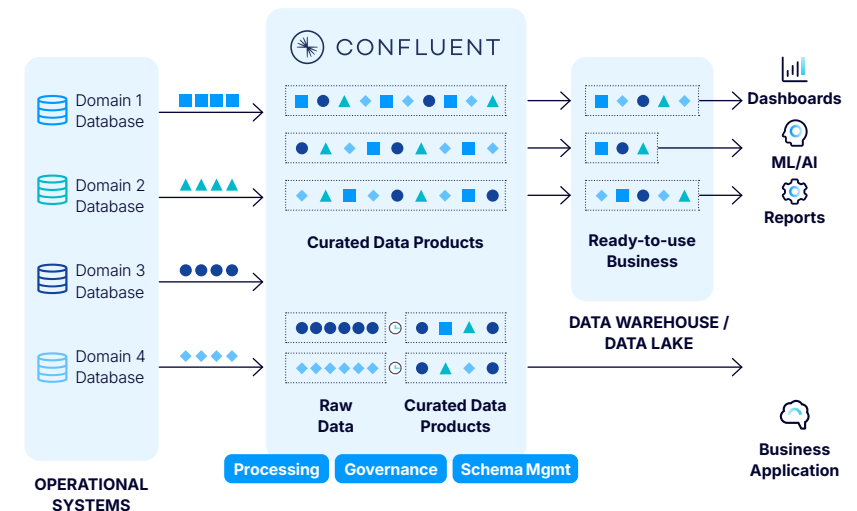
1.6 The Shift-Left Architecture

The shift-left architecture brings data processing, quality checks, and governance earlier into the data lifecycle. This means validating and shaping data closer to the point of origin, whether at the edge, in operational systems, or within integration pipelines, instead of waiting until data lands in storage or analytics platforms.

By moving these tasks upstream, organizations detect issues earlier, reduce reprocessing costs, and enable faster decisions. Shifting left

also improves data quality from the start, creating reliable inputs for both operations and analytics.

This architectural style separates data storage from processing and consumption. It prioritizes real-time action over delayed insights. Streaming data platforms are a key enabler, allowing continuous processing, transformation, and validation as events arrive.



The shift-left architecture for real-time data, better data quality, and reduced compute cost

The result is reusable, high-quality data products. These products, standardized and governed, can serve multiple purposes across the organization. Operations, analytics, and AI workloads all benefit from the same consistent, real-time information.

A shift-left mindset spans the full data journey. Data flows from edge systems like sensors and devices into local or cloud platforms for immediate processing. It is then routed to multiple targets such as cloud applications, AI models, and data lakes without losing quality or increasing latency.

Filtering and transforming data early reduces noise, lowers storage and compute costs, and simplifies downstream processing. This approach supports both agility and governance at scale, helping organizations stay responsive while keeping control.

"Shift-left means fixing data closer to the source for faster insights, lower costs, and better decisions."

1.7 Data Streaming vs Lakehouse

Integrating data streaming with traditional analytic platforms such as data warehouses, data lakes, lakehouses, or AI platforms has always been complex.

Kafka produces continuous events, often in small files and with evolving schemas, while lakehouses expect large, structured datasets for analytical queries. Traditional connectors leave teams managing fragile pipelines, duplicating data, and spending heavily on file compaction and

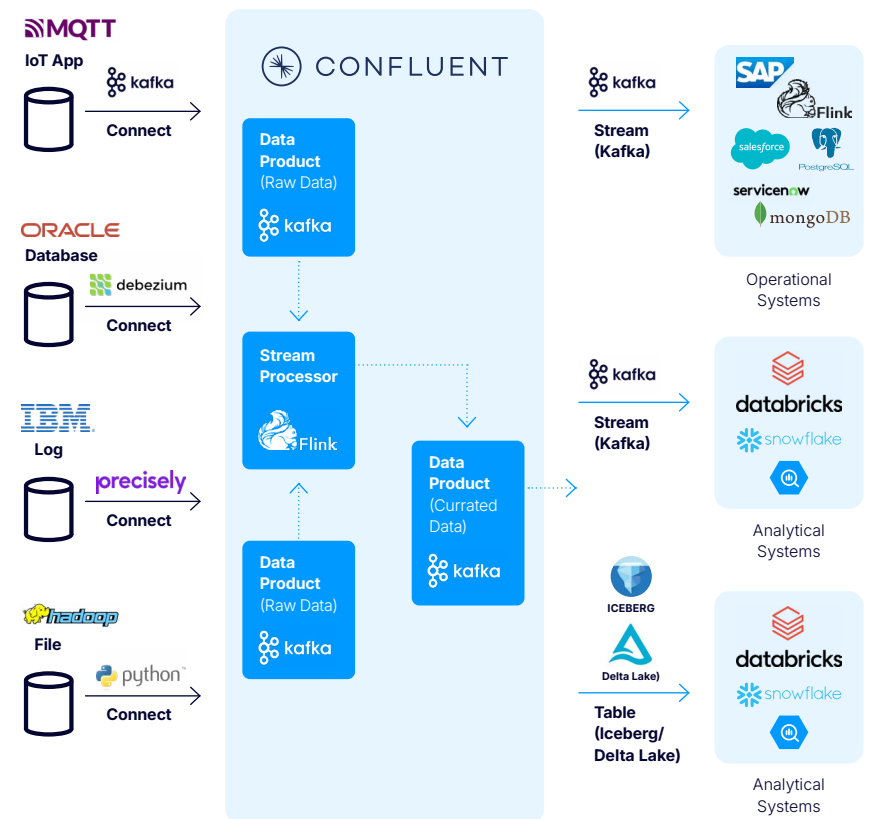
schema handling. Open table formats such as Apache Iceberg and Delta Lake bring reliable table structures to cloud object storage. They make it possible to store data once and use it for both operational and analytical workloads. But the real challenge is getting streams of events into these formats in a clean, consistent, and cost-efficient way.

Confluent's Tableflow solves this by representing Kafka topics directly as Iceberg or Delta Lake tables without custom pipelines or ETL jobs. It handles

schema mapping, file compaction, and catalog synchronization automatically, so the same data is instantly available for streaming applications, analytical queries, and AI models. Flink stream processing can further enrich and prepare the data before it reaches the table, ensuring good data quality from the start.

The result is a unified, governed data layer where operational speed and analytical depth coexist across the data streaming platform, lakehouse, and operational systems. Users can consume the data in the form that best fits their needs, without worrying about how it was moved or transformed.

Shift-Left Architecture With Confluent, Databricks and Delta Lake



Example of preprocessing of data before pushing into the operational and analytical systems

Data Streaming Use Cases

02

Each organization begins its data streaming journey in its own way. The initial use cases for adopting data streaming are as varied as the industries they appear in. However, the core technical use cases often remain the same, showing up differently across business contexts depending on the industry. This chapter highlights the most common horizontal technical use cases seen over recent years which is relevant to both those just starting and those already progressing in their data streaming adoption.

The following 10 use cases don't cover all of data streaming's technical capabilities, but they aim to provide a clear overview of the options available for starting and expanding the use of a data streaming platform.

2.1 Data Products

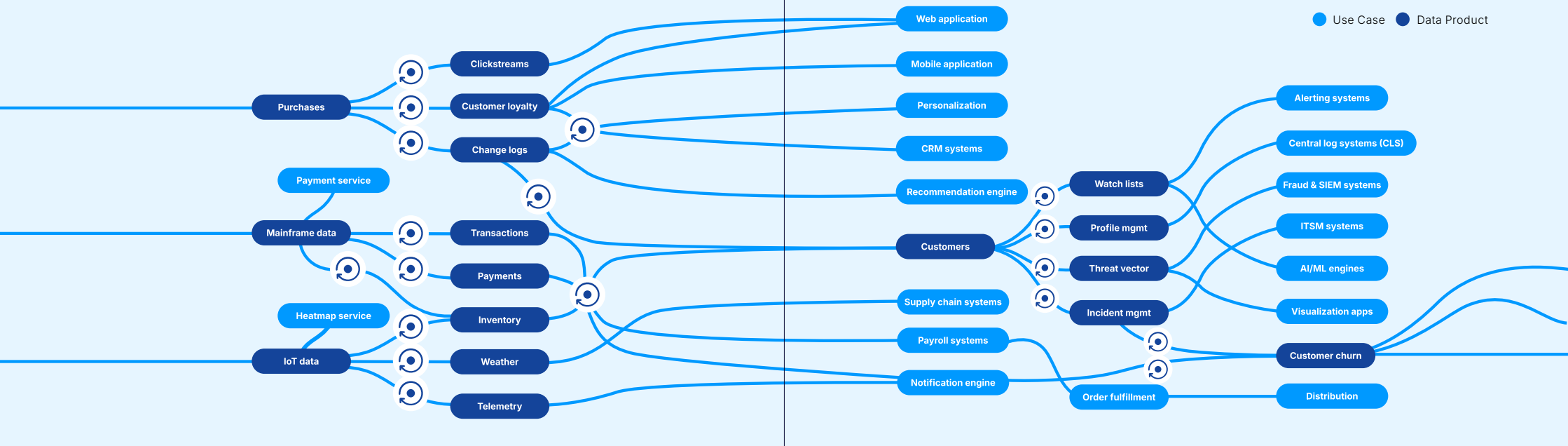
While the term "data products" may appear too technical, perhaps even abstract, we find ourselves surrounded by them. We grumble about the recommendation engines for our product portfolios or proudly show off our health data on smartwatches. We also rely on GPS as decision support on our daily commutes. Data products are an essential cornerstone for a multitude of applications in our daily routines.

Data products can take many forms, but at their core, they're curated, reusable datasets created from data pipelines that provide specific, valuable insights or functionality to applications. Data products deliver targeted, actionable insights that allow data-driven integration, decision-making, and innovation, while data mesh enhances business agility and scalability within the organization. Many organizations are espousing the benefit of data products. While core systems of record, such as consumer resource management (CRM) for customer data or enterprise

resource planning (ERP) for operational data remain the primary sources, curated data products make it far easier to share, reuse, and combine that information across an organization.

According to McKinsey and Harvard Business Review, the benefits of the data product approach can be significant:

- **New business use cases** can be delivered as much as 90% faster.
- **The total cost of ownership**, including technology, development, and maintenance, can decline by 30%.
- **The risk and data governance** burden can be reduced.



Relation of Data Products to Data Streaming

In an event-driven architecture, data is constantly ingested, processed, and delivered across the data pipeline to downstream consumers to ensure that information is always up to date. This real-time capability is essential for maintaining high data quality and consistency across various applications, whether the producers and/or consumers are real time, batch, or API-based. Data streaming improves the quality of data for all downstream consumers by providing a consistent and reliable flow of information. This is achieved through the continuous ingestion and processing of data, which ensures that any changes or updates are immediately reflected across the entire data pipeline. By integrating data sources and sinks in real time, the data

streaming platform enables seamless data flow and reduces the risk of data inconsistencies.

Examples for Data Products and Data Streaming

Example 1: Real-time inventory and sales management in retail

Imagine a retail company that uses data streaming to manage its inventory and sales data. By leveraging data streaming, the company can continuously ingest data from various sources such as point-of-sale systems, online stores, and supplier databases. This real-time data is then processed and enriched to create data products like inventory reports and sales forecasts, which are consumed by different departments within the organization.

Example 2: Fraud detection in financial services

Consider a financial institution that uses data streaming to monitor transactions and detect fraudulent activities. By integrating data from multiple sources such as transaction logs, customer profiles, and risk models, the institution can create a comprehensive, real-time view of each transaction in a data product. This data is then used to identify suspicious patterns and trigger alerts, enabling the institution to respond quickly and mitigate risks.

Data Products Build With Data Streaming for Data Consistency and Continuous Processing

Since data products are usually consumed by multiple downstream applications, data quality and

Reuse of data products across multiple use cases

governance are even more critical for compliance and observability. Depending on their use case, they also need to be kept up-to-date all the time.

Think of a global supply chain system that tracks inventory across warehouses, retailers, and distribution centers. You wouldn't want it to keep showing available stock for a product that was sold out minutes ago, would you? All these requirements are best met by adopting an event-driven architecture along with data streaming to ensure that data products are always accurate, consistent, and readily available to all business units. That way, organizations can make the most of their data by enhancing data quality, improving operational efficiency, and enabling real-time decision-making across the data pipeline.

2.2 IT Modernization

IT modernization involves updating and transforming an organization's IT infrastructure, applications, and processes to leverage modern technologies and practices.

This process often includes migrating legacy systems to the cloud, adopting microservices, introducing cutting-edge AI platforms, and implementing advanced data architectures. Data integration plays a crucial role in IT modernization by enabling seamless connectivity and data consistency between old and new systems to ensure that data flows smoothly across various platforms, whether unidirectional or bidirectional.

Data consistency is essential in this context, as it ensures that data remains accurate, reliable, and synchronized across the modernized IT landscape to support informed decision-making and operational efficiency no matter what technology, programming language, API, or communication paradigm each system uses.

Relation of IT Modernization to Data Streaming

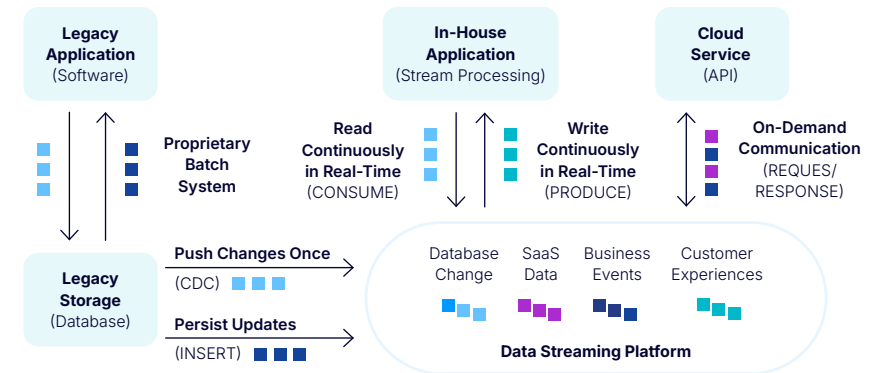
By providing an event-driven architecture with real-time data integration and processing capabilities, data streaming supports the transition from legacy systems to modern, event-driven applications. This shift is crucial to

maintain data consistency and ensure that information is always up to date and accurate. In a hybrid cloud environment, data streaming platforms provide seamless integration and synchronization between on-premises systems and cloud-native services. This capability is particularly important for organizations that need to modernize their IT infrastructures while maintaining compatibility with legacy technologies, such as mainframes, and integrating with modern applications, both self-developed and commercial products or SaaS.

Bidirectional Integration With Legacy and Cloud Systems

Modernization is not just about offloading data from legacy platforms; it's also about ensuring bidirectional synchronization between old and new environments. Data streaming enables real-time exchange with mainframes, transactional databases, and modern SaaS platforms through change data capture (CDC), message queues (MQ), or even lightweight flat-file ingestion when APIs are unavailable. This approach guarantees that legacy and cloud-native systems stay consistent, so updates in either direction are reflected without delay. By supporting a wide range of integration patterns, organizations avoid one-way silos and keep both historical and modern platforms aligned during the transition.

IT Modernization: Lift and Shift With Data Streaming



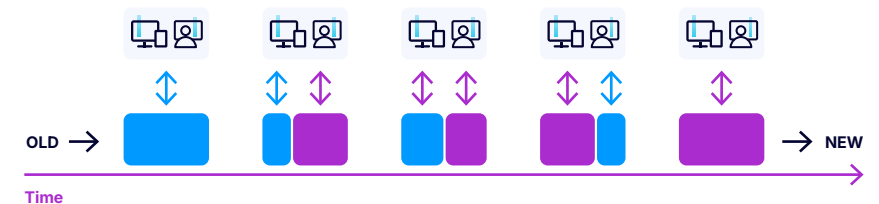
IT modernization through an event-driven architecture with decoupled applications and a variety of integration options

Gradual Rip and Replace of Legacy Systems

Replacing legacy systems does not need to be a risky "big bang" rewrite. A step-by-step modernization approach allows organizations to surround existing applications with new services that gradually take over functionality. When combined with an event-driven architecture, a data streaming platform becomes the backbone for synchronizing old and new components. This ensures consistent

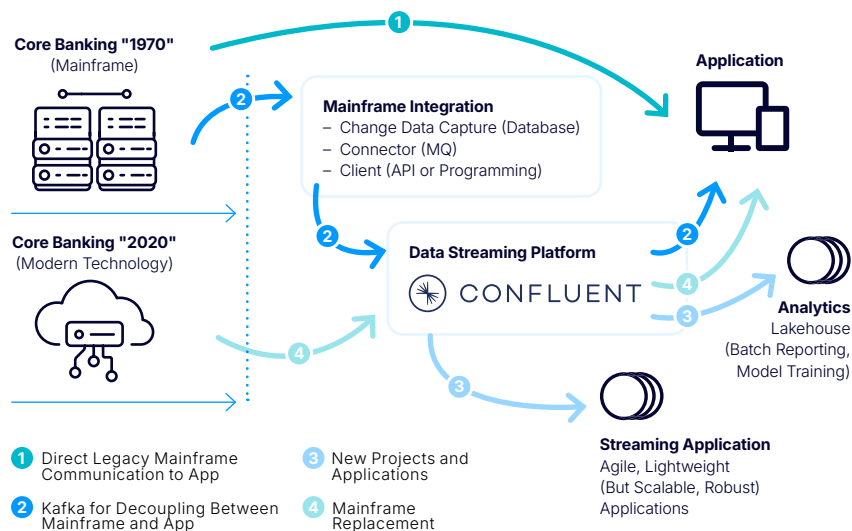
data across both worlds. Some systems can process data in real time, while others consume updates in near-real-time or batch. This flexibility enables legacy and modern systems to coexist smoothly until the older environment is fully retired. The incremental model minimizes disruption, reduces risk, and gives organizations the freedom to modernize at their own pace while still delivering immediate business value.

Strangler Fig Design Pattern: Integrate, Migrate, Replace



Gradually integrate streaming with legacy systems, migrate workloads and then replace the old architecture

IT Modernization Example: Mainframe Offloading and Replacement With Data Streaming



Transform legacy banking cores into agile, data-driven platforms with streaming-based modernization

Examples for IT Modernization With Data Streaming

Example 1: Real-time data integration in healthcare

Imagine a healthcare organization that is modernizing its IT infrastructure to improve patient care and operational efficiency. By leveraging data streaming, the organization can integrate data from various sources such as real-time electronic health records, near-real-time high-volume logs from medical devices, and transactional data from external databases in batch. The event-driven data integration ensures that healthcare providers have access to consistent and

up-to-date information, enabling better decision-making and improved patient outcomes.

Example 2: IT modernization in financial services

Consider a financial services company that is transitioning from a legacy mainframe system and monolithic middleware to decoupled, cloud-native applications. By using a data streaming platform, the company can continuously ingest and process data from its mainframe, ensuring that the information is always consistent and CDC. This real-time data integration enables the company to leverage agentic AI and

streaming agents for use cases such as automated fraud detection, personalized investment advice, and intelligent customer service assistants.

Example 3: Modernizing SIEM systems for security analytics

Traditional security information and even management (SIEM) solutions often struggle with the increasing volume and velocity of security log events. By leveraging data streaming, organizations can complement their SIEM solutions to preprocess data continuously and feed only relevant information into the SIEM. This approach helps handle backpressure, maintain supported throughput, and significantly reduce overall SIEM costs. Data streaming can ingest and process security logs and events from various sources, providing a unified and comprehensive view of the organization's security posture.

that data is always accurate and reliable.

The adoption of a data streaming platform is a strategic investment in the future of an organization's IT infrastructure. Adopting a holistic approach to IT modernization through data streaming improves operational efficiency and reduces costs. It enables true decoupling of systems. This makes it easier to integrate legacy and modern technologies seamlessly. The flexibility empowers each business unit to adopt and modernize at its own pace, ensuring that technological advancements align with specific business needs and objectives. As a result, organizations can optimize resource allocation, minimize disruptions, and drive continuous innovation, ultimately delivering greater business value and competitive advantage.

It means nothing less than this: By leveraging the potential of data streaming, organizations position themselves to foster a culture of innovation and continuous improvement.

IT Modernization With Data Streaming for Seamless Integration to Ensure Data Accuracy

The modernization of IT infrastructure through data streaming based on event-driven architectures offers significant business value by improving the scalability, flexibility, and data consistency of data pipelines. By leveraging data streaming, organizations can seamlessly integrate and synchronize legacy systems with modern cloud-native services to ensure

2.3 Observability

As IT infrastructures grow more complex, observability is becoming increasingly important. In the software context, this refers primarily to the ability to monitor or measure the internal state of a system by examining its outputs, such as logs, metrics, and traces.

- **Log aggregation** centralizes logs from various system components into a single platform, allowing for unified analysis and troubleshooting by correlating events across the system.

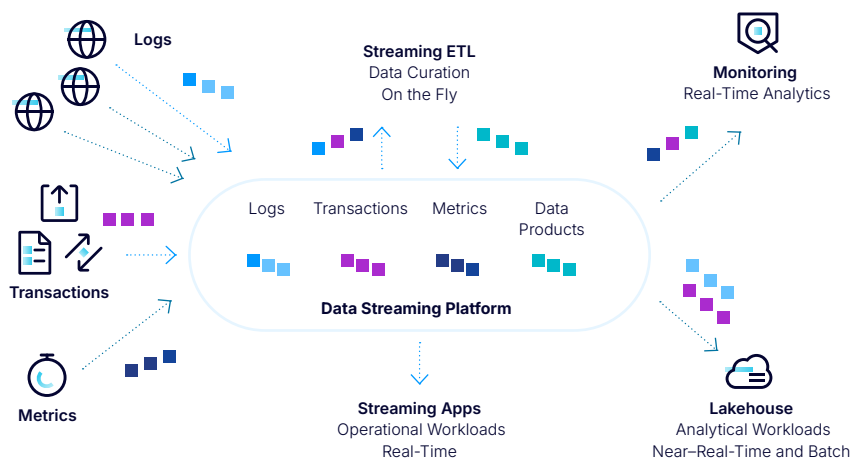
- **Metrics** involve the collection and monitoring of quantitative data points, such as CPU usage or request latency, providing real-time insights into the health and performance of a system.

- **Distributed tracing** tracks requests as they traverse through different services in a distributed system, offering a detailed view of the execution flow to identify performance issues and dependencies.

Observability enables developers and operators to understand, diagnose, and improve system performance and behavior in real time, which is crucial for maintaining reliable and efficient software systems.

Observability With Data Streaming:

Understand, Diagnose, and Act Continuously



Continuously understand, diagnose, and act with real-time visibility across all systems and data products

Relation of Observability to Data Streaming

A data streaming platform is a foundational element for modern observability efforts. In an event-driven architecture, logs, metrics, and traces are continuously ingested, processed, and analyzed in real time, enabling immediate insights and actions using stream processing. This real-time capability is essential for handling the high volume and high throughput data generated by modern systems.

Data streaming is complementary to batch-based log analytics tools and distributed tracing. A data streaming platform handles the real-time data pipeline to ensure that logs, metrics, and traces are processed and filtered before being ingested into an analytics platform for long-term storage and detailed analysis through indexing, searching, and visualizing the events.

Stream processing as part of the data streaming platform is best suited for time-sensitive and operational critical events because it provides fault-tolerant continuous processing in low latency, even at large scale with gigabytes-per-second throughput. Stream processing is also excellent for preprocessing data before it reaches downstream systems. This reduces storage and compute costs, improves data quality early in

the pipeline, and enables a shift-left approach to catching issues faster. For other use cases, the curated data products can be reused by simply connecting another data sink. The event-driven architecture enables reusability of data products, data pluggability with policy enforcement regarding the data contract, and ease of implementation with fast time to market.

"Stream processing enriches and refines observability data before it reaches downstream systems."

Examples for Observability With Data Streaming

Example 1: Real-time observability in technology operations

Consider a technology company that uses observability to monitor the performance and health of its systems. By integrating logs and metrics from various sources such as application servers, databases, and network devices, the company gains a comprehensive view of its infrastructure. This real-time data is then used to identify and resolve issues before they impact users, ensuring high availability and performance. A log analytics platform complements data streaming to index log data for long-term storage, dashboard analysis, and forensics.

Example 2: Clickstream analysis in ecommerce

Imagine an ecommerce company that uses clickstream analysis to understand user behavior and optimize its website and mobile app. By leveraging data streaming, the company can continuously ingest and process clickstream data from its website to provide real-time insights into user interactions. This information is then used to personalize the user experience, improve website performance, and increase conversion rates.

Example 3: Cybersecurity monitoring in financial services

In the context of cybersecurity, a financial institution might use data streaming to monitor and analyze log data for potential threats. By continuously ingesting and processing logs from firewalls, intrusion detection systems, and other security devices, the institution can detect and respond to threats in real time. This proactive approach helps mitigate risks and protect sensitive data.

Reliability, Security, and User Satisfaction With Data Streaming

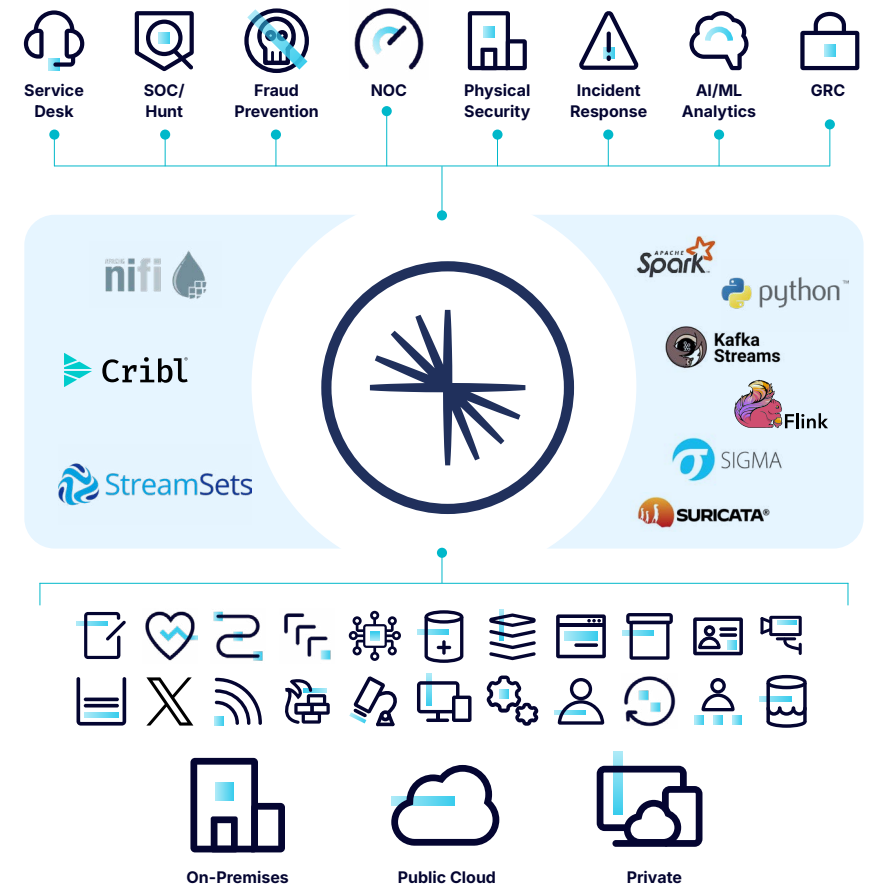
Security requires a data-centric approach in which source data is collected from any environment where it's created. It's published once, enriched into data products with the necessary context and metadata, and made available for any security function. Each team can then use the tools it prefers to work with that data.

Observability through data streaming provides significant business value by enabling real-time monitoring and analysis of system events, allowing organizations to detect and respond to issues instantly, minimizing downtime and improving operational efficiency. This approach enhances visibility across complex IT environments, supports proactive maintenance, and strengthens security. Results are better service reliability and user satisfaction by enabling rapid identification of anomalies or threats.

Streaming aggregation of events also enables real-time user behavior analysis by capturing and processing clickstream data instantaneously, allowing businesses to personalize user

experiences, track and trace business events across the supply chain, and respond swiftly to changing user preferences.

Data-Centric Security



Unify security, fraud, and operations data streams across cloud and on-premises environments for faster response

2.4 Hybrid and Multicloud

Data replication is a technique used to synchronize data across multiple regions and data centers. These can span edge, on-premises, hybrid, and multicloud environments in different geographic locations. This ensures that data is synchronized, allowing each data center or cloud region to have an up-to-date copy of the data, which is crucial for disaster recovery and high availability.

In an **active-active setup**, data replication allows all data centers to be operational simultaneously, providing access to data for users regardless of their locations. This setup enhances system reliability by enabling data continuity even if one data center or cloud provider experiences an outage.

Additionally, cross-data center replication supports load balancing to improve performance, resilience, and cost efficiency in global applications.

Relation of Hybrid and Multicloud to Data Streaming

While databases store data at rest and replicate in batch processes, data streaming provides the backbone for real-time data replication and synchronization across multiple data centers and clouds.

In an event-driven architecture, data is continuously ingested, processed, and replicated across multiple data centers

to ensure that information is always up to date and consistent. This real-time synchronization is crucial for maintaining data quality and enabling seamless integration between different environments.

Data streaming is often better for cross-data center replication than traditional databases or data lakes because it enables real-time, continuous replication of data in motion with lower latency and greater flexibility.

"Data streaming enables real-time, consistent replication across hybrid and multicloud environments."

A data streaming platform efficiently handles large volumes of data and can replicate not just static data but also dynamic, time-sensitive event streams to ensure that all data centers are always in sync. This approach is particularly convenient for applications that require immediate updates and real-time processing across distributed locations, where traditional database or data lake replication might struggle with latency or scalability.

The data streaming platform facilitates this unidirectional or bidirectional



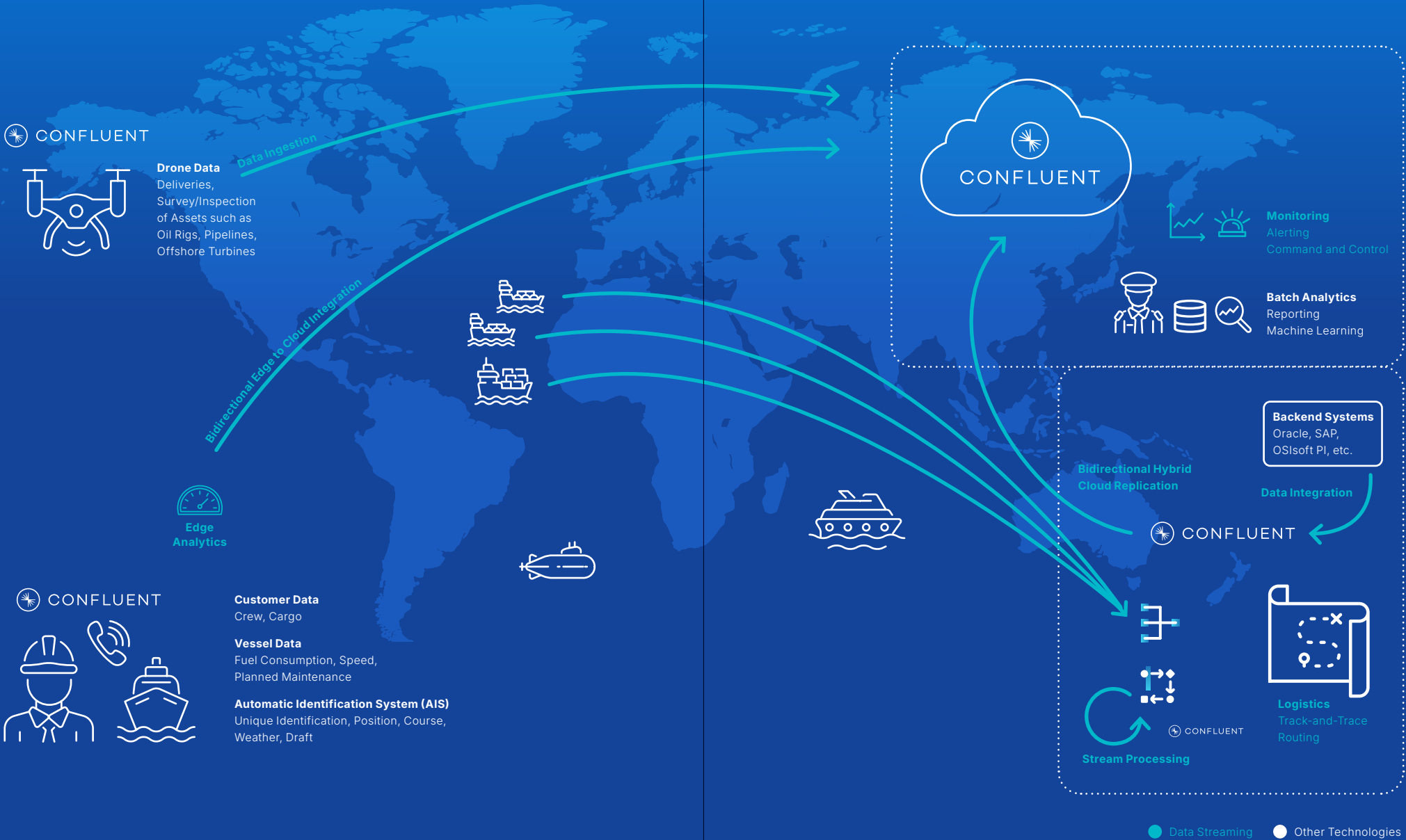
Replicate data in real time across regions, clouds, and data centers for maximum resiliency

replication and enables seamless data flow between on-premises, edge, and cloud environments by natively using the streaming technology to link clusters together for the real-time synchronization. No additional tools or infrastructure are needed. End-to-end monitoring and governance ensure that replicated data remains secure, compliant, and fully observable across all environments.

Stream processing further enhances data replication by enabling the preprocessing of data before replication. This includes filtering, aggregating, and anonymizing data, ensuring that the data replication cost is minimized and that only relevant and compliant data is replicated to other regions, clouds, or specific applications.

Hybrid Cloud Example: Logistics

Real-Time Operations, Predictive Maintenance, and Security Across the Supply Chain From Edge to Hybrid and Multicloud



Examples for Hybrid and Multicloud With Data Streaming

Example 1: Edge-to-cloud aggregation in manufacturing

Imagine an edge-to-cloud aggregation scenario in the manufacturing process across the logistics supply chain. By using data streaming, the factories and transportation vehicles can continuously ingest data from various sensors and equipment at the edge (e.g., temperature measurements), aggregate it every few minutes, and replicate only the relevant information to a cloud-based analytics platform. This enables real-time monitoring and analysis of supply chain operations for continuous optimization.

Example 2: Multicloud disaster recovery

In a multicloud disaster recovery scenario, a company might need to replicate data between two cloud providers to ensure business continuity even if a cloud vendor has an outage. By leveraging data streaming, the company can continuously replicate data between these cloud environments to ensure that information is always available and up to date, even in the event of failure of a cloud provider. Clients can fail over seamlessly to the secondary cloud without interruption.

Example 3: Migration from on-premises ERP to SaaS ERP

For a migration scenario, consider a company transitioning from a monolithic on-premises ERP system to a cloud-based elastic and flexible SaaS ERP. Data streaming helps the company to continuously replicate data between these systems to ensure a smooth and seamless migration process without time pressure. The final cutover (i.e., shutting down the on-premises ERP application) happens after all applications are migrated successfully.

Example 4: Global deployment

In a global deployment scenario, a software company might need to ensure regional data processing for worldwide availability of its application. By leveraging data streaming, the company can continuously replicate relevant data between different regions to ensure that information is always up to date and consistent, regardless of the user's location. Compliance requirements such as data privacy are addressed by curating data products so that only the necessary data is shared across regions rather than replicating all data without control.

Example 5: Synchronous multi-region replication with zero data loss

Some workloads can't tolerate the loss of even a single event. In these cases, synchronous multi-region replication ensures that every write is acknowledged only after it has been durably stored in multiple regions. This guarantees a recovery point objective (RPO) of zero, meaning no data is lost. It also enables automated failover if an entire region becomes unavailable. Typical examples include financial transactions that must remain accurate to the cent, operational telemetry in which every signal impacts system behavior, and compliance-related event logs in which auditability requires a complete and unbroken record. Unlike asynchronous replication, which may lose data in transit during an outage, synchronous replication provides the strongest durability and continuity guarantees. While this approach introduces additional latency and cost, it's essential for applications in which data integrity and uninterrupted availability are paramount.

Hybrid and Multicloud With Data Streaming for Global Availability and Scalability

The implementation of a data streaming platform for cross-data center replication across hybrid and multicloud environments is a strategic investment in the future of an organization's IT infrastructure. By enabling real-time data replication and processing, project teams can ensure that their organizations remain agile, responsive, and well-positioned to capitalize on emerging opportunities. This holistic approach to data replication drives operational efficiency, reduces costs, and fosters a culture of innovation and continuous improvement.

"One of the key advantages of Confluent is that it brings Kafka to the cloud, such as AWS. Alternatively, you can also leverage Confluent to run Kafka on-prem and connect with multicloud and hybrid clouds."

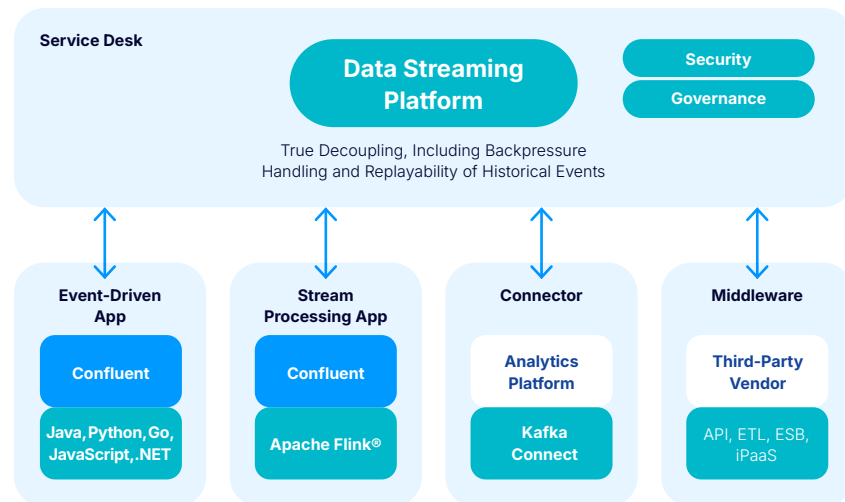
Sainsbury's

2.5 Event-Driven Microservices

A microservices architecture structures a monolithic application as a collection of small, independently deployable services, each focused on a specific business function. These services communicate over lightweight protocols and share data via data products. Microservices enable greater flexibility, scalability, and resilience compared to monolithic applications.

Building microservices with domain-driven design as a software development approach combined with an event-driven architecture is perfect for writing decoupled applications and services. This allows each business unit to choose its own product stack. From a technology perspective, an event-driven microservice can be a business service, legacy monolith, external SaaS service, or an API interface.

Event-Driven Decoupled Microservices (Real Time, Batch, and Request-Response)



Create a unified data layer where microservices communicate asynchronously and independently

Relation of Event-Driven Microservices and Data Streaming

While the first microservices architectures were built with point-to-point APIs, the software community realized that an event-driven architecture was ideal as the foundation of microservices because it enables real-time communication and data sharing between truly decoupled services. The event broker ensures that each microservice can independently process and respond to events as they happen. This approach supports decoupled, asynchronous interactions, which enhances the scalability and resilience of microservices architectures by allowing services to operate and evolve independently.

Additionally, data streaming ensures that microservices can maintain a consistent and up-to-date view of shared state, which is crucial for coordinated processing in distributed systems.

While the data sharing in a microservices architecture is event-based, each business unit can choose its own technology stack, whether it's a lightweight microservice, a legacy monolith, an external SaaS, or a request-response API. The data streaming platform enables seamless data sharing between these diverse technologies and communication paradigms and ensures that information is always accurate and up to date.

"Data streaming keeps microservices architectures flexible, scalable, and aligned with emerging business needs."

Examples for Event-Driven Microservices and Data Streaming

An underestimated additional benefit of the true decoupling of applications using a data streaming platform: The event persistence layer of the data streaming platform allows travel back in time to replay and process historical events with guaranteed ordering. This is a key differentiator from tightly coupled traditional integration platforms such as message queues, ETL, enterprise service bus (ESB), integration-platform-as-a-service (iPaaS), and API management.

Example 1: Retail company

A retail company uses an event-driven architecture to manage its inventory and sales data. By leveraging data streaming, the company can continuously ingest data from various sources, such as point-of-sale systems, ecommerce stores, and supplier databases.

This data is shared continuously between different microservices such as inventory management, salesforce casting, and CRM. The goal is to ensure that information is always up to date and consistent across the business. A worker in the warehouse can access it through a mobile app using an API request-response call. An automated order management system can rely on it in real time to process transactions. A batch analytics platform can use it to generate reports at the end of the day.

Example 2: Financial institution

A financial institution uses the event-driven architecture to monitor transactions and detect fraudulent activities. By integrating data from multiple sources, such as transaction logs, customer profiles, and risk models, the institution can create a comprehensive view of each transaction. This data is then shared in real time between different microservices used for business applications, such as fraud detection, risk management, and customer service, enabling the institution to respond quickly and mitigate risks. In parallel, another microservice for regulatory reporting can replay and analyze historical events with a stream processor for root cause analysis if an incident happens.

Event-Driven Microservices Using Data Streaming for an Evolving Architecture and Technology-Independent Integration

Organizations can remain agile, responsive, and well-positioned to capitalize on emerging opportunities by adopting data streaming for building and integrating microservices. This holistic approach to event-driven applications drives operational efficiency, reduces costs, and fosters a culture of innovation and continuous improvement. Event-based microservices allow organizations to build highly scalable, flexible, and resilient systems that can respond quickly to changing business needs. Additionally, they allow teams to develop, deploy, and update services independently, accelerating innovation and reducing time to market for new features.

2.6 Data Sharing

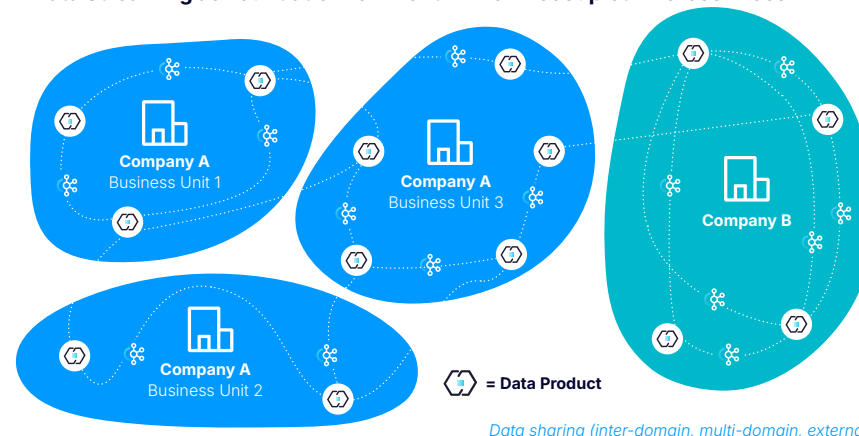
Data sharing is the process of exchanging or providing access to data among different individuals, organizations, or systems. Within an organization, data sharing facilitates collaboration and decision-making by ensuring that all departments have access to the necessary information. Externally, data sharing can enhance partnerships and drive innovation by allowing organizations to collaborate with external entities. The OpenAPI approach enables new business models by providing standardized interfaces for data exchange, allowing businesses to easily integrate and share data with partners and third-party applications for innovation and monetization.

time data exchange between producers and consumers. With a data streaming platform, data sharing becomes seamless—whether the application is real time, batch, or request-response and whether the application uses the streaming protocol, an API, or a connector. Consumers always have access to consistent data, regardless of the data's origin, the time it was produced, or what interface was used. Data streaming allows the creation of data products that can be easily shared across different applications that can use any technology to provide a unified view of data. This is particularly beneficial in event-driven architectures, where data is continuously generated and consumed by various services and applications. A data streaming platform helps implement efficient data sharing strategies that support both internal and external data exchange.

Relation of Data Sharing and Data Streaming

Data streaming plays a crucial role in modern data sharing by enabling real-

Data Streaming as Foundation for Event-Driven Decoupled Microservices



Examples for Data Sharing

Example 1: Multicloud or hybrid environments

Data sharing can evolve to data streaming clusters across regions, multiple clouds, or hybrid environments. Unlike traditional data sharing methods, which often involve complex and time-consuming processes, data streaming allows for seamless data exchange across different environments. In a multicloud or hybrid setup, data streaming enables organizations to maintain data consistency and availability, regardless of the underlying infrastructure.

Example 2: IoT in automotive supply chain

In the IoT space, a real-time data exchange between a car manufacturer and suppliers, as well as mobility services, allows better collaboration to publish and receive information about current orders, demand forecasts, and building new business models via partnerships. By leveraging data streaming, all stakeholders have access to the most up-to-date information, enabling timely and informed decision-making.

Example 3: Risk management in financial services

In the financial services industry, data sharing is critical for risk management and fraud detection. By using a data streaming platform, financial institutions can share data across different departments in real time. This enables them to quickly identify and respond to potential threats to improve overall security and compliance.

Example 4: B2B data marketplace

Modern B2B data marketplaces enable organizations to exchange and monetize real-time data streams with partners and customers. Instead of relying on static file exchanges or batch feeds, a marketplace powered by data streaming allows providers to publish live data products such as market insights, IoT telemetry, or risk signals that consumers can subscribe to instantly. Features like self-service discovery, usage-based billing, and secure access controls ensure that the process is scalable and compliant. This model unlocks new revenue streams for providers while giving consumers immediate access to fresh, contextual data they can use for analytics, AI models, or operational decisions.

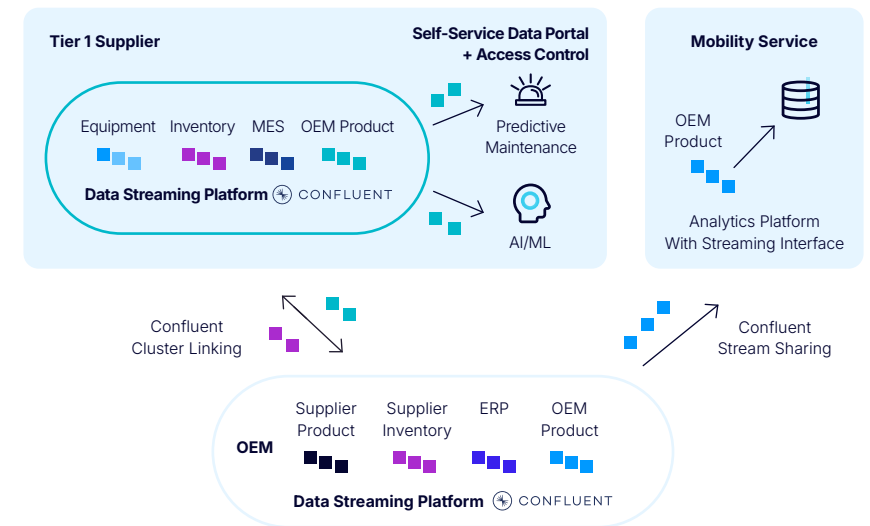
Data Sharing Through Data Streaming for Consistent Data Accessibility in Real Time

The business value of data sharing lies in its ability to enhance collaboration, drive innovation, and improve decision-making. By enabling seamless data exchange inside and outside the organization, businesses can unlock new opportunities and create more efficient processes or better monetization of existing datasets.

"Data streaming makes data sharing real time, reliable, and ready to use."

Data Sharing Example:

Internal and Cross-Company in Automotive



Continuous data sharing between multiple organizations

2.7 Artificial Intelligence

Artificial intelligence (AI) and machine learning (ML) are transformative technologies that empower systems to learn from data and make informed decisions. ML is a subset of AI that enables systems to automatically learn from data and improve without being explicitly programmed.

Predictive AI leverages historical data to anticipate future occurrences, enabling businesses to optimize operations and enhance decision-making processes. By analyzing patterns and trends, predictive models can forecast outcomes such as customer behavior, market trends, and potential risks like equipment failure.

Generative AI, on the other hand, concentrates on crafting new content by assimilating existing data to generate outputs such as text, images, and designs. Large language models (LLMs) are at the core of many of these applications, enabling systems to understand context, generate coherent responses, and interact naturally with users.

Both predictive and generative AI play crucial roles in advancing automation and enhancing human capabilities across various industries. As these technologies continue to evolve, they offer unprecedented opportunities for innovation—creating new business models and improving the efficiency of existing business processes.

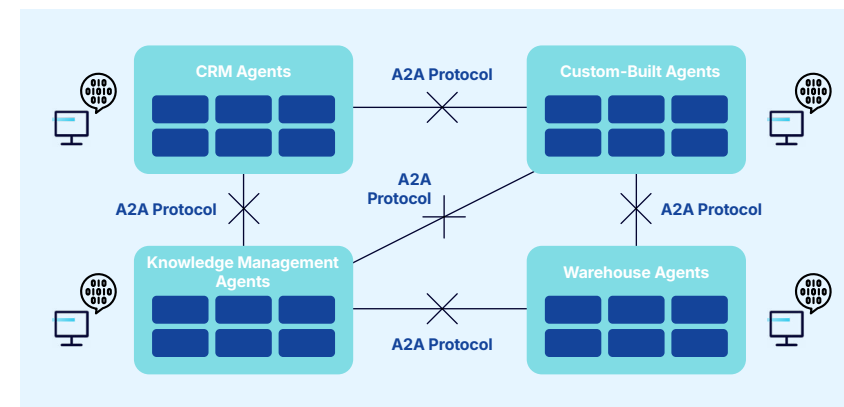
Beyond predictive and generative approaches, a new paradigm is emerging: agentic AI. These are autonomous, goal-driven systems that can make decisions, use tools, and coordinate tasks with minimal human intervention. Unlike reactive assistants, agents maintain context and operate continuously within business processes.

To support interoperability and collaboration, standards are beginning to form. The Model Context Protocol (MCP) defines how agents access context and tools, while the Agent-to-Agent (A2A) protocol specifies how agents discover each other and exchange information. Together, these efforts lay the foundation for scalable, interoperable ecosystems of AI agents.

Challenges of Agentic AI Development

While agentic AI introduces powerful new capabilities, current implementations face significant challenges. Many frameworks rely on point-to-point connections between agents, which quickly leads to "spaghetti architectures" that are fragile and hard to scale. Most are designed for prototyping rather than enterprise-grade operations, lacking robust fault tolerance, security, and observability. Without a consistent backbone for communication and context management, these systems struggle to deliver reliability, transparency, and governance.

Agentic Spaghetti Architecture



Tightly coupled agent-to-agent integrations create complexity and hinder scalability

This challenge mirrors the early days of microservices, when tight coupling and direct integrations created complexity and fragility. As with microservices, true decoupling through event-driven architectures and data streaming is essential to achieve scalability and resilience.

data, maintain context, and take autonomous actions, enabling always-on intelligence. Event-driven architectures take this further by decoupling producers and consumers, allowing multiple AI agents and services to share a common backbone rather than being constrained by point-to-point communication.

Relation of AI and Data Streaming

Data streaming forms the backbone of modern AI by ensuring that information is ingested, processed, and acted upon in real time. It provides the foundation for continuous learning and online training, allowing models to adapt dynamically as fresh events arrive instead of relying on stale batch updates. An online feature store built on streaming pipelines ensures that features are always current and consistent across training, inference, and monitoring. Streaming agents powered by stateful processing engines can consume live

This shift transforms isolated models into collaborative, agentic systems where agents can perceive, reason, and act in real time across distributed environments. Predictive AI benefits through more accurate forecasting from real-time data. Generative AI leverages RAG to ground outputs in fresh data. And agentic AI relies on streaming infrastructure to coordinate autonomous decision-making. Together, Kafka-style event-driven backbones and Flink-style stream processing engines create the nervous system for AI systems that are scalable, reliable, and adaptive.

Examples for AI using Data Streaming

Example 1: Financial institution, fraud prevention, and adaptive response

A financial institution applies AI to prevent fraud and optimize customer interactions. Data streaming continuously ingests and processes transactions so predictive models can flag anomalies before a payment is finalized. At the same time, agentic AI operates on this live data, autonomously escalating cases, requesting additional verification, or even blocking a transaction in coordination with external systems. Generative AI enhances the customer experience by drafting personalized explanations or next steps, ensuring that both prevention and customer communication happen instantly and consistently.

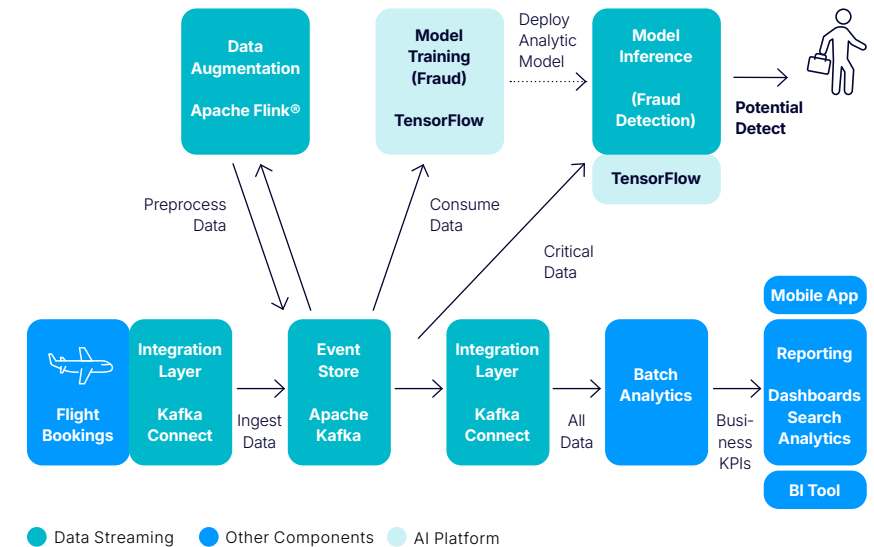
Example 2: Manufacturing, predictive maintenance, and self-optimizing operations

In manufacturing, AI is used to reduce downtime and improve efficiency. Sensor data flows continuously into a streaming backbone where predictive models detect potential equipment failures. Agentic AI agents consume the same event streams and can autonomously trigger work orders, reroute production schedules, or adjust machine parameters without human intervention. Generative AI supports workers by providing contextual maintenance instructions or creating real-time summaries of plant status. Together, this ecosystem enables factories not just to predict issues but to act on them immediately.

Example 3: Airline, customer service, and operational resilience

An airline leverages AI to improve both customer service and operations. When disruptions occur, data streaming integrates booking engines, flight schedules, and passenger profiles in real time. Generative AI powers chatbots that offer rebooking options and personalized recommendations. AI agents extend this further by autonomously interacting with operational systems to reserve seats, issue vouchers, or trigger downstream processes like payments or crew scheduling. Predictive models forecast passenger flow or delays, while streaming ensures that all systems stay synchronized so customers receive timely, accurate service without bottlenecks.

Data Streaming as Orchestration Layer for Predictive AI



Stream data seamlessly between operational systems and AI platforms for predictive, real-time insights

Data Streaming for More Powerful and Accurate Results in AI Use Cases

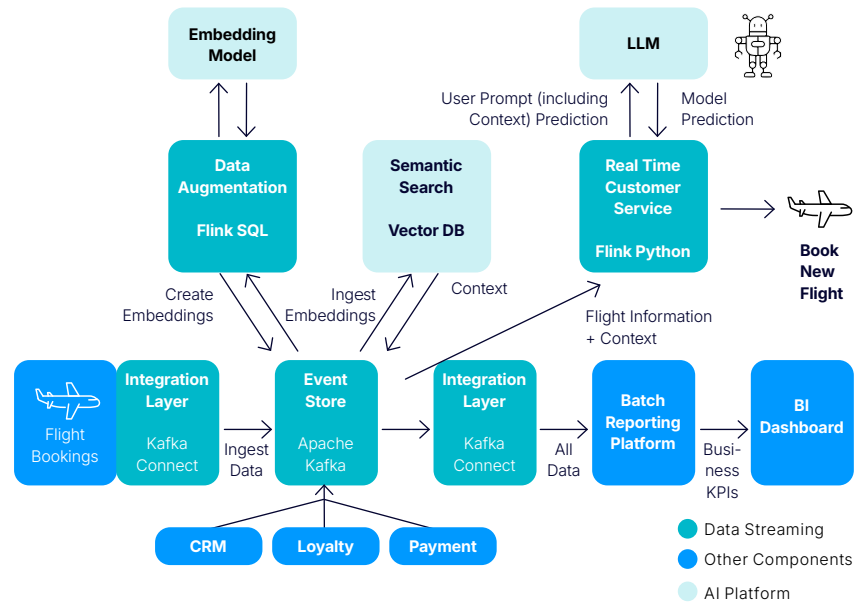
Using a data streaming platform tailored for AI is not just a technological upgrade; it's a foundational strategy for building intelligent, always-on applications. Accurate, timely decisions depend on continuously updated, high-quality data flows. This is something traditional batch pipelines or static data lakes can't provide.

Adopting data streaming for predictive, generative, and agentic AI offers significant business value by:

- **Enhancing** data quality at the point of ingestion and throughout processing
- **Improving** both model training and low-latency inference with live data
- **Enabling** real-time, autonomous decision-making across complex environments

This approach ensures that AI applications can seamlessly integrate operational and analytical data while maintaining consistency, governance, and reliability.

Data Streaming and RAG for Generative AI



Bridge enterprise systems with Generative AI through streaming-driven RAG for real-time predictions and decisions

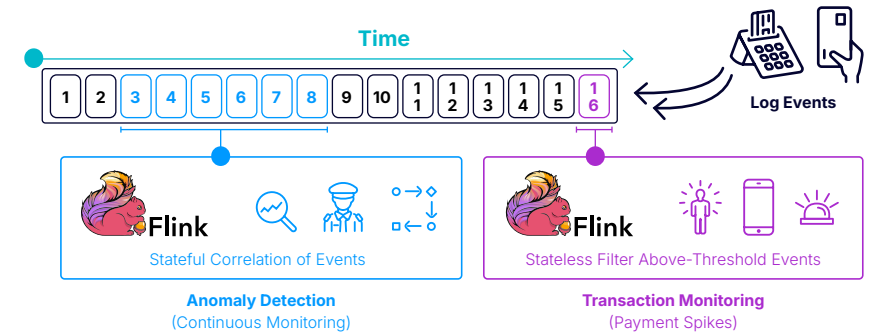
It enables continuous training and retraining of models with fresh data, keeping them relevant and accurate. For generative AI (GenAI), streaming provides the contextual grounding needed for RAG. For agentic AI, it delivers the real-time event-driven backbone that supports autonomous, goal-driven agents operating with memory, tool access, and collaboration.

Emerging standards such as the MCP and A2A show how interoperability between agents will evolve—but without a streaming foundation, these interactions risk becoming brittle and siloed. With event-driven integration, enterprises avoid these pitfalls and instead enable a scalable, resilient, and

governed AI ecosystem. Ultimately, adopting a data streaming platform enables organizations to realize the full potential of predictive, generative, and agentic AI. Confluent Intelligence delivers the essential AI infrastructure with three integrated capabilities: Built-in ML functions for anomaly detection, fraud prevention, and forecasting; Streaming Agents that run natively on Kafka and Flink to automate workflows in motion; and the Real-Time Context Engine, which serves high-quality, structured context to any AI system through MCP. Together, these capabilities turn real-time data into real-time intelligence, enabling trusted, context-aware, and continuously improving AI across the enterprise.

Predictive AI Example: Fraud Detection

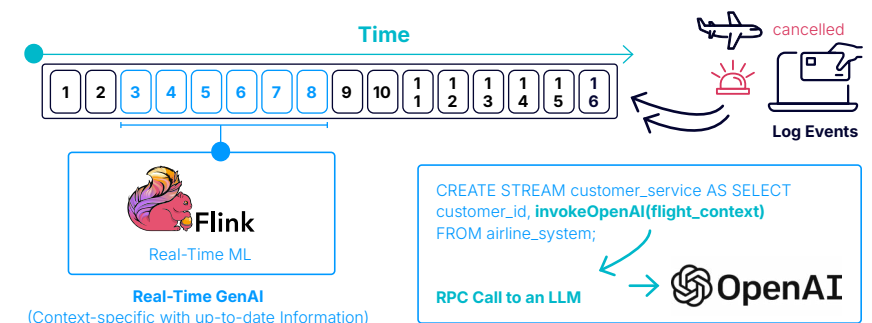
Stateless and stateful stream processing for real-time data correlation with Apache Flink®



Anomaly detection for fraud prevention with continuous stream processing

Generative AI Example: Context-Specific Customer Service in Real Time With Stream Processing

Stateless and stateful stream processing for real-time data correlation with Apache Flink® (Flink SQL)



Automated and context-specific customer service in real time by using GenAI

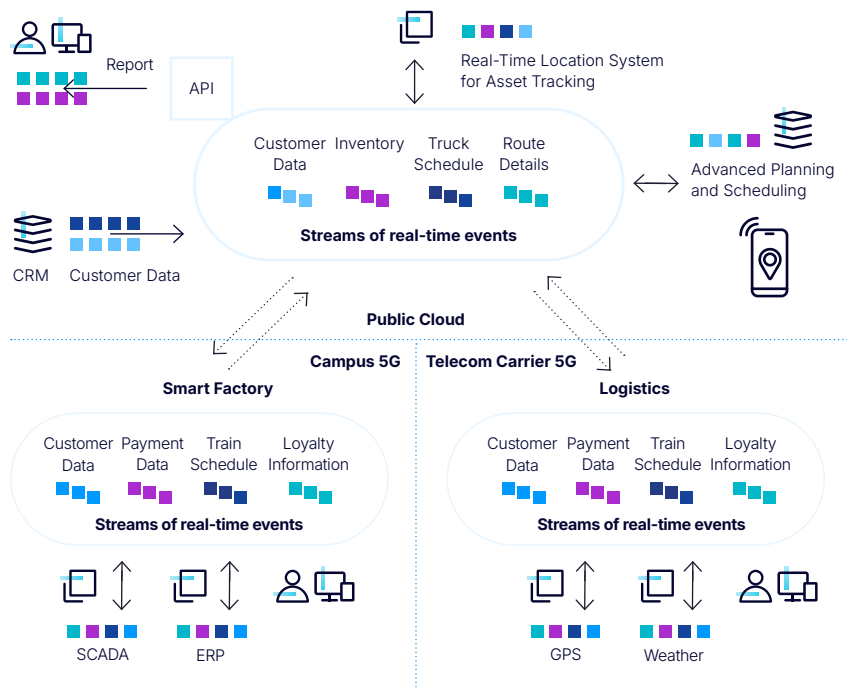
2.8 Internet of Things

IoT refers to the network of inter-connected devices that communicate and share data over the internet, enabling automation and smarter decision-making across various domains.

Industrial IoT (IIoT) connects sensors, equipment, and systems in sectors like manufacturing or energy and utilities to optimize operations through real-time

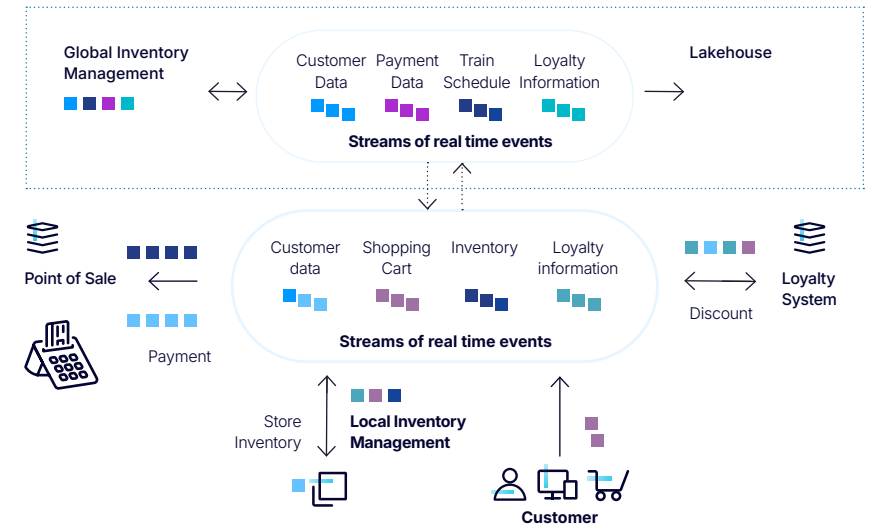
data and analytics. The integration of operational technology (OT) with IT in IIoT allows for seamless data flow and advanced process control, significantly improving the overall equipment effectiveness (OEE) by reducing downtime, enhancing productivity, and enabling predictive maintenance.

Example: IIoT



Improvement of OEE with data streaming across edge and hybrid cloud environments

Data Streaming at the Edge in the Smart Retail Store



Integration and synchronization between smart retail stores and global inventory management in the cloud

Consumer IoT includes everyday devices such as smart home appliances, wearables, and connected cars to enhance convenience, personalization, and efficiency for users. These devices seamlessly integrate into daily life, offering real-time data and insights that empower users to make informed decisions and automate routine tasks. As a result, CIoT not only improves the quality of life but also fosters a more connected and intelligent living environment.

Relation of IoT and Data Streaming

IoT naturally generates data continuously to be ingested, processed, and analyzed in real time. Data streaming is essential for managing the vast amounts of data generated by IoT. Stream processing ensures that information is

continuously aggregated and actionable. Data streaming technologies such as Kafka and Flink are not hard real time; they can't guarantee deterministic communication with zero spikes and zero latency. For this reason, they're not suitable for building safety-critical embedded systems such as a robotic operating system or advanced driver assistance systems in a car. However, they're highly valuable for processing non-safety-critical data. Data streaming also connects safety-critical systems with the rest of the IT landscape. This makes it possible to share operational insights, enrich data products, and integrate critical systems into broader business processes.

Data streaming fits into the enterprise architecture for IoT use cases by providing a scalable and reliable

platform for continuous data processing. A hybrid edge-cloud architecture offers the flexibility to deploy applications and services wherever they're needed most. The flexibility comes from a consistent data pipeline that can operate across both edge and cloud environments. With this approach, organizations can optimize their infrastructures based on specific use cases and requirements.

Many organizations choose a cloud-first approach for new projects, in which latency, security, and cost are the main factors in deciding if a service can be deployed in the cloud or needs to be at the edge. In some cases, services must also run in fully air-gapped environments where external connectivity is not allowed or in setups where only unidirectional data flows are permitted.

Examples for Data Streaming in the IoT World

Example 1: IIoT in manufacturing

A manufacturing company can use a data streaming platform on the factory shop floor. This includes logistics, communication with programmable logic controllers (PLCs) and robotics, monitoring of equipment, and transactional decisioning using a manufacturing execution system (MES), ERP, and related systems. By leveraging data streaming, the company can create a bridge between the OT world (usually at the edge) and the traditional IT world (usually in a private data center or the public cloud), enabling real-time monitoring and analysis of factory operations.

Example 2: Smart retail store

In a smart retail store, data streaming at the edge connects inventory management, location-based services, and point-of-sale (POS) systems to provide real-time updates and personalized customer experiences. Customer actions are quickly captured and sent to a hybrid cloud, where a loyalty platform and CRM system use the data to offer customized promotions and rewards. This setup improves customer satisfaction, keeps inventory levels just right, and boosts sales, helping the store grow its customer base.

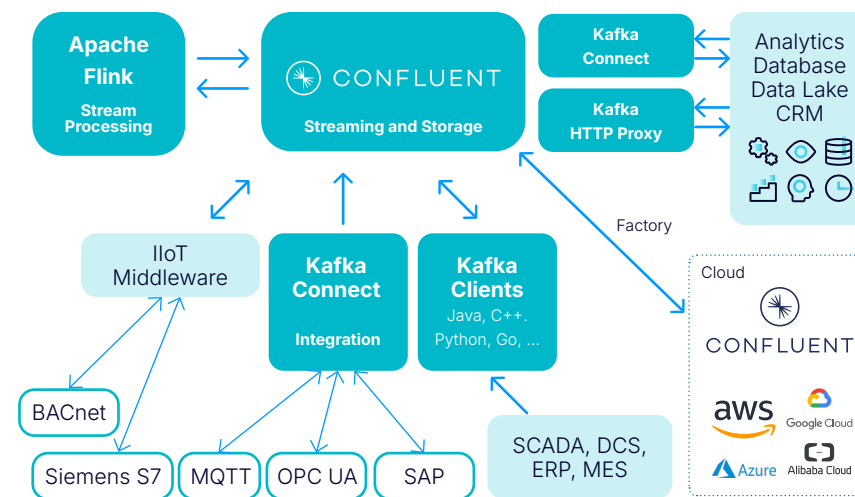
IIoT With Data Streaming to Optimize the OEE and Provide Innovative Consumer Experiences

Data streaming in IIoT projects provides significant business value by enabling continuous data processing and decision-making. The near-real-time capability allows organizations to monitor and respond to events as they happen, improving operational efficiency and reducing downtime.

By leveraging data streaming, organizations can seamlessly integrate and process:

- **High-volume** IIoT data from logs, sensors, robotics, equipment, and other interfaces
- **Low-volume** transactional data from operational systems such as MES, ERP, and CRM

Data Streaming Platform as Foundation for an Open and Scalable Data Historian



IT/OT integration between the smart factory at the edge and analytics in the cloud

In industrial settings, data streaming enhances predictive maintenance by continuously analyzing sensor data to prevent equipment failures, optimize maintenance schedules, and improve the OEE.

A unified namespace (UNS) can structure live OT data from sensors, PLCs, and machines into a consistent hierarchy for operational use. When integrated with data streaming in IT, this data becomes governed, reusable, and available in real time across enterprise systems—ensuring that predictive insights are both accurate and scalable.

For CIIoT, it supports dynamic personalization and rapid response to user behaviors, enhancing customer experiences. Data from wearables, smart appliances, and connected cars is

streamed continuously into IT systems, where real-time processing enriches it with context from CRM, loyalty, or recommendation engines. This integration ensures that personalization happens instantly and securely, while governed data flows prevent fragmentation and enable reuse across multiple digital services. Additionally, data streaming enables scalable and flexible analytics, allowing businesses to process large volumes of data from numerous devices without delays. This is critical for gaining insights and driving informed decisions in dynamic, fast-paced operating environments.

Overall, it accelerates innovation, reduces costs, and enhances competitiveness by ensuring that data is actionable as soon as it's generated.

2.9 Data Governance

Data governance refers to the comprehensive management of data's availability, usability, integrity, and security within an organization. It involves establishing processes, roles, policies, standards, and metrics to ensure that data is properly managed throughout its life cycle .

The primary goal of data governance is to ensure that data is accurate, consistent, secure, and compliant with regulatory requirements and organizational policies.

In the age of generative and agentic AI, governance becomes even more important because these systems rely on high-quality, trustworthy data. Poor governance leads to biased, insecure, or unreliable outcomes, while strong governance ensures that AI delivers safe, compliant, and valuable results.

Key Aspects of Data Governance

- **Compliance** in data governance ensures that data management practices adhere to relevant laws, regulations, and standards. This includes maintaining data privacy, security, and integrity to avoid legal and financial repercussions.
- **Data sovereignty** refers to the concept that data is subject to the laws and governance structures of the nation in which it's collected. It

emphasizes the need for organizations to understand and comply with the legal requirements of the jurisdictions in which they operate.

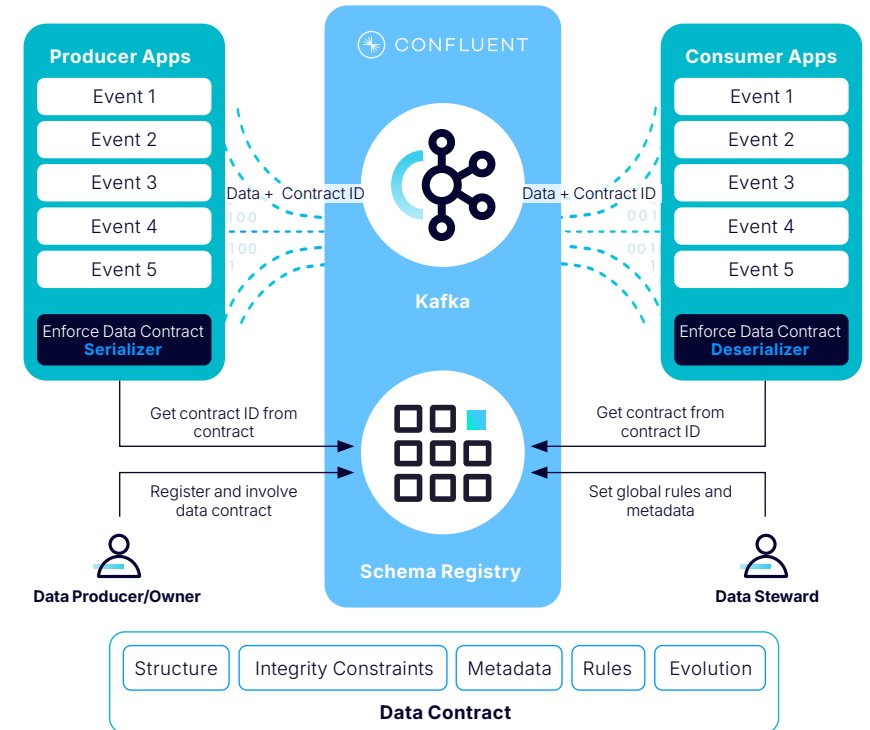
- **Data privacy and security** focus on protecting sensitive information from unauthorized access and ensuring that personal data is handled in compliance with privacy laws. This involves implementing robust security measures and privacy policies to safeguard data.
- **Data quality** involves ensuring that data is accurate, complete, reliable, and timely. High-quality data is essential for effective decision-making and operational efficiency, as it directly impacts the outcomes of data-driven processes.

Relation of Data Governance and Data Streaming

A data streaming platform enables organizations to process and analyze data continuously as it's generated, allowing for immediate actionability and decision-making. This real-time processing capability on top of data contracts ensures that data governance policies and controls can be applied instantaneously to address issues such as data quality, security breaches, or compliance violations as they occur.

In an **event-driven architecture**, a data catalog and data lineage built on top of

Data Governance for Compliance, Data Sovereignty, and Quality



Govern data at the source with enforceable contracts that guarantee accuracy, lineage, and policy adherence

data streaming facilitates the continuous flow of data across various systems. This ensures that data governance mechanisms are consistently enforced.

Data streaming provides the scalability and resilience needed to handle high volumes of data to ensure that governance processes can keep pace with the speed and scale of streaming data environments.

Built on **data products with data contracts** for structure, metadata,

policies, integrity constraints and schema evolution, this integration allows for the automation of governance processes, enabling timely detection, notification, and remediation of governance issues.

AI further strengthens governance by automating policy enforcement, anomaly detection, and compliance monitoring in real time. Agentic AI can act as a streaming agent within the governance process, continuously monitoring data flows, identifying risks, and taking corrective actions instantly.

Examples for Data Governance in Streaming Data Architectures

Example 1: Financial services, compliance, and fraud monitoring

A financial services company uses data streaming to monitor transactions in real time. By leveraging a data streaming platform, the company can implement data governance policies that detect fraudulent activities as they occur, ensuring compliance with financial regulations and protecting customer data.

Example 2: Healthcare, patient data management

A healthcare organization uses data streaming to manage patient data continuously. With real-time data processing on top of data contracts, the organization can guarantee data quality and privacy by applying governance controls that validate data accuracy and enforce privacy policies. This ensures compliance with healthcare regulations such as HIPAA (Health Insurance Portability and Accountability Act).

Business Value of Data Governance With Data Streaming

The adoption of data streaming offers significant opportunities for improving data governance across an entire enterprise architecture. By enabling continuous data processing and analysis, organizations can enhance their ability to:

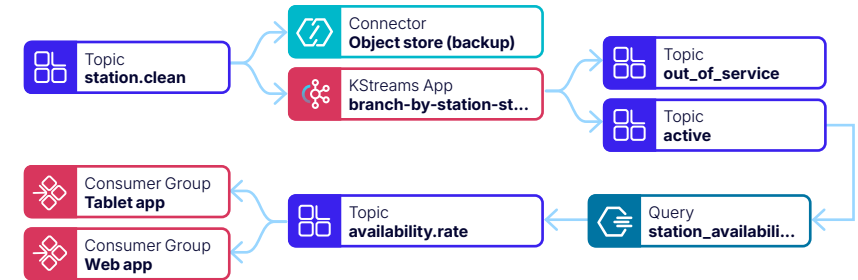
- **Manage** data effectively
- **Ensure** compliance, data sovereignty, privacy, and quality
- **Mitigate** risks associated with data management
- **Enhance** operational efficiency and decision-making

Furthermore, data streaming facilitates real-time visibility into data flows, allowing organizations to quickly identify and address potential issues before they escalate.

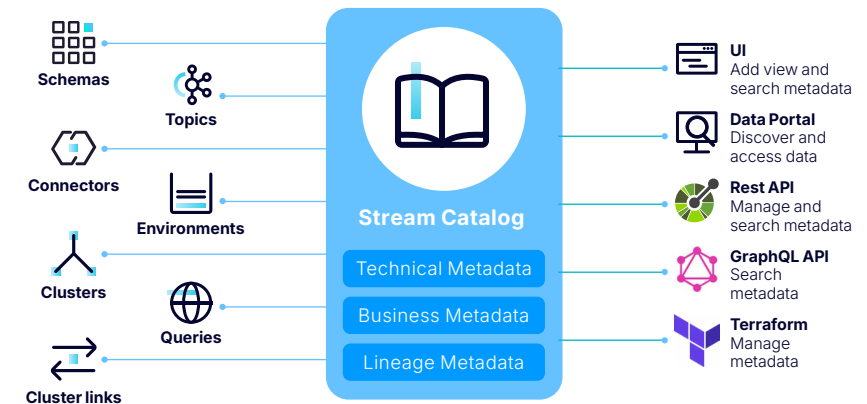
This proactive approach supports a culture of transparency and accountability, fostering trust among stakeholders.

A data streaming platform can be integrated with enterprise-wide governance frameworks. This ensures that data policies are consistently enforced across both internal systems and external platforms. The result is a stronger and more reliable data governance posture for the entire organization.

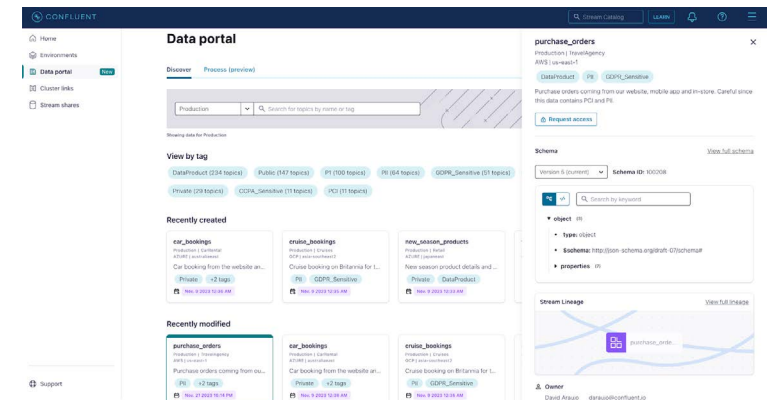
Stream Lineage



Data Catalog



Data Portal



2.10 Environmental, Social, and Governance

Environmental, social, and governance (ESG) criteria are a set of standards for a company's operations that socially conscious investors use to screen potential investments. ESG is important across all industries, as it addresses critical issues like climate change, social responsibility, and corporate governance.

Data-driven software plays a significant role in measuring, monitoring, and improving ESG performance, including regulatory requirements that are being implemented now, for instance, the **Corporate Sustainability Reporting Directive (CSRD)**.

AI amplifies the impact of ESG initiatives by automating data collection, monitoring, and reporting at scale. Advanced capabilities such as generative and agentic AI also help organizations identify risks, uncover improvement opportunities, and ensure compliance with evolving ESG regulations. At the same time, AI introduces new ESG risks, as increasing use of large-scale models drives higher energy consumption and environmental impact. This makes responsible AI adoption a key part of any ESG strategy.

Relation of ESG and Data Streaming

Data streaming plays a crucial role in advancing ESG initiatives by providing real-time insights and enabling proactive management of sustainability efforts.

- **Environmental monitoring:**

Continuous data streaming allows organizations to track key metrics such as air and water quality, energy consumption, and emissions in real time. This immediacy helps identify and mitigate environmental risks while optimizing resource usage and reducing carbon footprints.

- **Supply chain transparency:**

Data streaming ensures that companies can monitor their supply chains continuously and with good data quality, verifying that suppliers adhere to ethical and sustainability standards. This transparency improves the accuracy of ESG metrics and reporting.

- **Social impact monitoring:**

Streaming data enables real-time monitoring of working conditions, employee well-being, and community effects, allowing companies to promptly address any social issues and ensure compliance with labor and human rights standards.

- **Governance:**

Data streaming provides up-to-date information on regulatory compliance to help organizations adapt quickly to changing regulations and align their operations with ESG goals.

Finally, data streaming enhances the accuracy and timeliness of ESG reporting and offers stakeholders transparent, current information on a company's ESG performance. This strengthens investor confidence, supports regulation compliance, and improves corporate reputation.

If a portfolio company suddenly faces a social or environmental controversy, the bank's system instantly flags it, enabling quick responses such as adjusting investment strategy or engaging with the company to resolve the issue. Real-time data also helps ensure that lending practices align with ESG criteria (e.g., avoiding financing for projects that harm the environment or violate human rights).

Examples for ESG Use Cases With Data Streaming

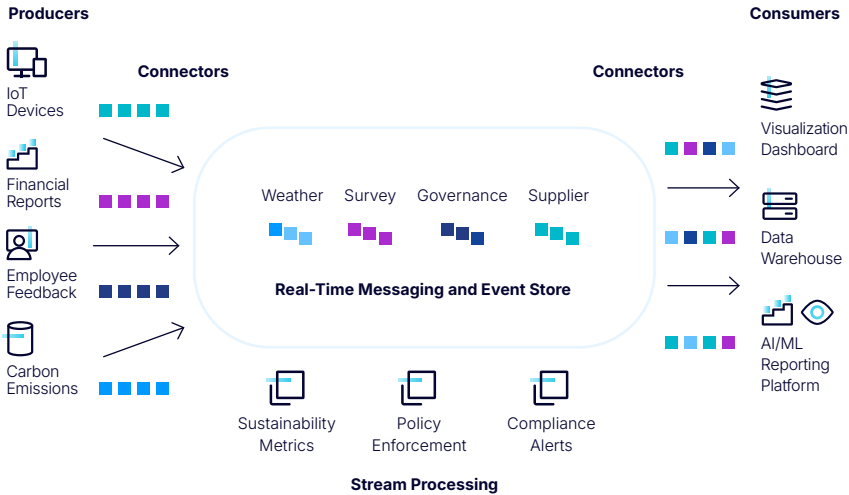
Example 1: Manufacturing, environmental performance

A manufacturing company uses data streaming to monitor and analyze emissions data from its industrial plants. By continuously ingesting and processing emissions data in real time, the company can identify and address potential issues before they become significant problems. This is crucial for reducing its carbon footprint and improving environmental performance.

Example 2: Financial services, ESG risk monitoring

A bank uses data streaming to monitor and analyze real-time transactions and activities across its investment portfolio. By continuously streaming data on the environmental impact, social practices, and governance structures of the companies in which it invests, the bank can better assess ESG risks and opportunities.

Event-Driven Architecture With Data Streaming to Improve ESG



Correlate IoT, financial, and supplier data streams to power continuous ESG monitoring and reporting

Business Value of ESG With Data Streaming

Using data streaming for ESG initiatives delivers substantial business value by enabling:

- **Real-time monitoring and risk management**, reducing the likelihood of ESG-related incidents that could lead to financial penalties, reputational damage, or regulatory scrutiny.
- **Proactive compliance, ensuring** adherence to evolving regulations before problems escalate.
- **Cost savings**, by preventing issues early.

- **Investor confidence**—Attracting ESG-focused investments and lowering the cost of capital with transparent ESG metrics
- **New opportunities for sustainability**—Optimizing resource usage or identifying sustainable investments for long-term growth

Overall, integrating data streaming into ESG strategies strengthens a company's ability to manage ESG factors effectively, aligning business practices with societal and regulatory expectations while creating value for both shareholders and stakeholders.

Industry Success Stories for Data Streaming

03

The previous chapter showcased the wide array of technical use cases in which data streaming can add significant value, regardless of industry. This book aims to demonstrate that these applications are already being implemented across various sectors. While not all technical use cases were covered in Chapter 2 and not every scenario in every industry can be addressed, the focus remains on some areas with strong data streaming adoption, illustrated through real-world examples to provide maximum value to readers.

This section offers a comprehensive overview of trends across all major industries, along with concrete examples of data streaming usage by customers highlighting the business value they gain. The aim is for both experienced users and streaming newcomers to explore

practical applications of data streaming through specific scenarios and industry examples. All industry trends are based on current research and discussions with existing Confluent customers. The scenarios are directly linked to these trends and are evident in real customer examples.

The following subchapters focus on real world scenarios and examples. Each section highlights typical challenges that companies face in their industry, such as financial services, manufacturing, or retail. The use cases show how organizations respond to digitalization and adapt their business models.



This chapter is just a snapshot of the many success stories and business results that are coming from data streaming. We aim to extend this book with more examples in every release.

For those interested in being featured in the next edition of The Ultimate Data Streaming Guide, contact the authors via [✉ use-case-book@confluent.io](mailto:use-case-book@confluent.io).

3.1 Financial Services

The financial services industry is undergoing a profound transformation driven by digitalization and innovation. The industry is experiencing a constant need to balance customers' high expectations for ease of use, service orientation, and availability with the traditionally high standards of security and system reliability. A rapid adoption of digital technologies is reshaping how financial institutions operate, interact with customers, and deliver services.

One of the most significant trends is the transition to cloud-based solutions, which offers unparalleled scalability, flexibility, and cost efficiency. Hybrid cloud integration, in particular, is gaining traction as it allows financial institutions to leverage the best of both public and private clouds. This approach ensures data security and compliance while enabling seamless access to advanced analytics and real-time processing capabilities.

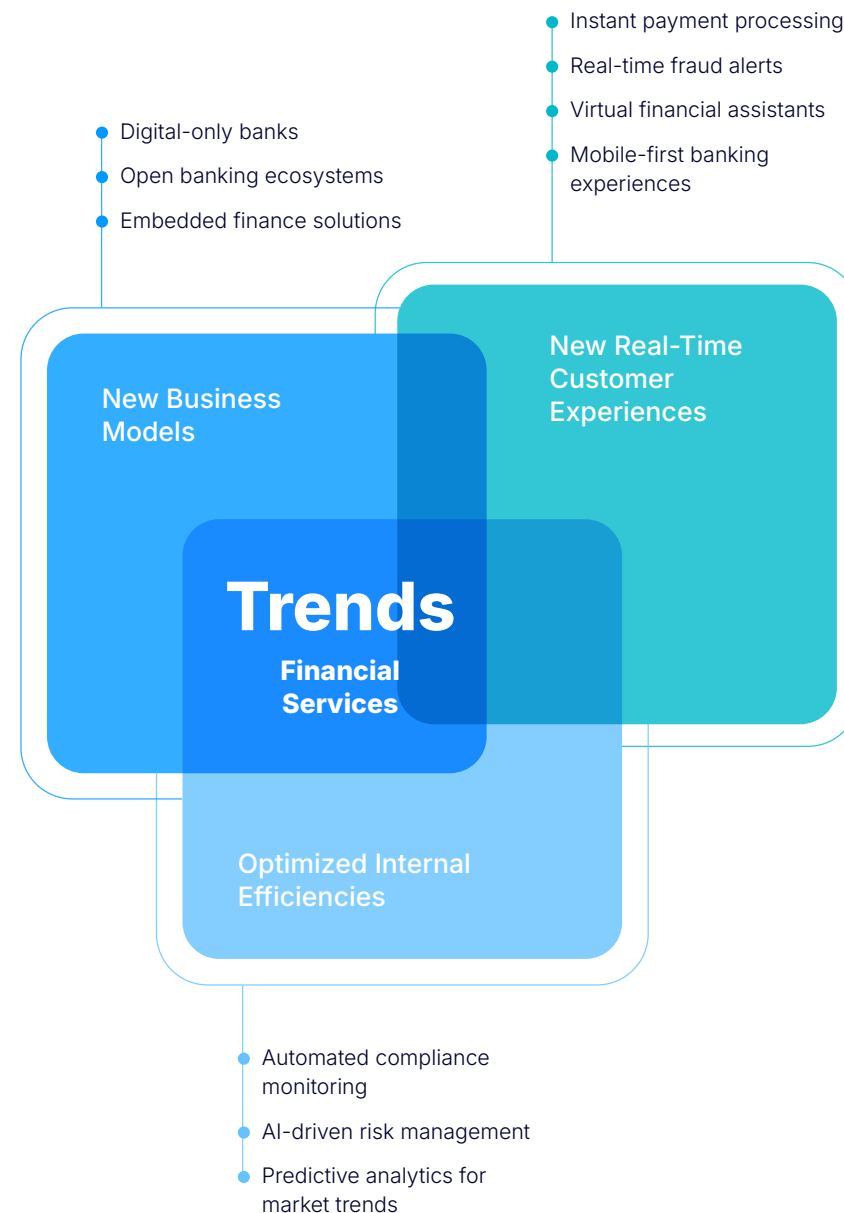
Additionally, the rise of fintech companies and digital-first banks is pushing traditional financial institutions to innovate and modernize their legacy systems. At the same time, open banking ecosystems and embedded finance models are reshaping how institutions collaborate and deliver services. Real-time data streaming and processing have become

essential for delivering personalized customer experiences, detecting fraud, and optimizing operations.

The integration of AI and ML into financial services is further enhancing decision-making and risk management. Together with real-time streaming, these technologies power instant payments, predictive risk scoring, and proactive fraud prevention.

As regulatory requirements continue to evolve, financial institutions must also ensure that their digital transformation initiatives comply with stringent standards.

Overall, the convergence of cloud computing, data streaming, and AI/ML is driving a new era of innovation and efficiency in the financial services industry. Institutions that embrace this convergence can unlock new revenue streams, reduce costs, and deliver customer trust at scale.



Scenario 1: Legacy Offloading

Legacy systems, particularly mainframes, have been the backbone of financial institutions for decades, handling critical transactional workloads and storing vast amounts of data. However, the high operational costs and inflexibility of these legacy systems have become a significant burden. Mainframe offloading, using data streaming platforms, offers a compelling solution to this challenge. By migrating data and workloads to a more flexible and scalable cloud-native environment, financial institutions can drastically lower their operational expenses and enhance the agility of their IT operations. Data streaming enables data consistency between legacy batch infrastructure and modern real-time cloud streaming applications and APIs, ensuring seamless integration and real-time data availability.

Business Value

- **Cost reduction:** The transition allows banks to reduce the number of reads on the mainframe, significantly cutting down on fixed infrastructure costs (OPEX).
- **Compliance and innovation:** Financial institutions maintain compliance with regulatory requirements and business logic while enabling the creation of new applications using the same event-based architecture.
- **Operational efficiency:** The new architecture enhances the agility of IT operations, allowing for faster deployment of new services and ensuring data consistency across mainframes, databases, and customer-facing applications.

"Banking systems are typically a mix of cloud-based applications and legacy databases. In scenarios where our developers don't have access to the underlying codebase for systems we want to connect to, we really like using pre-built connectors from Confluent. We can directly connect to databases, capture changes that are happening, create events from those changes, and keep these systems in sync with our downstream systems."

Badr Alduaiji, Head of Integration and Architecture, Arab National Bank

Scenario 2: Core Banking

Core banking systems are essential for processing transactions, facilitating payments, and managing customer accounts. Modernizing these systems to leverage data streaming platforms can significantly enhance their capabilities, providing real-time processing and greater scalability. By integrating data streaming technologies, financial institutions can build resilient and compliant infrastructures that handle high volumes of transactions with zero data loss. This modernization also enables faster time to market for new use cases and services, ensuring that banks can stay ahead in a competitive landscape.

Business Value

- **Resilient and compliant infrastructure:** The platform provides a resilient infrastructure that can handle high volumes of transactions with zero data loss.
- **Elastic scalability:** The platform's scalability ensures that banks can efficiently manage peak loads without compromising performance.
- **Faster time to market:** The ability to rapidly deploy new use cases and services accelerates time to market, enabling banks to stay ahead in a competitive landscape.

"Not only does Confluent provide lower latency and higher throughput than our legacy system; it is also more resilient."

Sunarto Rahardja, SVP and Head of Software Development, Singapore Exchange (SGX)

Scenario 3: Real-Time Payments

In financial services, instant payments and high-speed trading are crucial to meet customer demand for immediate transaction processing and to keep pace with modern markets. Financial institutions must ensure that transactions, asset transfers, and trades are executed without delay, especially during peak trading periods.

Traditional batch processing systems are insufficient for today's needs, leading to bottlenecks, latency problems, and operational risks.

Data streaming enables real-time processing and analysis of massive transaction volumes. It also allows continuous monitoring, instant transaction validation, anti-fraud checks, and risk analysis during the transfer process.

By ingesting data across trading systems, customer accounts, and compliance systems, data streaming enables near-instantaneous clearing and settlement. This digitizes electronic funds transfers (EFTs), covering, payments such as direct deposits, wire transfers, Automated Clearing House (ACH) payments, automated teller machine (ATM) transactions, and point-of-sale (POS) transactions. High-frequency trading operations with millisecond latency allow trading platforms to update risk profiles in real time and adjust strategies dynamically based on market data.

Business Value

- **Reduced latency:** Execute trades and process EFTs in real time, meeting the demands of modern high-frequency trading and immediate funds transfers.
- **Improved system resilience:** Achieve high availability and seamless failover, ensuring that operations continue uninterrupted, even during high-volume periods.
- **Scalability:** Scale dynamically to accommodate peak trading times or surges in EFT activity, maintaining performance without compromising data integrity.
- **Regulatory compliance and risk mitigation:** Meeting T+0 reporting requirements ensures immediate compliance with financial regulations, reducing the risk of penalties and enhancing trust with regulators and stakeholders.
- **Faster time to market:** Deploy new features and compliance updates quickly with microservices, allowing for rapid response to regulatory changes and market demands.

Scenario 4: Fraud Detection

Fraud detection is a critical priority for financial institutions, demanding robust, real-time analysis to detect, prevent, and respond to suspicious activities swiftly. Data streaming platforms empower institutions to implement real-time risk and anti-money laundering (AML) compliance models that continuously monitor and analyze transaction data for unusual patterns, behaviors, and anomalies that could indicate fraud or money laundering.

With advanced data streaming technologies, institutions can aggregate data from multiple sources, including transactions, customer profiles, and regulatory feeds, to generate a holistic view of potential risks. This real-time, cross-functional analysis allows for instant alerts, rapid investigation, and compliance with AML regulations, significantly reducing the window of opportunity for fraudulent actions and safeguarding both financial assets and institutional integrity.

Business Value

- **Reduced risk:** Real-time fraud and AML detection minimizes financial losses, protecting both customer and institutional assets.
- **Regulatory compliance:** Meeting AML standards in real time reduces penalties and ensures compliance with stringent regulatory requirements.
- **Enhanced customer trust:** Proactive fraud prevention strengthens customer confidence, loyalty, and brand reputation.
- **Operational efficiency:** Real-time analysis and rapid response streamline operations, improving accuracy in decision-making and resource allocation.
- **Scalable risk management:** Scalable data streaming enables efficient monitoring as transaction volumes grow, ensuring resilience and adaptability in fraud detection processes.

Scenario 5: Regulatory Reporting

Regulatory reporting is a critical function for financial institutions, particularly those involved in the derivatives market. It involves the use of trade data repositories and reporting service providers to ensure compliance with regulatory requirements. The accuracy and timeliness of these reports are essential for maintaining transparency and operational efficiency.

Financial services regulations increasingly demand near-instantaneous, T+0 (trade date) reporting for payments and trades, moving from the previous T+1 (next-day) standard to ensure real-time transparency and compliance.

Data streaming platforms play a vital role in enhancing the regulatory reporting process. By enabling real-time data validation and processing, these platforms ensure that the data submitted to regulators is accurate and up to date. This capability not only improves compliance but also reduces the operational burden associated with manual data processing and reporting.

Business Value

- **Transparency and real-time data validation:** Real-time processing ensures that all data is accurate and validated before submission, enhancing transparency.
- **Enhanced compliance with regulatory requirements:** Continuous data validation and processing ensure that all regulatory obligations are met promptly.
- **Operational efficiency:** Automating the data processing and reporting functions reduces the need for manual intervention, saving time and resources.
- **Competitive advantage:** By leveraging real-time data streaming, financial institutions can set themselves apart from competitors that rely on slower traditional methods.

Scenario 6: Embedded Finance

Embedded finance integrates banking, payments, lending, and insurance directly into non-financial platforms, allowing consumers to transact, access credit, or purchase insurance without leaving the app or service they're using. This could be a marketplace offering instant credit, a mobility app processing in-app payments, or a subscription platform bundling insurance seamlessly into checkout.

For this model to work, payment systems, credit scoring services, fraud detection tools, and compliance platforms must exchange information instantly and securely. Even a brief delay can cause a failed payment, rejected credit request, or missed fraud alert. A modern data streaming foundation connects these diverse systems, continuously moving financial events in real time, applying business logic as they flow, and safeguarding sensitive information through policy-driven controls.

By combining streaming integration, in-flight processing, and built-in governance, organizations can enrich transaction streams with contextual data, detect anomalies within milliseconds, and enforce regulatory rules without slowing down customer interactions. This also ensures that AI-driven agents and credit risk models operate on fresh, accurate information to avoid the flawed outputs that occur when relying on stale or incomplete datasets.

Business Value

- **Frictionless transactions:** Instant payments, transfers, and credit decisions are within the user experience.
- **Proactive fraud defense:** Immediate anomaly detection to prevent losses.
- **Security and compliance:** Policy-driven controls protect sensitive data and meet financial regulations.
- **Continuous insight:** Unified real-time and historical data improves decision-making and transparency.
- **Scalable growth:** Increasing volumes are handled without downtime.
- **Revenue expansion:** Embedding finance creates new recurring revenue streams.

Scenario 7: FinOps

FinOps aligns finance, engineering, and operations to manage cloud costs in real time. This is a necessity in finance departments across all industries, where trading, payments, analytics, and customer applications run on large-scale cloud workloads. Traditional batch-based cost reports arrive too late to prevent runaway expenses or misconfigurations.

A streaming data architecture enables continuous ingestion of cost and usage data from billing APIs, infrastructure telemetry, and application logs. These streams are enriched with business context, validated for accuracy, and governed for both security and compliance with regulations such as the Payment Card Industry Data Security Standard (PCI DSS) and the EU General Data Protection Regulation (GDPR).

AI-enabled streaming agents can forecast cost trends, detect anomalies, and trigger automated policies, such as pausing underused workloads, before costs escalate. This approach ensures that AI and analytics tools base decisions on current, contextual, and trusted financial data, not stale snapshots. The result is accurate forecasting, faster audits, and proactive cost optimization without compromising compliance.

Business Value

- **Instant visibility:** Detect spikes or anomalies within seconds.
- **Proactive compliance:** Ensure privacy and audit readiness with built-in governance.
- **Automated cost control:** Trigger workload adjustments when thresholds are reached.
- **Cross-team alignment:** Get a shared, accurate view of cloud spending across functions.
- **Predictive optimization:** Anticipate costs and identify efficiency opportunities.
- **Reliable oversight:** Ensure that cost monitoring never lags with streaming backed by service level agreements (SLAs).

Scenario 8: Central Bank Digital Currencies (CBDCs)

Central bank digital currencies (CBDCs) represent a major shift in the way money moves. Issued directly by central banks, they offer fast, secure, and programmable payments for both retail and wholesale use. Unlike traditional systems that rely on batch processing and intermediaries, CBDCs are designed for real-time settlement and continuous availability.

Traditional financial infrastructure often cannot meet the speed, transparency, and reliability CBDCs demand. Batch processing creates latency, reconciliation errors, and risk. A data streaming platform solves this by capturing and validating transactions as they happen across central banks, intermediaries, and applications. This enables instant visibility, lowers risk, and ensures full auditability.

Data streaming also allows central banks to enforce policy in real time. Spending limits or liquidity thresholds can be applied dynamically. Streaming agents monitor the network, detect anomalies, and trigger automated actions to prevent wider disruption. Built-in governance and lineage provide full transparency for regulators.

The architecture integrates legacy systems, wallets, gateways, and analytics into a single, always-on data layer. This ensures interoperability and resilience at national scale.

Business Value

- **Instant settlement:** Real-time processing ensures that CBDC transactions are executed and confirmed immediately, removing clearing delays.
- **Programmable money:** Event-driven logic allows dynamic policy enforcement, such as transaction limits or rate adjustments.
- **Regulatory compliance:** Continuous data lineage and governance provide transparency for regulators and auditors.
- **Interoperability:** Seamlessly integrate across central bank systems, commercial banks, and digital wallets.
- **Resilience and scalability:** Streaming architectures handle nationwide transaction volumes with zero downtime and high reliability.

Scenario 9: Stablecoins

Stablecoins are becoming a key link between traditional finance and the digital asset economy. Pegged to fiat currencies like the US dollar or euro, they offer the stability of conventional money with the flexibility of digital assets. This makes them ideal for instant payments, liquidity transfers, and automated settlements.

Banks and fintechs are using stablecoins for cross-border payments, 24/7 liquidity, and integration with digital commerce. A data streaming platform underpins these efforts by processing every transaction event, like minting, transfers, redemptions, and reserve checks, in real time. This ensures consistent, traceable, and compliant data across blockchain ledgers, reserve systems, and regulatory tools. To maintain full reserve backing, stablecoin systems must sync on-chain and off-chain data. The streaming platform guarantees reliable processing and supports continuous analytics for fraud detection, reserve tracking, and compliance.

As stablecoins move beyond crypto markets into regulated finance, institutions are building their own digital currencies. Industry partnerships are working on tokenized deposits and programmable payments powered by real-time data. A streaming backbone connects these innovations with legacy banking, risk, and compliance systems to build a resilient payment fabric.

Business Value

- **Real-time transparency:** Continuous reconciliation between on-chain and off-chain records ensures that every token is fully backed and auditable.
- **Instant settlement:** Payments, remittances, and liquidity transfers execute instantly without end-of-day reconciliation.
- **Regulatory compliance:** Built-in data governance and lineage support Anti-Money Laundering (AML), Know Your Customer (KYC), and reserve reporting requirements automatically.
- **Interoperability:** A common streaming backbone connects stablecoins, tokenized deposits, and traditional banking systems, ensuring consistent data and integration.
- **Innovation and programmability:** Real-time event data enables programmable payments, AI-driven risk management, and smart-contract automation.

Real-Life Examples

- DNB ASA
- Erste Group Bank
- KOR

"Confluent Platform and Apache Kafka, by enabling us to build and deploy real-time event-driven systems for credit scoring, have helped BRI become the most profitable bank in Indonesia. With event streaming and the ability to capitalize on real-time analytics in merchant services, we tripled agent banking sales. And, thanks to Confluent and Kafka, our fraud detection systems are now able to detect anomalies immediately, so we can take action to shut down fraudulent activities as they are attempted."

Kaspar Situmorang, Executive Vice President, Digital Center of Excellence at BRI

Real-Life Example: DNB ASA

Use Cases: Data Products | IT Modernization | Hybrid and Multicloud | Event-Driven Microservices

Company Overview

DNB ASA is Norway's largest financial services group and one of the leading banks in the Nordics. Headquartered in Oslo, DNB provides retail, corporate, investment banking, insurance, and asset management services. The group is also behind Vipps, the dominant mobile payment solution in Norway, now merged with MobilePay to form a major Nordic payments player.

DNB's long-term vision is to act as a "technology company with a banking license." This means shifting from batch-oriented, siloed IT to a modern digital architecture centered on customer data. The bank is investing in cloud, data streaming, and AI to deliver real-time services and maintain a competitive advantage against fintech startups and global tech companies.

Implementation

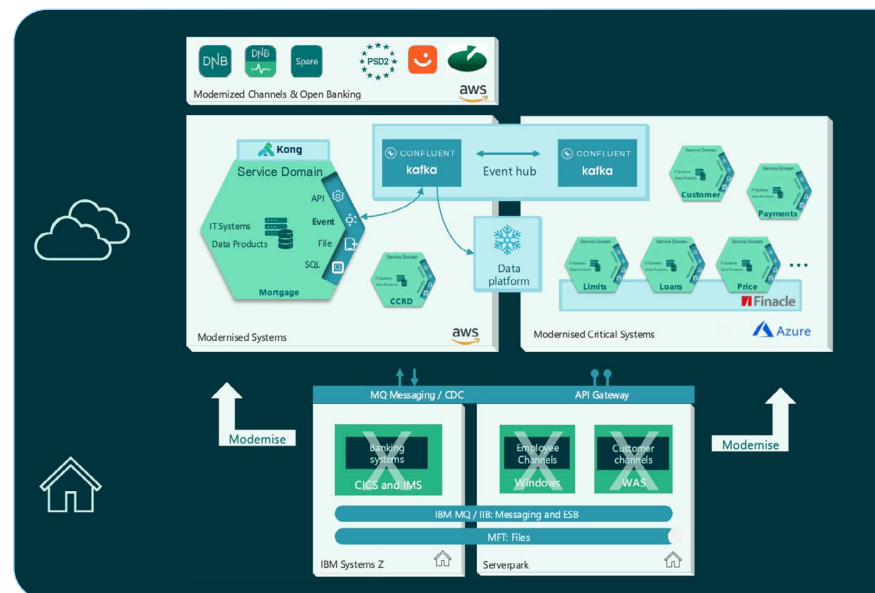
DNB's modernization journey started with an ambitious core transformation program. The bank faced an extensive legacy landscape: more than 60 legacy routines, 11,000 batch jobs, and 500+ microservices across banking, customer, and payments systems. These were primarily hosted on mainframes and tightly coupled architectures such as Citrix, AppNet, and BizTalk.

To overcome these challenges, DNB launched a strategy of standardization and event-driven modernization, anchored by Confluent's data streaming platform. Key elements include:

- 1 **Event Hub Platform:** Built on Confluent, it serves as the backbone for streaming data across operational and analytical estates. It connects mainframes, IBM MQ, Snowflake, and other critical systems.
- 2 **Service Domain Architecture:** It replaces monolithic applications with logical domains that expose capabilities via APIs and Kafka events. This allows teams to decompose legacy workloads into composable services.
- 3 **Real-Time Integrations:** It moves from batch file transfers to continuous event streaming, enabling near-zero RPO/RTO for critical value chains such as mortgages, payments, and customer data.

- 4 **Multicloud Infrastructure:** It uses AWS for digital channels and Microsoft Azure for critical banking systems, with Confluent providing replication and resilience across environments.
- 5 **Data Governance and Metadata Management:** Leveraging schema registry, domain models, and a federated data contract approach ensures discoverability, lineage, and compliance across distributed systems.
- 6 **AI and Analytics Enablement:** It streams operational data into Snowflake and AI platforms, supporting use cases from real-time fraud detection to generative AI-powered customer interactions.

DNB applied a "slice the elephant" approach: modernizing one service domain at a time. For example, in mortgage lending, the bank moved from COBOL-driven batch processes to event-driven workflows. Customer, loan, and payment events are now published in real time, consumed by downstream services, and synchronized with Snowflake for analytics.



Business Value

DNB's new event-driven foundation is delivering both business and IT benefits.

- **Reduced legacy costs:** Gradual decommissioning of mainframe workloads and ETL processes lowers operational costs and reduces reliance on scarce COBOL skills.
- **Improved customer experience:** Real-time personalization and faster loan processing enhance digital engagement and loyalty.
- **Resilience and compliance:** Multicloud deployment with replication ensures high availability and regulatory compliance with near-zero recovery objectives.
- **Faster innovation:** Self-service access to streaming data enables business domains to build new AI, ML, and fraud detection use cases without central bottlenecks.
- **Operational simplicity:** Standardized platforms and tooling, combined with event-driven principles, streamline integration and reduce complexity.
- **Future-proof architecture:** By embedding event streaming at the core, DNB can support generative AI and streaming agents that rely on fresh, contextual data.

Through this transformation, DNB is not only modernizing its IT landscape but also laying the foundation for its evolution into a data-driven, AI-enabled bank.

Real-Life Example: Erste Group Bank

Use Cases: Data Products | IT Modernization | Data Sharing | Data Governance

Company Overview

Erste Group Bank, a leading financial services provider in Central and Eastern Europe, faced escalating costs associated with maintaining its mainframe infrastructure. By offloading data from the mainframe to a cloud-native, microservice-based architecture powered by data streaming, the bank achieved substantial cost savings.

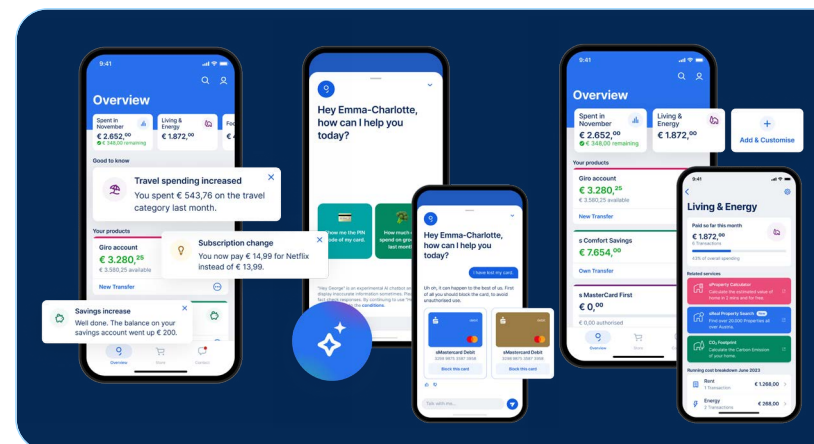
Implementation

Erste's mainframe offloading strategy involved the following steps:

- 1 Data extraction:** Data was extracted from the mainframe using Confluent.
- 2 Data streaming:** The extracted data was streamed in real time to a cloud-native environment.
- 3 Microservices integration:** The data was then processed and managed by a suite of microservices, ensuring seamless integration with existing applications and services.

Erste faced the challenge of maintaining and operating expensive mainframe systems while needing to support modern applications and services. To address this challenge, the bank decided to offload workloads from its mainframe systems to a hybrid cloud environment. The implementation involved leveraging data streaming platforms such as Confluent built on top of Apache Kafka to ensure data consistency between the legacy mainframe systems and the new cloud applications.

The data streaming platform facilitated the real-time integration of data from the mainframe systems with the cloud environment. This enabled the bank to process and analyze data in real time, ensuring that cloud applications had access to the most up-to-date information. The platform also supported the development of APIs that allowed modern applications to interact seamlessly with the legacy systems.



Banking app George by Erste Group Bank

Business Value

- Cost reduction:** The transition allowed the bank to reduce the number of reads on the mainframe, significantly cutting down on fixed infrastructure costs.
- Compliance and innovation:** The bank maintained compliance with regulatory requirements and business logic while enabling the creation of new applications using the same event-based architecture.
- Operational efficiency:** The new architecture enhanced the agility of IT operations, allowing for faster deployment of new services and ensuring data consistency across the mainframe, databases, and customer-facing applications.

"We currently have 214 different projects on our production cluster—1,691 topics, with approximately 50,000 messages per second incoming at around 75 MB per second and outgoing at approximately 100 MB per second. This last metric is our best indicator of the rising adoption of a data sharing platform: Data is read more often than written, i.e., data is produced for multiple consumers."

Mathias Frey, Chief Engineering Technology Officer, Erste Group Bank

Real-Life Example: KOR

Use Cases: IT Modernization | Observability | Data Governance

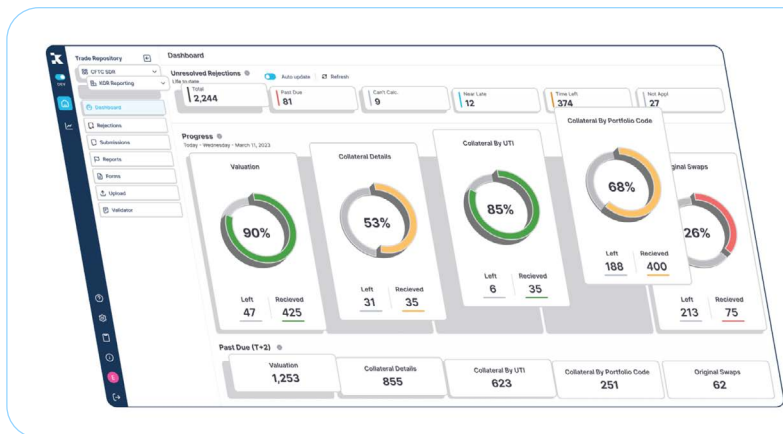
Company Overview

KOR, a leading trade data repository and reporting service provider for the global derivatives markets, has successfully implemented data streaming technologies to enhance its regulatory reporting capabilities. By using Confluent, KOR has created a robust and efficient regulatory reporting environment that supports real-time data processing and analytics.

Implementation

KOR's regulatory reporting platform continuously streams data from client transaction and life cycle data, third party trade activity, and reference data to its cloud-based platform.

Advanced algorithms process the data in real time to ensure accuracy, identify discrepancies, and generate regulatory reports. Real-time data validation with analytics and support of complex scenario processing set KOR apart from competitors using traditional methods.



Real-time dashboards at KOR

Dashboards and alerts allow operations and compliance to monitor data streams, issues, and rejections to ensure timely and accurate reporting. The data streaming platform also supports ongoing maintenance and updates, ensuring that KOR can focus on innovation and customer service.

Business Value

- **Transparency:** Real-time data validation and transparency improve the accuracy and reliability of regulatory reports.
- **Enhanced compliance:** Continuous monitoring and real-time analytics ensure compliance with regulatory requirements.
- **Operational efficiency:** Complex trading and reporting scenarios are efficiently handled, along with their remediation actions.
- **Scalability and flexibility:** Cloud-based data streaming platform provides the scalability and flexibility needed to handle large volumes of data and support business growth.
- **Reduced risk:** Real-time data processing and advanced analytics reduce the risk of non-compliance and associated penalties.
- **Improved customer trust:** Enhanced regulatory reporting capabilities improve customer trust and satisfaction.

"At KOR, we have a very specific problem that we are trying to solve, which is collecting trading information for regulators. And we decided to do it differently to the way that most people are doing it. Where others would be using Kafka merely as a message bus and using a data lake as the primary data store, we decided to use Kafka as our system of record. We are building our system to store petabytes in Confluent Cloud and maintain the flexibility to materialize projections based on that. So it's a long retention use case."

Andreas Evers, Chief Technology Officer, KOR

Conclusion

The financial services industry is experiencing a profound transformation, with data streaming, AI, and cloud-native solutions driving innovation across customer experiences, business models, and operational efficiencies.

Key trends are reshaping the landscape: Open banking ecosystems, digital-only banks, and embedded finance solutions are creating new opportunities for revenue and collaboration. Customers now expect mobile-first experiences, instant payment processing, real-time fraud alerts, personalized wealth management, and virtual financial assistants, requiring financial institutions to operate at the cutting edge of technology.

Real-world examples illustrate the power of these trends. Erste Group Bank modernized its operations by offloading mainframe workloads to a hybrid cloud environment, achieving significant cost savings and operational agility. KOR has transformed regulatory reporting by integrating real-time data validation and AI-driven analytics into its cloud-based platform, ensuring transparency, compliance, and operational efficiency. Other institutions are extending these capabilities into emerging areas such as embedded finance, digital currencies, and real-time cloud cost management, showing the breadth of possibilities unlocked by data streaming.

Data streaming also optimizes internal efficiencies through automated compliance monitoring, AI-driven risk management, and predictive analytics for market trends. Fraud detection and risk monitoring systems built on data streaming platforms now provide continuous protection at scale, minimizing financial losses and strengthening institutional integrity.

Streaming agents take this further by applying contextual intelligence directly to live financial data. They can detect anomalies, recommend credit adjustments, and automate compliance actions in milliseconds. This shifts institutions from reacting to yesterday's reports to actively shaping outcomes in real time.

These advancements demonstrate the critical role of data streaming in addressing industry challenges. By adopting scalable, resilient, and innovative architectures, financial institutions can reduce costs, maintain compliance, and enhance customer satisfaction. Transparency, scalability, and flexibility enable institutions to handle vast data volumes while driving rapid innovation and operational excellence.

The convergence of these trends highlights a broader imperative: Financial institutions must leverage technologies such as data streaming, AI, and hybrid cloud architectures to remain competitive. They can redefine their market leadership by creating real-time customer experiences, optimizing internal processes, and embracing new business models such as embedded finance, programmable money, and AI-powered real-time decisioning. Companies such as Erste Group Bank and KOR are already exemplifying what's possible, showing that the future of financial services is built on a foundation of real-time insights, operational agility, and relentless innovation.

3.2 Manufacturing and Automotive

Few industries integrate the physical and digital worlds as effectively as industrial manufacturing and the automotive sector. The potential of digitalization is significant. Industry representatives have been discussing "Industry 4.0" for more than a decade, which now comes to life. And the transformation is accelerating, driven not only by smart factories but also by new business models such as equipment-as-a-service (EaaS) and mobility-as-a-service (MaaS).

Green manufacturing focuses on reducing environmental impact through sustainable practices and energy-efficient technologies. Digital twins, which are virtual replicas of physical assets, enable real-time monitoring and optimization of manufacturing processes. Cybersecurity and data governance are becoming increasingly important as manufacturers adopt more connected and automated systems, making them vulnerable to cyber threats and regulatory risks.

The integration of IoT devices and advanced analytics is revolutionizing the

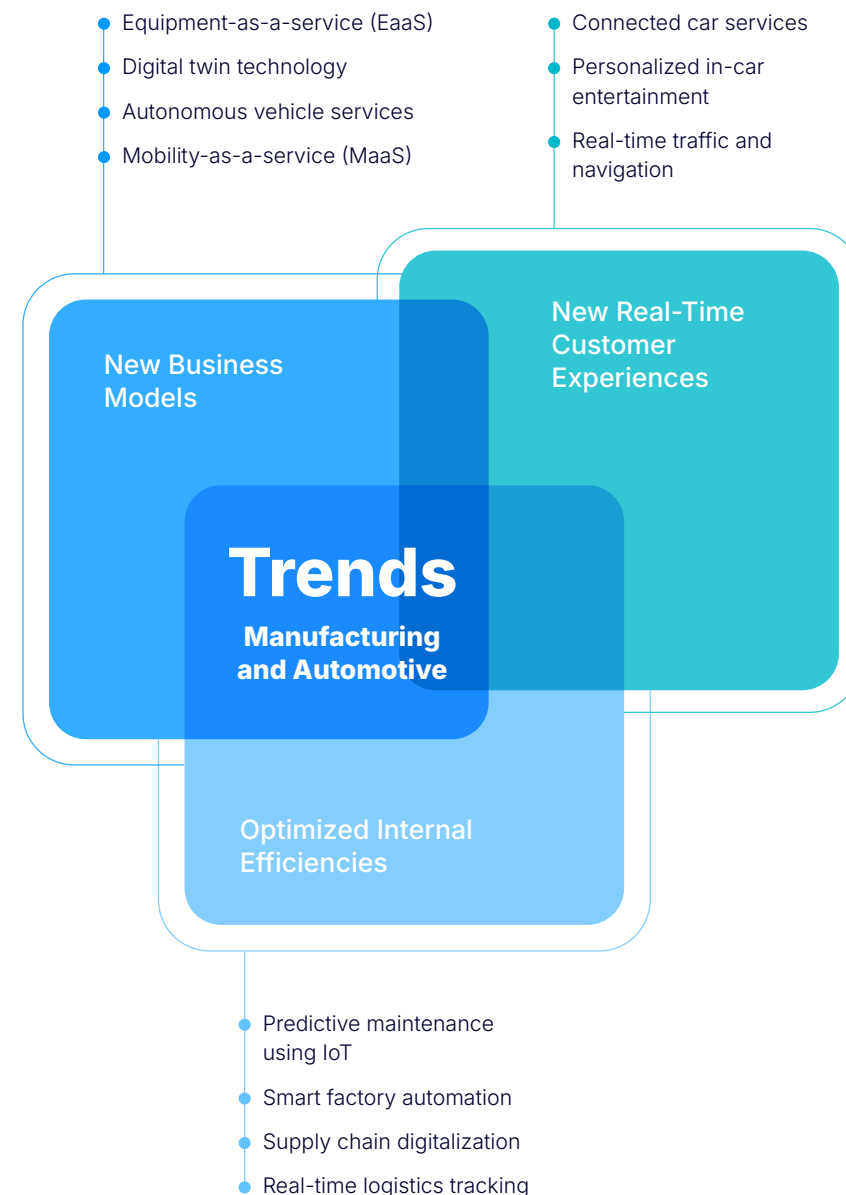
way manufacturers monitor and maintain their equipment on the shop floor. Predictive maintenance, often powered by AI, allows manufacturers to predict equipment failures before they occur, reducing downtime and improving OEE.

Beyond the factory floor, data streaming is transforming how products are designed, operated, and serviced. Real-time integration between engineering systems accelerates product development, while connected vehicles and equipment generate continuous streams of data that open new revenue models, from usage-based insurance to predictive servicing.

Supply chain management is also being transformed through the use of data pipelines and IT modernization, providing end-to-end visibility and enabling more efficient operations. Track-and-trace solutions and real-time locating systems leveraging IoT and advanced applications offer real-time insights into the entire supply chain, enhancing efficiency and transparency.

"We chose Confluent Cloud not only because of the best cost/benefit ratio but also because it best fits our needs, which were data store capability ... near-real-time data provisioning ... and easy connectivity."

Stefan Baer, IT Architect and Solution Integrator, Siemens



Scenario 1:

Condition Monitoring and Predictive Maintenance

Condition monitoring and predictive maintenance involve continuously tracking shop floor equipment to predict and prevent failures. IoT sensors and advanced analytics collect real-time data on equipment performance, identify patterns, and predict when maintenance is needed.

If this data is then streamed and analyzed in real time on a central platform, it can be used to detect anomalies and automatically alert the maintenance team to address the issue before it leads to a breakdown. This proactive approach minimizes downtime, lowers maintenance costs, and enhances OEE.

Business Value

- **Minimized unplanned downtime:** By predicting potential equipment failures before they occur, manufacturers can schedule maintenance proactively, reducing unexpected breakdowns and associated downtime.
- **Reduced maintenance costs:** Transitioning from reactive to predictive maintenance allows for more efficient use of resources, reducing the frequency and cost of maintenance activities.
- **Improved OEE:** Continuous monitoring and timely maintenance ensure that equipment operates at optimal efficiency, enhancing productivity and maximizing output.
- **Extended equipment lifespan:** Regular monitoring and maintenance based on actual equipment condition help extend the lifespan of machinery, delaying the need for costly replacements.
- **Enhanced safety and compliance:** Predictive maintenance ensures that equipment is always in safe working condition, reducing the risk of accidents and ensuring compliance with safety regulations.
- **Increased production quality:** Well-maintained equipment operates more reliably, leading to consistent production quality and reducing the likelihood of defects.
- **Sustainability goals:** By optimizing equipment performance and reducing waste, predictive maintenance contributes to sustainability initiatives and reduces environmental impact.

Scenario 2:

Supply Chain

Supply chain management involves coordinating and optimizing various processes such as transportation, inventory management, and order fulfillment. Data pipelines and IT modernization enable end-to-end visibility across the entire supply chain for more efficient operations and better decision-making. A truly decoupled, event-driven choreography brings more flexibility and scalability than a central orchestration.

Business Value

- **Enhanced visibility and transparency:** Achieve real-time, end-to-end visibility across the entire supply chain, allowing stakeholders to monitor and track the movement of goods and materials seamlessly.
- **Improved decision-making:** Access to real-time data enables more informed and timely decision-making, reducing the risk of errors and enhancing strategic planning.
- **Increased operational efficiency:** Streamlined processes and automated workflows reduce manual intervention, leading to faster order fulfillment and reduced operational costs.
- **Reduced lead times:** Real-time data processing and analytics help in predict demand and optimize inventory levels, thereby reducing lead times and improving customer satisfaction.
- **Cost reduction:** Optimizing transportation routes and inventory levels significantly reduces logistics and warehousing costs.
- **Enhanced collaboration:** Improved data sharing and communication across different departments and partners foster better collaboration and alignment of supply chain activities.
- **Sustainability and compliance:** Efficient resource management and reduced waste contribute to sustainability goals, while real-time data ensures compliance with regulatory requirements.
- **Customer satisfaction:** Faster and more reliable order fulfillment enhances the overall customer experience, leading to increased loyalty and repeat business.

Scenario 3: Track-and-Trace

Track-and-trace solutions involve the real-time monitoring of tools, equipment, and products throughout the supply chain. By leveraging IoT devices and advanced applications, manufacturers can gain visibility into the location and status of their assets, ensuring transparency and accountability. This real-time visibility helps prevent losses, improve inventory management, and enhance overall supply chain efficiency.

Business Value

- **Enhanced visibility and transparency:** Real-time tracking of tools, equipment, and products provides complete visibility across the supply chain, ensuring transparency and accountability at every stage.
- **Improved inventory management:** Accurate, real-time data on asset location and status helps optimize inventory levels, reducing overstock and stockouts and improving inventory turnover.
- **Loss prevention:** Continuous monitoring helps prevent losses due to theft, misplacement, and damage by providing timely alerts and insights into asset movements.
- **Increased supply chain efficiency:** Streamlined tracking processes reduce delays and inefficiencies, leading to faster order fulfillment and improved supply chain responsiveness.
- **Better customer satisfaction:** Real-time updates on product location and status enhance customer communication and satisfaction by providing accurate delivery estimates and reducing uncertainty.
- **Regulatory compliance:** Track-and-trace solutions ensure compliance with industry regulations and standards by maintaining accurate records of product movements and status.

Scenario 4: Engineering Tool Integration for Product and Equipment Design

Manufacturers depend on many engineering and business systems, such as product life cycle management (PLM) systems, computer-aided design (CAD) for product design, ERP for procurement, and MES for production. These systems are often connected through fragile point-to-point integrations, batch file exchanges, or custom plugins. This creates data silos, drives up integration costs, and causes delays and inconsistencies across departments.

A data streaming platform provides a unified, decoupled backbone that connects all engineering and manufacturing tools in real time. Instead of waiting for batch jobs, design changes, bill of material updates, and simulation results are streamed instantly into downstream systems like PLM, ERP, and MES. Data streaming also allows for in-flight processing for filtering irrelevant data, transforming formats, detecting anomalies, and correlating updates before publishing.

For example, a CAD update that changes a component dimension can trigger a PLM update to adjust documentation, an ERP signal to update supplier requirements, and an MES alert to reschedule production. This keeps every team aligned with the latest, validated data - without relying on fragile custom integrations.

Business Value

- **Faster time to market:** Reduce delays in the product life cycle with immediate synchronization between CAD, PLM, ERP, and MES.
- **Improved product quality:** Minimize errors and design mismatches with consistent, validated data across all systems.
- **Lower integration costs:** Replace fragile point-to-point connections with a scalable backbone that reduces maintenance overhead.
- **Operational agility:** Support both legacy batch workflows and real-time, event-driven processes, enabling smooth coexistence during modernization.
- **Risk and compliance management:** Maintain auditability and traceability across engineering data exchanges, reducing regulatory and quality risks.
- **Cross-team collaboration:** Ensure that all stakeholders (design, procurement, production, and suppliers) have access to the latest data to make better decisions.

Scenario 5: Connected Vehicles and Real-Time Car APIs

Vehicles today are no longer standalone machines. They are connected, software-defined platforms that continuously generate diagnostic, performance, and contextual data. Automotive APIs expose valuable information such as vehicle specifications, service history, and telemetry, but traditional API-driven architectures rely on periodic calls and siloed integrations. This creates delays, fragmented datasets, and limited reusability, making it difficult to scale modern mobility applications.

By combining car APIs with a data streaming platform, original equipment manufacturers (OEMs) and mobility providers can transform raw API calls into governed, real-time data products. Streaming ensures that every diagnostic update, location signal, or ownership change is captured instantly, enriched with context and distributed consistently to all relevant systems. This enables fleet managers, insurers, dealerships, and smart city platforms to operate on fresh, accurate, and reusable data streams rather than rely on stale snapshots or one-off integrations.

Business Value

- **Real-time insights:** Continuous vehicle telemetry supports live fleet tracking, predictive servicing, and usage-based insurance.
- **Reduced integration effort:** Shared data products eliminate redundant point-to-point connections across teams and partners.
- **Scalability:** Elastic streaming infrastructure supports millions of vehicles and billions of events per day.
- **Improved customer experience:** Personalized services, instant diagnostics, and proactive maintenance notifications optimize the customer experience.
- **Operational efficiency:** A unified, event-driven backbone replaces fragmented API pulls with continuous, high-quality streams.
- **Revenue growth:** Standardized vehicle data products can be securely shared with ecosystem partners to unlock new revenue streams and monetization opportunities.

Scenario 6: Real-Time Quality Intelligence and Plant Standardization

Manufacturers must detect production and equipment issues early to avoid rework, waste, and compliance failures. Traditional batch-based processes delay visibility, letting defects move downstream and increasing validation effort.

A data streaming platform delivers real-time visibility by connecting sensors, MES, ERP, and quality systems into a single backbone. Quality inspections, compliance checks, and defect tracking rely on timely data. Streaming pipelines process events in seconds, detect anomalies, trigger alerts, and enable corrective actions before defective products leave the line. Instead of waiting for batch reports, supervisors can monitor live dashboards showing units produced, machine status, and inspection results.

Manufacturers often struggle with inconsistent data across plants, ranging from legacy air-gapped systems to modern cloud platforms. This fragmentation makes it hard to get a clear, real-time view across different production sites. A data streaming platform solves this by standardizing communication and logic into a unified layer, enabling global benchmarking and clearer performance visibility.

Business Value

- **Early defect detection:** Identify quality issues within seconds to prevent rework, scrap, and production delays.
- **Cost reduction:** Minimize waste and manual effort through real-time monitoring and automated alerts.
- **Standardized operations:** Establish a single data and communication layer across plants, regardless of differences in MES, ERP, PLM, supervisory control and data acquisition (SCADA), or other production and quality systems.
- **Real-time performance tracking:** Monitor production output, machine status, and inspection results instantly instead of waiting for batch updates.
- **Simplified validation:** Replace fragmented systems and spreadsheets with a single, governed stream of trusted quality data.
- **Faster modernization:** Support cloud-connected plants worldwide with consistent analytics and process control, enabling manufacturers to modernize at their own pace. Regional sites can meet local requirements while contributing real-time data to a unified global view of production performance.

Real-Life Examples

- PAUL Tech AG
- Knauf
- Siemens
- BMW Group

"With Confluent, we are transforming this very batch-based, very monolithic information system into one in which data is constantly in motion. It really helps us to decouple our monolith into autonomous systems, to help them evolve independently of each other, and therefore, to be more agile for our business to drive real-time, data-driven decisions and operations. It is data in motion serving business agility."

Olivier Jauze, IT Architect and CTO, Mastero Marketplace, a service line in Michelin

Real-Life Example: PAUL Tech AG

Use Cases: Hybrid and Multicloud | IoT | ESG

Company Overview

PAUL Tech AG, a pioneering force in the realm of smart building technology, is committed to creating a sustainable future by optimizing energy consumption in buildings. With a focus on reducing carbon emissions, PAUL Tech AG has positioned itself as a leader in the industry by leveraging cutting-edge technology to drive significant environmental impact. In 2023 alone, the company is projected to save approximately 21,500 tons of CO₂, a feat equivalent to the carbon absorption capacity of 10,000 square meters of forest. This achievement underscores the company's dedication to sustainability and its innovative approach to energy management.

Implementation

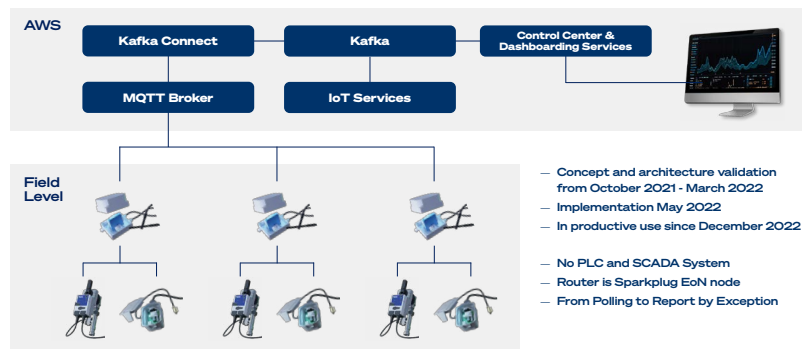
In today's world, where energy conservation and sustainability are paramount, PAUL Tech AG identified a critical challenge: the need to reduce energy consumption in buildings by up to 40% without compromising the comfort of occupants. Traditional methods of energy optimization often require significant physical modifications to existing infrastructure, which can be costly and disruptive. PAUL Tech AG sought a solution that allows for seamless integration with existing systems to leverage data-driven insights to enhance efficiency and sustainability.

To address this challenge, PAUL Tech AG turned to data streaming technology with Confluent. The company implemented an adaptive hydraulic balancing system in drinking water and heating systems, powered by AI to analyze and optimize the current state of the entire system.

This AI-driven approach considered various factors such as consumption behavior, vacancy, weather conditions, sun orientation, and time of day or season, combining billions of data points from other buildings to deliver optimal results.

The integration process involved connecting IoT devices via MQTT from the edge to Confluent Cloud, facilitating the flow of IoT data into the data streaming platform. The right configuration ensured that data from the same edge node was stored together, maintaining the true sequence of events for subsequent processing. Once the data was ingested, stream processing applications leveraged the company's machine learning models to optimize actuator values. These optimized values were then sent back to the IoT devices.

New Architecture 2022



Business Value

- **Significant energy savings:** Optimizing energy consumption through real-time data processing and machine learning enabled PAUL Tech AG to achieve up to 40% energy savings in buildings.
- **Enhanced ESG and sustainability:** Reduced energy consumption improved ESG ratings, reinforcing PAUL Tech AG's commitment to sustainability and responsible corporate practices.
- **Lower operational costs and financial advantage:** Energy efficiency and improved ESG ratings contributed to reduced operational costs and favorable refinancing rates.
- **Future-proofed infrastructure:** Data-driven precision in heating system sizing and adaptability for alternative heat sources prepared buildings for future energy innovations.

Real-Life Example: Knauf

Use Cases: Data Products | Hybrid and Multicloud | Event-Driven Microservices | Data Sharing

Company Overview

Knauf is a family-owned manufacturing group headquartered in Iphofen, Germany. Founded in 1932, the company has grown into one of the world's leading providers of building materials and construction systems, with more than 43,000 employees and operations in more than 90 countries. While Knauf is known for high-quality products in drywall, insulation, and plaster systems, its global scale also demands strong digital engagement to support marketing, customer communication, and brand visibility.

Implementation

Knauf's marketing and communications teams faced a challenge: content updates in their content management system were not instantly visible across systems, especially in the global search functionality on knauf.com. For a manufacturer operating worldwide, where campaigns, product launches, and safety updates must reach customers, partners, and employees without delay, waiting minutes or hours for updates was no longer acceptable.

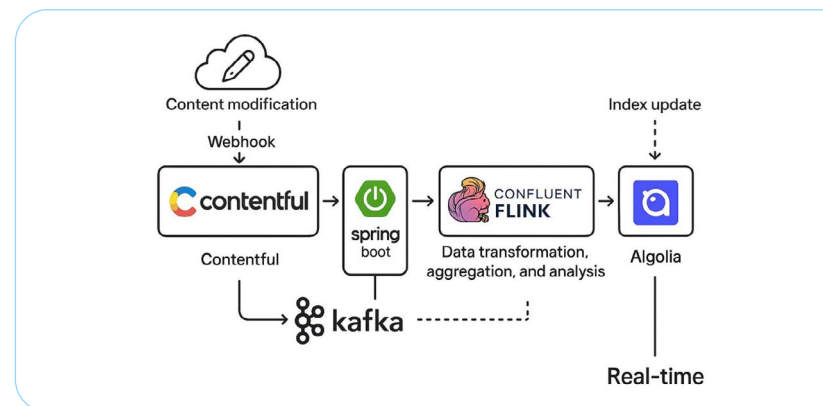
Each day, around 200 editors work on content across Knauf's digital properties, generating up to 3.3 million connections per day. To ensure consistency, Knauf built a real-time data streaming solution using Confluent.

The system connects Knauf's content management platform with a real-time streaming backbone. Whenever editors make updates, changes are captured instantly and passed through a custom application into the data streaming platform. From there, the data is processed and refined in real time before being pushed into the global search and AI-driven platforms. This ensures that information is always fresh and consistent across Knauf's digital channels, so users worldwide see the latest updates without delay.

The approach transformed content operations from a slow, batch-driven process into a seamless, near-real-time experience. Marketing campaigns

"We went from ~20,000 visitors a day to ~3M visitors a day, and the content did not suffer lags or delays between creation and content consumption."

Eliei Lima Oliveira, DevSecOps Lead, Knauf Digital



Data transformation, aggregation, and analysis in Flink at Knauf

can now be executed and reflected in digital channels without delay, keeping pace with modern expectations and supporting Knauf's global brand presence in the manufacturing industry.

Business Value

- **Faster marketing campaigns:** Real-time streaming enables immediate updates to digital channels, ensuring that global campaigns launch without delays.
- **Seamless global search:** Integration with the internal search platform provides consistent, up-to-date content visibility across Knauf's websites.
- **Operational efficiency:** Zero-touch integration between CMS and the data streaming platform reduces manual intervention and improves editor productivity.
- **Scalability for global operations:** The system supports ~200 editors and millions of daily connections, ensuring reliability at enterprise scale.
- **New standard for speed:** Millisecond response times raise expectations across teams and establish real-time operations as the baseline for Knauf's digital engagement.

Real-Life Example: Siemens

Use Cases: Data Products | Hybrid and Multicloud | Event-Driven Microservices | Data Sharing | AI | IoT

Company Overview

Siemens is one of the world's leading technology companies—focused on industry, infrastructure, transport, and healthcare—that transforms the everyday lives of billions of people. By combining the real world of buildings, grids, factories, and trains with the digital world of technologies such as AI and digital twins, Siemens helps customers accelerate their digital transformation and become more competitive, resilient, and sustainable.

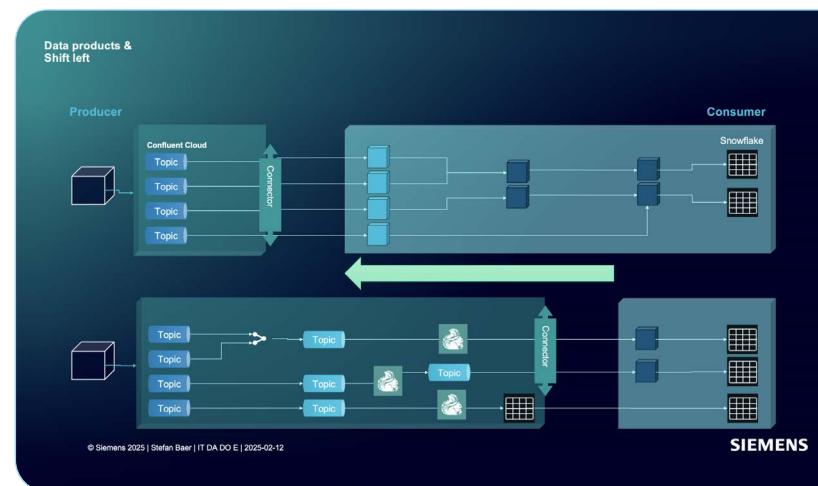
As part of this journey, Siemens Digital Industries is reinventing its approach to data. Fragmented deployments of open source Apache Kafka and Amazon MSK had created inefficiencies, high costs, and governance gaps. Data updates took weeks to reach critical systems, hampering innovation and agility. To address these challenges, Siemens turned to Confluent Cloud to establish a unified data backbone and accelerate its digital transformation.

Implementation

Siemens replaced its fragmented Kafka deployments with Confluent Cloud, creating a unified streaming backbone that cut costs and simplified governance. Teams no longer manage separate clusters; instead, Siemens offers topic-as-a-service giving engineers self-service access to data streams without dealing with infrastructure.

A central design principle is the shift-left architecture. Instead of waiting until data lands in warehouses to apply transformations, Siemens processes and enriches data directly in the streaming layer using Confluent. This ensures higher-quality, reusable data products that feed both operational systems and analytics in real time.

One example is the modular intralogistics platform, which orchestrates material flows in factories. Previously tied together with rigid, point-to-point integrations, logistics, ERP, and manufacturing systems now interact through Kafka events. Stream processing with Apache Flink® reacts instantly to changes for rerouting deliveries, triggering alerts, and adjusting workloads with AI.



"Data streaming creates a single point of truth between the physical and virtual worlds. It's transforming our factory digitalization efforts and unifying operational and information technology into a single source of information. Confluent, with its extensive connector ecosystem, plays a crucial role in making these integrations possible."

Jose Carlos Mendes Neves, Service Manager, Siemens

Business Value

Siemens' transformation with Confluent delivered tangible results across cost, agility, and innovation.

- **Significant cost savings:** Consolidating fragmented clusters into a managed backbone reduced total cost of ownership and eliminated inefficiencies.
- **Faster time to market:** Product updates that once required four weeks are now distributed in two days, accelerating responsiveness to customer needs.
- **Operational efficiency:** Real-time logistics orchestration, quality inspection feedback, and automated job execution boost productivity across factories.
- **Connected IT and OT systems:** Seamless integration between industrial devices and enterprise platforms enables predictive maintenance and digital twins.
- **Scalable, future-proof platform:** Confluent Cloud handles millions of daily events with minimal resource use, supporting Siemens' AI and Industry 4.0 ambitions.

By combining real-time data streaming with shift-left principles, Siemens has not only optimized its current operations but also created a robust foundation for the next generation of digital factories.

Real-Life Example: BMW Group

Use Cases: Hybrid and Multicloud | AI | IoT

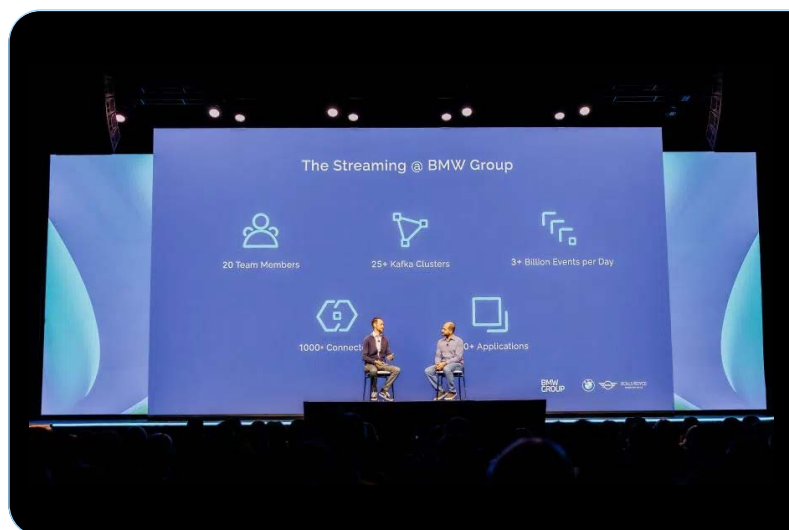
Company Overview

Founded as Bayerische Motoren Werke GmbH in 1916, today, the BMW Group is the world's leading premium manufacturer of automobiles and motorcycles. The Group operates 31 plants in 15 countries, with a global sales network in 140 countries

Implementation

Inside BMW Group's state-of-the-art plants, data is collected from all the critical minutiae of manufacturing, including machine temperatures, machine performance, and location of parts on the assembly line. Data is also continuously produced and consumed by sales and the research and development (R&D) operation.

Making the abundance of real-time Internet of Things (IoT) data available to all teams that want to put it to use is a critical role for BMW Group's Integration Platforms organization to drive innovation through use cases such as enabling operational applications that streamline factory production and providing real-time data to the R&D team as they create and test new products.



Data flows continuously through Confluent via a combination of commercial solutions and custom code, using streaming ETL—the process of continuously transforming, filtering, and enriching data as it moves. As Confluent's Infinite Storage persists the incoming events, BMW Group can also replay historical data later to enable additional use cases. Any asset that produces data is connected to Kafka, with multiple different application types using the same data to trigger different, specific processes.

BMW Group operates a complex infrastructure, with a large number of SAP systems, a lot of legacy software, plus third-party developed connectors and tools. The company now has dozens of clusters, hundreds of connected applications, and over one billion events produced per day—all with Confluent functioning as the central nervous system for this continuous river of critical business data. Highly scalable teams at BMW Group are onboarding multiple new applications and use cases every week.



"BMW Group's successful implementation of Confluent has already resulted in increased speed, flexibility, cost savings, and simplicity. Yet the company is ambitious and keen to expand the use of the data processing platform."

BMW Group

Business Value

The implementation of data streaming technology has provided significant business value for BMW Group.

- **Enhanced operational efficiency:** Real-time data processing and analytics enable BMW Group to optimize production processes, reduce downtime, and improve overall equipment effectiveness (OEE).
- **Cost and risk reduction:** By leveraging fully managed cloud service for data processing and eliminating the need for manual infrastructure management, BMW Group has reduced operational costs and risks.
- **Improved innovation and flexibility:**
The scalable nature of Confluent allows BMW Group to accelerate project timelines and increase opportunities for innovation. The ability to process vast amounts of data in real time supports continuous IoT innovation and enhances customer engagement.
- **Ubiquitous access to real-time data:** Confluent provides BMW Group with a powerful platform for accessing, storing, and managing data. Self-service access to data products enables teams to make informed decisions quickly, driving business growth, efficiency, and faster time to market.

Conclusion

The manufacturing and automotive industries are leading a new wave of digital transformation in which the physical and digital worlds operate as one. Real-time data streaming connects machines, systems, and people across the shop floor, creating an intelligent ecosystem that powers continuous optimization and insight. On this foundation, new service models such as EaaS and MaaS open new revenue streams and strengthen customer relationships.

This transformation extends far beyond factory walls. Data streaming unites engineering design, production, supply chains, and connected products into a continuous digital loop. Smart factory workflows, predictive maintenance, and supply chain visibility deliver efficiency, resilience, and sustainability at scale. AI enhances this loop with predictive intelligence that detects anomalies, optimizes machine performance, and improves product quality in real time.

Agentic AI now builds on these capabilities by acting autonomously. Streaming Agents monitor sensor data, flag anomalies, and trigger actions such as adjusting equipment settings or rerouting production tasks. This evolution turns factories and mobility platforms into self-optimizing systems that adapt instantly to change, improving safety, uptime, and productivity.

Real examples show the scale of this transformation. BMW Group processes more than a billion events each day

across its global plants. With Confluent as its backbone, BMW Group achieves faster innovation, lower costs, and greater flexibility, connecting IT and OT to realize the vision of Industry 4.0. BMW Group has unified its fragmented data landscape into a governed streaming platform that powers digital factories. Material flows, logistics, and quality data move in sync, improving efficiency and collaboration across operations.

PAUL Tech AG proves that sustainability and profitability can align. Its AI-driven analytics platform optimizes energy consumption in buildings, reducing costs and emissions by up to 40%.

Knauf demonstrates that data streaming goes beyond the factory floor by transforming digital engagement. Real-time integration between content systems and search platforms ensures instant updates and a consistent customer experience across global digital channels.

Together, these stories show how manufacturers and mobility providers are redefining their industries. Data streaming, AI, and agentic AI enable continuous collaboration, automation, and learning across every layer of the value chain. Organizations that invest in this foundation today will gain the agility, sustainability, and resilience needed to lead in tomorrow's connected economy.

3.3 Retail and eCommerce

The retail industry is experiencing a wave of digital transformation driven by the need for omnichannel experiences and unified commerce, automation through AI, retail media networks, and the influence of social media. Unified commerce integrates various shopping channels, providing a seamless customer experience across online and offline platforms. Automation through AI enables retailers to optimize operations, personalize customer interactions, and improve decision-making. Retail media networks allow companies to monetize their digital platforms by offering advertising opportunities to brands.

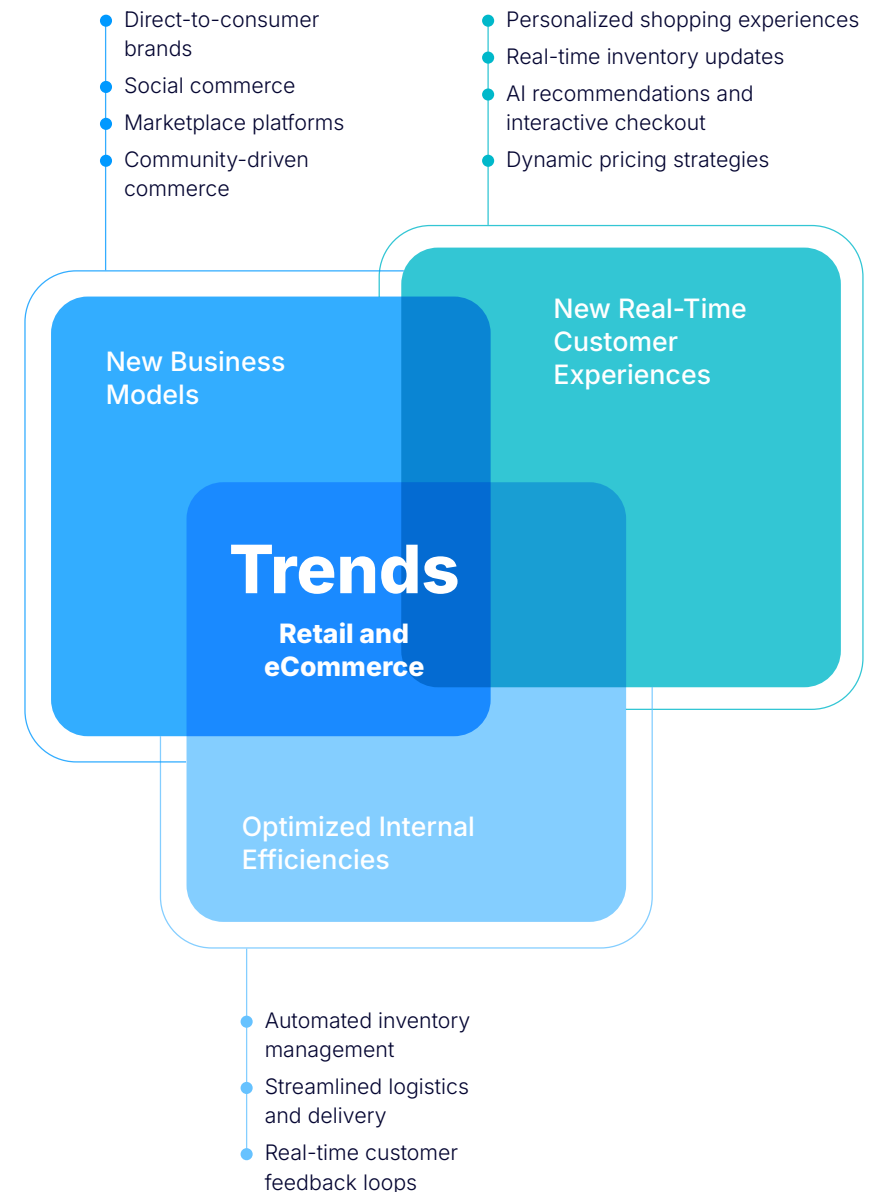
Social media plays a crucial role in shaping consumer behavior and driving engagement through interactive and personalized content.

Data streaming has become the foundation for this transformation by enabling real-time insights, personalized offers, and instant interactions across every channel. From inventory forecasting to live commerce and retail media networks, streaming connects fragmented systems and allows retailers to react instantly to customer signals.

Generative AI extends personalization further by powering conversational shopping assistants and dynamic product campaigns, while agentic AI enhances operational agility by autonomously adjusting promotions, replenishing stock, or detecting fraud as events unfold. Together, these technologies create a retail environment that is more adaptive, intelligent, and customer-centric than ever before.

"At Victoria's Secret, data is key for empowering agile decision making, we receive data from various systems, including logistics, stores, and digital systems"

Harikumar Venkatesan, Platform Engineering Lead, Victoria's Secret



Scenario 1: Inventory Management

Inventory management ensures optimal availability while reducing costs through tracking and replenishment. With data pipelines and IT modernization, retailers can use real-time data from point-of-sale systems, supplier databases, and demand patterns to enhance forecasting and replenishment processes.

Real-time data streaming helps retailers reduce excess inventory, minimize stockouts, and improve overall operational efficiency, resulting in significant cost reductions.

Business Value

- **Improved forecasting accuracy:** Real-time data from various sources enhances forecasting models, allowing retailers to predict demand more accurately and adjust inventory levels accordingly.
- **Optimal stock levels:** By maintaining the right balance of inventory, retailers can reduce excess stock and minimize stockouts, ensuring product availability without overstocking.
- **Reduced inventory costs:** Efficient inventory management reduces holding costs and minimizes the financial impact of unsold stock, leading to significant cost savings.
- **Enhanced replenishment processes:** Real-time data processing enables more efficient and timely replenishment, ensuring that products are available when and where they're needed.
- **Increased operational efficiency:** Streamlined inventory processes reduce manual intervention and errors, improving overall operational efficiency and productivity.
- **Better supplier collaboration:** Real-time data sharing with suppliers enhances collaboration and coordination, leading to more efficient supply chain operations.
- **Improved customer experience:** By ensuring that products are consistently available and reducing stockouts, retailers can enhance customer satisfaction and loyalty, leading to repeat business and positive brand perception.

Scenario 2: Loyalty Platforms

Retailers implement loyalty platforms powered by data streaming to significantly enhance customer engagement and drive incremental revenue. Data streaming technologies play a crucial role in this process by continuously collecting and processing real-time data from various sources, such as transaction history, membership activity, and feedback channels. This enables companies to create dynamic and responsive loyalty programs that offer points, rewards, and exclusive benefits tailored to individual customer behaviors and preferences. With real-time transaction monitoring, retailers can instantly update loyalty points and rewards balances, ensuring that customers receive timely recognition for their loyalty. Additionally, the integration of instant feedback allows retailers to quickly adapt loyalty program features to better meet customer expectations, enhancing overall satisfaction.

Business Value

- **Enhanced customer loyalty:** Real-time data processing allows retailers to offer timely and relevant rewards, fostering a sense of appreciation and loyalty among customers.
- **Increased customer engagement:** Dynamic loyalty programs encourage active participation, as customers are motivated to earn and redeem rewards, leading to higher engagement levels.
- **Improved brand perception:** Offering exclusive benefits and personalized rewards enhances the brand's image, making it more attractive to both existing and potential customers.
- **Data-driven insights:** Real-time data collection and analysis provide valuable insights into customer behaviors and preferences, enabling retailers to refine their loyalty strategies and offerings.
- **Competitive advantage:** A well-executed loyalty platform differentiates the brand in a crowded market, attracting more customers and retaining existing ones.
- **Additional revenue:** Effective loyalty programs can drive additional spend and can be monetized through business partnerships.

Scenario 3: Personalization

Data streaming technologies empower retailers to offer personalized shopping experiences to their customers. By processing real-time data from diverse sources such as purchase history, browsing behavior, and demographic information, companies can customize product recommendations, marketing messages, and shopping experiences to align with individual preferences. This capability in real-time data processing enables retailers to boost customer satisfaction, elevate conversion rates, and foster stronger customer relationships.

Business Value

- **Personalized customer experience:** Real-time data processing enables retailers to offer tailored product recommendations and marketing messages, creating a more engaging and relevant shopping experience.
- **Increased conversion rates:** Personalized offers and recommendations drive higher conversion rates, as customers are more likely to purchase products that align with their interests and needs.
- **Enhanced customer satisfaction:** By delivering a shopping experience that resonates with individual preferences, retailers can improve customer satisfaction and encourage repeat visits.
- **Stronger customer relationships:** Personalization fosters a deeper connection between the brand and its customers, leading to long-term relationships and increased loyalty.
- **Optimized marketing strategies:** Data-driven insights allow retailers to refine their marketing strategies, ensuring that campaigns are targeted and effective in reaching the right audience.

"Real-time data is important to AO.com because we deliver real-time personalized banners to our users as they navigate our site. Confluent unlocks these types of use cases that are very important to our business."

Amelia McVennon, Developer, AO.com

Scenario 4: Live Commerce

Live commerce involves the use of social commerce for live auctions and interactive selling, providing a dynamic and engaging shopping experience. By leveraging data pipelines, log aggregation, and advanced applications, retailers can enable real-time interactions between sellers and buyers, driving engagement and sales. This real-time data processing capability helps retailers innovate and create new revenue streams through interactive and personalized shopping experiences.

Business Value

- **Dynamic and engaging shopping experience:** Real-time interactions between sellers and buyers create a lively and interactive shopping environment, enhancing customer engagement and enjoyment.
- **Increased sales and conversion rates:** The immediacy and interactivity of live commerce encourage impulse purchases and drive higher conversion rates, boosting sales.
- **New revenue streams:** By innovating with live auctions and interactive selling, retailers can tap into new revenue opportunities and diversify their sales channels.
- **Improved brand visibility and reach:** Live commerce events can attract a wide audience, increasing brand visibility and expanding reach to new customer segments.

Scenario 5: Retail Media

Retail media networks allow brands to deliver personalized ads across online, mobile, and in-store channels using retailers' first-party customer data. Effectiveness depends on delivering these promotions instantly, optimizing in real time, and measuring results across channels.

A streaming platform unifies customer interactions, inventory status, and ad performance metrics into a continuously updated, secure environment. Promotions can be triggered the moment a shopper browses an item online, walks into a specific store aisle, or interacts with a kiosk. Security-hardened data flows ensure customer privacy, while governed datasets give advertisers confidence in the accuracy of the performance metrics they receive.

AI-powered streaming agents adjust bids, suggest relevant products when stock runs low, and block fraudulent ad activity—all in real time. By feeding both live and historical data into attribution models, retailers can show exactly how campaigns drive sales, allowing AI and analytics platforms to act on fresh, complete information rather than outdated reports.

Business Value

- **New revenue streams:** Monetize first-party data with targeted advertising.
- **Higher conversions:** Serve timely, relevant ads at peak decision moments.
- **Real-time optimization:** Adapt campaigns instantly to shopper behavior.
- **Trusted metrics:** Deliver accurate, compliant, and auditable performance data.
- **Fraud protection:** Block invalid or suspicious ad activity instantly.
- **Unified experience:** Integrate promotions seamlessly across channels.

Real-Life Examples

- Migros

"Our ability to preserve the timing of our ERP program launch saved millions of dollars in consulting and resources that would have been needed to keep our old system going. We would never have been able to keep that timing if we did not have the new, flexible event-driven architecture enabled by Confluent."

Beth Zuke, North America Technology Director, Amway

Real-Life Example: Migros

Use Cases: Data Products | Observability | Event-Driven Microservices | IoT

Company Overview

Migros is a cooperative owned by over two million members and one of the two leading retail companies in Switzerland. With a revenue of \$35B+ and over 90.000 employees, Migros is a cornerstone of the Swiss economy, operating a diverse portfolio of businesses: supermarkets, online retail (Migros-Online, Digitec Galaxus), gastronomy, healthcare and financial services (Migros Bank). In addition, Migros Industry produces a wide range of products.

Implementation

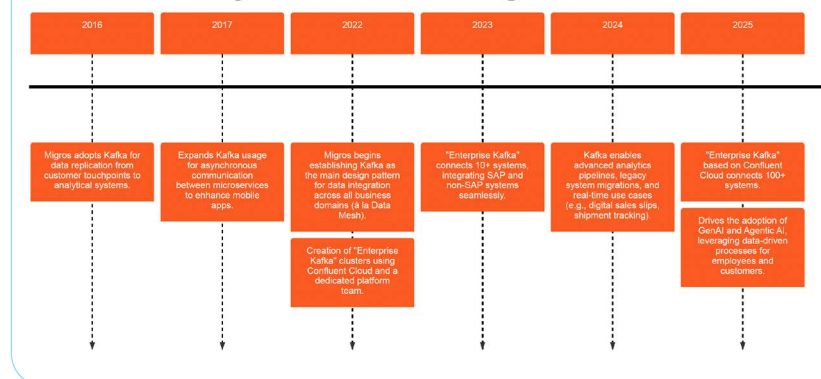
Migros has adopted Kafka and data streaming already in 2016, starting with the replication of data from customer touchpoints to analytical systems. The use of data streaming had been rapidly expanded, e.g. for asynchronous communication between microservices bolstering customer-facing mobile apps.

In 2022, Migros embarked on an even more ambitious journey to start establishing Kafka as the main design pattern for data integration across all business domains. The main drivers for this endeavor were de-coupling of systems and decentralization of operational responsibilities. This seismic shift was inspired by the ideas of Data Mesh (or "Data Product Thinking") and led to the creation of a set of consolidated Enterprise Kafka clusters based on Confluent Cloud, and a dedicated platform team.

Since then, the Enterprise Kafka Platform has become a central hub for data integration at Migros, connecting about 100+ systems. Most beneficial is the seamless integration between SAP and non-SAP systems. Based on the write once read many paradigm data-replication is as simple as it gets, enabling powerful analytics pipelines as well as supporting migrations away from legacy systems. Kafka real-time use-cases reach from digital sales slips on our customers' mobile phones to shipment tracking and delivery timeline forecasts.

Our biggest challenges for using Kafka as the main design pattern for data integration were to establish a common understanding of streaming semantics and schema evolution, as well as end to end business process monitoring. The education of IT and non-IT personal is an ongoing task. Nevertheless, being a data-driven corporation allowed us to quickly adopt GenAI and Agentic AI in our processes (e.g. RAG), from which employees and customers alike already benefit.

History of Kafka at Migros



History of Kafka at Migros

Business Value

- **Optimized supply chain efficiency:** A single streaming pipeline enabled Migros to achieve a global optimum, enhancing supply chain efficiency and reducing operational costs.
- **Improved decision-making with real-time insights:** Real-time transportation and logistics data allowed for prompt responses to events, preventing lost opportunities.
- **Enhanced customer experience:** Consolidating customer touchpoints and integrating key services streamlined operations, providing customers with timely and relevant information.
- **Increased agility and reduced delays:** Transitioning to faster choreography processes improved system scalability, reduced information propagation delays, and increased supply chain agility.

Conclusion

The retail and ecommerce industry is evolving quickly as technology, data, and customer expectations converge. Data streaming drives this transformation by enabling real-time insights, instant personalization, and seamless experiences across all channels. Unified commerce connects stores, marketplaces, mobile apps, and social platforms into one continuous ecosystem where customers can shop and engage without friction.

Retailers use data streaming to break down silos between marketing, logistics, and operations. Inventory updates, order confirmations, and personalized offers flow instantly across systems. Real-time analytics improve forecasting, optimize pricing, and ensure availability. AI adds automation to predict demand, manage replenishment, and enhance engagement at scale.

Generative AI adds conversational shopping assistants, smart product recommendations, and dynamic retail media. Agentic AI goes further by autonomously adjusting promotions, triggering restocks, and preventing fraud. These technologies create adaptive retail environments that respond to customer behavior and market conditions.

Migros shows the power of streaming at scale. One pipeline links logistics and IoT systems, enabling full supply chain visibility and synchronization. Accurate tracking improves efficiency, reduces delays, and supports sustainability. Digital receipts deliver instant offers and personalized recommendations, building loyalty and turning each purchase into a relationship.

Across the industry, streaming enables real time personalization, live commerce, and dynamic loyalty platforms. Retail media networks combine inventory, shopper behavior, and ad data into a single trusted platform. Promotions trigger as soon as someone browses a product or enters a store, turning every interaction into an opportunity.

The convergence of streaming and AI is reshaping modern retail. A data streaming foundation connects supply chains, powers personalization, and accelerates innovation. Retailers adopting these technologies move beyond static omnichannel models toward intelligent ecosystems that adapt continuously, enabling smarter decisions and better customer experiences.

3.4 Telecom and Media

The telecom and media industry is at the forefront of digital transformation, driven by the rapid adoption of 5G, IoT, and advanced data analytics. The deployment of 5G networks is revolutionizing connectivity, offering ultra-low latency and high-speed data transfer, which is essential for real-time applications and services like multi-device video streaming. IoT is enabling the creation of smart environments, from connected homes to smart cities, providing new opportunities for telecom and media companies to offer innovative services.

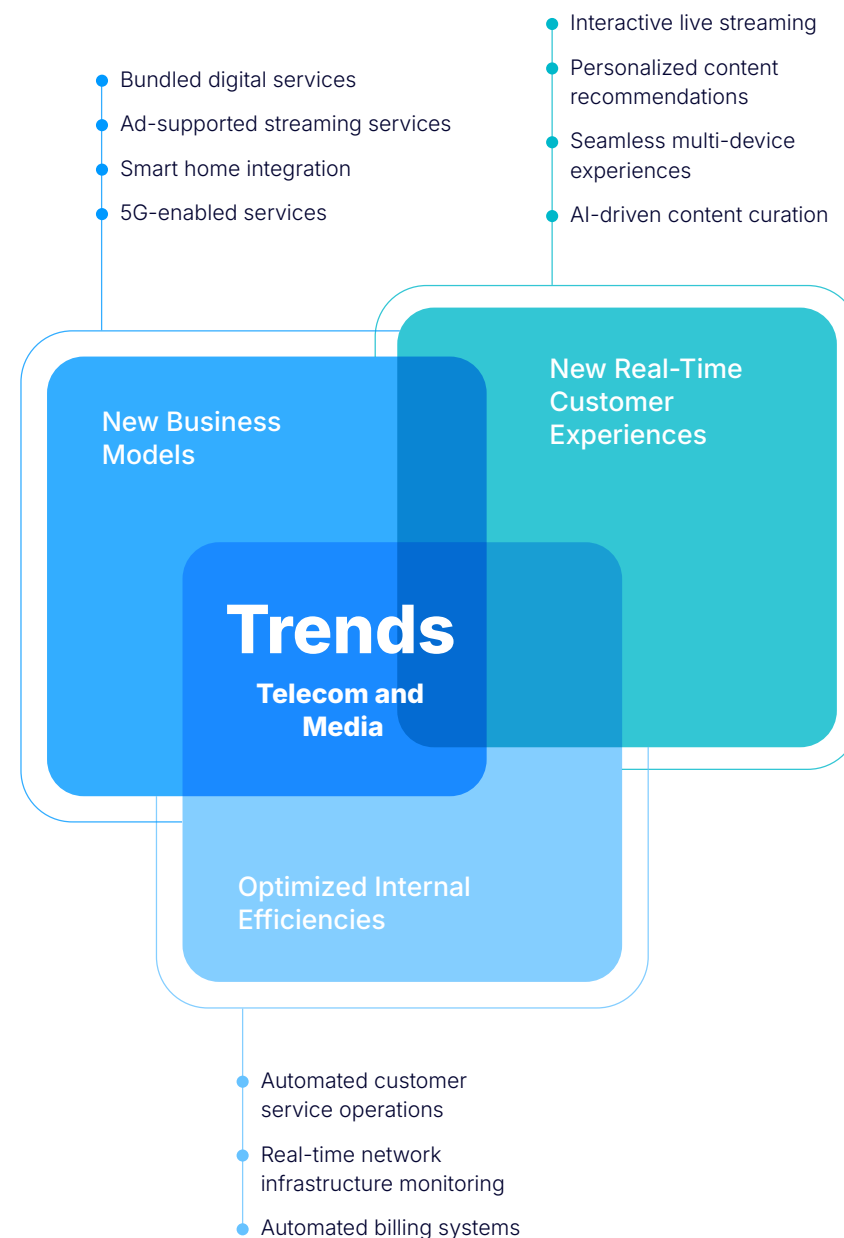
Personalization is another significant trend, with consumers expecting tailored content and experiences. Advanced analytics and AI/ML are being used to analyze user behavior and preferences, enabling companies to deliver personalized content and advertisements. The impact of interactive gaming is also noteworthy, as it drives demand

for high-performance networks and real-time data processing capabilities. The integration of data streaming platforms is transforming the way telecom and media companies manage their operations and deliver services. These platforms enable real-time data processing, providing insights that drive decision-making and enhance customer experiences. From live event analytics and real-time advertising to fraud prevention, digital rights management, and intelligent production pipelines, the use cases now extend well beyond connectivity. These capabilities also open new monetization opportunities through 5G APIs, partner data sharing, and open radio access network (RAN) operations, making telecom and media a hub for ecosystem innovation.

As the industry continues to evolve, the adoption of data streaming technologies will be crucial in maintaining competitiveness and driving innovation.

"Data at telecom scale is massive. We measure in terms of hundreds of petabytes per day. But to start extracting and understanding that data, we partner with companies that understand what the data can mean, how to transact on it, and how to realize the value. And for those communications to happen and for us to share data between different clouds, we want to have a broker of information. Confluent is our partner for this."

DISH Wireless



Scenario 1: Network Monitoring and Alerting

Network monitoring and alerting involve continuously monitoring network performance to detect issues before they impact customers. IT modernization, log aggregation, and multiple clusters offer real-time network insights and help ensure high availability and reliability.

Telecom companies can implement network monitoring and alerting systems that process real-time data from various sources, including network devices, log files, and performance metrics. For example, data streaming is used to fully automate site provisioning, service ordering, and service assurance monitoring and reporting for their television networks. This helps telecom companies improve network performance, reduce downtime, and enhance customer satisfaction.

Business Value

- **Real-time network visibility:** Continuous monitoring provides telecom companies with real-time insights into network performance, enabling proactive management and optimization.
- **Improved network performance:** By detecting and addressing potential issues promptly, telecoms can maintain optimal network performance and ensure high availability and reliability.
- **Reduced downtime:** Real-time alerting allows for quick identification and resolution of network issues, minimizing downtime and service disruptions for customers.
- **Cost reduction:** Reduce TCO by filtering and enriching telemetry data in-stream, cutting the volume and cost of data sent to expensive observability systems.
- **Proactive issue detection:** The ability to detect potential issues before they impact customers allows telecoms to take preventive measures, reducing the likelihood of major outages.
- **Enhanced customer satisfaction:** Reliable network performance and reduced service interruptions lead to higher levels of customer satisfaction and loyalty.

Scenario 2: OTT Services

Over-the-top (OTT) services deliver personalized on-demand content to consumers via the internet, bypassing traditional channels. Using log aggregation, advanced applications, and AI/ML, media companies can analyze user behavior, content metadata, clickstream data, and viewing history in real time. This in turn allows them to provide personalized content recommendations and improve the overall user experience, helping to drive engagement, increase revenue, and enhance customer satisfaction.

Business Value

- **Personalized content recommendations:** Real-time data processing allows media companies to deliver tailored content suggestions based on individual user preferences, enhancing the overall viewing experience.
- **Increased user engagement:** Personalized recommendations and on-demand content keep users engaged and encourage longer viewing sessions, boosting platform usage.
- **Enhanced customer satisfaction:** By providing content that aligns with user interests, OTT services can improve customer satisfaction and foster loyalty.
- **Data-driven insights:** Access to real-time user behavior and content performance data supports more informed decision-making, allowing for strategic content acquisition and marketing efforts.
- **Improved content discovery:** Personalized recommendations help users discover new content, increasing the visibility and consumption of a broader range of offerings.
- **Competitive advantage:** OTT platforms that offer personalized and engaging experiences can differentiate themselves in a crowded market, gaining a competitive edge.

Scenario 3: Advertisements

Targeted advertisements deliver personalized and relevant messages to consumers based on their preferences and behavior. With advanced applications and AI/ML, media companies can analyze consumer data like behavior, app usage, and demographic information in real time to deliver ads that are more likely to resonate with their audiences.

Data streaming helps companies optimize ad spend through automation, behavioral analytics, audience targeting, and rewards programs, resulting in improved effectiveness of advertising campaigns and increased ad revenue.

Business Value

- **Enhanced ad relevance:** Real-time data processing allows media companies to deliver personalized and relevant ads based on individual user preferences and behaviors, increasing the likelihood of engagement.
- **Improved campaign effectiveness:** Targeted advertisements are more likely to resonate with audiences, leading to higher conversion rates and improved overall campaign performance.
- **Increased ad revenue:** By delivering more effective ads, media companies can attract higher ad spend from advertisers, boosting ad revenue.
- **Optimized ad spend:** Real-time analytics and automation enable more efficient allocation of ad budgets, ensuring that resources are directed toward high-performing campaigns and audiences.
- **Behavioral analytics:** Access to real-time user behavior data supports more accurate audience segmentation and targeting, enhancing the precision of advertising efforts.
- **Audience targeting:** Leveraging demographic and behavioral data allows for more precise audience targeting, ensuring that ads reach the most relevant users.
- **Real-time engagement:** Delivering ads in real time ensures that they're timely and contextually relevant, increasing the chances of user interaction and engagement.

Scenario 4: Network Security and Threat Detection

Modern telecom networks face constant security threats from DDoS attacks and data breaches to rogue configurations and insider misuse. Traditional, batch-based security monitoring often reacts too late, allowing threats to spread before mitigation begins.

By streaming telemetry data from routers, firewalls, gateways, and edge devices, security teams gain continuous visibility into network behavior. Real-time processing detects anomalies such as traffic spikes, command-and-control communication patterns, and policy violations. Automated actions, such as isolating affected endpoints or blocking malicious IPs, can be triggered instantly to contain threats.

For example, if abnormal outbound traffic suddenly appears from a specific cell site, the system can correlate logs across routers and security appliances, confirm the pattern matches known exfiltration behavior, and automatically quarantine the site for further investigation. AI-driven security agents enhance this process by learning from historical and live data, identifying new attack signatures, and adapting defenses before they cause customer impact.

Business Value

- **Continuous protection:** Stream telemetry data in real time to identify threats such as DDoS attacks or malware infections the moment they appear, reducing exposure time.
- **Rapid containment:** Automatically block suspicious IP ranges or isolate infected edge devices within seconds to prevent attacks from spreading.
- **Anomaly detection:** Correlate traffic patterns across regions to uncover coordinated intrusion attempts or data exfiltration events before they escalate.
- **Compliance and governance:** Maintain detailed, real-time audit trails of all network and security events to support telecom regulations such as GDPR and ISO 27001.
- **Operational efficiency:** Replace manual log review with automated event correlation and enrichment, freeing analysts to focus on higher-value investigations.
- **Proactive defense:** Use AI-driven streaming agents to detect evolving threats, such as zero-day exploits or botnet traffic, by continuously learning from both historical and live data streams.

Scenario 5: Fraud Detection and Prevention

Telecom fraud, such as SIM swaps or international revenue share fraud, moves fast, often within minutes. Batch-based detection is too slow to protect revenue and customer trust.

A streaming approach continuously correlates call records, payment activity, authentication events, and network logs in real time. Processing pipelines detect anomalies instantly and trigger protective actions such as blocking suspicious calls or freezing compromised accounts. This enables millisecond detection and prevention of scams such as SIM swap fraud and account takeovers before the next billing cycle.

Data flows are secured and governed to meet regional privacy laws, allowing safe information sharing with fraud teams and regulators. AI-enhanced detection models evolve continuously as they process fresh patterns, keeping prevention efforts accurate, adaptive, and proactive rather than reactive.

Business Value

- **Instant action:** Automatically block suspicious transactions such as abnormal international call bursts or repeated SIM change attempts within seconds.
- **Comprehensive coverage:** Correlate multiple data sources, including calls, billing, network activity, and user authentication to detect various fraud types in one streaming architecture.
- **Adaptive models:** Continuously retrain AI models with live data to recognize new schemes such as device spoofing and synthetic identity attacks.
- **Regulatory compliance:** Maintain secure, auditable data pipelines that meet telecom privacy and financial regulations, such as GDPR and Payment Services Directive 2.
- **Customer trust:** Prevent account takeovers and fraudulent billing to protect users and preserve brand reputation.
- **Operational savings:** Reduce manual case reviews through automated fraud scoring and alert prioritization, enabling teams to focus on the highest-risk incidents.

Scenario 6: Data Sharing With MVNOs and Partners

Telecoms increasingly share network analytics and operational insights with mobile virtual network operators (MVNOs), IoT providers, and enterprise customers. But batch data delivery is too slow for managing SLAs, detecting issues, or optimizing services.

Continuous event streaming delivers up-to-date network performance, usage metrics, and customer insights to partners. Security-hardened, policy-driven governance ensures that only authorized, compliant data is shared.

AI-enabled pipelines tailor feeds to each partner, detecting anomalies for fraud teams, predicting churn for marketing, or delivering location-aware analytics for enterprise applications. By ensuring that partner-facing analytics are powered by current, contextual data, telcos avoid the inaccuracies and delays of static reports.

Business Value

- **New monetization models:** Create revenue streams by offering real-time data products, such as network-quality dashboards and predictive analytics feeds, to MVNOs and enterprise partners.
- **Improved partner performance:** Provide live SLA metrics so that MVNOs can detect service degradation instantly and adjust routing or capacity before customers are affected.
- **Efficiency gains:** Automate secure data delivery, governance, and policy enforcement, reducing the effort of manually validating partner reports.
- **Regulatory alignment:** Ensure that data sharing complies with privacy laws and contractual terms by applying automated access policies and lineage tracking.
- **Scalable sharing:** Stream millions of network events per second to multiple partners without impacting core operations, supporting continuous ecosystem growth.
- **Faster service launches:** Enable partners to act on live usage trends—for example, launching targeted roaming offers or IoT service upgrades in real time.

Scenario 7: Open RAN Operations

The modular architecture of Open RAN offers flexibility but it requires real-time coordination across distributed network components. Delayed insights lead to degraded service quality.

Streaming telemetry from radio units, controllers, and base stations provides instant visibility into performance. Processing pipelines detect interference, optimize handovers, and adjust configurations on the fly. Security and compliance are built into the architecture to protect subscriber data and meet telecom standards.

AI agents use this fresh operational data to predict failures, fine-tune spectrum use, and guide capacity planning—avoiding the missteps of outdated or incomplete datasets.

Business Value

- **Rapid innovation:** Roll out new RAN features or vendor integrations faster by validating performance instantly across distributed components.
- **Proactive optimization:** Detect and fix cell interference, signal degradation, or failed handovers in real time before customers experience disruptions.
- **Cost efficiency:** Scale resources dynamically based on traffic demand, reducing overprovisioning and hardware costs across remote radio sites.
- **AI-driven adaptability:** Use predictive analytics to forecast failures, optimize spectrum allocation, and maintain peak network performance under changing load conditions.
- **Compliance assurance:** Protect subscriber data with encryption, policy enforcement, and full lineage tracking to meet telecom and privacy regulations.
- **Scalable operations:** Support thousands of distributed radios and controllers by processing millions of telemetry events per second, enabling continuous, automated optimization.

Scenario 8: Monetizing Network APIs

5G APIs enable partners to access real-time network functions such as network slicing, dynamic quality of service (QoS), and device location. To monetize them effectively, responses must be fast, accurate, and secure.

A streaming data layer delivers live network events and performance metrics into API gateways. Real-time processing ensures that API calls return the most current context, while governance enforces privacy and contractual policies. AI-enhanced capabilities offer predictive congestion alerts, fraud detection, and personalized network services.

By keeping API outputs fresh and compliant, telcos avoid service failures and enable partners to build trustworthy, high-value applications.

Business Value

- **New revenue streams:** Monetize 5G capabilities through premium APIs that real-time functions such as dynamic QoS adjustment and precise device geolocation.
- **Faster partner innovation:** Give developers instant access to live network insights so they can create latency-sensitive services such as autonomous delivery, telemedicine, and cloud gaming.
- **Enhanced experiences:** Enable applications to deliver smooth, personalized performance—for example, adapting bandwidth allocation to support high-definition streaming during live events.
- **Automated operations:** Track usage, SLAs, and billing in real time to simplify partner settlements and reduce manual reconciliation.
- **Compliance-ready:** Guarantee privacy and data integrity by enforcing consent policies and telecom-grade encryption across all API interactions.
- **Ecosystem growth:** Build a thriving developer community by offering reliable, event-driven APIs that accelerate partner integration and new service launches.

Scenario 9: Live Event Analytics and Dynamic Ad Placement

Live sports, concerts, and esports generate enormous amounts of real-time data, from ticket scans and video feeds to engagement metrics, betting activity, and mobile app interactions. Traditional batch workflows can't keep up with second-by-second audience shifts or synchronize experiences across multiple channels.

A data streaming platform ingests and processes these signals instantly to enable dynamic ad insertion, real-time scoreboard updates, and seamless content distribution across TV, mobile, and in-venue displays. AI and streaming agents can personalize fan experiences with push notifications, targeted offers, and gamified interactions during the event.

Business Value

- **New revenue opportunities:** Sell premium ad inventory dynamically based on real-time audience behavior. For example, displaying targeted ads during a surge in viewership or fan engagement.
- **Fan engagement:** Deliver personalized messages and instant rewards, such as app-based trivia games and seat upgrades, to boost loyalty and participation.
- **Operational agility:** Update live content, ads, and sponsor overlays across broadcast and digital platforms instantly without interrupting streams.
- **Cross-platform consistency:** Synchronize live video, betting data, and scoreboards across TV, mobile, and in-venue screens to ensure a unified, real-time fan experience.
- **Data-driven insights:** Provide advertisers and event organizers with continuous performance analytics to refine campaigns, optimize engagement, and maximize ROI.

Scenario 10: Intelligent Media Production Pipelines

Media production involves multiple disconnected systems for editing suites, asset libraries, subtitling and dubbing tools, marketing platforms, and distribution services. Batch-based integrations slow down collaboration, introduce errors, and drive up costs.

Data streaming connects all production stages with continuous, event-driven pipelines. As assets move through the workflow, streaming agents can enrich metadata, automate subtitling, support AI-driven language dubbing, and optimize ad slot placements. This ensures that all stakeholders, from editors to distributors, work with the most current, verified data.

Business Value

- **Faster production cycles:** Synchronize editing, rendering, and publishing in real time so that new content can be distributed minutes after approval rather than hours or days later.
- **Cost savings:** Automate repetitive processes such as metadata tagging, subtitling, and ad scheduling, reducing manual labor and post-production overhead.
- **Higher quality:** Enable real-time quality control—for instance, detecting missing subtitles or audio sync issues before final render—to minimize rework.
- **Revenue growth:** Optimize ad placements and distribution timing by analyzing live engagement data, ensuring that content and ads reach audiences when viewership peaks.
- **Collaborative agility:** Allow editors, producers, and distributors to work from a single source of live data, improving coordination and reducing version conflicts.

"By setting aside the need to oversee platform operations, our data engineering team can spend more time on things that matter, such as building data products that can support our goal for personalization at scale."

John Bibal II, Director of Data Engineering, Globe Group

Scenario 11: Digital Rights and Asset Management

Consumers increasingly purchase or stream content across multiple platforms, but fragmented content libraries and inconsistent rights management create frustration. Batch reconciliation between retailers and platforms introduces errors and delays, undermining trust.

A data streaming backbone enables a centralized, always-up-to-date digital asset registry. Rights and entitlements are updated in real time, ensuring that users can seamlessly access purchased content across services. Governance and traceability are built in, ensuring compliance and reducing disputes between partners.

Business Value

- **Seamless customer experience:** Guarantee instant access to owned or licensed content across devices and services. For example, a movie purchased on one platform becomes immediately available to stream on a smart TV or mobile app.
- **Fraud prevention:** Verify digital rights in real time to block unauthorized streaming or playback, preventing piracy and lost revenue.
- **Compliance and auditability:** Maintain detailed data lineage and time-stamped transactions that meet legal, contractual, and regional content distribution requirements.
- **Partner efficiency:** Streamline rights synchronization and settlement between studios, streaming services, and retailers, reducing manual reconciliation and contractual disputes.
- **Operational transparency:** Give all partners a single, trusted view of asset ownership and usage, fostering accountability and smoother collaboration.

Scenario 12: Autonomous Network Operations

As 5G, IoT, and edge networks grow, manual operations can't keep up. Traditional OSS/BSS systems respond only after problems occur, delaying recovery and increasing costs. Autonomous networks bring intent-driven automation. A data streaming platform ingests telemetry from radios, routers, and devices, analyzing it in real time to detect anomalies, optimize performance, and trigger automated actions.

For example, data streaming pipelines can predict outages, manage congestion, and adjust routing across RAN and core networks. In Open RAN, the RAN intelligent controller fine-tunes spectrum use and handovers within milliseconds. Governance features keep actions secure and auditable. AI agents learn from historical and live data, turning reactive operations into proactive, self-healing networks.

Business Value

- **Closed-loop automation:** Continuous streaming of telemetry and events enables real-time detection and resolution of issues. Without streaming, autonomous operations would be delayed by batch-driven insights.
- **Improved service reliability:** Data streaming provides the always-fresh context needed for autonomous adjustments, minimizing downtime and ensuring consistent quality of experience.
- **Cost reduction:** Real-time pipelines replace manual monitoring and ticket-driven processes, reducing operational costs in network operations centers and field maintenance.
- **Scalability:** Streaming architectures make it possible to manage growing device density, traffic, and services at scale without linear increases in staff.
- **Proactive optimization:** Only with continuous streaming data can AI agents predict and prevent issues before they affect customers.
- **Governed intelligence:** A streaming data platform enforces lineage, security, and policy in real time, ensuring that autonomous decisions are traceable, compliant, and trusted.
- **Strategic differentiation:** Operators that combine autonomous networks with streaming capabilities deliver premium SLAs and accelerate innovation, setting themselves apart in a competitive market.

Conclusion

The telecom and media sectors are undergoing profound changes driven by the deployment of 5G networks, the expansion of IoT, and the integration of real-time data analytics. Emerging business models such as bundled digital services, ad-supported streaming platforms, and 5G-enabled smart home integrations are creating new revenue streams and reshaping customer expectations. Real-time customer experiences, including interactive live streaming, personalized content recommendations, and seamless multi-device connectivity, are now essential for engaging audiences. Internally, automated systems for customer service, billing, and network monitoring improve efficiency and scalability.

OTT services showcase the potential of personalized content delivery. By analyzing real-time data from user behavior and preferences, platforms deliver tailored recommendations that enhance customer satisfaction and increase engagement. Targeted advertising, enabled by real-time behavioral analytics, improves ad relevance, boosts campaign effectiveness, and increases ad revenue through precision targeting.

Network monitoring powered by data streaming ensures real-time infrastructure visibility, proactive issue detection, and reduced downtime. These capabilities maintain high network performance and reliability, enhancing customer satisfaction and loyalty. Fraud

detection, software-defined wide area network (SD-WAN) optimization, open RAN operations, and autonomous networks highlight how telecoms use streaming to protect revenue, secure networks, and adapt quickly to dynamic demand. Live event analytics, intelligent production pipelines, and digital rights management further demonstrate how media organizations use real-time insights to engage audiences, automate workflows, and reduce disputes.

These examples highlight the pivotal role of data streaming in telecom and media operations. Continuous real-time data processing enables businesses to deliver personalized experiences, optimize internal workflows, and adapt to evolving customer demands.

Companies like DISH Wireless demonstrate how integrating technologies such as AI, IoT, and real-time analytics drives operational excellence and customer-centric innovation. Agentic AI extends this by forecasting congestion, enriching production pipelines, and tailoring partner-facing insights in real time, turning networks and platforms into adaptive, self-optimizing ecosystems. As the industry evolves, the adoption of data streaming will be critical for telecom and media companies to stay competitive, meet consumer expectations, and unlock new growth opportunities. From improving network reliability to delivering engaging, personalized content, the future of telecom and media is being shaped by real-time insights and scalable, data-driven solutions.

3.5 Gaming

Two of the most notable trends the gaming industry has undergone in the past few years is the rise of mobile gaming and gaming-as-a-service (GaaS). Mobile gaming continues to dominate the market, with smartphones having become the primary gaming device for many users. This shift has led to the development of more sophisticated and engaging mobile games, supported by advanced data analytics and AI/ML technologies. GaaS allows players to access games on demand via cloud platforms, eliminating the need for high-end hardware.

Monetization innovation is another critical trend, with game developers exploring new ways to generate revenue through in-game purchases, advertisements, and subscriptions. The use of data streaming platforms is revolutionizing the gaming industry by enabling real-time data processing and analytics. This capability allows game developers to deliver personalized experiences, optimize game performance, and enhance player engagement.

GaaS platforms now let players stream full games instantly to mobile devices, PCs, and consoles without the need for downloads or high-end hardware. This approach enables seamless, on-demand access to high-quality experiences across any device, powered by real-time cloud and streaming technology.

The importance of real-time data can't be overstated, as it enables live monitoring, troubleshooting, and making server-side changes while a game is in progress.

Beyond core gameplay, streaming also underpins esports tournaments, live social interactions, anti-cheat enforcement, and cross-platform synchronization, all of which depend on low-latency, highly scalable infrastructures. Agentic AI is increasingly woven into these systems to adapt content dynamically, moderate live communities, and ensure fair play in real time.

The rise of esports and interactive gaming has further increased the demand for robust and scalable infrastructure to support millions of concurrent players. As the gaming industry continues to evolve, the adoption of data streaming technologies will be crucial in maintaining competitiveness and driving innovation.



Scenario 1: Infrastructure Operations

Infrastructure operations in the gaming industry involve the continuous monitoring and management of game servers and networks to ensure optimal performance and availability. By leveraging data pipelines, log aggregation, and multiple clusters, gaming companies can gain real-time visibility into their infrastructure and quickly identify and resolve issues. This real-time data processing capability helps gaming companies provide 24/7 service health, perform root cause analysis in case of issues, and support cloud-native elasticity for traffic spikes, such as during a new game release or esports tournament.

Data streaming allows gaming companies to serve millions of gamers worldwide without outages, ensuring a seamless gaming experience. Cloud-native infrastructure operations automate site provisioning, service ordering, and service assurance monitoring and reporting. Hence, companies can process vast amounts of data in real time, ensuring business continuity without outages.

Business Value

- **24/7 service health:** Continuous monitoring and real-time data processing ensure that game servers and networks maintain optimal performance and availability, providing uninterrupted service to gamers worldwide.
- **Seamless gaming experience:** By preventing outages and minimizing latency, gaming companies can deliver a smooth and enjoyable gaming experience, enhancing player satisfaction and retention.
- **Rapid issue resolution:** Real-time visibility into infrastructure allows for quick identification and resolution of issues, reducing downtime and maintaining service quality.
- **Cloud-native elasticity:** The ability to scale infrastructure dynamically supports traffic spikes during events like new game releases or esports tournaments, ensuring consistent performance under varying loads.
- **Cost efficiency:** Optimized resource allocation and automated processes reduce operational costs, maximizing the return on infrastructure investments.

Scenario 2: Game Telemetry

Game telemetry involves collecting and analyzing data generated by players during gameplay. Gaining insights into player behavior and preferences through large-scale observability and advanced applications allows gaming companies to deliver personalized experiences and improve game performance.

This real-time data helps gaming companies make server-side changes while the player is playing the game, such as time-limited events that give rewards or real-time updates to improve the game or align with audience needs.

Business Value

- **Personalized player experiences:** Real-time analysis of player behaviors and preferences allows gaming companies to deliver tailored experiences, enhancing player engagement and satisfaction.
- **Dynamic game updates:** The ability to make server-side changes in real time enables the introduction of time-limited events, rewards, and updates that keep the game fresh and exciting for players.
- **Improved game performance:** Continuous monitoring and analysis of game telemetry data help identify performance bottlenecks and optimize game mechanics, ensuring a smooth and enjoyable gaming experience.
- **Increased player retention:** By offering personalized and engaging experiences, gaming companies can increase player retention and foster long-term loyalty.
- **Data-driven game development:** Insights gained from telemetry data inform game design and development, allowing companies to align their offerings with player preferences and market trends.
- **Real-time feedback loop:** Immediate access to player data creates a feedback loop that enables rapid iteration and improvement of game features and content.
- **Enhanced monetization strategies:** Understanding player behaviors and preferences allows for the development of targeted monetization strategies, such as in-game purchases and premium content offerings.

Scenario 3: Monetization Network

Monetization networks in the gaming industry generate revenue via in-game purchases, advertisements, and micro-transactions. Leveraging AI/ML and real-time data streaming, gaming companies can analyze player behaviors and preferences in real time to deliver targeted advertisements and personalized in-game offers. This helps them optimize ad spend, increase player engagement, and drive revenue growth through upselling and cross-selling opportunities.

Business Value

- **Increased revenue generation:** Real-time data processing enables the delivery of targeted advertisements and personalized in-game offers, maximizing upselling and cross-selling opportunities to drive revenue growth.
- **Optimized ad spend:** By analyzing player behaviors and preferences in real time, gaming companies can optimize their advertising strategies, ensuring that ad spend is directed toward the most effective campaigns and audiences.
- **Enhanced player engagement:** Personalized in-game offers and advertisements resonate more with players, increasing engagement and interaction with monetization features.
- **Data-driven monetization strategies:** Access to comprehensive player data supports the development of informed and effective monetization strategies, aligning offers with player preferences and behaviors.
- **Real-time adaptability:** The ability to process and analyze data in real time allows gaming companies to quickly adapt their monetization tactics to changing player behaviors and market conditions.
- **Sustainable business growth:** By maximizing monetization opportunities and driving revenue growth, gaming companies can achieve sustainable business success and long-term profitability.

Scenario 4: Social Live Experiences and eSports

Live-streamed gaming events and esports attract millions of players and spectators worldwide, often across multiple platforms. Delivering seamless, interactive experiences requires continuous ingestion of match telemetry, player statistics, audience engagement metrics, and live-stream feeds. Data streaming enables real-time leaderboard updates, instant highlights, and interactive fan features such as polls, rewards, and live chats.

At the same time, cross-platform synchronization ensures that progress, purchases, and game states are consistent whether a player is on a PC, console, or mobile device. Streaming agents can moderate live chat for toxicity, detect fraudulent activity in wagering systems, and surface personalized fan experiences such as targeted rewards and premium ad placements. By bringing together esports operations and cross-platform ecosystems, game publishers and streaming platforms create dynamic, engaging, and monetizable live experiences.

Business Value

- **Immersive fan engagement:** Real-time overlays, replays, and interactive features enhance audience participation.
- **Seamless cross-platform play:** Progress and purchases are synchronized across devices.
- **New monetization models:** Dynamic ad placement, sponsorships, and real-time betting integrations are available.
- **Healthy communities:** Streaming agents moderate chats, reduce toxicity, and encourage safe interactions.
- **Operational resilience:** Real-time synchronization prevents mismatched scores, downtime, and streaming errors.

Scenario 5: Anti-Cheat and Fair Play

Cheating and fraud are persistent challenges in multiplayer gaming, undermining player trust and damaging brand reputation. Traditional detection methods, which rely on post-game audits or static rule sets, are too slow to prevent harm during live sessions. A data streaming approach continuously processes gameplay telemetry, log-in data, account activity, and transaction histories to detect anomalies in real time.

Streaming agents can respond instantly by flagging unusual aim precision, suspicious latency manipulation, or irregular account behavior. Offending sessions can be blocked mid-game, with alerts automatically routed to fraud teams or moderators. This ensures fairness not only in casual multiplayer experiences but also in competitive esports, where integrity is paramount. By protecting gameplay environments, streaming-based anti-cheat systems preserve long-term player loyalty and safeguard revenue.

Business Value

- **Fair competition:** Ensure integrity across casual and competitive play.
- **Real-time prevention:** Block cheaters during gameplay rather than after the fact.
- **Trust and retention:** Build community confidence and long-term engagement.
- **Operational savings:** Reduce costly manual investigations with automated detection.
- **Brand protection:** Preserve the reputation of developers, publishers, and tournament organizers.

Scenario 6: Dynamic Content and Live Ops

Live service games depend on continuously evolving content to keep players engaged. Traditional release cycles, which require large updates or downtime, struggle to deliver the level of personalization and freshness that players expect. With a data streaming foundation, developers can push server-side changes instantly, tailoring experiences to specific player segments based on real-time telemetry.

Skill-based matchmaking, a cornerstone of competitive gaming, becomes smarter and more adaptive by combining historical performance with live player data. Using rolling windows and stateful processing, the system continuously updates rankings and match quality, ensuring fair, balanced, and dynamic gameplay as player behavior evolves in real time.

Streaming agents analyze engagement data, skill levels, and spending patterns to dynamically adapt game mechanics, introduce time-limited events, and serve personalized offers. Players can be shown different challenges, rewards, or seasonal content aligned with their behaviors and preferences without needing to update the client. This continuous, data-driven approach not only boosts engagement but also creates new monetization opportunities through targeted micro-transactions and premium content delivery.

Business Value

- **Personalized engagement:** Adaptive content increases motivation and immersion.
- **Higher retention:** Fresh, evolving gameplay reduces churn.
- **Faster innovation:** Developers can test and release new features instantly.
- **Revenue growth:** Targeted offers and events increase conversion rates.
- **Operational agility:** No downtime or patching is required for live updates.
- **Competitive advantage:** Games evolve dynamically, staying relevant in fast-moving markets.

Conclusion

The gaming industry is at the forefront of digital innovation, driven by trends such as free-to-play models with in-game purchases, cloud gaming subscriptions, and play-to-earn platforms. The rise of mobile gaming and GaaS has further expanded accessibility, allowing players to enjoy sophisticated experiences on demand. Real-time personalized recommendations, multiplayer gaming, and seamless social experiences are transforming player engagement, while continuous analytics, elastic server infrastructures, and fraud detection are ensuring operational efficiency.

Game telemetry is another critical application of data streaming, providing continuous insights into player behavior and enabling real-time adjustments. Developers use telemetry data to introduce time-limited events, optimize game mechanics, and deliver personalized rewards, keeping games fresh and engaging. This approach improves player retention, informs data-driven game development, and supports enhanced monetization strategies.

Monetization networks in gaming rely on real-time data streaming to analyze player preferences and deliver targeted in-game advertisements and offers. This capability drives revenue growth through optimized ad spend and personalized upselling opportunities. By quickly adapting monetization tactics to player behaviors, gaming companies can sustain business

growth and maximize profitability. The demand for robust infrastructure is further amplified by the rise of esports and interactive gaming, which require low latency and scalable architectures. Continuous server monitoring ensures 24/7 service health, rapid issue resolution, and seamless player experiences.

Anti-cheat detection and fair play enforcement showcase how streaming can protect trust, while dynamic content delivery and live operations demonstrate how games can evolve continuously without downtime. Live fan engagement, esports, and cross-platform ecosystems illustrate the broader potential of streaming to create immersive, monetizable, and community-driven experiences.

These examples underscore the transformative role of data streaming in the gaming industry. It enables real-time insights, personalized player experiences, and scalable operations to ensure that gaming companies remain agile and innovative in a competitive market. With agentic AI analyzing gameplay, moderating communities, and adapting live content in milliseconds, gaming platforms are shifting from reactive support to proactive, intelligent ecosystems. As the industry continues to evolve, continuous data processing will be pivotal in delivering engaging content, optimizing infrastructure, and achieving sustainable growth.

3.6 Government and Public Sector

Governments and public institutions are increasingly adopting digital public infrastructure to improve service delivery and enhance citizen experiences. The rise of smart cities and buildings is another critical trend, with IoT and data analytics enabling the creation of intelligent environments that improve quality of life and drive economic growth.

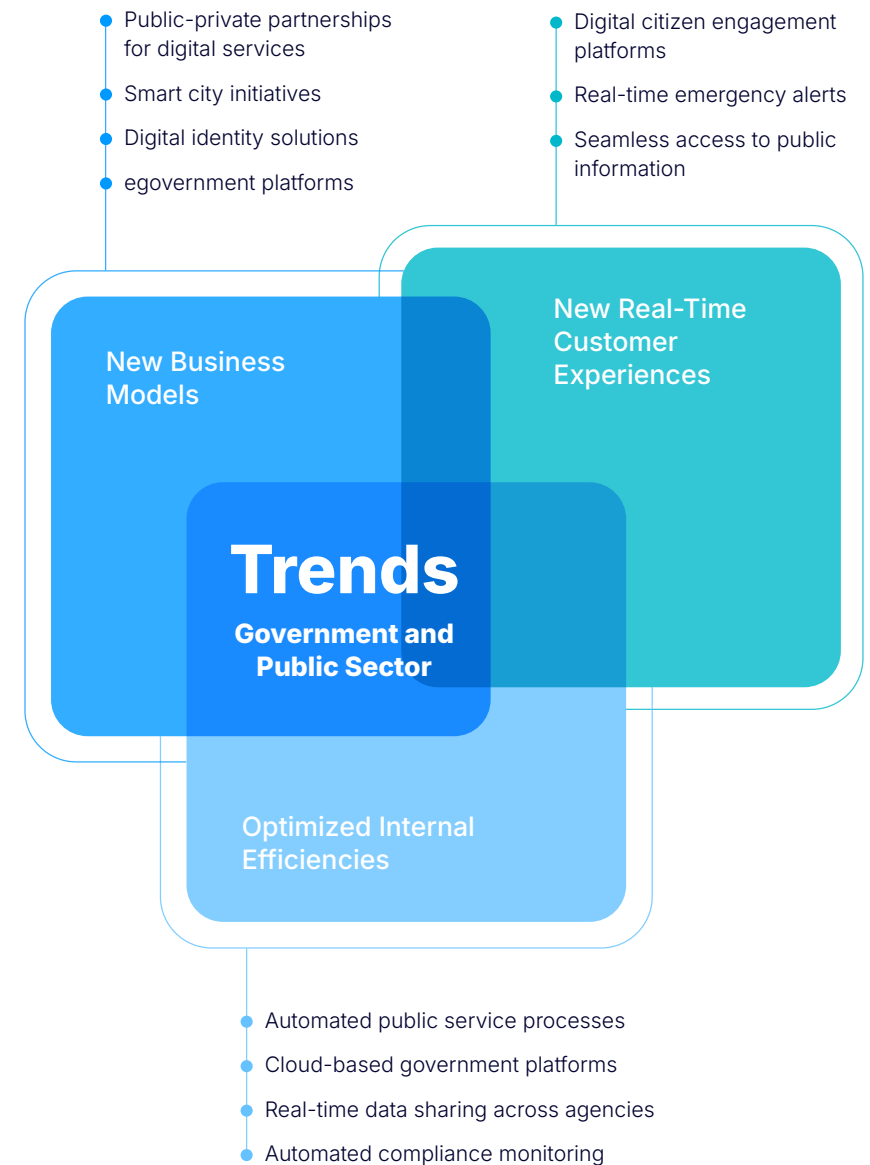
Environmental, social, and governance (ESG) considerations are also becoming more important, with public sector organizations focusing on sustainability and social responsibility.

Real-time capabilities in the public sector cannot be overstated, as they enable continuous monitoring, analysis, and sharing of data across institutions. As public sector digitalization accelerates, the demand for scalable, secure, and intelligent streaming architectures continues to grow.

Modernizing public sector IT is not optional - it is foundational. An event-driven architecture enables a step-by-step migration from legacy systems, allowing agencies to lift and shift critical workloads without disruption.

Beyond urban services, data streaming is being applied to emergency response, healthcare, public safety, and national security. In these domains, seconds can make the difference between resilience and disruption. Agentic AI builds on these foundations by prioritizing alerts, automating triage, and coordinating responses across agencies in real time.

The rise of digital public infrastructure and smart cities has further increased the demand for robust and scalable data streaming solutions.



Scenario 1: Citizen Services

Citizen services are available through various channels, such as online portals, mobile apps, and in-person interactions. Using real-time data from government, healthcare providers, and social services, organizations can create a single view of citizens, providing personalized and efficient services through all phases of life: work, family, health, retirement, and social security. Data streaming provides consistency that helps the public sector improve inter-agency data sharing and enhance citizen experiences.

Business Value

- **Comprehensive citizen view:** Real-time data integration from multiple sources provides a single, comprehensive view of each citizen, enabling personalized services.
- **Improved inter-agency collaboration:** Enhanced data sharing between government agencies facilitates better coordination and collaboration, leading to more cohesive and effective public services.
- **Enhanced citizen experience:** Personalized and efficient service delivery improves the overall citizen experience, fostering trust and satisfaction with public sector organizations.
- **Streamlined service delivery:** Real-time data processing enables quicker response times and more efficient service delivery, reducing wait times and improving accessibility for citizens.
- **Data-driven decision-making:** Access to comprehensive citizen data supports informed decision-making, allowing public sector organizations to tailor services to meet the specific needs of their communities.
- **Increased operational efficiency:** By modernizing IT infrastructure and leveraging data streaming, public sector organizations can optimize their operations and reduce redundancies.
- **Proactive citizen support:** Real-time insights into citizen needs and circumstances allow for proactive support and intervention, improving outcomes in areas such as health, social security, and family services.
- **Cost savings:** Streamlined processes and improved data sharing reduce administrative costs and resource waste, leading to significant cost savings for public sector organizations.

Scenario 2: Smart Cities

Smart cities use IoT and data analytics to create intelligent environments that improve quality of life and drive economic growth. Modern IT infrastructures and applications allow the use of real-time data from sources such as traffic sensors, cameras, and connected vehicles to reduce traffic and road accidents through intelligent transportation systems. Data streaming enables smart cities to make data-driven decisions and enhance public safety.

Business Value

- **Improved traffic management:** Real-time data processing enables cities to monitor and manage traffic flow efficiently, reducing congestion and improving overall traffic conditions.
- **Enhanced public safety:** Intelligent transportation systems that leverage real-time data can reduce road fatalities and accidents by providing timely alerts and updates to drivers.
- **Data-driven decision-making:** Access to real-time traffic data supports informed decision-making, allowing city planners and officials to implement effective traffic management strategies.
- **Reduced environmental impact:** By optimizing traffic flow and reducing congestion, smart city initiatives can lower vehicle emissions, contributing to a cleaner and more sustainable urban environment.
- **Economic growth:** Improved transportation systems enhance overall quality of life, attracting businesses and residents and driving economic growth in the city.
- **Proactive infrastructure management:** Real-time insights into traffic patterns and infrastructure usage allow for proactive maintenance and management, extending the lifespan of city assets.
- **Innovation and modernization:** Implementing smart city technologies positions public sector organizations as leaders in urban innovation and modernization.

Scenario 3: National Security

National security involves protecting a country's citizens, infrastructure, and interests from various threats, including cyberattacks, terrorism, and natural disasters. Collection disciplines, such as IoT sensors, signals, and human intelligence, can continuously collect, analyze, and share data from vehicles, devices, apps, and infrastructure in real time. Leveraging these capabilities through data streaming allows government organizations to provide comprehensive, accurate, and current insights into the operating environment and enhance national security.

Business Value

- **Enhanced threat detection and response:** Real-time data processing enables the continuous monitoring and analysis of potential threats, allowing for rapid detection and response to cyberattacks, terrorism, and other security risks.
- **Comprehensive situational awareness:** By collecting and analyzing data from various sources, public sector organizations can gain a fuller picture of the operating environment, improving situational awareness and decision-making.
- **Proactive risk management:** Continuous data collection and analysis enable proactive risk management, reducing the likelihood of security breaches and enhancing overall resilience.
- **Inter-agency collaboration:** Enhanced data sharing between government agencies and departments facilitates better coordination and collaboration, leading to more cohesive and effective national security efforts.
- **Innovation and modernization:** Leveraging advanced technologies and real-time data analytics positions public sector organizations as leaders in security innovation and modernization.

Scenario 4: Emergency Response and Disaster Management

When natural disasters, accidents, or public health crises occur, fast information sharing is critical. Emergency response agencies rely on data from IoT sensors, satellites, weather services, geospatial mapping, and citizen reports to coordinate rescue and recovery.

A data streaming platform allows real-time correlation of these feeds to identify hotspots, optimize resource allocation, and push alerts to first responders and citizens. Streaming agents can prioritize communications based on urgency, dynamically reroute traffic data for evacuation planning, and triage citizen requests automatically to reduce bottlenecks. With this, governments can move from reactive crisis handling to proactive, coordinated disaster management.

Business Value

- **Faster response times:** Real-time data streaming enables immediate detection of emergencies and accelerates the flow of critical information to decision-makers and first responders. This significantly shortens the time between incident detection and response, helping save lives and reduce harm.
- **Improved coordination:** Broken down silos allow different agencies, such as fire departments, police, emergency medical services, and civil protection authorities, to share and act on the same data stream in real time. This ensures that all teams are aligned and can coordinate actions without delay.
- **Public safety:** Citizens receive instant alerts, evacuation instructions, and safety recommendations through connected apps, digital signage, or emergency broadcast systems. This timely communication improves public awareness and helps people take the right actions.
- **Reduced impact:** Early detection and real-time insights allow authorities to act before a situation escalates. Whether it's rerouting traffic to avoid blocked roads or deploying resources to high-risk zones, proactive responses help minimize casualties, infrastructure damage, and economic disruption.
- **Trust and transparency:** Data-driven actions backed by real-time evidence help public agencies communicate more clearly and credibly during crises. Transparent communication builds public trust and fosters a sense of security, even in high-pressure situations.

Scenario 5: Public Health and Pandemic Monitoring

Public health agencies increasingly need real-time visibility into disease spread, vaccination uptake, and hospital capacity. During the COVID-19 pandemic, dashboards often relied on batch updates, creating lag in critical decision-making. An event-driven architecture enables continuous ingestion of case reports, hospital telemetry, pharmacy stock levels, and mobility data to provide an up-to-date view of public health.

Embedded AI can detect early outbreak signals, correlate trends with mobility or social media chatter, and trigger targeted interventions before crises escalate. Public-facing dashboards powered by live data improve transparency and guide citizen behavior, while inter-agency data pipelines ensure that governments can coordinate across borders.

Business Value

- **Early warning:** Real-time data ingestion from clinics, testing centers, hospitals, and mobility systems enables faster detection of outbreaks. AI-driven pattern recognition helps identify anomalies or clusters of concern before they become widespread, allowing for quicker containment strategies and targeted public health actions.
- **Resource optimization:** With real-time visibility into hospital bed availability, intensive care unit (ICU) capacity, vaccine inventory, and staffing levels, healthcare providers and agencies can better manage resources. This prevents overloading specific facilities and supports smarter allocation of equipment and personnel.
- **Informed citizens:** Public-facing dashboards powered by live data provide clear, timely information about infection rates, local health measures, vaccination availability, and safety guidelines. This transparency improves public trust and encourages compliance with health interventions.
- **Cross-border collaboration:** Streaming architectures enable real-time data sharing across municipal, regional, and national boundaries. This is critical in managing pandemics that don't respect borders. Coordinated responses and shared early warnings lead to better containment and international cooperation.

Scenario 6: Transportation and Public Infrastructure Operations

Railways, airports, and public transit systems carry millions of passengers daily. Batch-based monitoring of schedules, usage, and disruptions often leads to delays in communicating issues. A streaming data approach ingests real-time signals from trains, buses, planes, and supporting infrastructure (e.g., ticketing systems, sensors, weather feeds) to provide live operational intelligence.

Governments and public operators can send instant updates about delays, reroute passengers, and adjust capacity dynamically. For freight and logistics, data streaming supports just-in-time operations by integrating with customs, port authorities, and supply chains. Intelligent applications powered by AI models can also monitor anomalies for predictive maintenance to detect failing equipment before it causes disruptions.

Business Value

- **Improved passenger experience:** Real-time updates on delays, arrivals, and service changes reduce uncertainty for passengers and improve satisfaction. Passengers can make informed travel decisions through apps, station displays, and notifications, leading to a smoother journey.
- **Operational resilience:** Streaming data supports predictive maintenance by monitoring vehicle and infrastructure health in real time. Early detection of equipment failure prevents breakdowns and service interruptions, keeping systems running reliably.
- **Economic efficiency:** By integrating freight data with customs, port systems, and logistics partners, just-in-time delivery becomes more reliable. Real-time coordination across the supply chain reduces bottlenecks, inventory costs, and delays in goods movement.
- **Transparency:** Instant communication about disruptions, route changes, or emergency protocols builds trust between public operators and citizens. Consistent updates reduce confusion and improve public confidence in transportation systems.
- **Sustainability:** Data-driven traffic and route optimization lowers fuel use, reduces idle times, and helps manage congestion. This cuts emissions and supports climate goals while improving the overall flow of passengers and goods.

Real-Life Examples

- PNCC

Real-Life Example: PNCC

Use Cases: Data Products | IT Modernization | Hybrid and Multicloud | IoT | Data Governance

Company Overview

Palmerston North City Council (PNCC) is the local authority for Palmerston North, New Zealand. The council's mission is to improve quality of life for citizens through efficient public services, sustainable city development, and community engagement. As part of its smart city strategy, PNCC has invested in modern data-driven technologies to transform how local government services are delivered.

Implementation

PNCC previously relied on batch-based, siloed systems to manage data across citizen services, infrastructure, and utilities. These systems were not able to deliver the real-time insights needed for proactive service delivery, predictive maintenance, or responsive citizen engagement.

To overcome these limitations, PNCC adopted Confluent Cloud as the backbone of its smart city data platform. By leveraging Confluent's fully managed services, the council connected diverse data sources including IoT devices, sensor networks, and operational systems across hybrid and multicloud environments.

Streaming pipelines now support a variety of use cases, such as monitoring water, using IoT telemetry, analyzing financial information, empowering AI insights, and improving citizen-facing digital services. With stream processing, PNCC can act on data in real time—for example, detecting anomalies in water infrastructure, optimizing consent workflows, and providing timely updates to residents and staff about cityservices. Stream governance ensures that all data is secure, auditable, and compliant with regulations.

"Good AI needs good data. Confluent is our trusted source of truth, streaming high-quality data to our data lakes and AI platforms to train models in real time. It provides context and orchestration for our agents to automate workflows, accelerating our smart city transformation."

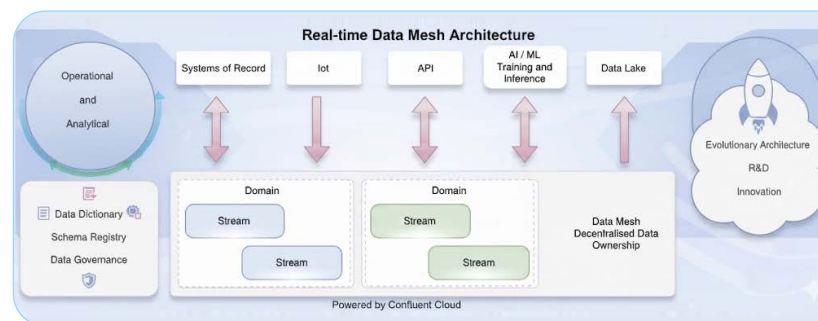
Atilio Ranzuglia, Head of Data and AI, Palmerston North City Council

Business Value

- **Improved public services:** Real-time data enables faster response times for citizen requests, while AI-powered financial processing ensures smarter resource allocation.
- **Infrastructure reliability:** Predictive monitoring of utilities such as water and energy systems reduces downtime and improves service continuity.
- **Operational efficiency:** Automated data flows cut down manual processes and free up council staff to focus on higher-value community initiatives.
- **Transparency and trust:** Secure and governed data pipelines provide citizens with confidence in the reliability and accountability of council services.
- **Future-Ready smart city:** A scalable and flexible data platform positions PNCC to adopt new smart city applications and innovations in areas such as sustainability, mobility, and digital citizen engagement.

"Confluent Cloud has significantly accelerated our capabilities—not only enabling us to reach this goal faster but also positioning us at the forefront of innovation."

Ryan Eames, Chief Information Officer, Palmerston North City Council



Real-time data mesh architecture at PNCC

Conclusion

The public sector is undergoing a profound digital transformation. Governments and institutions are building smart cities, improving service delivery, and strengthening resilience through real-time data. Data streaming has become the foundation of this shift, connecting systems, citizens, and infrastructure into a single, intelligent network.

Smart city initiatives show how this works in practice. IoT sensors, connected vehicles, and energy systems stream live data to monitor traffic, manage resources, and improve safety. Instead of reacting to static reports, cities can now respond instantly—rerouting traffic, adjusting lighting, and optimizing energy use. This real-time intelligence creates safer, cleaner, and more efficient environments that support economic growth and sustainability.

Citizen services are also being redefined. Data streaming unifies healthcare, social, and administrative systems to create a complete citizen view. This enables personalized support, faster processing, and proactive assistance across life stages. Real-time integration between agencies reduces duplication, increases transparency, and lowers operational costs while improving citizen satisfaction.

The same data-driven foundation powers national security, emergency response, and public health. The data streaming platform correlates live feeds from IoT devices, weather systems, and communication networks to detect threats, guide first responders, and monitor disease outbreaks. Agentic AI extends these capabilities by prioritizing alerts, coordinating resources, and triggering automated interventions in real time.

A strong example comes from PNCC in New Zealand. The council adopted Confluent Cloud as its data streaming backbone, integrating data from IoT devices, utilities, and citizen services across hybrid and multicloud environments. Real-time analytics now support water and energy monitoring, transport optimization, and infrastructure maintenance. Automated pipelines improve efficiency and transparency while governed data ensures compliance and public trust. The platform positions PNCC for future innovation in sustainability, mobility, and digital citizen engagement.

Across the public sector, data streaming enables continuous operations and informed decision-making. Real-time data improves predictive maintenance, automates workflows, and enhances service quality. Combined with AI and agentic AI, these event-driven systems can anticipate needs, optimize resources, and coordinate responses across agencies.

Governments that embrace this model move from reactive administration to proactive governance. Real-time insight improves efficiency, strengthens trust, and creates more transparent and resilient institutions. The convergence of data streaming, AI, and automation is building the next generation of digital public infrastructure—responsive, sustainable, and centered on the needs of citizens.

3.7 Energy and Utilities

One of the most significant trends in the energy and utilities sector is the push toward decarbonization and the integration of renewable energy sources. As countries and companies strive to meet ambitious climate goals, the adoption of solar, wind, and other renewable energy sources is accelerating. This shift necessitates the modernization of the grid to handle the intermittent nature of renewable energy and ensure reliable power delivery.

Grid modernization and the development of smart grids are critical components of this transformation. Smart grids leverage IoT, data analytics, and advanced communication technologies to provide real-time monitoring and control. This capability enhances grid reliability, optimizes energy usage, and enables the integration of distributed energy resources.

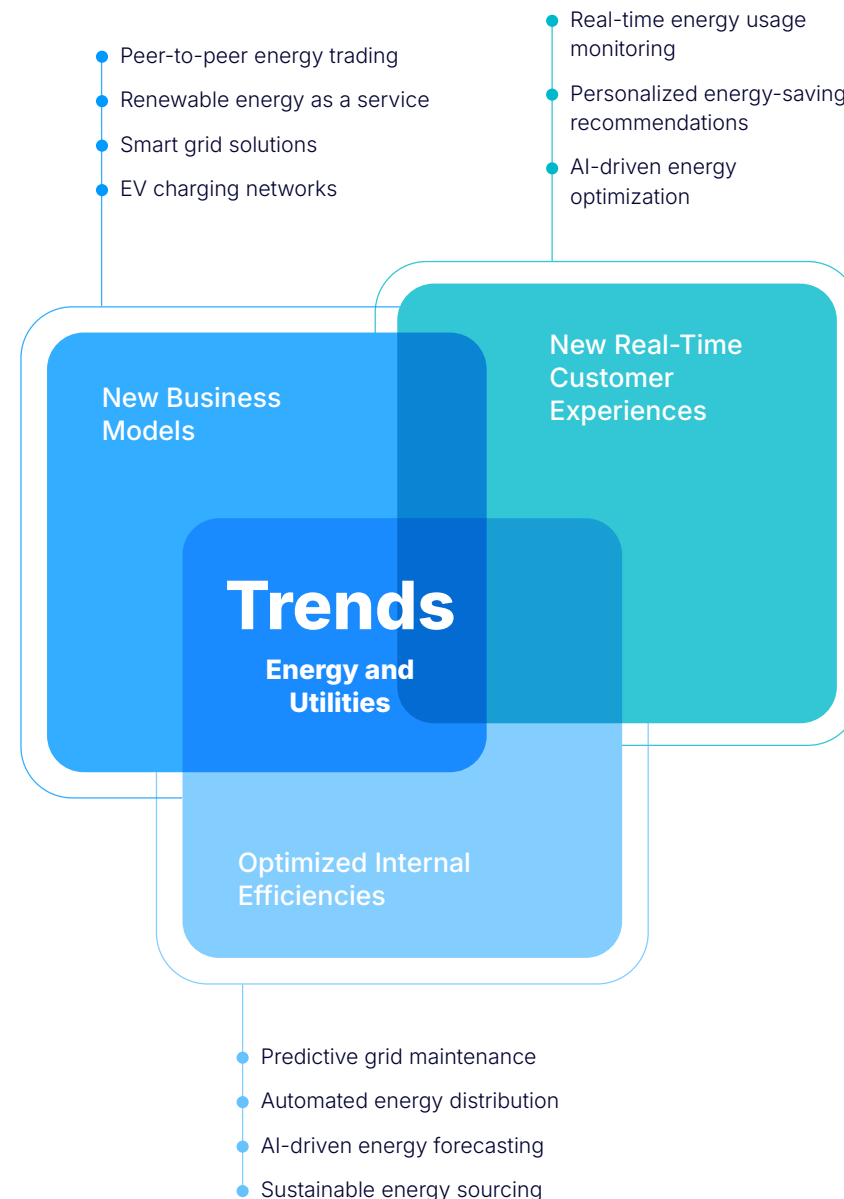
The digital customer experience is also becoming increasingly important in the energy and utilities sector. Customers now expect seamless, personalized interactions with their energy providers, similar to their experiences with other digital services.

Beyond customer-facing interactions, utility companies are also deploying real-time capabilities for trading, distributed energy networks, and electric vehicle (EV) charging infrastructure. Streaming platforms unify signals from markets, assets, and grid systems to make millisecond-level adjustments. Agentic AI builds on these foundations by predicting demand surges, balancing distributed generation, and orchestrating automated responses across the grid.

Data streaming plays a crucial role in enabling these trends by providing real-time data processing and analytics capabilities.

"[Grid Operators] still need to balance energy production and energy consumption, and they need to do that at the millisecond level. The importance of accurate, real-time forecasting has never been more important."

Dr. Dora Simroth, Head of Data and AI Engineering, E.ON Digital Technology



Scenario 1: Smart Grid Monitoring

Smart grid monitoring ensures optimal performance and reliability by monitoring and managing the electrical grid. Data pipelines and IoT offer real-time insights into the grid to quickly identify and resolve issues. Data streaming allows working with real-time data from various sources, such as weather conditions, solar panel performance, and energy usage, to quickly react to customer situations such as weather changes, tree shade, or component failures that impact power generation. This capability helps energy companies optimize energy usage, improve grid reliability, and enhance customer satisfaction.

Business Value

- **Optimized energy usage:** Real-time data processing allows energy companies to monitor and manage energy consumption efficiently, optimizing energy usage and reducing waste.
- **Improved grid reliability:** Continuous monitoring of the electrical grid enables quick identification and resolution of issues, enhancing grid reliability and minimizing downtime.
- **Enhanced customer satisfaction:** By providing reliable energy services and timely responses to customer situations, energy companies can improve customer satisfaction and foster loyalty.
- **Comprehensive grid visibility:** Access to real-time data from various sources provides a comprehensive view of the grid, supporting informed decision-making and strategic planning.
- **Proactive maintenance:** Real-time insights into grid performance allow for proactive maintenance and management, reducing the likelihood of component failures and extending the lifespan of grid assets.
- **Sustainable energy management:** By optimizing energy usage and integrating renewable energy sources, smart grid monitoring supports sustainable energy management and environmental responsibility.
- **Competitive advantage:** Energy companies that leverage real-time data for smart grid monitoring can differentiate themselves by offering reliable and efficient energy services, gaining a competitive edge.

Scenario 2: Industrial Control Systems

Industrial control systems monitor and control industrial processes and equipment to ensure optimal performance and safety. Modern IT and IoT systems enable open, flexible, and elastic supervisory control and data acquisition (SCADA) systems that stream data in real time from sources such as legacy OT gateways, ERP systems, and monitoring applications. These capabilities help energy companies ensure business continuity and enhance operational efficiency.

Business Value

- **Business continuity:** Continuous data processing and monitoring capabilities guarantee business continuity by quickly identifying and resolving issues before they escalate.
- **Improved safety and compliance:** Real-time insights into industrial processes enhance safety measures and ensure compliance with industry regulations and standards.
- **Seamless integration:** Modernized SCADA systems that connect to legacy OT gateways, ERP systems, and modern cloud-native applications provide seamless integration and data flow across the organization.
- **Proactive maintenance:** Access to real-time data allows for proactive maintenance and management of industrial equipment, reducing the likelihood of failures and extending asset lifespan.
- **Scalability and flexibility:** Open SCADA systems provide the scalability and flexibility needed to adapt to changing industrial demands and technological advancements.
- **Cost savings:** Improved operational efficiency and reduced downtime lead to significant cost savings, maximizing the return on investment in industrial control systems.
- **Innovation and modernization:** Leveraging IT modernization and IoT technologies positions energy companies as leaders in industrial innovation and modernization.

Scenario 3: Energy Trading

Energy trading involves the buying and selling of energy commodities in wholesale markets. By leveraging data pipelines and IoT, energy companies can implement algorithmic (algo) trading systems, energy trading and risk management (ETRM) systems, and external trading markets that are connected with correlated data. This real-time data processing capability helps energy companies process data faster, even with increased trading volumes and connected measurement devices.

By leveraging real-time data from various sources, including algo trading systems, ETRM systems, and external trading markets, energy companies can process and analyze trading data in real time. This real-time data processing capability allows organizations to process data faster and make data-driven trading decisions.

Business Value

- **Faster data processing:** Real-time data processing enables energy companies to handle increased trading volumes efficiently, ensuring timely and accurate data analysis.
- **Data-driven trading decisions:** Access to real-time data from various sources supports informed and strategic trading decisions, optimizing trading outcomes and profitability.
- **Enhanced risk management:** Integration of ETRM systems allows for continuous monitoring and assessment of market risks, improving risk mitigation strategies.
- **Increased trading efficiency:** Automated algo trading systems streamline trading operations, reducing manual intervention and increasing overall efficiency.
- **Improved market responsiveness:** Real-time insights into market conditions and trading data enable quick adaptation to market changes, enhancing competitiveness.
- **Seamless integration:** Connected data pipelines and IoT devices ensure seamless integration and data flow between trading systems and external markets.

Scenario 4: Decarbonization

Decarbonization is a critical goal for the energy and utilities sector, driving the adoption of cleaner energy sources and more efficient energy management practices. This use case involves the deployment of IoT devices and data pipelines to collect and analyze data on energy usage, emissions, and environmental impact. By leveraging a data streaming platform, utility companies can process and analyze this data in real time, enabling more efficient energy usage and improved ESG reporting.

Business Value

- **Optimized energy usage:** Real-time data analysis allows utility companies to monitor and manage energy consumption efficiently, reducing waste and optimizing energy usage.
- **Reduced emissions:** Continuous monitoring of emissions data enables utility companies to identify and implement strategies to reduce carbon emissions, contributing to environmental sustainability.
- **Improved ESG reporting:** Access to comprehensive and real-time data supports accurate and transparent ESG reporting, enhancing corporate responsibility and reputation.
- **Enhanced regulatory compliance:** Real-time insights into energy usage and emissions ensure compliance with environmental regulations and standards, reducing the risk of penalties and fines.
- **Public trust and confidence:** By demonstrating a commitment to sustainability and environmental responsibility, utility companies can build public trust and confidence in their operations.

Scenario 5: Virtual Power Plants and Distributed Energy Networks

The rise of renewable energy is driving a shift from centralized power generation to decentralized, distributed energy resources. Thousands of home batteries, rooftop solar panels, wind farms, and grid-scale storage systems can now be connected into a virtual power plant. This creates a coordinated, intelligent energy network that acts as a single flexible power source.

Unlike traditional batch-driven architectures, a data streaming platform continuously ingests, processes, and distributes real-time telemetry from these distributed assets. Every watt of production, every state-of-charge update, and every grid signal is streamed as an event. Stream processing enriches this data with grid demand, weather forecasts, and market conditions to trigger millisecond-level actions, such as supply and demand, redistributing stored power, and participating in wholesale energy markets.

This event-driven approach transforms millions of small, intermittent sources into a reliable, resilient, and profitable power grid. Homes and businesses not only consume electricity but can also contribute back, selling surplus energy into the grid while strengthening sustainability goals.

Business Value

- **Grid stability:** Dynamic balancing of distributed energy resources prevents outages and smooths renewable fluctuations.
- **Energy monetization:** Households and businesses can sell surplus power, creating new revenue opportunities.
- **Resilience:** Provides backup power during grid failures, extreme weather, or peak load conditions.
- **Sustainability:** Maximizes the use of renewables, reducing reliance on fossil fuels.
- **Regulatory alignment:** Provides real-time traceability and reporting support compliance with energy market and ESG requirements.
- **Scalability:** A single streaming backbone can manage millions of distributed energy devices with resilience and reliability.

Scenario 6: Electric Vehicle Charging Networks

As EV adoption accelerates, charging infrastructure faces new challenges: fluctuating grid demand, real-time availability updates, complex billing models, and the need for seamless customer experiences. Traditional batch-based or monolithic systems struggle to keep pace with the scale and responsiveness required.

A modern data streaming platform connects thousands of distributed charging stations, continuously ingesting telemetry such as charging status, energy consumption, and fault signals. Each charging event becomes a governed stream of data that multiple services can consume independently. Billing, mobile apps, roaming networks, and fraud detection all act in real time without tight coupling.

Streaming agents add intelligence by dynamically adjusting pricing, predicting maintenance needs, or coordinating with grid operators for load balancing. Instead of reactive operations, the network evolves into an adaptive system that balances supply, demand, and customer expectations autonomously.

Business Value

- **Seamless customer experience:** Drivers see accurate, real-time station availability and transparent billing.
- **Scalability:** Event-driven architecture supports millions of charging sessions without bottlenecks.
- **Operational efficiency:** Decoupled services simplify maintenance, reduce downtime, and accelerate new feature rollout.
- **Dynamic pricing and energy optimization:** Streaming agents balance grid demand, prevent overloads, and optimize costs.
- **Predictive maintenance:** Early detection of charger faults reduces failures and service interruptions.
- **New revenue opportunities:** Usage-based pricing and value-added energy services create monetization models beyond charging fees.
- **Sustainability impact:** Integration with renewable sources and bidirectional charging vehicle-to-grid supports decarbonization goals.

Real-Life Examples

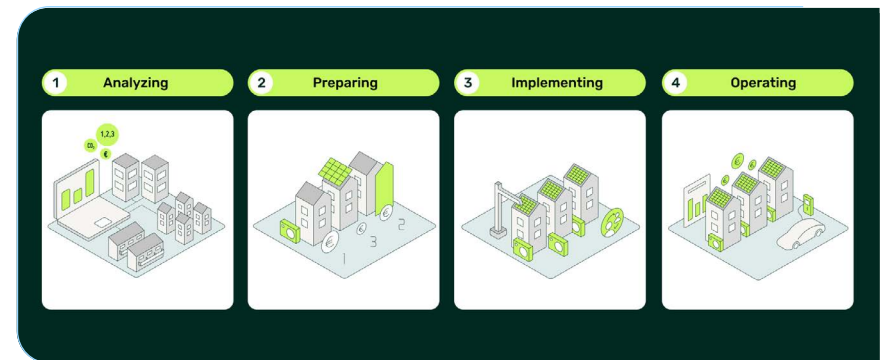
- AMPEERS ENERGY
- aedifion
- Uniper

Real-Life Example: AMPEERS ENERGY

Use Cases: Hybrid and Multicloud | IoT

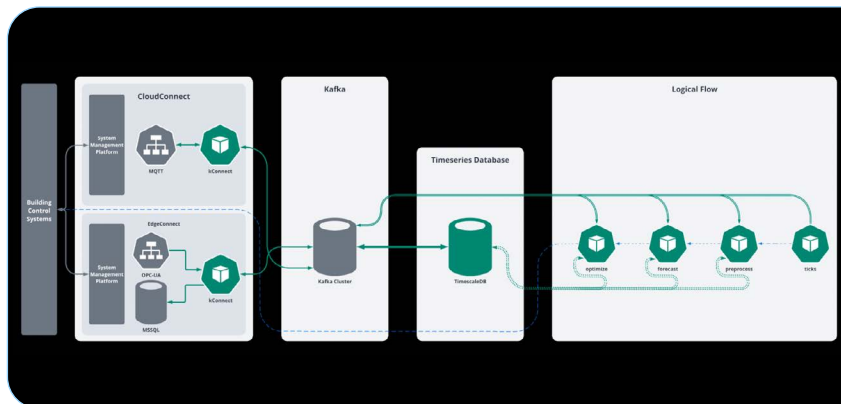
Company Overview

AMPEERS ENERGY is a leading provider of energy management solutions, specializing in the optimization of energy usage and the integration of renewable energy sources. The company offers a range of innovative solutions designed to help organizations achieve their decarbonization goals, improve energy efficiency, and enhance ESG reporting. AMPEERS ENERGY solutions leverage advanced IoT technologies and data streaming platforms to provide real-time insights and optimization capabilities.



Implementation

AMPEERS ENERGY implemented a comprehensive decarbonization solution for a large real estate portfolio, focusing on district management and local supply optimization. The solution involved the deployment of IoT sensors and devices across multiple buildings to collect real-time data on energy usage, temperature, humidity, and other relevant parameters. This data was then streamed to a central platform powered by Confluent with Apache Kafka, where it was processed and analyzed in real time.



Architecture overview of AMPEERS ENERGY's Kafka Implementation

The district management component of the solution involved the use of advanced forecasting and optimization algorithms to predict energy usage patterns and identify opportunities for optimization. By analyzing historical data and real-time inputs, the system could generate accurate forecasts of energy demand and implement strategies to reduce energy usage during peak periods. This included adjusting heating and cooling systems, optimizing lighting schedules, and implementing demand response measures.

The local supply management component focused on accounting for local energy usage and ensuring that energy was sourced from renewable or low-carbon sources. This involved integrating data from energy meters, weather forecasts, and renewable energy generation systems to provide a comprehensive view of energy supply and demand. By leveraging data streaming platforms, AMPEERS ENERGY was able to process and analyze this data in real time, enabling more informed decision-making and better management of energy resources.

"The biggest benefits for us to use Confluent Cloud are that the management goes into someone that founded Kafka and then operates it on a well-known level. This takes responsibility away from us, and we can simply say, 'This works!'"

Lucas Recknagel, CTO, AMPEERS ENERGY

Business Value

- More efficient energy usage and reduced carbon emissions:** The implementation of the decarbonization solution allowed AMPEERS ENERGY to monitor and manage energy usage in real time, enabling more efficient energy consumption across multiple buildings. The district management component optimized energy usage, leading to substantial cost savings and reduced environmental impact. By leveraging advanced data analytics, AMPEERS ENERGY could identify inefficiencies and make data-driven decisions to further reduce energy consumption and emissions.
- Improved renewable energy sourcing and support for decarbonization goals:** The technology ensures that energy is sourced from renewable or low-carbon sources, aligning with the decarbonization objectives of AMPEERS ENERGY and its clients. This not only reduces reliance on traditional energy sources but also helps AMPEERS ENERGY's clients meet sustainability goals and comply with increasing regulatory demands for renewable energy usage.
- Local supply management:** This feature ("Mieterstrom" in German) can make it possible to refinance investments via a local electricity tariff. The tariff chooses locally produced electricity from solar panels over grid electricity. This offers landlords and tenants the economic benefits of quick and easy solar power generation.
- Enhanced energy management system (EMS) reporting and regulatory compliance:** By using real-time data collection and analysis, AMPEERS ENERGY provides clients with accurate and comprehensive EMS reporting. The solution enabled precise tracking of energy usage and emissions, ensuring compliance with regulatory requirements and demonstrating client commitment to sustainability. The use of data streaming platforms allowed for the real-time processing of large data volumes, driving continuous improvement in energy management and long-term decarbonization efforts.

Real-Life Example: aedifion

Use Cases: Observability | IoT | ESG

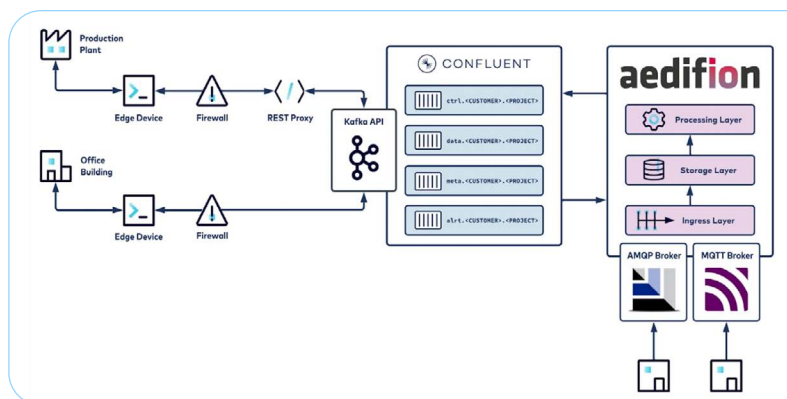
Company Overview

Industry leader aedifion provides cloud-based energy management solutions for optimizing building portfolios. The company's platform leverages IoT sensors, data streaming technologies, and advanced analytics to monitor and optimize building performance in real time. Its solutions are designed to help property managers reduce energy consumption, lower operational costs, and achieve sustainability goals.

Implementation

aedifion implemented a comprehensive energy management solution for a large real estate portfolio, integrating with the existing building management systems (BMS) to provide real-time data and insights. The solution involved deploying IoT sensors throughout the buildings to monitor various parameters, including temperature, humidity, occupancy, and energy usage. These sensors generated vast amounts of data that were streamed in real time to aedifion's cloud platform using Confluent.

The data streaming platform facilitated the real-time processing and analysis of the sensor data, enabling aedifion to monitor building performance continuously. Advanced analytics and machine learning algorithms were applied to identify inefficiencies, detect anomalies, and recommend optimization measures. The platform also provided property managers with real-time dashboards and alerts, allowing them to take immediate action to address issues and optimize energy usage.



Confluent-based data pipeline at aedifion

Business Value

- **Energy savings, reduced operational costs, lower carbon footprint:** The decarbonization initiative provided significant business value for aedifion's clients, like for example 17.1% CO₂ savings. By leveraging real-time data streaming and advanced analytics, the energy management solution enabled property managers to optimize building operations and enhance energy efficiency.
- **Preventing energy waste and improving overall sustainability:** The real-time visibility into building performance allowed property managers to identify and address inefficiencies promptly, preventing energy waste and improving overall sustainability. The solution also supported compliance with environmental regulations and sustainability certifications, enhancing the reputation and marketability of the real estate portfolio.
- **Improved comfort and satisfaction of building occupants:** Furthermore, the integration of IoT sensors and data streaming technologies enabled continuous monitoring and optimization, ensuring that buildings operated at peak efficiency at all times. This proactive approach to energy management not only contributed to decarbonization efforts but also improved the comfort and satisfaction of building occupants.



Significant reductions achieved in the Kaiser Hof Cologne

Real-Life Example: Uniper

Use Cases: Event-Driven Microservices | Data Sharing

Company Overview

Düsseldorf-based Uniper is a European energy company with global reach and activities in more than 40 countries. With approximately 7,400 employees, the company makes an important contribution to security of supply in Europe, particularly in its core markets of Germany, the UK, Sweden, and the Netherlands. Uniper's operations encompass power generation in Europe, global energy trading, and a broad gas portfolio. Uniper procures gas, including liquefied natural gas (LNG), and other energy sources on global markets. The company owns and operates gas storage facilities with a total capacity of more than 7 billion cubic meters.

Uniper intends to be completely carbon-neutral by 2040. It aims for its installed power-generating capacity to be more than 80% zero-carbon by the early 2030s. To achieve this, the company is transforming its power plants and facilities and investing in flexible, dispatchable power-generating units. Uniper is already one of Europe's largest operators of hydropower plants and is helping further expand solar and wind power, which are essential for a more sustainable and secure future. The company is progressively expanding its gas portfolio to include green gases such as hydrogen and biomethane and aims to convert to these gases over the long term.

Uniper is a reliable partner for communities, municipal utilities, and industrial enterprises for planning and implementing innovative, lower-carbon solutions on their decarbonization journeys. As a hydrogen pioneer, Uniper is active worldwide along the entire hydrogen value chain and is conducting projects to make hydrogen a mainstay of the energy supply.

Implementation

In the volatile and interconnected world of energy trading, Uniper recognized the need to leverage advanced technologies to optimize its trading operations and maintain a competitive edge. The wholesale trading market involves a myriad of interactions between energy producers, utility companies and traders, all working to balance energy supply and demand. To navigate this intricate ecosystem, Uniper implemented data streaming technologies to enable real-time data processing and analytics, thereby enhancing its trading capabilities.

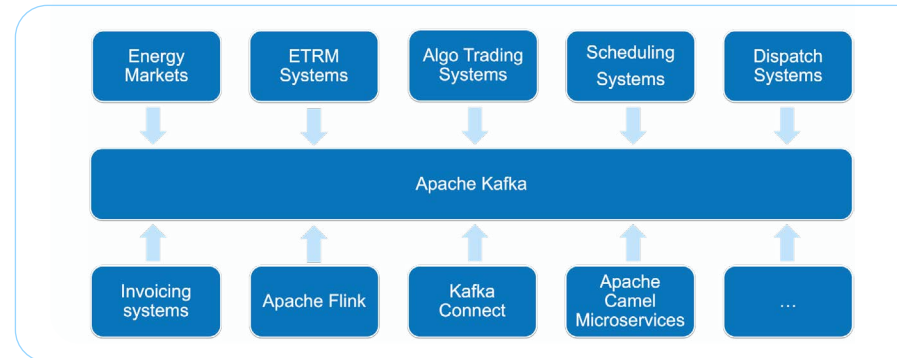
Uniper's energy trading operations are underpinned by Confluent's data

streaming platform, which facilitates the continuous streaming of data from a variety of sources, including market prices, weather forecasts, and grid conditions. By processing this data in real time, Uniper supports its algo trading systems, ETRM systems, and external trading markets.

The data streaming platform enables seamless integration and correlation of data, providing traders with real-time insights and actionable information. This integration is crucial for creating a seamless trading environment where algo trading systems, ETRM systems, and scheduling/dispatch/invoice systems are interconnected.



Uniper's sales and trading focus market areas



Decoupled producers and consumers at Uniper

Uniper applies advanced analytics and machine learning algorithms to the streamed data to optimize trading strategies, manage risks, and enhance decision-making. Real-time dashboards and alerts are employed to allow traders to continuously monitor market conditions and respond swiftly to changes. This capability ensures that Uniper's trading operations are not only profitable but also efficient.

Business Value

- **Faster data processing:** The ability to process data in real time enables Uniper to respond quickly to market changes, optimizing trading strategies and capturing opportunities as they arise.
- **Increased trading volumes:** The data streaming platform supports increased trading volumes and the integration of several trading systems. This allows Uniper to expand its trading activities.
- **Enhanced decision-making:** Real-time insights and advanced analytics improve trading decisions and risk management, giving Uniper competitive advantage in the energy market.
- **Operational efficiency:** The seamless integration of trading systems and data sources enhances operational efficiency to reduce the complexity and cost of managing trading operations.
- **Scalability and flexibility:** The cloud-based data streaming platform provides the scalability and flexibility needed to handle large volumes of data and support Uniper's business growth. This ensures that Uniper can adapt to changing market conditions and continue to innovate in its trading operations.

"We saw the demand in our business rising; trading volumes have been increasing significantly, and this put more pressure on our systems. More messages were coming in daily, and we required faster processing of data. We did not want to miss any opportunity in the market just because our IT processes were not fast enough. The decision to go with data streaming based on Apache Kafka and Apache Flink® was both business- and architecture-driven."

Alexander Esseling, Platform & Security Architect, Uniper

Conclusion

The energy and utilities sector is undergoing one of the most significant transformations in its history. Decarbonization, renewable integration, and digital innovation are reshaping how energy is produced, distributed, and consumed. Data streaming is at the core of this shift, connecting assets, markets, and customers in real time to balance performance, reliability, and sustainability.

Smart grids now combine IoT, analytics, and automation to monitor and control energy distribution with near-real-time responsiveness. Continuous data flows from sensors, weather models, and grid systems help forecast demand, balance renewables, and prevent outages. Static infrastructure becomes a dynamic, data-driven network that adapts to change.

From generation to consumption, data streaming creates intelligence across the energy value chain. Distributed networks and virtual power plants link decentralized assets—like solar panels, wind turbines, and home batteries—into unified systems. Each event, from production changes to local demand shifts, is processed instantly to maintain grid stability and enable new business models such as peer-to-peer trading.

Real-world innovators show measurable results. AMPEERS ENERGY optimizes building portfolios with IoT and streaming to cut emissions and improve efficiency. Aedifion uses real-time analytics to reduce energy waste in property operations. Uniper applies streaming data in energy trading to correlate prices, weather, and grid signals in real time for better risk management.

AI amplifies these capabilities by analyzing grid telemetry, forecasting consumption, and triggering automated actions for pricing, load balancing, and maintenance. On the trading side, AI acts on live market data. For consumers, data streaming powers personalized energy experiences via smart meters, EV charging, and dynamic pricing.

This convergence of data streaming, AI, and distributed energy marks a decisive shift. Continuous data processing enables predictive maintenance, optimized usage, and seamless renewable integration. Organizations embracing event-driven architectures are building more resilient, intelligent, and sustainable energy systems for the future.

3.8 Transportation and Travel

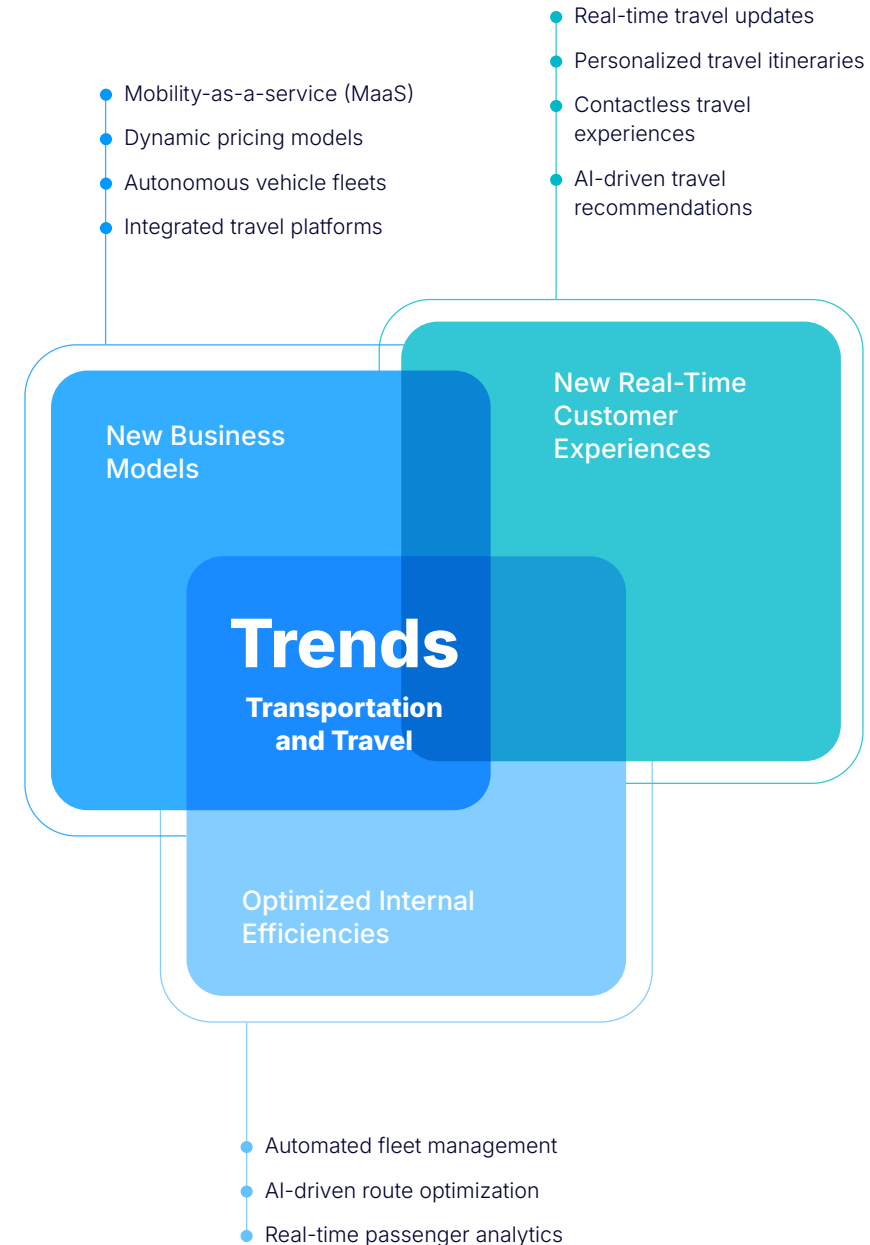
The transportation and travel industry is experiencing a wave of digitalization and innovation, driven by advancements in technology and changing consumer expectations.

One of the most significant trends is the adoption of robotics and autonomous vehicles, which are revolutionizing the way goods are transported and delivered. These technologies enhance operational efficiency, reduce costs, and improve safety. Biometric and digital identity solutions are also becoming increasingly important in the travel industry. These technologies enable seamless and secure travel experiences, from check-in to boarding, by leveraging biometric data for identity verification.

Improving the B2B and B2C customer experience is another critical trend in the logistics and travel industry. Customers now expect personalized, seamless interactions with logistics and travel providers that are to their experiences with other digital services. Data streaming plays a crucial role in enabling these trends by providing real-time data processing and analytics capabilities.

At the same time, on-demand mobility services, smart logistics, and border operations require instant coordination of vehicles, goods, and travelers across complex ecosystems. A data streaming platform unifies signals from IoT devices, booking engines, customs systems, and traffic networks to provide live visibility and adaptability. Generative and agentic AI extend these capabilities by personalizing travel recommendations, optimizing tariffs and routes, and even orchestrating autonomous fleets in real time.

The rise of robotics, autonomous vehicles, and digital identity solutions has further increased the demand for robust and scalable data streaming solutions. As the logistics and travel industry continues to evolve, the adoption of data streaming technologies will be crucial in maintaining competitiveness and driving innovation.



Scenario 1: Transport Management

Transport management involves the planning, execution, and monitoring of the movement of goods and people. IoT and ESG principles can help logistics companies transform their transport management systems into transport visibility and management systems. Data streaming can be used to monitor the movement of vehicles and goods in real time and provide real-time updates to customers, providing end-to-end visibility for logistics processes, optimization of operations, and reduction of environmental impact.

Business Value

- **End-to-end visibility:** Real-time data processing provides comprehensive visibility into the logistics process, allowing companies to monitor the movement of goods and people effectively.
- **Optimized logistics operations:** Continuous monitoring and analysis of transportation data enable companies to optimize routes, reduce delays, and improve overall logistics efficiency.
- **Reduced environmental impact:** By integrating ESG principles and leveraging real-time environmental data, companies can implement strategies to minimize their carbon footprints and promote sustainability.
- **Enhanced customer satisfaction:** Real-time updates and transparency in logistics operations improve customer satisfaction by providing timely and accurate information on the status of shipments.
- **Data-driven decision-making:** Access to real-time data from IoT sensors, GPS tracking, and environmental sources supports informed decision-making, allowing companies to implement effective transport management strategies.

Scenario 2: On-Demand Mobility and Delivery Services

On-demand mobility and delivery services such as ride-hailing, food delivery, and last-mile logistics rely on real-time coordination between supply and demand. Data streaming connects driver or courier availability, customer requests, traffic conditions, and order details to optimize operations at scale. Providers can also enrich these streams with payment data, location history, and external signals such as weather and events to improve decision-making and create more personalized customer experiences.

Business Value

- **Optimized pricing strategies:** Real-time data enables dynamic pricing models that adjust based on demand, availability, and traffic, maximizing revenue and competitiveness.
- **Enhanced customer satisfaction:** Faster, more reliable rides and deliveries improve convenience and loyalty.
- **Maximized resource utilization:** Real-time insights balance supply and demand, ensuring that drivers and couriers are allocated where they're needed most.
- **Effective fraud detection:** Continuous monitoring flags anomalies in orders, payments, and user activity, reducing losses.
- **Targeted engagement:** Data streaming powers real-time marketing campaigns, promotions, and loyalty rewards during peak times.
- **Competitive advantage:** Reliable, transparent, and personalized services help companies differentiate in a crowded market.

Scenario 3: Travel Booking

Travel booking involves the planning and booking of travel arrangements, including flights, hotels, and activities. Data pipelines and AI/ML allow travel companies to integrate chatbots and generative AI to support customer service functions such as rebooking, refunds, and itinerary changes. With data streaming, these systems can access real-time data from various sources, including customer interactions, booking data, and travel information. This makes it possible for travel companies to provide automated self-service options, enhance customer satisfaction, and increase net promoter scores.

Business Value

- **Enhanced customer satisfaction:** By providing efficient and responsive customer service through chatbots and generative AI, travel companies improve customer satisfaction and foster loyalty.
- **Automated self-service options:** Real-time data processing and AI integration enable automated self-service options, allowing customers to manage bookings, rebook, request refunds, and make itinerary changes independently.
- **Increased net promoter scores:** Streamlined and personalized customer interactions contribute to higher net promoter scores, reflecting improved customer experiences and brand advocacy.
- **Comprehensive operational visibility:** Access to real-time data from customer interactions, booking data, and travel information provides a comprehensive view of vacation booking operations, supporting informed decision-making.
- **Improved operational efficiency:** Automated processes and efficient resource management reduce operational costs and improve the overall efficiency of vacation booking services.
- **Data-driven personalization:** Real-time insights into customer preferences and behaviors enable personalized service offerings, enhancing customer engagement and satisfaction.

Scenario 4: Baggage and Cargo Tracking Services

Baggage and cargo tracking ensures end-to-end visibility of goods and luggage during transport. By streaming real-time data from IoT tags, radio-frequency identification scanners, and baggage handling systems, airlines and logistics providers can monitor assets across airports, warehouses, and transit hubs. A streaming platform integrates data from multiple stakeholders, enabling passengers to receive live updates on baggage location and logistics companies to ensure that freight arrives on time. Streaming agents also detect anomalies, such as misrouted luggage and excessive delays, and trigger corrective workflows before problems escalate.

Business Value

- **Reduced losses:** Continuous, real-time tracking of baggage and cargo across the entire transport chain significantly reduces incidents of lost items, misplaced freight, or delayed shipments. This lowers operational costs and improves insurance claim accuracy.
- **Enhanced customer satisfaction:** Live updates via mobile apps or notifications keep passengers and logistics clients informed of the exact location and status of their baggage or shipments. This transparency improves trust and reduces frustration during delays or disruptions.
- **Proactive issue resolution:** Streaming agents detect anomalies such as misrouted luggage, excessive wait times, or stalled shipments in real-time. Early alerts allow staff to intervene before issues escalate, improving recovery time and avoiding compensation claims.
- **Operational efficiency:** Real-time data sharing automates handoffs between baggage handlers, freight operators, airlines, and logistics hubs. This improves coordination, reduces manual errors, and speeds up handling across complex global operations.
- **Increased trust:** Providing end-to-end visibility and timely communication fosters confidence among passengers, business customers, and partners. This reliability builds long-term loyalty and strengthens the brand's reputation in a competitive market.

Scenario 5: Autonomous Fleet Operations

Autonomous trucks, drones, and delivery robots are becoming a reality in logistics and urban mobility. Data streaming enables the real-time coordination of autonomous fleets by processing telemetry data from vehicles, GPS, traffic systems, and weather feeds. With continuous monitoring, fleets can dynamically reroute vehicles to avoid delays, optimize delivery schedules, and detect anomalies such as unsafe driving conditions or equipment failures. Streaming applications with embedded AI models enhance operations by providing predictive insights, ensuring uptime, and supporting sustainable logistics strategies.

Business Value

- **Increased efficiency:** Real-time route optimization and traffic-aware navigation reduce idle time, fuel consumption, and overall delivery costs. Autonomous fleets can make continuous adjustments to maximize throughput and asset utilization.
- **Improved safety:** Continuous monitoring of telemetry and environmental data enables early detection of hazards, mechanical failures, or unsafe driving conditions. Automated alerts and interventions help prevent accidents and equipment damage.
- **Faster delivery:** Autonomous vehicles adapt in real-time to road closures, congestion, and weather changes. This flexibility ensures faster, more reliable delivery times and higher service levels in dynamic urban and regional environments.
- **Sustainable operations:** Optimized routing, smoother driving patterns, and reduced vehicle downtime contribute to lower emissions. Real-time data supports greener logistics strategies aligned with sustainability goals and regulations.
- **Competitive advantage:** Leveraging data streaming and AI for autonomous logistics enables faster innovation cycles, higher automation, and lower operating costs. This gives early adopters a strong position in an increasingly automated market.

Scenario 6: Border Control, Customs, and Tariff Management

Border control and customs processing are essential for secure and efficient international trade and travel. Data streaming integrates biometric systems, digital identities, cargo manifests, customs declarations, and tariff databases to verify travelers and goods in real time. Authorities can continuously analyze shipments to detect anomalies such as mismatched identities, unauthorized cargo, or incorrect tariff classifications that impact duties and trade compliance. With a data streaming platform, customs officers and trade agents gain real-time visibility into goods movement, enabling faster tariff calculations, smoother customs clearance, and immediate identification of risks. Agentic AI enhances decision-making by suggesting optimal trade routes, flagging compliance risks, and automating routine tariff checks.

Business Value

- **Reduced wait times:** Automated verification of identities, cargo, and documentation speeds up customs clearance, reducing bottlenecks at border crossings, airports, and ports.
- **Enhanced security and compliance:** Real-time anomaly detection across cargo manifests, biometric data, and declarations strengthens defenses against fraud, smuggling, and trade violations.
- **Optimized trade flows:** Continuous monitoring of cargo and tariffs streamlines logistics, reducing bottlenecks at borders.
- **Improved inter-agency collaboration:** Seamless data sharing across customs, immigration, and trade authorities ensures consistency and compliance.
- **Revenue protection:** Accurate, real-time tariff enforcement prevents revenue leakage and ensures fair taxation.
- **Balanced priorities:** This accelerates legitimate trade while upholding security, regulatory, and fiscal responsibilities.

Real-Life Examples

- Austrian Post
- Deutsche Bahn
- Flix
- LKW Walter
- Lufthansa
- Swiss Post
- Schiphol Group

"Confluent enables an event-driven architecture that improves efficiency for software engineering. Adopting data streaming opens up numerous valuable use cases by bringing insights closer to the analysts."

Maxim Foursa, Senior Engineering Manager, Booking.com

Real-Life Example: Austrian Post

Use Cases: Data Products | Observability | Event-Driven Microservices

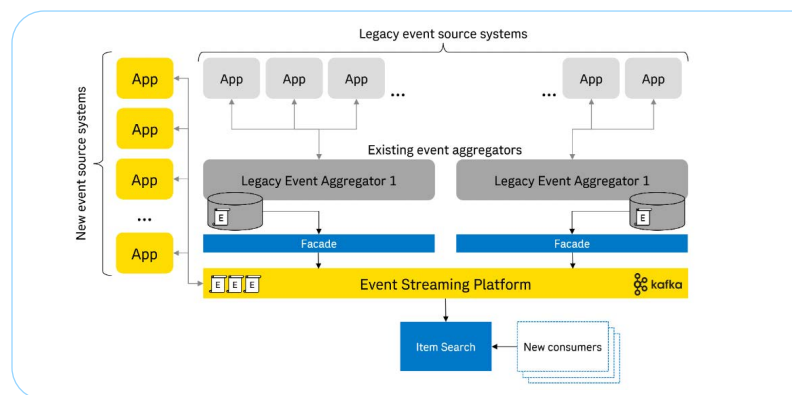
Company Overview

Austrian Post is an international postal, logistics, and service provider which is central to Austria's economy. The company is firmly focused on the very highest quality and offers a comprehensive product and service portfolio to provide the best possible match for current customer needs. Austrian Post groups its operations into three divisions: Mail, Parcel & Logistics, and Retail & Bank. The company also has an international presence, operating in the markets of Germany, Southeast and Eastern Europe, Turkey, and Azerbaijan.

Implementation

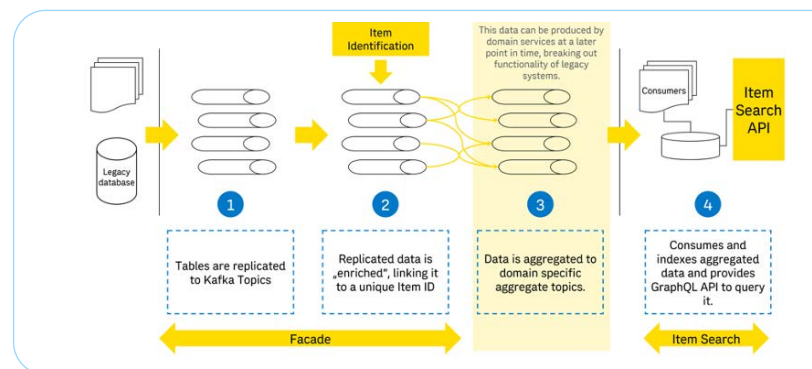
Austrian Post faced challenges with its existing shipment data management system ('track-and-trace'). The data was centralized in a large and conventional relational database system. It was difficult to maintain the system due to its monolithic structures. This, in turn, led to scaling limitations. Additionally, since many other systems were dependent on the shipment data meant an ever-increasing load was placed on the core system.

There was no conceivable migration scenario beyond a "big bang" replacement, which, for obvious reasons, was not an option. To address these issues, Austrian Post implemented a comprehensive solution involving its Confluent-based data streaming platform, a newly built façade around their legacy core system, and a GraphQL-based item search to provide easy access to the shipment data.



The facade wraps the legacy monolith(s)

The solution involves continuously pulling data from the legacy database, enhancing it with unique IDs, organizing it into domain-specific aggregates, and making it easily searchable and accessible through an API. This solution leverages Kafka-based technology, whereby a couple of dozen complex stream applications are employed to handle high-volume data, reduce monolith dependencies, allow access to important data without scaling issues, and provide a way to merge data for specific domains that are currently spread over several legacy applications. Additionally, it marks the first important step in a sweeping transformation strategy that will allow Austrian Post to decompose the monolith(s) in the future.



From legacy database to scalable item search

Business Value

- **Increased transparency:** The real-time availability of shipment data increases internal transparency and enables full item visibility with additional scans.
- **Increased customer satisfaction:** Rapid access to their own shipment data at scale enhances the customers' experience, resulting in higher satisfaction levels.
- **Reduced time to market:** The solution provides a shortcut to access data without waiting for all the individual systems to connect to the event streaming platform.
- **Improved cross-product visibility:** The ability to merge domain data from legacy applications increases cross-product visibility and simplifies access to data.

Real-Life Example: Deutsche Bahn

Use Cases: Data Products | IT Modernization | Event-Driven Microservices

Company Overview

Deutsche Bahn (DB) is Germany's national railway company and a vital part of Europe's transport infrastructure, responsible for the journeys of over 10 million passengers daily. The company operates thousands of trains and manages a complex network of stations, making access to timely and accurate information a mission-critical component of the customer experience.

Implementation

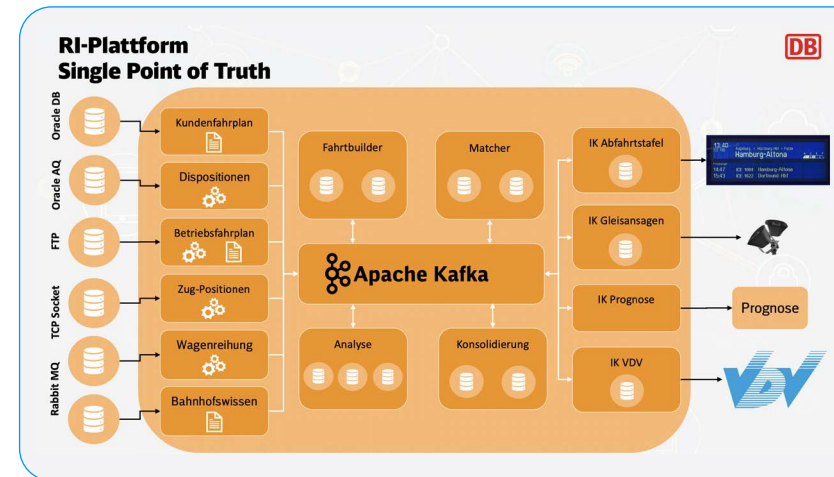
Deutsche Bahn introduced 2018 RI-Plattform-short for Reisendeninformationsplattform (Passenger Information Platform), a real-time, event-driven data streaming architecture to provide reliable and comprehensible information to passengers. Acting as a central system for passenger data, it aggregates and processes inputs from a wide range of sources, including live train positions, schedules, and station sensor data.

Stream processing was introduced early on to enable stateful real-time and batch processing. With this, the RI team reduced the time for a major data import process from 20 minutes just to 2 minutes and validated the technical foundation for a highly scalable platform. From the beginning, DB built RI-Plattform as an open hub. It integrates seamlessly with over 16 external partners, including Lufthansa and Google, to power a unified, multimodal travel experience across systems.

The platform, which went live in 2018, has since grown from a pilot at 80 stations to nationwide coverage across more than 11,000 stations. Today, it handles 1.5 billion events every 24 hours (an eightfold increase from the early days) and supports more than 400 microservices. These microservices are developed and operated by 12 agile DevOps teams under a "you build it, you run it" model. Continuous deployment is standard practice: New features go live up to 10 times per day without downtime, thanks to a resilient, cloud-native architecture and dedicated 24/7 support teams.

"For our passengers, trust in real-time information is essential. They need to know that the platform number in their app is the exact same one they'll see on the station display. By creating a single, real-time source of truth on Confluent, we've eliminated the data discrepancies that cause confusion. The result is a more seamless and predictable travel experience for millions of people every day."

Alexander Herrmann, Product Owner, Scaled DevOps Team, DB Systel



Reiseninformationsplattform as the single source of truth for Deutsche Bahn based on Apache Kafka and Confluent

Business Value

- **Trusted passenger experience:** A single source of truth ensures that all channels from mobile apps to station signs and external partners deliver the same reliable data in real time. This data consistency builds passenger confidence and reduces stress during travel.
- **Increased operational agility:** The move to an event-driven microservices landscape has enabled DB's teams to iterate quickly, ship features rapidly, and respond to change with flexibility. Time to market for new functionality has dropped significantly.
- **Open ecosystem integration:** RI-Plattform acts as a central hub for real-time data exchange with more than 16 external partners, including Lufthansa and Google. It enables collaboration beyond organizational boundaries and supports the vision of end-to-end journey planning across providers.
- **Future-proof scalability:** From 80 to 1,000+ stations, from millions to billions of daily events, the platform has scaled reliably to match growing demand and evolving complexity. This provides a solid foundation for long-term innovation.
- **Foundation for continuous innovation:** By treating real-time data as a product, DB has unlocked the ability to create new, competitively differentiating services, such as automated station announcements triggered by sensor data.

Real-Life Example: Flix

Use Cases: Data Products | Hybrid and Multicloud | Event-Driven Microservices

Company Overview

Flix is a global travel-tech company reinventing long-distance bus and train travel. Founded in Germany, the company now operates in Europe, North America, South America, and India. In 2021, Flix expanded its footprint in the United States through the acquisition of Greyhound, the largest provider of long-distance bus travel in North America. This move strengthened Flix's global presence and positioned the company as a leader in affordable, sustainable, and connected mobility. To support rapid international growth and deliver reliable services, Flix relies heavily on real-time data to power operations, bookings, and customer engagement.



Implementation

As the business scaled, Flix struggled with the growing complexity of managing self-hosted Apache Kafka. Engineering teams spent significant time fixing infrastructure issues instead of building new features. Data bottlenecks slowed application performance, delaying innovation and affecting customer experience.

To overcome these challenges, Flix migrated to Confluent Cloud. The fully managed platform allowed the company to offload infrastructure management, streamline data flows, and improve governance across its global operations. With Confluent, Flix established resilient streaming pipelines that processed booking data, vehicle telemetry, payment transactions, and customer interactions in real time. These streaming pipelines now power a wide range of mission-critical use cases, from near-real-time

fraud detection (identifying fare evasion, payment manipulation, and bots attempting to buy up high-demand tickets) to optimizing seat availability to predicting vehicle arrivals. This enables both smoother customer journeys and stronger financial protection for the business.

Key improvements included reliable event streaming across regions, zero-downtime migration planning, and built-in observability to monitor metrics. This foundation enabled Flix to focus engineering resources on advanced digital services and data-driven features instead of infrastructure maintenance. In 2025, Flix's AI-driven systems managed more than 542 million seat combinations daily and 0.9 million rides per day and re-optimized pricing for more than 47 million seats each day. These real-time insights only became possible with a modern, reliable streaming infrastructure.

Business Value

- **Accelerated feature delivery:** New product features are rolled out 4–6x faster, improving time to market and competitiveness.
- **Engineering productivity:** 60% of engineering time was reallocated from infrastructure maintenance to innovation and customer-focused development.
- **Improved customer experience:** An 80% drop in service desk requests reflects smoother, more reliable booking processes.
- **Stronger fraud prevention:** Real-time data pipelines enable proactive fraud detection, reducing financial risks.
- **Global scalability:** Confluent Cloud provides the resilience and flexibility needed to support Flix's international expansion in multiple continents.

"It was our first Kafka migration, and Confluent brought in the experience and structure we needed. They helped us map everything out—how to make it zero downtime, how to prepare our apps, even what metrics to monitor. That guidance made a huge difference."

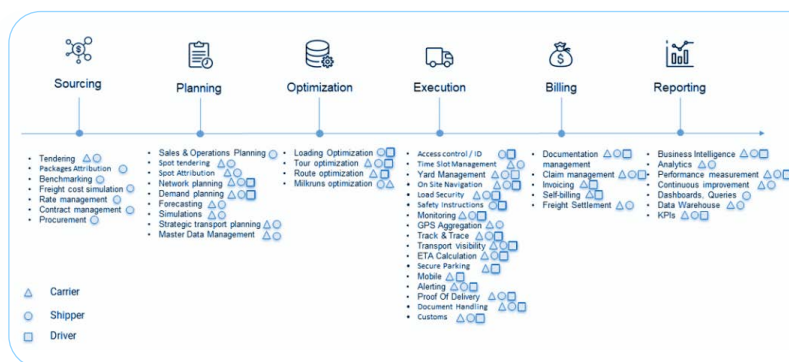
Ashish Kumar, Senior Software Engineer, Flix

Real-Life Example: LKW Walter

Use Cases: Observability | Event-Driven Microservices | Data Sharing

Company Overview

LKW Walter, a leading European transport organization, implemented a comprehensive transportation management system leveraging Confluent as a data streaming platform. The solution involved deploying IoT sensors on trucks and throughout the supply chain to monitor their locations and statuses in real time. These sensors generated vast amounts of data that were streamed in real time to LKW Walter's cloud platform using Confluent.



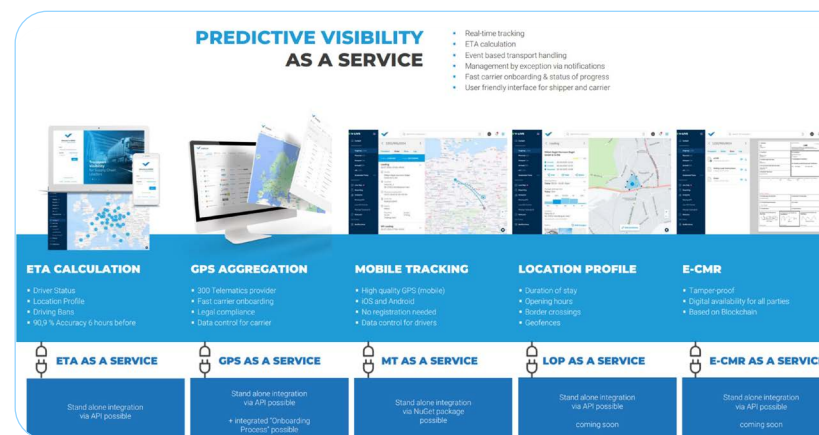
The end-to-end logistics process at LKW Walter that is now fully event-driven

Implementation

The data streaming platform facilitated the real-time processing and analysis of the sensor data, enabling LKW Walter to monitor and optimize truck logistics operations continuously. Advanced analytics and machine learning algorithms were applied to optimize routes, reduce fuel consumption, and ensure that deliveries were made on time and within budget. The platform also provided real-time dashboards and alerts, allowing logistics managers to take immediate action to address issues and optimize operations.

"An event-driven architecture lets us respond to events in real time, dramatically boosting the efficiency of our business processes. By automating decisions and notifications instead of relying on manual intervention, we keep customers updated on the status of their deliveries in real time. This level of agility and transparency gives us a distinct competitive edge."

Leo Hintersteiner, CIO, LKW Walter



Predictive visibility as a service at LKW Walter

Business Value

- **Reduced transportation costs:** By optimizing routes and reducing fuel consumption, LKW Walter significantly lowered transportation costs.
- **Enhanced delivery reliability:** Real-time monitoring and optimization ensured that deliveries were made on time, enhancing reliability.
- **Improved operational efficiency:** Continuous monitoring and real-time data processing enabled efficient management of logistics operations.
- **Increased customer satisfaction:** Timely deliveries and improved reliability led to higher customer satisfaction.
- **Competitive advantage:** Efficient truck logistics operations gave LKW Walter a competitive edge in the market.

As a concrete measurable example, LKW Walter receives double the stall fees compared to before, thanks to real-time visibility that allows the company to charge accurate penalties.

Real-Life Example: Lufthansa

Use Cases: IT Modernization | Data Sharing | AI

Company Overview

Lufthansa is a major German airline and one of the largest in Europe, renowned for its extensive network of domestic and international flights. As a member of the Star Alliance, Lufthansa offers a wide range of services, including passenger transportation and cargo logistics.

Implementation

Traditionally, airlines have used message queuing, enterprise service buses (ESB), and extract-transform-load (ETL) tools for system integration. However, these methods often lead to tightly coupled systems, limiting flexibility and scalability. Lufthansa has adopted data streaming as a cloud-native middleware, capitalizing on its real-time messaging, persistence layer, data integration, and stream processing capabilities. This approach decouples systems, manages backpressure, and enables historical event replay, enhancing both operational and analytical data integration.

Lufthansa's KUSCO (Kafka Unified Streaming Cloud Operations) project underscores its commitment to data streaming for mission-critical operations. By choosing Confluent over traditional messaging queues, Lufthansa achieved a more cost-effective and scalable infrastructure for real-time data processing. The KUSCO platform (now known as the One Integration Platform) has streamlined project onboarding and provides real-time, high-quality data feeds across business units.

Using data streaming, Lufthansa drives real-time analytics and AI/ML applications. By consolidating and processing data from various sources, the airline enables real-time alerting and model scoring, delivering timely and accurate predictions for operational decision-making.

Use cases include:

- 1 Anomaly detection:** Lufthansa uses stream processing to aggregate data and detect anomalies in real time, enabling proactive alerts and maintenance.
- 2 Fleet management:** The airline leverages machine learning models embedded within the data streaming applications to optimize aircraft operation and ensure efficient fleet management with low latency and critical SLAs.



Machine learning for real-time operations steering at Lufthansa with a high-level architecture

Business Value

The integration of data streaming technologies has delivered substantial business value to Lufthansa.

- **Operational efficiency:** By decoupling systems and enabling real-time data processing, Lufthansa has enhanced its operational efficiency, reducing costs and improving scalability.
- **Innovation and competitiveness:** Data streaming has empowered Lufthansa to innovate rapidly, offering personalized customer experiences and maintaining a competitive edge in the aviation industry.
- **Enhanced customer experience:** Real-time analytics and AI applications have enabled Lufthansa to provide superior customer service, from predictive maintenance to personalized travel experiences.
- **Scalability and flexibility:** The cloud-native architecture of the One Integration Platform allows Lufthansa to scale its operations efficiently, accommodating growing data volumes and diverse use cases.
- **Data-driven decision-making:** Access to real-time data feeds has facilitated data-driven decision-making across Lufthansa's business units, driving continuous improvement and strategic growth.

Real-Life Example: Swiss Post

Use Cases: IT Modernization | Event-Driven Microservices | Data Sharing

Company Overview

Swiss Post, Switzerland's national postal service, is a backbone of the country's infrastructure, moving millions of parcels and letters every day. To stay ahead in an increasingly competitive logistics market, the company is driving a digital transformation to create a fully transparent, real-time, and data-driven logistics network.

Implementation

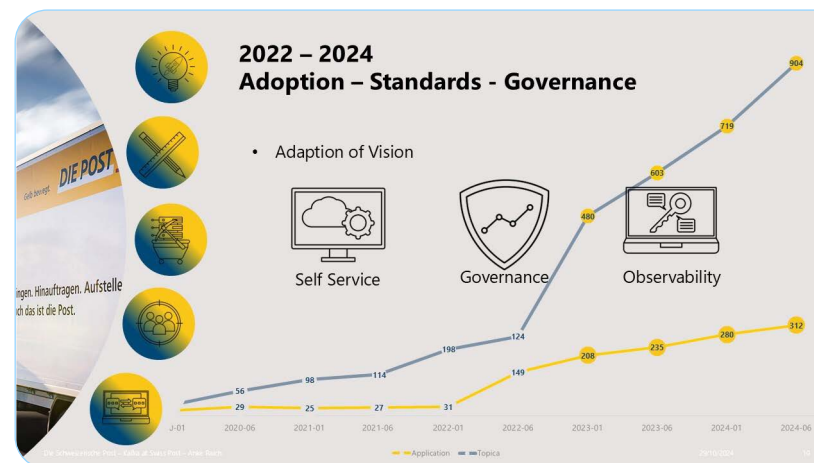
One of Swiss Post's biggest challenges was the fragmentation of data across siloed, monolithic applications, which limited real-time visibility across the logistics chain. To overcome this, Swiss Post set out to build a digital backbone that could connect thousands of operational systems, stream events from every point in the process, and process them instantly while applying strong governance for quality and compliance.

The new architecture integrates data from thousands of IoT devices, from hand scanners in sorting centers to sensors in delivery vehicles, feeding a continuous stream of operational events. More than 200 applications are now connected, processing around 40 million messages per day and enabling step-by-step tracking of every parcel's journey.

Business units can tap directly into these streams to build their own event-driven microservices, launching new products, optimizing workflows, and responding to market needs independently. Moving from a centralized, monolithic model to a decentralized architecture around sharable data products has unlocked faster innovation while ensuring data consistency, security, and regulatory compliance across the organization.

"With our data streaming platform based on Confluent, we are creating the digital backbone for the logistics of the future. We are moving away from a monolithic to a decentralized world, which allows us to act faster and more flexibly on the market and to develop new, data-driven services for our customers."

Anke Raich, Cluster Lead Technology Enablement, Swiss Post



Apache Kafka adoption at Swiss Post 2022-2024

Business Value

- **End-to-end transparency:** Complete, real-time visibility from the first scan to final delivery for every parcel.
- **Accelerated innovation:** Discoverable, secured, and governed data products enable teams to develop new services and analytics without central bottlenecks.
- **Operational excellence:** Live data streams power proactive monitoring, routing optimization, and early issue detection.
- **Trusted data foundation:** Integrated governance ensures accuracy, security, and compliance while data flows freely across the organization.
- **AI/ML enablement:** A continuous, high-quality event feed supports advanced use cases, such as predictive delivery times and automated anomaly detection.

Real-Life Example: Schiphol Group

Use Cases: IT Modernization | Data Sharing | IoT | ESG

Company Overview

Schiphol Group, a prominent Dutch company, is renowned for its management and operation of airports in the Netherlands and internationally, with Amsterdam Airport Schiphol as its flagship. As one of Europe's busiest and most significant airports, Schiphol is a hub of activity, facilitating millions of passengers and tons of cargo annually. Schiphol Group's operations extend beyond aviation services to encompass a wide range of non-aviation activities, making it a comprehensive airport management entity.

In its quest to become a leading autonomous airport by 2050, Schiphol Group has embarked on a digital transformation journey. This is driven by the need to enhance operational efficiency, improve passenger experiences, and ensure seamless integration of various airport functions. The adoption of data streaming has been central to this transformation, enabling real-time data processing, data exchange, and data analysis across multiple facets of airport operations.

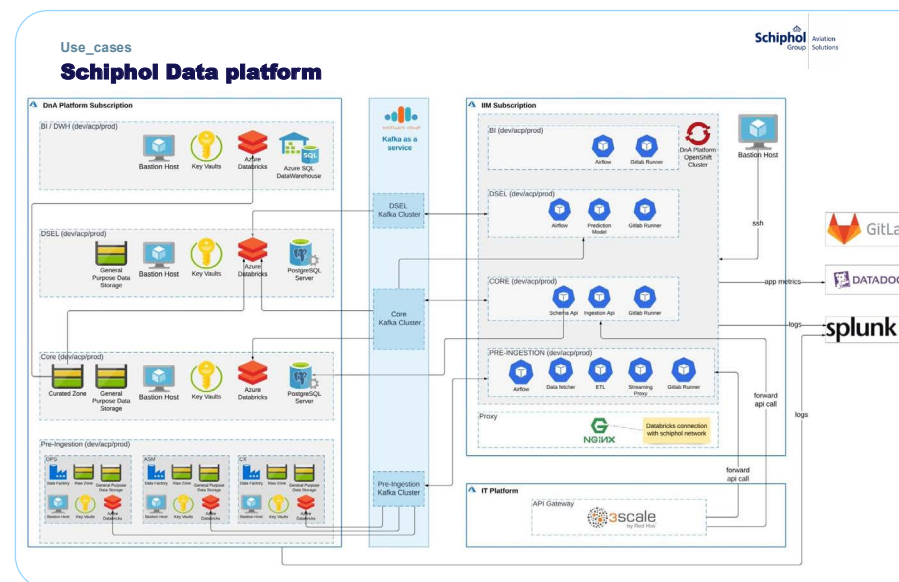
Implementation

The digitalization of Schiphol Airport is a multifaceted endeavor that leverages data streaming to revolutionize airport operations and passenger experiences. At its core is the implementation of data streaming, which facilitates the continuous flow and immediate processing of critical data from diverse sources such as flight information, passenger movements, and physical asset performance.

Event-driven is becoming more and more the basic principle for everyday data processing operations within Schiphol. It enables them to process incoming events in real time, providing transparency and context-specific decision-making capabilities not only within the organization but also with APIs to the outside world.

This architecture supports a wide range of use cases coming to life in the future, from predictive maintenance and personalized passenger services to enhanced security measures.

One of the key challenges of Schiphol's digitalization efforts is correlating trusted, qualitative data from different sources and departments. This will enable Schiphol to optimize operational excellence and realize the integration of Internet of Things (IoT) scenarios. Data sources can be events



Schiphol data platform

from various interfaces, including gates, retail stores, and passenger mobile apps. By continuously correlating data in real time, Schiphol can implement predictive forecasting, planning, and maintenance as well as develop advertisement and recommendation engines to improve customer experience and increase revenue.

The implementation of data streaming technologies has also facilitated significant improvements in operational efficiency. Automation with IoT sensors, paperless processes, and software innovations have enabled more cost-effective and reliable airport operations and cargo processes.

Schiphol Group has modernized its legacy IT infrastructure by complementing traditional systems, such as message queues, ETL processes, and ESBs, with a data streaming platform. This transition has enabled scalable and cloud-native data processing and seamless integration across diverse airport systems, significantly reducing scalability and data availability issues and enhancing operational efficiency.

Business Value

- **Enhanced operational efficiency:** Reduced end-to-end latency in data processing and automation of key processes streamline airport operations (e.g., electronic cargo documents, reduced turnaround times), minimizing disruptions and optimizing coordination.
- **Improved passenger experience:** Mobile apps and boarding provide passengers with a seamless, efficient, and satisfying journey.
- **Elevated customer service and communication:** Real-time updates enhance passenger satisfaction and loyalty engagement.

Conclusion

The transportation and travel industry is being reshaped by real-time connectivity and intelligent automation. Data streaming forms the digital backbone connecting fleets, vehicles, airports, and logistics networks. It enables organizations to optimize routes, reduce delays, and deliver seamless customer experiences with greater precision and transparency.

At the core, streaming and AI work together to improve reliability and efficiency. Predictive analytics detect maintenance needs and prevent disruptions before they occur. Real-time insights help balance resources, reduce emissions, and maintain punctuality in high-volume environments.

Swiss Post uses data streaming to transform logistics at national scale, connecting thousands of devices for full visibility from parcel scan to delivery. Real-time data products enable proactive monitoring, faster innovation, and strong governance.

LKW Walter applies streaming and IoT to modernize pan-European logistics, reducing costs and fuel use while improving fleet utilization and customer satisfaction.

Flix, a global travel technology company, relies on streaming to manage millions of bus and train journeys worldwide. By migrating to Confluent Cloud, Flix eliminated infrastructure bottlenecks, scaled seamlessly, and accelerated feature delivery. Its AI systems process

bookings, payments, and telemetry in real time to optimize seat availability, detect fraud, and personalize pricing, creating reliable and sustainable travel experiences.

In aviation, Lufthansa uses streaming analytics for predictive maintenance and operational efficiency. Schiphol Group leverages event-driven architecture to coordinate airport operations, optimize passenger flow, and improve service quality. Together, these examples show how streaming and AI enable the seamless movement of people and goods while improving safety, sustainability, and customer experience.

Agentic AI extends these capabilities by predicting delays, flagging compliance risks, and dynamically balancing resources across fleets, airports, and borders. Streaming agents transform travel and logistics providers from reactive operators into proactive orchestrators of efficient, personalized, and sustainable journeys.

Real-time data streaming enables transportation and travel providers to scale globally, stay resilient, and lead the next era of connected, intelligent mobility.

3.9 Insurance

With the mention of insurance and digitalization, the topic of regulation quickly follows. This impacts the use of data and the adoption of certain trends at every stage. One of the most prominent trends is the adoption of telematics and IoT, which enable insurers to collect real-time data on customer behavior and asset conditions. This data is crucial for developing personalized insurance products and improving risk assessment. Digital customer engagement is another critical trend, as customers now expect seamless, personalized interactions with their insurers that are similar to their experiences with other digital services.

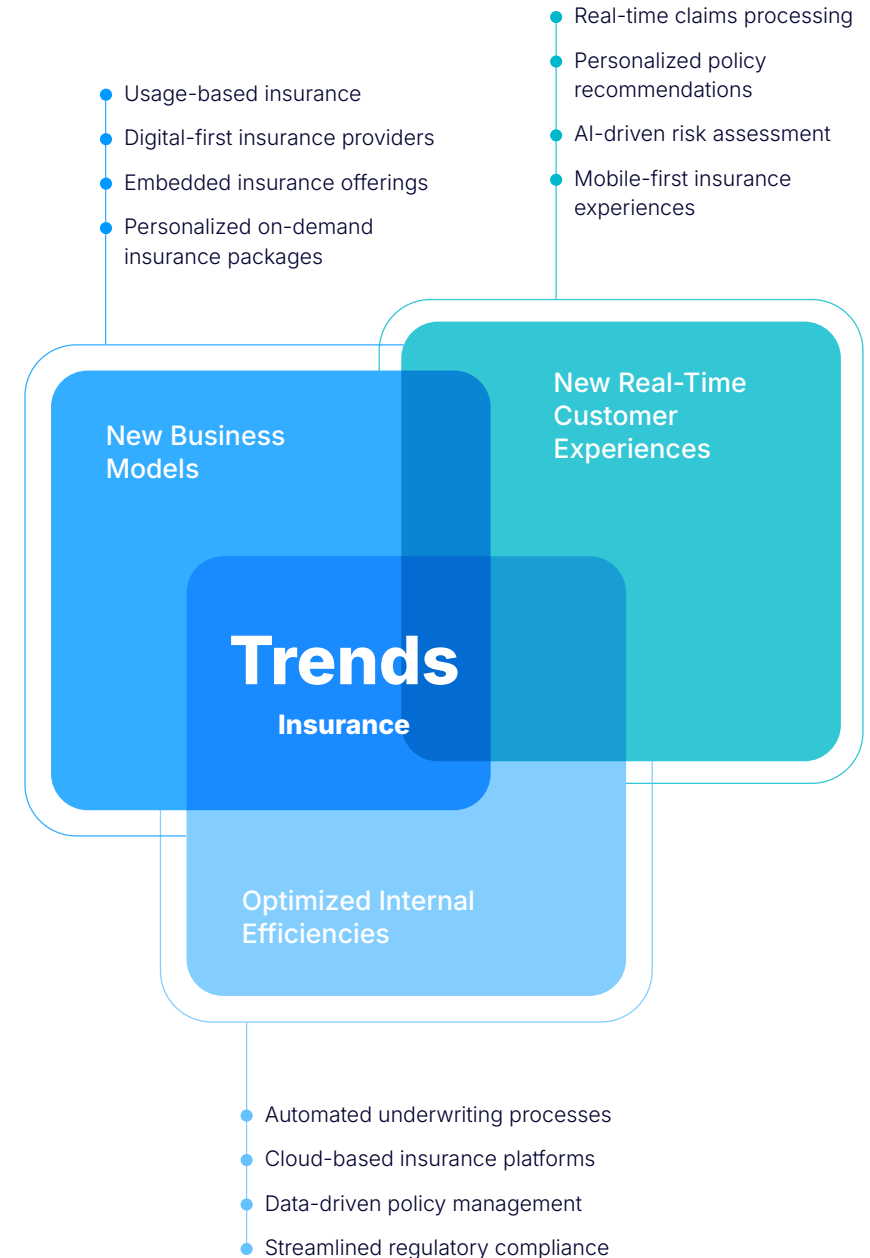
Usage-based insurance is gaining traction, allowing insurers to offer policies based on actual usage rather than traditional risk factors. This approach is particularly relevant in the automotive sector, where telematics data can be used to monitor driving behavior and adjust premiums accordingly. Embedded insurance, which integrates insurance products into the purchase process of other goods and services, is also becoming popular. This trend simplifies the insurance buying process and enhances customer

convenience.

Data streaming platforms play a crucial role in enabling these trends by providing real-time data processing and analytics capabilities. The ability to process and analyze data in real time is essential for developing innovative insurance products, improving customer engagement, and optimizing operational efficiency.

Beyond efficiency, insurers are experimenting with generative and agentic AI to enhance underwriting, personalize micro policies, and detect fraud as it unfolds. By combining continuous event data with AI-driven insights, insurers can move from static, risk-averse operations toward adaptive, customer-centric services.

As the insurance industry continues to evolve, the adoption of data streaming technologies will be crucial in maintaining competitiveness and driving innovation.



Scenario 1: Workflow Orchestration

Workflow orchestration in the insurance industry involves automating and coordinating various processes, such as claim processing, policy management, and customer service. Insurers can streamline their workflows by using real-time data from various sources, including customer interactions, policy details, and external data providers. Data streaming ensures data consistency and accuracy, allowing insurers to automate large portions of their processing workflows. More efficient operations lead to more accuracy, ensuring compliance with policy terms and improved customer satisfaction.

Business Value

- **Streamlined workflow automation:** Real-time data processing enables the automation and coordination of various insurance processes, such as claims processing and policy management, improving overall workflow efficiency.
- **Improved data consistency and accuracy:** Continuous data validation and processing ensure data consistency and accuracy, reducing errors and enhancing the reliability of insurance operations.
- **Enhanced claims processing:** Automated review and validation of claims allow for accurate and timely processing, ensuring compliance with policy terms and improving customer satisfaction.
- **Increased operational efficiency:** Transitioning from legacy batch processing to real-time workflows reduces processing times and operational costs, enhancing overall efficiency.
- **Comprehensive operational visibility:** Access to real-time data from customer interactions, policy details, and external data providers provides a comprehensive view of claims processing operations, supporting informed decision-making.
- **Improved customer satisfaction:** Efficient and accurate claims processing enhances customer satisfaction by providing timely and reliable service, fostering customer loyalty.
- **Compliance and risk management:** Real-time insights into insurance operations ensure compliance with industry regulations and standards, reducing the risk of penalties and enhancing risk management.

Scenario 2: Full-Stack Contract Handling

In the insurance industry, full-stack contract handling means managing insurance policies from start to finish. Advanced applications and AI bring a direct-to-consumer approach to life insurance, speeding up the signing process and activating policies within minutes.

Real-time data from various sources, such as health and driving records, offers comprehensive insights into signing and policy activation processes. Data streaming automates the signing process by quickly contacting third-party data providers and gathering necessary information, with the customer's permission. This significantly streamlines the customer experience, cuts operational costs, and improves customer satisfaction.

Business Value

- **Faster underwriting process:** Real-time data processing and automation streamline the underwriting process, reducing the time needed to assess and approve applications.
- **Quick policy activation:** Policies can be activated within minutes, enhancing the customer experience by providing immediate coverage and service.
- **Improved customer satisfaction:** A seamless and efficient process from application to activation leads to higher customer satisfaction and loyalty.
- **Direct-to-consumer approach:** Offering a direct-to-consumer model simplifies the purchasing process, making it more accessible and appealing to customers.
- **Comprehensive data utilization:** Access to real-time data from various sources, such as health and driving records, provides a complete view of the underwriting process, improving decision-making accuracy.
- **Competitive advantage:** Companies that implement full-stack contract handling can differentiate themselves by offering faster, more efficient services, gaining a competitive edge in the market.

Scenario 3:

Claims and Policy Fraud Prevention

Fraudulent claims and policy applications are a major challenge for insurers. With real-time data streaming, insurers integrate customer interactions, geolocation, medical and repair records, and payment histories to instantly detect anomalies. AI models continuously scan for suspicious activity such as duplicate claims, unusual billing patterns, and identity mismatches, and flag them for immediate review. This enables proactive prevention rather than reactive investigation.

Business Value

- **Fraud prevention:** Instant detection reduces fraudulent payouts and protects revenue.
- **Faster claims settlement:** Processing legitimate claims without delay improves customer satisfaction.
- **Cost reduction:** Data streaming technology minimizes investigation costs and operational inefficiencies caused by fraud.
- **Regulatory compliance:** Transparent fraud monitoring supports adherence to industry standards.
- **Stronger trust:** Protecting against fraud enhances customer confidence in the insurer.

Scenario 4:

Catastrophe Modeling and Risk Management

Natural disasters, extreme weather, and geopolitical events create sudden surges in claims. With data streaming, insurers continuously integrate data from weather services, IoT devices, geospatial mapping, and government alerts to model potential impacts as events unfold. Streaming analytics enables insurers to predict claims volume, optimize reinsurance coverage, and dynamically adjust risk exposure.

Business Value

- **Proactive response:** Predicting surges in claims allows mobilization of resources before customer demand spikes.
- **Improved risk modeling:** Incorporating live environmental and geospatial data enhances accuracy.
- **Faster payouts:** Real-time event monitoring accelerates claims validation and payments.
- **Stronger resilience:** Dynamic adjustments to risk exposure reduce financial vulnerability.
- **Customer trust:** Quick, transparent responses in crises foster loyalty and brand strength.

Scenario 5: Embedded Insurance at Scale

Embedded insurance integrates seamlessly into the purchase process of goods and services, for example, insuring electronics at checkout, ride-hailing trips, or making ecommerce deliveries. Real-time data streaming enables this by linking partner platforms, transaction systems, and policy engines in milliseconds. Streaming agents manage dynamic pricing, policy activation, and claims triggers directly within the customer's journey, without requiring a separate insurance interaction.

Business Value

- **Seamless experience:** Insurance becomes invisible yet always available, reducing customer friction.
- **New revenue streams:** Partnerships with retailers, travel companies, and mobility providers expand market reach.
- **Dynamic risk assessment:** Real-time integration ensures that policies reflect actual usage and purchase conditions.
- **Faster policy activation:** Coverage is confirmed instantly at checkout.
- **Scalability:** Event-driven architecture supports millions of micro policies across industries.

"Data is becoming increasingly important and we run everywhere. Data, it's actually money. As an insurer, at the end of the day, we have been always a data essentially industry because our specialty takes really using data in order to price risk. And we are pretty good at it. What now we can do more with data, it's again, what maybe we have not been so good in the past, which is the engagement that we can get with our customer."

Pietro Carnevale, Director of Strategy & Innovation, Generali Switzerland

Scenario 6: Micro Insurance

Micro insurance caters to specific short-term needs, such as insuring brief excursions with rental bikes or similar activities. Advanced applications and IoT technologies enable insurers to implement dynamic pricing models that adjust premiums based on a combination of internal CRM data and external factors.

Real-time data analysis made possible by data streaming allows swift adjustments to pricing models to reflect current conditions and individual risk profiles. Through this, insurers can optimize their pricing strategies and maintain a competitive edge in the market.

Business Value

- **Tailored insurance solutions:** With real-time data processing, insurers can offer personalized, risk-based insurance products that cater to specific, short-term needs, enhancing customer satisfaction.
- **Dynamic pricing models:** The ability to adjust premiums based on real-time data, including CRM insights and external factors like weather conditions, ensures competitive and fair pricing.
- **Enhanced customer experience:** By providing insurance solutions tailored to individual needs and circumstances, insurers can foster a more engaging and satisfying customer experience.
- **Agility and responsiveness:** Insurers can swiftly respond to changing conditions and customer behaviors, ensuring that their offerings remain relevant and attractive.
- **Competitive edge:** The ability to optimize pricing strategies in real time helps insurers maintain a competitive advantage in the market.
- **Increased customer loyalty:** Personalized and responsive insurance offerings can lead to higher customer retention and loyalty.

Real-Life Examples

- Generali Switzerland

"At Vitality Group, we also had the opportunity to build from the ground up so there was no limitation on legacy choices. By using Confluent, we found that it also alleviates the day-to-day admin burden. And this basically frees us up with time and allows us to optimize to build our continuous integration (CI)/continuous delivery (CD) build pipeline. And Confluent also allows for seamless scalability so there is no downtime incurred whenever we need to scale."

Ryan James, Chief Data Officer, Vitality

Real-Life Example: Generali Switzerland

Use Cases: IT Modernization | Hybrid and Multicloud | Data Governance

Company Overview

Generali Switzerland is a key part of the global Generali Group, one of the world's largest insurance providers. Known for its commitment to innovation, Generali Switzerland offers a broad range of insurance products and services, including life, health, property, and casualty insurance, to both individual and corporate clients. The company has continuously adopted new technologies to improve its operations and deliver exceptional customer experiences.

IT Modernization Initiative

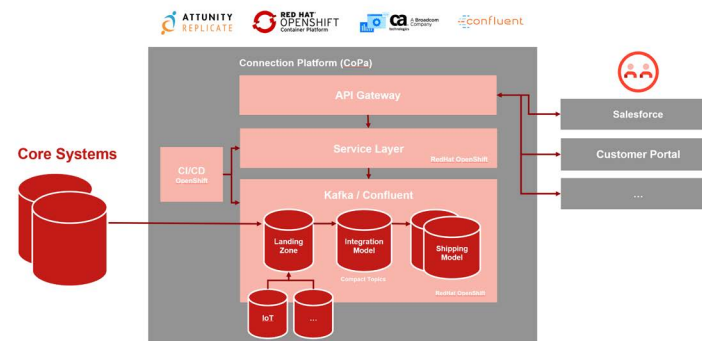
As the insurance industry evolves, Generali Switzerland recognized the need to modernize its core insurance systems to stay competitive and meet rising customer expectations. The company's essential functions—underwriting, policy administration, claims processing, and billing—are critical for assessing risk, managing policies, handling claims, and processing payments. Generali Switzerland embarked on a major IT transformation using Confluent's data streaming platforms to achieve real-time data processing and analytics, making these functions more responsive and efficient.

The transformation began by deploying a data streaming platform in Generali's data centers, close to its core relational databases. By leveraging real-time data streaming and change data capture, the company could continuously optimize core insurance processes, build new cloud-native applications, and integrate new SaaS solutions. This setup enabled a decoupled, stable architecture, reducing risks while integrating seamlessly with modern platforms that required core insurance data.

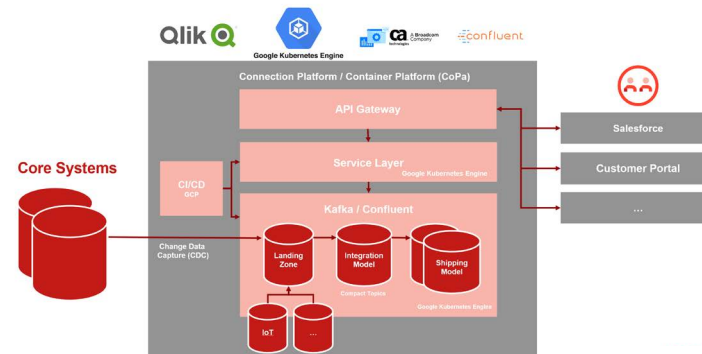
Moving to the Cloud

After establishing data streaming in its data centers, Generali Switzerland began migrating the entire platform to the public cloud. This migration was carefully managed to ensure business continuity, with data streaming support via Confluent Replicator enabling a smooth transition from on-premises to a cloud-based data streaming platform. For this first step into a public cloud, Generali moved to managed Kubernetes but retained the self-managed, containerized Confluent Platform. After successful migration and stabilization, Generali continued modernizing by moving to a fully managed Confluent Cloud service, which reduced infrastructure costs and allowed IT teams to focus on business challenges instead of maintaining systems. Again, Confluent Replicator enabled the smooth transition from the ceding Confluent Platform to the new Confluent Cloud.

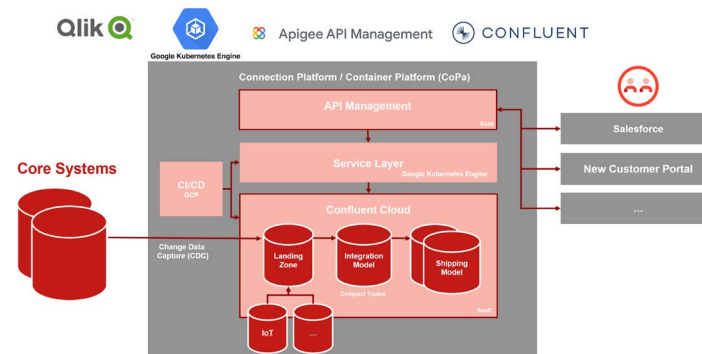
2018 – First Iteration : the “on-premise” platform



2021 – Second Iteration : the Google Cloud enabled platform



2024 – Third Iteration : Cloud All-In



Key Benefits of Modernization

Generali Switzerland's modernization strategy delivered substantial business value.

- **Speed of innovation for bringing new digital solutions to its customers:**

Digital solutions can be integrated and rolled out faster with minimal risk and minimal-to-no changes in core systems, thanks to a stable, decoupled architecture. They can either replace legacy core functions, such as implementing a partner master data management system that is built cloud- and streaming-native or integrated on top of core systems, such as Salesforce Services, Digital Signature or any other Insurtech SaaS services that are plugged into the business process. Decoupled means that the data streaming layer is decoupling the core and digital layer from each other.

- **Enhanced operational efficiency:** The enablement of new digital solutions through the streaming layer can increase customer experience, self-servicing, and straight-through processing to a new level that drives operational efficiency. Real-time data processing streamlines core insurance functions, making processes like underwriting, claims handling, and billing faster and more efficient, ultimately boosting operational performance. Integrating legacy databases with modern cloud services has cut operational costs. By reducing the need for maintaining outdated systems, Generali can allocate resources more effectively. New platforms can also be added with minimal risk, thanks to a stable, decoupled architecture.

Leadership Perspective

Generali Switzerland's Head of IT, Alexander Auner, states, "Our modernization journey has been about building a resilient, decoupled architecture that supports digital transformation without compromising the stability of our core systems, such as contract management." For the Lead of the COPA (Connection Platform) Nada Kammoun-Ksibi "moving to Confluent Cloud reduced infrastructure costs and allowed our IT teams to focus on business challenges instead of maintaining systems." Lead Architect Thomas Peter says, "This decoupling has empowered us to integrate high-end APIs and a robust data streaming layer, allowing digital initiatives to thrive while our foundational contract and claims management remains stable and secure."

Conclusion

The insurance industry is undergoing a major digital shift. Data streaming connects core systems, digital channels, and partner ecosystems, enabling faster, more transparent operations. This foundation powers usage-based insurance, on-demand coverage, and embedded policies that align with customers' daily lives.

Data streaming and AI are transforming the insurance value chain. Real-time claims processing, mobile-first services, and data-driven risk assessment allow instant responses to customer needs. Continuous intelligence supports fraud detection and catastrophe modeling, helping insurers anticipate claims surges and adjust exposure dynamically. Automated underwriting and micro-policies reduce cycle times and embed protection directly into customer journeys.

Generali Switzerland demonstrates how to modernize without disrupting core systems. Its event-driven streaming-first architecture decouples legacy platforms from modern digital tools. Powered by Confluent, it connects underwriting, claims, billing, and engagement workflows in real time. Change data capture keeps cloud apps and on-prem systems in sync, creating a unified data foundation.

After launching data streaming in its data centers, Generali Switzerland moved to hybrid and then to fully managed Confluent Cloud, reducing infrastructure costs and accelerating

digital service delivery. The platform integrates with a CRM in the cloud and e-signature solutions, supporting new customer-facing apps and self-service experiences. The decoupled architecture enables innovation without sacrificing stability, increasing both agility and resilience.

This approach captures a broader industry trend. Streaming and AI empower insurers to automate workflows, personalize offerings, and continuously optimize performance. Real time data becomes the shared foundation for risk modeling, claims automation, and customer analytics. Agentic AI now builds on this by orchestrating complex workflows, anticipating risks, and recommending policy adjustments autonomously, keeping insurers proactive in dynamic markets.

Together, these innovations mark a decisive shift from static, reactive operations to adaptive, customer-centric ecosystems. Data streaming, AI, and agentic AI enable insurers to deliver real-time protection and personalized experiences at scale. Companies such as Generali Switzerland show how a modern data backbone unlocks faster innovation, lower costs, and long-term competitiveness in a rapidly evolving digital landscape.

3.10 Healthcare and Life Sciences

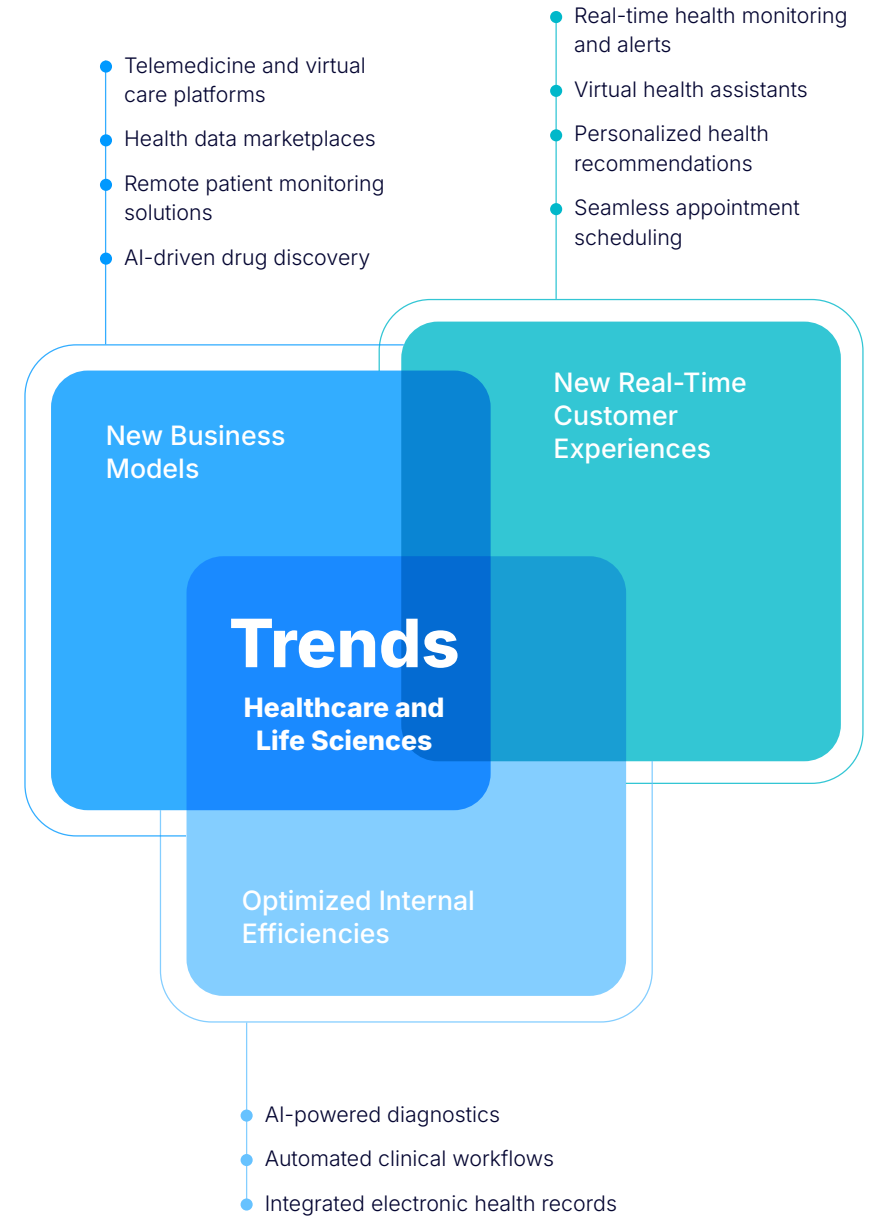
Debates about digitalization are common in the healthcare industry. However, unlike other sectors, these discussions tend to be more cautious, consistent, and understated. One of the most critical trends is the emphasis on data privacy and security since healthcare organizations handle sensitive patient information. Ensuring the confidentiality, integrity, and availability of this data is paramount, and robust data streaming platforms play a crucial role in achieving this.

Telemedicine and telehealth are also gaining traction, enabling healthcare providers to offer remote consultations and monitoring services. This trend was accelerated by the COVID-19 pandemic, which highlighted the need for remote healthcare solutions.

Remote patient monitoring, facilitated by IoT devices and data streaming technologies, allows healthcare providers to continuously monitor patients' health conditions and intervene when necessary.

The integration of AI in healthcare is another significant trend, enabling advanced diagnostics, personalized treatment plans, efficient drug discovery processes, and improved hospital operations. Increasingly, agentic AI is being applied to orchestrate workflows, anticipate patient risks, and automate actions such as treatment adjustments and resource allocation, ensuring that healthcare delivery becomes more adaptive and proactive.

Data streaming platforms are essential in enabling these trends by providing real-time data integration and analytics capabilities.



Scenario 1: Medical Diagnostics

In medical diagnostics, advanced technologies are used to detect and diagnose medical conditions. IoT and AI systems make real-time patient monitoring possible, helping to prevent the risk of deadly conditions. Processing real-time data from various sources, such as patient monitors, electronic health records, and laboratory results, with data streaming technologies enables healthcare providers to monitor patients' health conditions continuously and intervene when necessary, improving patient outcomes and saving lives. A concrete example of the use of data streaming is monitoring patients that undergo bone marrow transplants for signs of sepsis.

Business Value

- **Improved patient outcomes:** Real-time monitoring and data analysis enable early detection of medical conditions, so timely interventions can improve treatment and save lives.
- **Enhanced patient care:** Continuous monitoring of health conditions enables healthcare providers to offer more personalized care.
- **Early detection and prevention:** Detecting early signs of conditions such as sepsis allows for prompt intervention, reducing the risk of complications.
- **Comprehensive health insights:** Integrating data from patient monitors, electronic health records, and laboratory results offers a holistic view of a patient's health, supporting more accurate diagnoses and treatment plans.
- **Operational efficiency:** Automation of data collection and analysis reduces the need for manual monitoring, freeing up healthcare professionals to focus on direct patient care.
- **Risk management:** Real-time data processing enhances the ability to manage and mitigate risks associated with critical health conditions.
- **Innovation in healthcare:** Leveraging IoT and AI technologies positions healthcare providers at the forefront of medical innovation, offering cutting-edge diagnostic solutions.
- **Cost reduction:** Early detection and intervention can lead to reduced healthcare costs by preventing the escalation of medical conditions and minimizing the need for extensive treatments.

Scenario 2: Pharmaceutical R&D

Pharmaceutical research and development (R&D) is all about the discovery and development of new drugs and treatments. Data streaming and AI/ML technologies enable event-driven workflows and process orchestration, delivering faster data processing to analyze biological images, laboratory results, and clinical trial data more efficiently and with improved data quality. This speeds up the drug discovery process and aids in identifying potential drug candidates, thereby enhancing the efficiency and effectiveness of R&D operations. But while near-real-time processing is beneficial, maintaining data consistency remains paramount. Data streaming provides consistent access to multiple consumers in the end-to-end data pipeline.

Business Value

- **Accelerated drug discovery:** Continuous data processing and event-driven workflows enable faster analysis of biological images, laboratory results, and clinical trial data, significantly speeding up the drug discovery process.
- **Enhanced R&D efficiency:** By automating data analysis and process orchestration, pharmaceutical companies can improve the efficiency and effectiveness of their R&D operations, reducing time to market for new drugs.
- **Improved decision-making:** Access to consistent information across real-time and non-real-time applications allows researchers to make informed decisions quickly, identifying potential drug candidates and optimizing research strategies.
- **Increased innovation:** Leveraging IT modernization and AI/ML technologies fosters a culture of innovation, enabling the development of cutting-edge treatments and therapies.
- **Cost reduction:** Streamlined processes and efficient data management reduce operational costs associated with drug discovery and development.
- **Comprehensive data insights:** Real-time analysis of diverse data sources provides a holistic view of research progress, supporting more accurate and effective R&D efforts.
- **Regulatory compliance:** Real-time data processing supports compliance with regulatory requirements by ensuring accurate and timely reporting of research findings.

Scenario 3: Caregiving

Caregiving provides services such as senior care and housekeeping. Using data pipelines and applications, caregiving companies can create online marketplaces that connect caregivers with those in need. Real-time data processing capability helps companies offer a wide range of care services, improving customer satisfaction and operational efficiency.

Imagine an online marketplace using data streaming to match caregivers with clients based on real-time data. This makes the caregiving experience more seamless, leading to better customer satisfaction and business growth.

Business Value

- **Enhanced customer satisfaction:** Real-time data processing enables the delivery of personalized care services, ensuring that individuals receive the right care at the right time, thereby improving customer satisfaction.
- **Efficient caregiver matching:** By using data streaming, caregiving companies can quickly and accurately match caregivers with individuals in need of care, optimizing the allocation of resources and reducing wait times.
- **Comprehensive service offerings:** The ability to process real-time data allows companies to offer a wide range of care services, such as senior care and housekeeping, meeting diverse customer needs.
- **Operational efficiency:** Automation of data collection and analysis streamlines caregiving operations, reducing manual intervention and operational costs.
- **Scalability and flexibility:** Data streaming technologies provide the scalability needed to handle large volumes of service requests and adapt to changing market demands, even in extreme situations such as a pandemic.
- **Increased business growth:** By providing a seamless and efficient caregiving experience, companies can attract more customers and expand their market presence.
- **Improved care quality:** Real-time insights into caregiver performance and customer feedback enable continuous improvement of care quality and service delivery.

Scenario 4: Value-Based Care

Value-based care rewards outcomes, not service volume. Achieving it requires immediate awareness of patient status, treatment adherence, and cross-provider coordination. Traditional, siloed systems that rely on delayed batch updates make it impossible to act at the right moment.

By continuously streaming clinical events, device telemetry, and lab results, providers gain live visibility into patient progress. Streaming pipelines enrich and process this data to flag care gaps, trigger interventions, and keep AI-driven risk scoring models current. GenAI and streaming agents can then recommend personalized treatment adjustments, predict potential readmissions, or generate real-time alerts for care teams.

Integrated governance and security measures protect protected health information (PHI) under HIPAA, GDPR, and similar regulations, while lineage tracking supports transparent reimbursement audits. Combining real-time feeds with historical context enables providers to measure quality accurately, act proactively, and innovate new care models that consistently improve patient outcomes and meet value-based contract goals.

Business Value

- **Better outcomes:** Act on changes before conditions worsen. Real-time insights enable timely interventions that reduce complications, readmissions, and length of stay.
- **Accurate metrics:** Reliably track reimbursement performance. Continuous, auditable data feeds align quality measures with the latest patient events to prevent under- or over-reporting.
- **Regulatory compliance:** Achieve secure and compliant PHI handling. End-to-end governance, fine-grained access controls, and lineage simplify audits and ensure privacy by design.
- **Operational gains:** Automate alerts and workflows. Care teams spend less time chasing data and more time delivering care, improving throughput without adding headcount.
- **Innovation enablement:** Launch new care models quickly. Modular streaming and AI building blocks let teams pilot, iterate, and scale programs with lower risk and cost.

Scenario 5: Clinical Trials Monitoring

Clinical trials generate vast amounts of data from patients, labs, wearables, and remote monitoring devices. Traditionally, this information is collected in batches and analyzed after the fact, delaying the detection of safety issues or protocol deviations. With data streaming, trial sponsors and research organizations can continuously monitor patient responses, adherence, and adverse events across geographies.

Streaming pipelines ensure data integrity, enable adaptive trial designs, and provide regulators with transparent, auditable records. Embedded AI models can automatically flag anomalies (e.g., unexpected side effects) and suggest protocol adjustments to improve patient safety and trial efficiency.

Business Value

- **Improved patient safety:** Immediate detection of adverse events, as real-time alerts enable rapid dose adjustments or temporary holds that reduce event severity and protect participants.
- **Faster time to market:** Adaptive trials accelerate development cycles, as continuous data flows compress interim analyses and decision cycles, shaving weeks off protocol amendments and enrollment changes.
- **Enhanced compliance:** Transparent, auditable data streams support regulatory reporting.
- **Operational efficiency:** Acting in real time reduces costly trial delays through real-time action, with proactive site outreach and automated query resolution minimizing data queries, rescreens, and unplanned monitoring visits.
- **Higher success rates:** Data-driven insights optimize protocol design and execution, and early trend detection refines inclusion criteria, dosing strategies, and retention tactics to keep trials on track.

Scenario 6: Hospital Operations and Patient Flow Optimization

Hospitals operate under immense pressure to balance patient care with limited resources. Real-time data streaming integrates inputs from electronic health records, bed occupancy sensors, emergency department queues, and medical device telemetry. By analyzing this data continuously, hospitals can predict bottlenecks, optimize bed allocation, and dynamically manage staffing levels.

Agentic AI agents can trigger actions such as redirecting incoming patients to less crowded facilities, reprioritizing elective surgeries, and alerting care teams to delays in diagnostics. This ensures smoother operations, better patient outcomes, and more efficient use of limited resources.

Business Value

- **Reduced wait times:** Faster triage, bed visibility, and dynamic routing shorten queues and improve patient throughput.
- **Resource optimization:** Dynamic allocation of staff, beds, and equipment aligns capacity with demand across units and shifts.
- **Cost savings:** Eliminating overcrowding and diagnostic delays reduces overtime, diversion penalties, and excess length of stay.
- **Enhanced patient experience:** Shorter ER stays and smoother care transitions increase satisfaction and reduce complaints.
- **Proactive capacity management:** Predicting surges and preventing system overload minimizes diversions and cancellations while maintaining service levels.

Real-Life Examples

- City of Hope
- Siemens Healthineers

Real-Life Example: City of Hope

Use Cases: Data Products | Observability | Event-Driven Microservices | AI | IoT

Company Overview

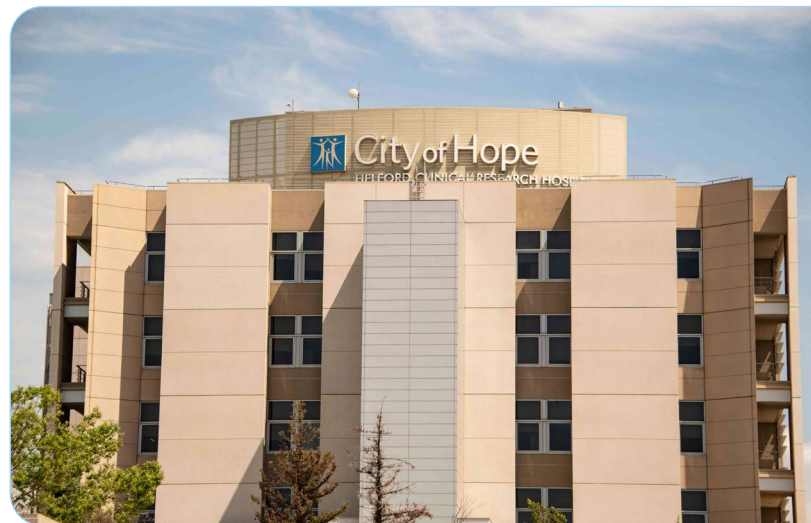
City of Hope is one of the largest and most advanced cancer research and treatment organizations in the United States. With more than 20,000 bone marrow transplants performed to date, it operates one of the largest and most successful transplant programs nationwide. The institution's mission is to transform cancer care and improve patient outcomes through leading-edge science, treatment, and innovation.

Implementation

Sepsis is a life-threatening infection that can escalate rapidly, causing organ failure or death. Bone marrow transplant patients are especially vulnerable due to chemotherapy, radiation, and temporary immune system suppression. City of Hope sought to reduce these potential risks with predictive, real-time clinical insights.

The organization developed an AI-powered predictive model to identify early signs of sepsis in high-risk patients. To ensure accuracy and timeliness, Confluent serves as the data streaming backbone. Vital signs, lab results, and clinical observations are streamed in real time, continuously enriching the AI model. If sepsis risk is detected, alerts are immediately delivered to clinicians, allowing for rapid intervention before the condition escalates.

This real-time pipeline ensures that predictive models move from research into actionable, clinical decision support. Data governance and reliability are essential, as models must comply with strict healthcare standards while maintaining the trust of clinicians and patients.



Business Value

Real-time data streaming and predictive models are expected to support improvements in clinical care environments.

- **Improved patient safety:** Early identification of sepsis risk reduces intensive care unit escalation rates for bone marrow transplant patients.
- **Faster clinical response:** Real-time alerts give clinicians the ability to intervene before infection becomes life-threatening.
- **Operationalized AI in healthcare:** Predictive models move seamlessly from development to clinical use, bridging research and patient care.
- **Trusted and compliant data:** Governed streaming pipelines ensure transparency, reliability, and adherence to healthcare regulations.
- **Foundation for future AI models:** The same real-time infrastructure can support predictive analytics for other treatment complications beyond sepsis.

Real-Life Example: Siemens Healthineers

Use Cases: Hybrid and Multicloud | AI | IoT | Data Governance

Company Overview

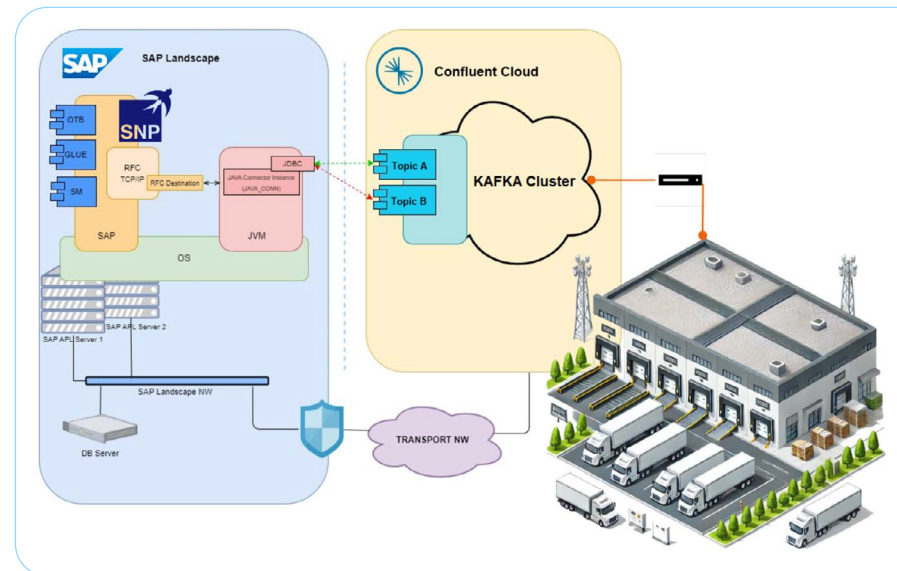
Siemens Healthineers is a global leader in medical technology, with more than 125 years of experience and more than 70,000 employees worldwide. The company pioneers breakthroughs in healthcare, ranging from imaging and diagnostics to advanced therapies and digital health services. Each year, Siemens Healthineers supports millions of patients globally through innovations that improve diagnosis, treatment, and efficiency in healthcare delivery.

Implementation

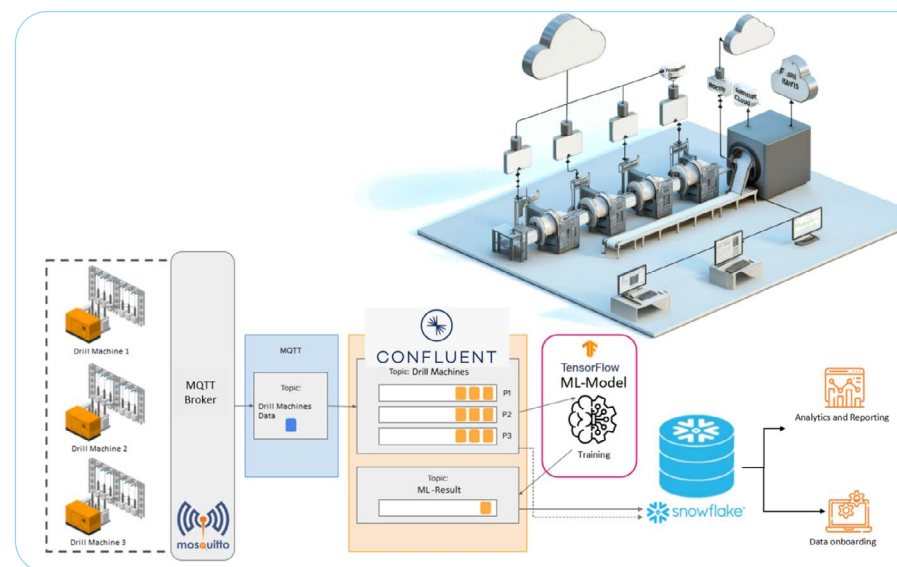
Modern healthcare demands real-time insights. Diagnostic imaging machines, lab equipment, and digital health applications generate vast amounts of data every second. Traditionally, this data was collected in silos, making it difficult to use it effectively across the enterprise. Siemens Healthineers needed a solution that could unify these data streams and make them instantly available to clinical applications, research teams, and healthcare providers.

With Confluent as the central data streaming platform, Siemens Healthineers integrated data from thousands of medical devices and systems across hybrid and multicloud environments. Data from imaging machines, laboratory diagnostics, and patient monitoring devices is streamed in real time through Confluent, where it's enriched, processed, and distributed to downstream systems.

Streaming ETL pipelines transform raw device data into structured information for AI models, which can then detect anomalies, optimize diagnostic workflows, and provide early clinical insights. Stream governance ensures compliance with strict healthcare regulations, including data lineage and security, while hybrid deployment allows sensitive workloads to remain on-premises and other workloads to scale in the cloud.



Seamless data flow from factory floor to healthcare facilities



Confluent as the central data streaming platform at Siemens Healthineers

Business Value

- **Faster and more accurate diagnoses:** Real-time access to diagnostic and patient data enables healthcare providers to shorten turnaround times and improve decision-making at the point of care.
- **Personalized treatment and preventive care:** Continuous data feeds power AI models that support predictive diagnostics, anomaly detection, and tailored treatment strategies for patients.
- **Streamlined clinical operations:** Breaking down data silos and automating pipelines reduces administrative overhead and improves interoperability across global healthcare facilities.
- **Trusted and compliant healthcare data:** Governance features ensure transparency, security, and compliance with strict healthcare regulations, building confidence among providers and patients.
- **Future-ready digital health services:** A hybrid and multicloud architecture supports innovation in connected medical devices and next-generation healthcare applications.

"Confluent has enabled us to replace manual processes with automated data flows. This allows our engineers to focus on equipment performance rather than data movement. The result is more reliable diagnostic equipment and better healthcare outcomes."

Scott Elfering Head of Data Ingestion, Siemens Healthineers

Conclusion

The healthcare and life sciences sector is rapidly moving toward a data-driven, intelligent future. AI, telemedicine, remote monitoring, and personalized care are transforming treatment delivery and patient experience. Data streaming is the backbone, connecting devices, clinical systems, and research platforms to deliver real-time insights and actions while protecting privacy and security.

In hospitals and clinics, continuous data streaming enables proactive, personalized care. Patient vitals, lab results, and electronic health records flow seamlessly across systems, giving clinicians up-to-date views of every patient. Real-time analytics support early detection of critical conditions, personalized treatment adjustments, and faster clinical decision-making. This reduces risk, improves outcomes, and enhances operational efficiency across departments.

A powerful example comes from City of Hope, one of the United States' leading cancer research and treatment institutions in the United States. Using an event-driven architecture and Confluent as the data streaming backbone, City of Hope developed an AI-powered model to detect early signs of sepsis in bone marrow transplant patients. Vital signs, lab data, and clinician notes are streamed continuously to enrich the predictive model continuously. When a risk signal appears, alerts are sent instantly to medical teams, enabling life-saving intervention before the

condition escalates. Governed streaming pipelines ensure transparency, compliance, and reliability to bridge research and clinical care while setting the stage for future predictive models across other treatments.

Across the industry, similar advancements are reshaping every stage of care. Pharmaceutical R&D uses event-driven workflows to process laboratory data, biological images, and clinical trial results continuously, accelerating drug discovery and reducing time to market. Value-based care relies on streaming to track outcomes, coordinate across providers, and trigger timely interventions. Remote patient monitoring and telemedicine connect patients and clinicians beyond hospital walls, delivering continuous insights and faster responses. Hospital operations benefit from real-time patient flow management, with agentic AI reallocating beds, staff, and resources dynamically to prevent bottlenecks.

These advancements are redefining the foundations of modern healthcare. Real-time data streaming combined with AI enables systems that think, learn, and act continuously, supporting precision medicine and faster discovery. Hospitals gain agility, researchers shorten cycles, and caregivers focus on what matters most: patient well-being. With every connected device and data stream, healthcare moves toward a future where insights are instant, decisions proactive, and care personalized and profoundly human.

3.11 Technology

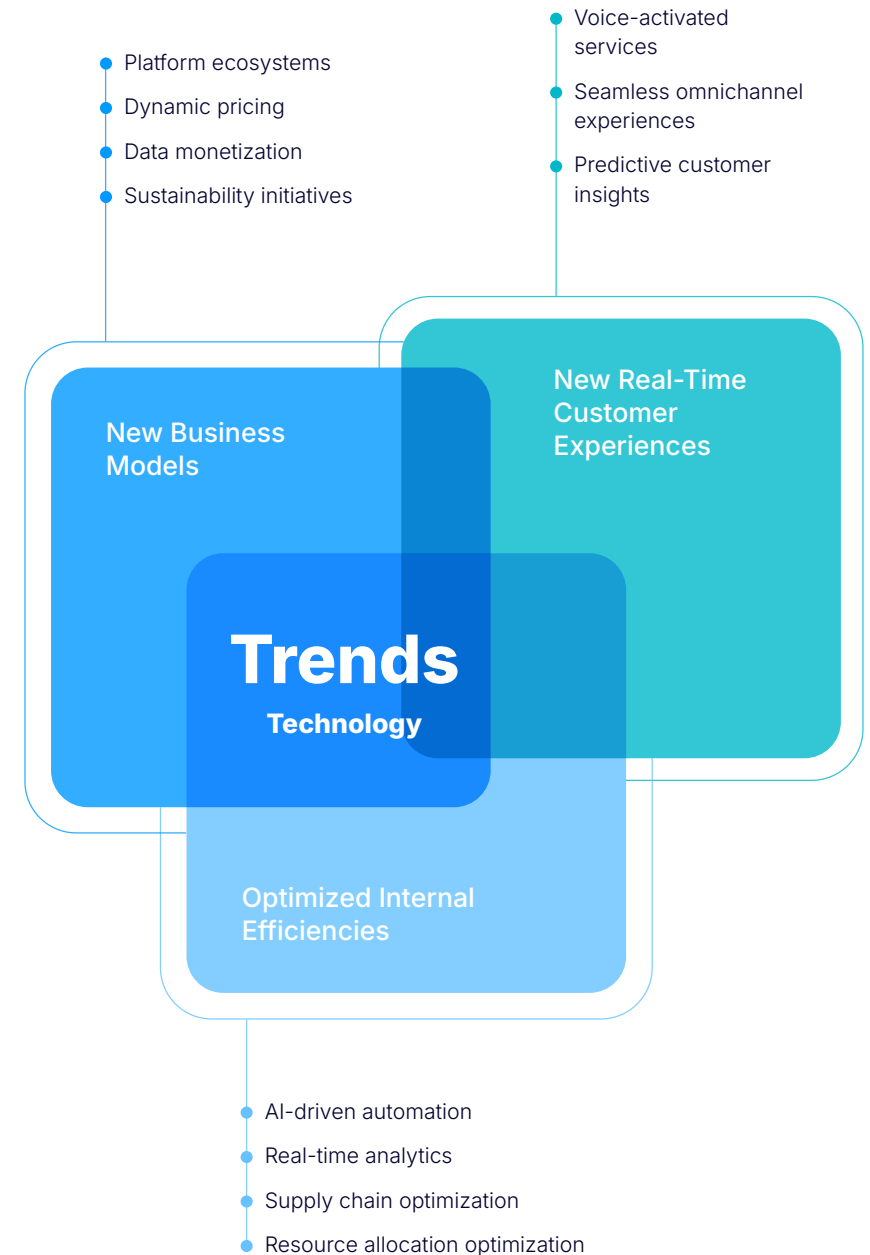
In today's world, technology companies—and increasingly all types of businesses—are transforming into software-defined organizations. This shift, as Confluent's CEO Jay Kreps stated in 2019, includes companies moving beyond simply using software as a productivity tool. Instead, software now forms the foundation of business operations, with core processes automated, connected, and executed in code. Accordingly, many of the examples in this chapter could be found in previous industry chapters. Technology is reshaping all types of business models, or, as Jay rightly said, "Every company is becoming software."

In the technology sector, data streaming has emerged as the critical platform enabling companies to react to data in real time, providing a seamless, instant flow of information across all functions. This is especially true of digital-native companies that are born in the cloud and adopt data streaming from the beginning at the core of their data architectures, running their complete businesses on data streaming platforms. Several trends are driving this transformation, making real-time data streaming not just beneficial but essential. Cloud computing and SaaS offer companies the scale and flexibility to operate globally, while data privacy and cybersecurity remain top priorities amid increasing threats.

AI is enabling more dynamic and personalized user experiences, and microservices architectures allow businesses to develop applications with unprecedented agility and scalability.

Increasingly, agentic AI plays a pivotal role by autonomously orchestrating workflows, detecting anomalies, and adapting services in real time. For technology companies, this means moving from reactive operations to adaptive, self-optimizing systems that continuously improve user experiences, infrastructure efficiency, and business outcomes.

This chapter explores how data streaming empowers technology companies to not only meet current demands but also build a future-proof foundation for continuous innovation. As companies shift toward event-driven architectures, they unlock the potential to stay competitive, responsive, and resilient in an ever-evolving digital economy. Far from being a passing trend, real-time data streaming is becoming the backbone of the modern enterprise, transforming how businesses operate, engage with customers, and deliver value at scale.



Scenario 1: Platforms and Mobile Apps

Platforms in the categories of on-demand service and gig economy software are designed to connect consumers with independent providers of real-time logistical services such as ride-hailing and food delivery.

These applications can be categorized as follows.

- **On-demand service platforms:**

This is the most common category, as these apps provide services on demand by connecting users to service providers based on real-time needs. Examples include ride-hailing, grocery delivery, and food delivery.

- **Marketplace platforms:**

These platforms function as digital marketplaces where consumers and providers meet to transact. They include features such as payment processing, customer service, and logistics, facilitating smooth transactions between parties.

- **Logistics and delivery software:**

Apps that manage inventory, delivery schedules, and driver routes also fall under logistics software.

- **Mobility-as-a-service (MaaS):**

Transportation services that offer an alternative to vehicle ownership by providing transportation as a flexible service fit within this category.

Data streaming is revolutionizing the

way platforms and mobile apps operate by delivering dynamic content, personalizing user interactions, and ensuring seamless performance across platforms. Real-time data streaming allows for the continuous flow of information, enabling platforms and apps to respond instantly to user actions and external events.

For instance, a social media platform can harness the power of data streaming to deliver real-time updates on user feeds. This ensures that users are constantly engaged with the freshest content, whether it's a friend's status update, a trending news article, or a live event broadcast. Similarly, for an on-demand service platform like a ride-sharing app, data streaming allows users to see the exact location of their drivers in real time, receive instant updates on estimated arrival times, and get notified of any route changes due to traffic conditions. This real-time interaction not only enhances the user experience by providing transparency and reliability but also builds trust and satisfaction, as users feel informed and in control of their service experiences.

Similarly, an ecommerce app can use data streaming for personalized product recommendations by analyzing a user's browsing history, purchase behavior, and current market trends. This allows the app to adjust recommendations in real time, for example, to showcase hiking gear for a user interested in outdoor equipment. This level of

personalization improves the shopping experience by making it more relevant for the user and boosts conversion rates. Real-time data streaming enables the app to offer timely discounts or suggest complementary products at the moment of consideration.

Business Value

- **Enhanced user engagement:** Real-time updates and dynamic content keep users constantly engaged, increasing the time spent on the platform and fostering a more interactive experience.
- **Personalized user experience:** By analyzing user behavior in real time, platforms can offer personalized interactions and recommendations, enhancing user satisfaction and loyalty.
- **Improved transparency and trust:** Real-time data streaming provides users with up-to-date information, such as driver locations and order statuses, building trust and reliability in the service.
- **Operational efficiency:** Automation of data processing and logistics management reduces manual intervention, streamlining operations and lowering costs.
- **Increased conversion rates:** Personalized recommendations and timely offers based on real-time data analysis can significantly boost conversion rates and sales.
- **Scalability and flexibility:** Data streaming technologies provide the scalability needed to handle large volumes of transactions and user interactions, supporting business growth and expansion.
- **Competitive advantage:** Companies that leverage real-time data streaming can differentiate themselves by offering superior user experiences and innovative features, gaining a competitive edge in the market.
- **Proactive issue resolution:** Continuous monitoring and real-time insights allow for the quick identification and resolution of potential issues, minimizing downtime and maintaining service quality.
- **Informed decision-making:** Access to real-time data enables companies to make informed decisions quickly, optimizing operations and strategic planning.

Scenario 2: Infrastructure Monitoring

Data streaming is transforming infrastructure monitoring, particularly for companies that require real-time insights for optimal performance and security. By leveraging data streaming, these companies can continuously collect, process, and analyze data from various sources, enabling proactive monitoring and rapid response to issues. This capability ensures that systems remain robust, efficient, and secure and enhances customer satisfaction.

For example, a cloud service provider can use data streaming to monitor server performance, addressing bottlenecks before they impact users. Similarly, a cybersecurity firm can use data streaming to detect anomalies in real time, allowing for immediate intervention to prevent breaches and safeguard sensitive information.

Business Value

- **Proactive issue resolution:** Real-time monitoring allows companies to identify and address potential issues before they escalate, minimizing downtime and maintaining service quality.
- **Enhanced security:** Continuous data analysis enables the rapid detection of anomalies and potential threats, allowing for immediate intervention to protect sensitive information.
- **Operational efficiency:** Automation of monitoring processes reduces the need for manual oversight, lowering operational costs and increasing efficiency.
- **Improved system performance:** By continuously optimizing system performance through real-time insights, companies can ensure high availability and reliability of their services.
- **Scalability:** Data streaming technologies provide the scalability needed to handle large volumes of monitoring data, supporting business growth and expansion.
- **Innovation and agility:** Data streaming fosters a culture of innovation, enabling companies to quickly adapt to changing market demands and technological advancements.

Scenario 3: Digital Marketplaces

Digital-native companies often operate global marketplaces that connect buyers, sellers, and creators. A data streaming platform allows these ecosystems to process millions of events in real time to match supply with demand, monitor fraud, and ensure instant payments. For example, a creator marketplace can adjust recommendations on the fly, highlight trending items, and prevent fake reviews by continuously analyzing transactions, clicks, and user behavior.

Business Value

- **Improved matching efficiency:** Real-time data ensures that buyers and sellers are connected instantly, increasing engagement and conversion rates.
- **Fraud protection:** Continuous monitoring of transactions prevents fake listings, scams, and duplicate accounts, protecting revenue and brand trust.
- **Personalized experiences:** Live recommendations and trending highlights keep content relevant and engaging for each user.
- **Scalability at global scale:** Marketplaces can expand to millions of users and transactions without latency issues.
- **Increased ecosystem trust:** Transparent and secure processes foster credibility with both users and partners.

Scenario 4: Community and Social Platforms

Streaming is core to the success of community-driven platforms such as gaming communities, creator networks, and social collaboration tools. These businesses need to deliver live updates, personalized feeds, and moderation at scale. By continuously streaming interactions, posts, and engagement metrics, platforms can surface the most relevant content, detect toxic behavior in seconds, and adapt recommendations in real time to keep communities safe, vibrant, and sticky.

Business Value

- **Higher engagement:** Personalized feeds and live updates keep users active and returning frequently by surfacing timely conversations through real-time ranking and notifications that increase session length and daily active usage.
- **Safer environments:** Real-time moderation tools detect harmful or toxic content immediately, and automated classifiers with human-in-the-loop review remove policy-violating content within seconds to maintain trust, ensure compliance, and reduce regulatory risk.
- **Continuous relevance:** Streaming interactions enable platforms to surface trending topics and conversations instantly, while freshness-aware recommendations adapt as signals change to prevent feed staleness and user churn.
- **Stronger network effects:** Vibrant, interactive communities drive organic growth and retention as live features like chat, co-streams, and events create shared moments that attract new members and re-engage lapsed users.
- **Revenue growth:** Personalized engagement increases the effectiveness of ads, premium content, and creator monetization as real-time propensity and pricing models optimize offers and inventory yield without degrading user experience.

Scenario 5: Subscription and Digital Experience Platforms

Many cloud-native companies run subscription-based businesses (e.g., SaaS, media, lifestyle apps). With data streaming, companies can monitor churn signals—such as declining usage, failed payments, and negative sentiment—in real time. They can respond immediately with personalized offers, retention campaigns, or proactive support. They can also experiment with new pricing models (usage-based, freemium, and dynamic) without disrupting existing services because the data is processed continuously.

Business Value

- **Reduced churn:** Real-time detection of risk signals—such as declining usage, failed payments, or negative sentiment—triggers personalized offers, targeted retention campaigns, and proactive support to re-engage at-risk customers.
- **Revenue optimization:** Usage-based and dynamic pricing models adapt instantly to demand, with streaming metering and real-time elasticity tuning offers and price points to maximize ARPU and LTV without service disruption.
- **Improved satisfaction:** Personalized offers and immediate, omni-channel support resolve issues in the moment and create frictionless experiences that lift NPS/CSAT and loyalty.
- **Faster innovation:** Continuous feedback loops power rapid product updates and feature experimentation, using streaming analytics, feature flags, and real-time A/B tests to iterate safely and roll back instantly when needed.
- **Accurate forecasting:** Live subscriber analytics provide reliable cohort health and churn propensity signals, strengthening demand planning and retention budgeting with up-to-the-minute visibility.

Real-Life Examples

- Character.AI
- Wix
- Grafana Labs

"In terms of scaling with Confluent Cloud, we utilize the possibility to increase or decrease the capacity with a click of a button. It means that when we get ready for Black Friday or something like that, we can easily scale up the capacity that is required. Also, when we understand that there is lower traffic, we can easily decrease it. So we take advantage of the easy auto-scaling capability of Confluent Cloud."

Natan Silnitsky, Senior Backend Engineer, Wix

Real-Life Example: Character.AI

Use Cases: Data Products | IT Modernization | Data Sharing | AI | Data Governance

Company Overview

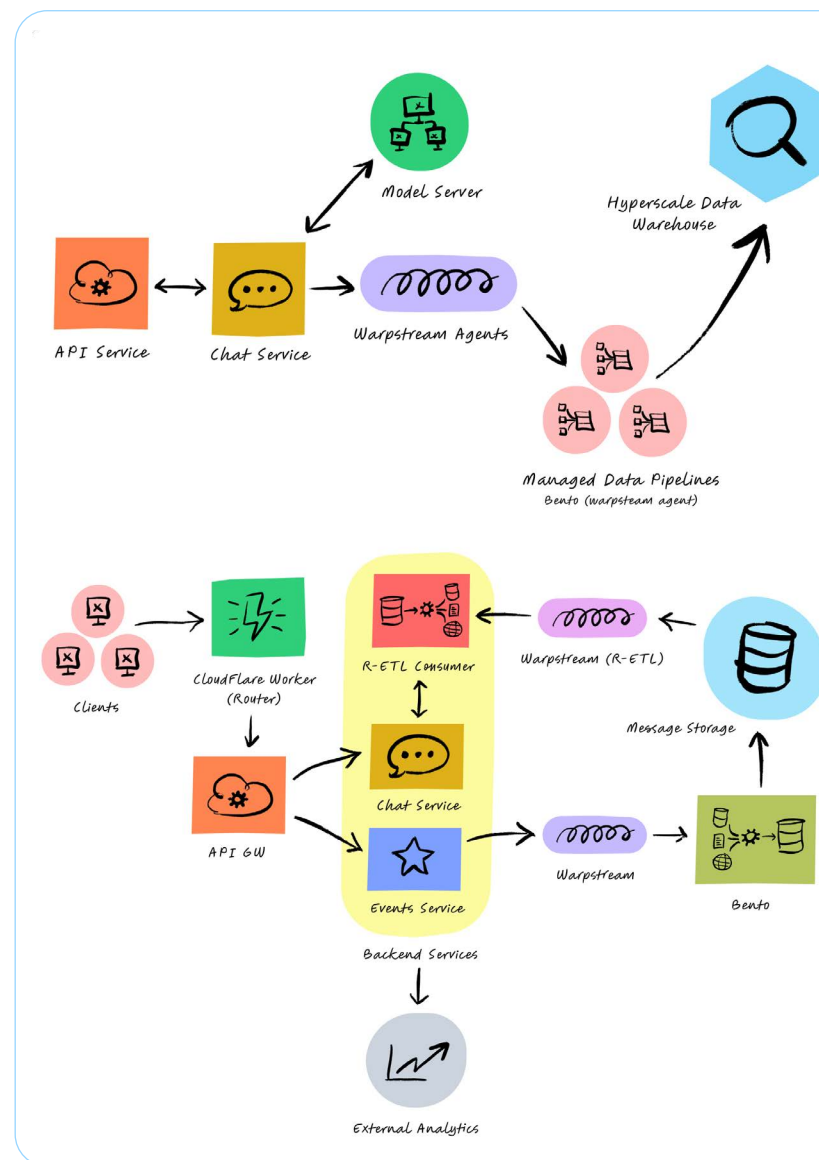
Character.AI is a leading conversational AI platform serving more than 20 million monthly active users. Its purpose is to empower people to connect, learn, and tell stories through interactive entertainment, blending storytelling, gaming, social connection, and creative expression. Despite being a relatively small company, it operates at global scale, handling complex data storage and processing requirements across both mobile and web applications. By combining advanced AI models with imaginative user experiences, Character.AI delivers personalized, interactive conversations while fostering a vibrant community of creators and fans. To sustain this growth, the company continuously evolves its data infrastructure to balance performance, scalability, and cost efficiency.

Implementation

Character.AI initially relied on batch-oriented ETL workflows powered by the Data Build Tool (DBT). While suitable for early research use cases, this model could not keep pace with the company's evolving need for real-time user insights. Certain use cases, such as capturing user prompts and analyzing in-app behaviors, required streaming pipelines capable of delivering low-latency data to downstream systems.

The company evaluated multiple technologies and concluded that Kafka compliance was a key requirement for its next-generation data platform. This would ensure compatibility with existing tools such as Spark and enable scaling across diverse workloads. At the same time, Character.AI sought a Bring Your Own Cloud (BYOC) approach to take advantage of its own compute resources, reduce costs, and avoid vendor lock-in.

After assessing alternatives, Character.AI selected WarpStream, a Kafka-compatible data streaming platform built directly on object storage. WarpStream's stateless agent architecture and lightweight operational model aligned with Character.AI's need for simplicity and cost-effectiveness. The platform's native integration with Google Cloud Storage further streamlined adoption.



Character.AI's high-level architecture, including WarpStream

Two primary use cases emerged.

- **Prompts pipeline:** Character.AI built a real-time pipeline to stream user prompts directly into WarpStream and then into its data warehouse for analysis. This setup now processes more than 3 GiBs of data per second, giving teams immediate access to user interaction data while also making it easy for other systems to tap into the same feed.
- **Reducing analytics costs:** With millions of users across mobile and web, third-party analytics costs were rising quickly. By moving analytics to WarpStream, Character.AI created a near-real-time engine that delivers faster insights while significantly lowering expenses. The new approach also gives the company greater flexibility and control over how analytics are managed.

This transition marked a fundamental shift from batch to real time, enabling Character.AI to continuously refine its understanding of user behavior and deliver richer, more responsive user experiences.

Business Value

- **Scalability:** WarpStream enables sustained throughput at multiple gigabytes per second, meeting the demands of a rapidly growing user base.
- **Cost savings:** Replacing third-party analytics pipelines with WarpStream reduces vendor fees and optimizes unit economics.
- **Operational simplicity:** Stateless agents and DevOps-driven configuration streamline deployment and reduce operational burden for a small engineering team.
- **Real-time insights:** Continuous data flows allow faster understanding of user interactions, improving personalization and engagement.
- **Flexibility:** Kafka compatibility supports integration with Spark, data warehouses, and other tools without architectural lock-in.
- **Future readiness:** The BYOC model ensures that Character.AI can scale on its own infrastructure while retaining flexibility to adopt new technologies.

"Confluent wants to offer data streaming to all customers with all requirements and workloads. I've been deeply impressed with WarpStream—it's BYOC done right. With this acquisition, we have a data streaming offering for everyone."

Jay Kreps, Co-founder and CEO, Confluent, in the press release on the WarpStream acquisition in September 2024

Real-Life Example: Wix

Use Cases: Observability | Hybrid and Multicloud | Event-Driven Microservices | AI

Company Overview

Wix is a leading global platform for creating, managing, and growing a complete digital presence. Founded in 2006, Wix empowers millions of users, including self-creators, agencies, enterprises, and more, with industry-leading infrastructure, performance, and security. The platform combines advanced AI, flexible design, and robust business and commerce solutions to help users build stronger brands, connect with their audiences, and scale their businesses online. Wix is shaping the future of how digital experiences are built, with its intuitive AI-powered website builder and no-code application creation through Base44, making sophisticated creation accessible to all.

Implementation

As Wix expanded globally, its self-managed Apache Kafka deployment in the public cloud created operational challenges. The complexity of managing thousands of microservices, combined with the need for real-time analytics and AI-driven personalization, strained internal engineering resources.

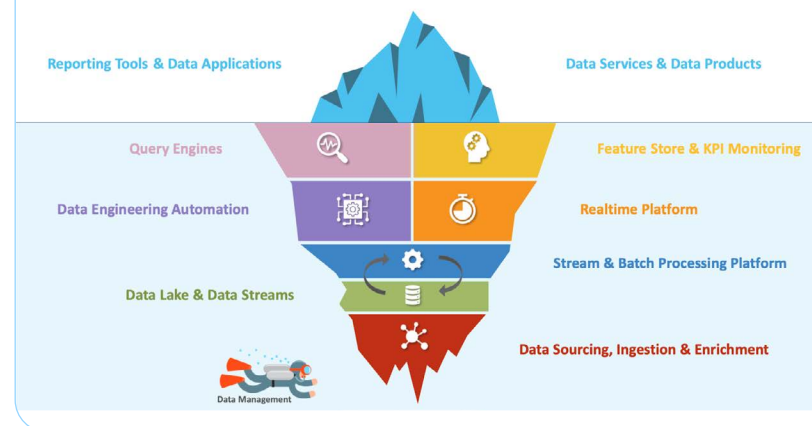
To address this, Wix migrated to Confluent Cloud as its central data streaming platform. Confluent now powers integrations across more than 4,500 microservices, connecting workflows for payments, orders, inventory, and analytics. Real-time pipelines process vast streams of event data, enriched with Apache Flink® to deliver actionable insights instantly.

This architecture allows Wix to build user-facing analytics that enable website developers to monitor performance, understand visitor behavior, and receive continuous feedback in real time. Stream governance ensures reliable data access across teams, while managed services free developers to focus on innovation instead of infrastructure maintenance.

"With Wix Analytics, the value of user feedback is tremendous for our customers. That's what differentiates us from other platforms, and Confluent makes it possible, in real time and at massive scale. We frequently hear that those real-time insights are what keep us the preferred platform for some of our longest-standing customers. The ability to know what is happening in your business—especially a virtual business—as near to real time as possible is invaluable for Wix and for its users."

Josef Goldstein, Head of R&D, Big Data Platform, Wix

Anatomy of Wix's Data Platform



Business Value

- **Improved customer experience:** Real-time web analytics empower Wix customers to act on visitor feedback instantly, enhancing engagement and website performance.
- **Higher developer productivity:** Pre-built integrations and managed services reduce operational overhead, accelerating feature development and innovation.
- **Operational reliability:** Confluent's fully managed services significantly reduced downtime, ensuring resilient operations across thousands of microservices.
- **Personalized user journeys:** Streaming data pipelines enable hyper-personalization at scale, tailoring ecommerce and content experiences to individual users.
- **Scalable global platform:** A flexible data foundation supports continuous growth, new services, and peak traffic demands across millions of global users.

Real-Life Example: Grafana Labs

Use Cases: IT Modernization | Observability | Hybrid and Multicloud | Data Sharing

Company Overview

Grafana Labs is the company behind Grafana, the open source visualization and observability platform used by millions of developers and enterprises worldwide. Among its core technologies is Grafana Mimir, a horizontally scalable, multi-tenant time series database (TSDB) designed to provide Prometheus-compatible metrics storage at global scale. Since its launch, Mimir has become a leading choice for organizations seeking high-performance, cloud-native observability, with thousands of community contributors and deployments across industries.

Implementation

As adoption of Grafana Mimir grew, the company faced challenges common to hyperscale observability systems: balancing massive volumes of writes and queries while guaranteeing reliability and cost efficiency across multi-availability zone (AZ) environments. In its original design, Mimir ingester nodes were stateful, combining in-memory TSDBs with local write-ahead logs (WALs). While this architecture ensured durability, it created operational trade-offs. Heavy query loads could disrupt live writes, and data ingestion required complex state management.

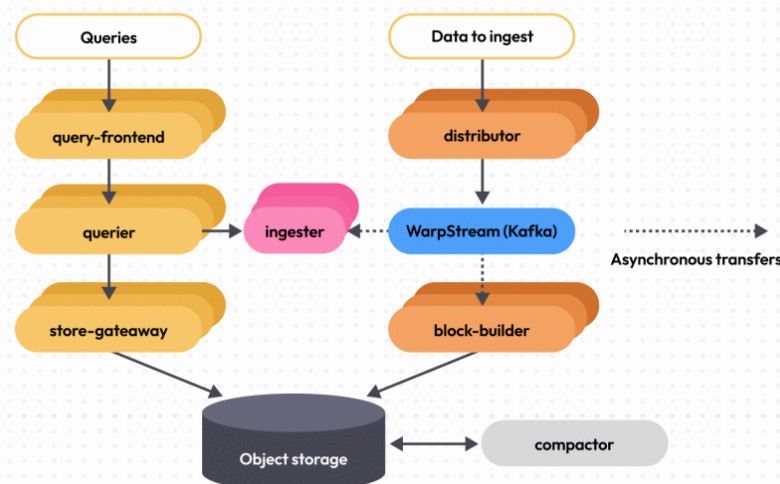
To overcome these constraints, Grafana Labs redesigned Mimir's architecture with the goal of decoupling read and write paths. The team selected Apache Kafka as the backbone of this new architecture, leveraging its distributed event streaming capabilities to isolate ingestion from query workloads. This shift increased resilience by ensuring that failures in one path would not impact the other while also simplifying the management of data ingestion.

For Grafana Cloud Metrics, the company pursued an even broader objective: enabling reliable, cost-efficient operations across multiple availability zones to meet customer SLA requirements. Traditional Kafka clusters incurred significant cross-AZ data transfer costs at scale. To address this, Grafana Labs partnered with WarpStream, the Kafka-compatible data streaming platform from Confluent built directly on object storage. WarpStream eliminated inter-AZ networking costs, removed the need for disk management, and offered stateless, auto-scaling clusters.

Stress testing demonstrated WarpStream's capacity to handle throughput at the scale Grafana Cloud Metrics required: up to **10 GiB/s** (compressed) written through a single cluster, consumed with **3x** fan-out for an aggregate of **40 GiB/s**. With these results, Grafana Labs validated that its new observability architecture could scale to meet future demand.

The redesigned Mimir, now gradually rolled out across Grafana Cloud Metrics, provides a foundation for future innovations in observability while remaining fully open source.

New Mimir architecture



Business Value

- **Improved reliability:** Decoupling reads and writes ensures that spikes in query or ingest traffic no longer affect system stability.
- **Cost optimization:** WarpStream's object storage foundation reduces cross-availability zone costs and simplifies infrastructure operations.
- **Scalability:** The new architecture supports massive throughput requirements, enabling cloud-scale observability for global customers.
- **Operational simplicity:** Stateless components and auto-scaling capabilities reduce management overhead and complexity.
- **Customer trust:** Higher SLA compliance strengthens confidence in Grafana Cloud Metrics for mission-critical observability.
- **Future-proof innovation:** The open source Mimir project gains a robust foundation to support next-generation use cases in metrics storage and analysis.

Conclusion

The technology sector is redefining how the world builds, operates, and innovates. As every organization becomes software-defined, real-time data streaming has become the foundation for agility, personalization, and resilience. It connects applications, systems, and users in a continuous flow of information, enabling instant insights and adaptive experiences at a global scale.

Data streaming architectures now underpin the digital economy. Platforms and apps deliver seamless experiences as events move instantly across services, while infrastructure teams monitor performance and security in real time. Marketplaces and community platforms match users, detect fraud, and personalize engagement as data flows continuously. Subscription and SaaS businesses use streaming to predict churn, optimize pricing, and power dynamic customer journeys. Across every use case, event-driven systems replace static operations with intelligent, responsive ecosystems.

This transformation is embodied by leading innovators. Grafana Labs, creator of one of the world's most popular observability platforms, rebuilt its architecture using WarpStream to separate reads from writes across multicloud environments. This reduced cost and complexity while delivering the scalability needed to support global observability workloads and open source innovation.

Character.AI, serving tens of millions of users, moved from batch analytics to real-time pipelines built on WarpStream, processing more than three gigabytes of payments. Real-time pipelines enable live insights and hyper-personalized web experiences, while managed services increase reliability and free developers to focus on product innovation.

These pioneers illustrate how data streaming combined with AI is transforming the technology landscape. Streaming agents detect anomalies, orchestrate workflows, and adapt services autonomously, turning digital platforms into self-optimizing systems. In this new era, software is no longer static. Software is living, learning, and evolving in real time. Companies that embrace streaming-first architectures are not just improving efficiency; they're shaping the next generation of intelligent, adaptive, and continuously innovating digital enterprises.

Creation and Positioning of a Data Streaming Organization and Community

04

Introducing a data streaming platform is more than just a technical shift. It requires a fundamental change in the business. As such, it must be managed like any other transformational initiative, aligning with key change management principles. This includes strategic internal communication and leadership buy-in, which requires awareness of the benefits of data streaming both vertically and horizontally within the organization. Community building is crucial in this process, serving as a hub for exchanging use cases and successes. A vibrant community not only educates and inspires but also ensures the platform's continued relevance and usage. Key change management principles supporting this approach include:

- **Communication:** Ensuring consistent, clear messaging across the organization to build awareness and understanding of the platform
- **Leadership alignment:** Securing executive sponsorship to champion the initiative, ensure resourcing, and guide teams through the transition
- **Engagement and involvement:** Involving all relevant teams early in the process, fostering cross-functional collaboration and encouraging shared ownership
- **Managing resistance:** Addressing potential resistance by timing the community's launch carefully and providing the right support structures
- **Education and training:** Equipping stakeholders with the knowledge and skills they need to adopt and implement the platform effectively.

- **Sustaining change:** Building momentum through ongoing communication and showcasing successful use cases to sustain engagement over time

Framing the adoption of a data streaming platform through these principles helps make the platform an integral part of operations, driving innovation and competitive advantage. An active data streaming community is the core of the Data Streaming Organization (DSO), a framework designed to guide organizations to meet increasingly complex customer needs with governed, shared, accessible, and real-time data. This is achieved by providing a unified technology platform, established ways of working, and a guiding data streaming strategy and vision.

If you want to read more about the holistic framework of the DSO, we highly recommend the white paper "[Maximize Data Value and Innovation Across the Organization](#)," which was written by the Confluent Strategic Advisory Team. In previous chapters, we shared several use cases that were first introduced within internal data streaming communities. By sharing the success stories, companies encouraged teams to achieve their business goals with data streaming. Successful use cases should be highlighted through various channels, which discuss in this chapter.

This book is one such channel for showcasing use cases. If you believe your use case would be a good fit to be featured in this book, reach out to us through use-case-book@confluent.io.

4.1 The Journey to a Data Streaming Organization

Data streaming revolutionizes how companies process and use their data. A data streaming platform can be a powerful component of any data architecture—as long as it's used and communicated correctly. For this to happen, it needs the buy-in of all stakeholders involved, from business to IT. While clear internal positioning and consistent, continuous, and emotionalized communication are all essential factors, they're all second to one crucial requirement: **timing**. Starting a community can easily do more harm than good if it doesn't happen at the right time with the right message. Companies need to have reached a certain level of maturity in data streaming before a communication strategy can be put into place.

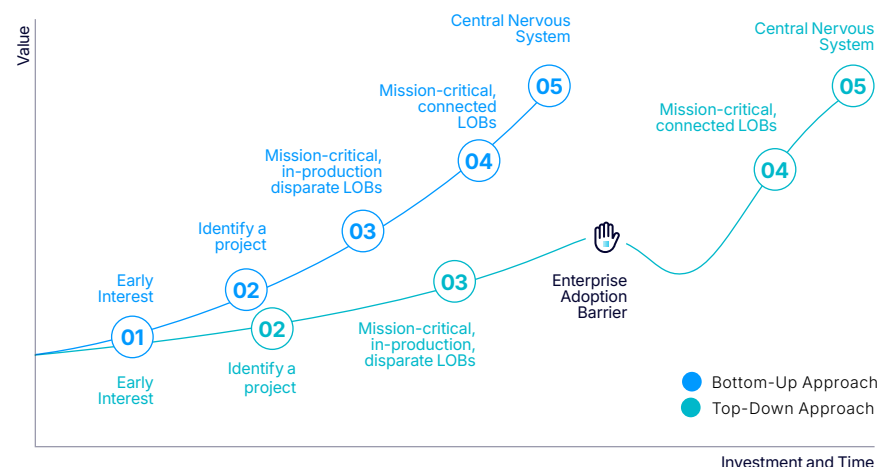
Back in 2018, Lyndon Hedderly set out a maturity model in his white paper, "[Five Stages to Streaming Platform Adoption](#)," which has lost none of its relevance. His guide describes the enterprise adoption process of Apache Kafka and a data streaming platform in five stages, from awareness-building all the way to full integration of the platform as the central nervous system of the company. The direction of the initial drive within the company varies greatly, but the **bottom-up** approach (initiation comes from the project team) appears to be more common than the **top-down** approach (initiative comes from management).

The **Enterprise Adoption Barrier**—as displayed in the next graphic—is a frequent phenomenon that occurs within organizations as they move up the maturity curve following the bottom-up approach. Moving from Stage 3 to 4 often requires a disruptive organizational and processual transformation—without it, companies hit the aforementioned barrier, effectively hindering further adoption.

Another indispensable prerequisite is a centralized team for enterprise services, even with a fully managed service like Confluent Cloud. Someone in the organization still needs to define and own governance, cost accounting/ attribution, and a common operating model for infrastructure while offering user-friendly self-service tools and APIs. These challenges often lead to a somewhat painful process of developing centralized coordination, strategy, and oversight after data streaming has been widely adopted.

We want to briefly highlight the individual stages of data streaming adoption, based on the most common bottom-up approach. You'll find that these phases often reflect the development steps and maturity stages of individual business units within a company as well as the entire organization.

Unified Data Streaming Strategy



In the data streaming maturity model, the difference between a top-down and a bottom-up approach lies in the Enterprise Adoption Barrier

Stage 1: Early Interest—pilot projects and initial successes

The journey often begins with small pilot projects initiated by technology teams running non-critical use cases. This phase makes it possible to test initial applications and experience the agility and flexibility of a data streaming platform. Teams usually benefit from increased development speed and the new possibilities of real-time data processing.

Stage 2: Identify a project—production-ready applications

The company is starting to implement the first applications in production. These applications offer both technical and business benefits, such as improved customer experiences and the

development of new business models. This phase marks the transition from experimental applications to mission-critical systems.

Stage 3: Mission-critical, in-production, disparate lines of business (LOBs)—integrated streaming and strategic alignment

In this phase, the data streaming platform is closely linked to the company's strategic goals. Different business units work together to integrate data streams and break down silos. The platform supports the security, reliability, and scalability of data processing. The establishment of a central streaming service and/or a Center of Excellence (CoE) is usually initiated in this phase, with the clear goal to move to Stage 4. Decisions made in this stage include core components of a DSO, which

also includes establishing an internal community around data streaming as well as creating an integrated communication and positioning concept that surrounds the central service.

Stage 4: Mission-critical, in-production connected LOBs—global streaming and cloud integration

In the global streaming stage, the platform is scaled across the enterprise. The company starts using the advantages of cloud services to process data efficiently and comply with more legal requirements. The platform enables the company to make data available in different geographical regions and consolidate global business processes at the same time. The internal community is established and growing, building on a multichannel communication strategy that ensures the continued buzz about successes and future strategies.

Stage 5: The central nervous system of the company

In the final stage, the company becomes a true DSO, in which the data streaming platform becomes the central nervous system. All data is available in real time, which significantly improves efficiency and the ability to innovate. Companies at this stage demonstrate the power of this architecture by processing large amounts of data in real time and continuously developing new applications.

"86% of IT leaders highlight investments in data streaming as a top strategic or important priority, and 51% now identify as being in Level 3, which is up from just 3% in 2024"

2025 Data Streaming Report

Transitioning an entire organization to produce and consume streaming data usually requires significant training and education to promote this new technology paradigm. There is always at least some technical infrastructure gap that necessitates homebrew solutioning. This can become an expensive endeavor and requires executive sponsorship to some extent. Once set up, other parts of the business must align—which comes with the significant burden of cross-functional execution.



[Check out the white paper on how to create a Data Streaming Organization](#)

4.2 Key Drivers for a Community Within a Data Streaming Organization

Creating a unified, reliable source of information for a particular technology often demands internal support and persuasion. Typical questions, such as "Is the platform/service ready?," "Is this topic important for our goals?," and "Do we have meaningful results to share yet?" are natural, and they should be confidently answered with a "Yes!" by the project team before getting started.

The motivations behind forming a communication strategy and community are as follows:



Internal (and external) positioning



Know-how transfer



Consolidation of projects and use of best practices



Hiring (internal and external)

Internal (and external) positioning

Apache Kafka, Apache Flink®, and Confluent are innovative technologies that open up new opportunities for companies. Project teams, owning the central data streaming service, get this and believe in the strategic relevance the paradigm of data streaming brings to the organization. For this paradigm to be understood across the organization, the concept, the vision, and the benefits need to be positioned and seen.

So it's not just about keeping your position in the market by executing a successful data strategy. It's also about positioning the internal data streaming service within the company in order to reach the next stage on the streaming maturity curve and create more added value through the platform. That means positioning the CoE and project team itself within the organization. The message should be clear, and it should be confident: **This department makes significant difference—if not THE difference—in the company-wide digitalization strategy! Don't hide your light under a bushel.**

Regardless of whether your data streaming strategy is initiated through a bottom-up or top-down approach, in both cases, the opposite party should be involved as quickly as possible. This is because no strategy works without implementation and vice versa (also see "Enterprise Adoption Barrier"). To achieve this, the strategic relevance, the actual functionalities, and the vision of the platform must be communicated all across the enterprise and, most importantly, to all different target groups.

Equally important as the internal communication is the external visibility. Because, spoiler alert: For some colleagues, the penny will drop only when they read about the company's internal data streaming project—to put it bluntly—in the daily news over breakfast. We'll see concrete examples for internal and external positioning later in this chapter.

Good news for the bottom-up project teams: **For 86% of IT leaders surveyed in the 2025 Data Streaming Report** data streaming is one of the top strategic priorities for IT investment in 2025, compared to 51% in 2024 and 44% in 2023. In addition, **87% expect data platforms will be increasingly used to feed AI systems with real-time, contextual, and trustworthy data.** But even if the adoption of data streaming increases, it's important to continuously engage all stakeholders.

Know-how transfer

The success of any technology implementation depends on how well it's understood internally and how effectively that knowledge is shared. Many service owners and CoEs are therefore in close partnership with the internal training or academy departments. These departments usually have the necessary structures in place to make training opportunities visible and accessible within the company.

"The decisive thing about knowledge is that you take it to heart and apply it."

Confucius

Many companies rely on the free on-demand video courses by Confluent for the basics on Kafka, Flink, and beyond, which offer a good introduction for developers and architects in videos that build on each other.

In addition to building up technical knowledge, be careful not to lose sight of the conceptual and, above all, strategic aspects. For that, consider running a variety of educational sessions in which you explain the core concepts of data streaming and an event-driven architecture to non-IT departments and stakeholders.

Consolidation of projects and use of best practices

Open source is a wonderful thing! Anyone can use it and test it out with new use cases. As wonderful as this may seem, it's sometimes a thorn in the side of many IT departments. Some may even hush conspiratorially about the infamous and unpleasant thing called "shadow IT." This makes it even more important for service owners to create a central point of contact for all matters in this area as quickly as possible. In other words: You want Kafka? We can do Kafka. And we can do Kafka in a structured and compliant way to your benefit. A central data streaming platform makes sense only when it's actually central and the whole company can publish and subscribe to its streams of data in a governed way.

Besides that, we noticed several occurrences with clients where different teams were working on similar use cases behind office door 1 as well as office door 3. The time saved if those teams met through an internal community is obvious. Over time, there's no need to reinvent the wheel, as the internal community also serves to distribute architectural blueprints and give guidance on regulations and compliance.

It's recommended to ask your internal communications team what the most common collaboration platforms and channels are so that you can choose an environment that your colleagues feel most at home with.

Hiring (internal and external)

We'll explore the options for presenting the project externally to attract new talent (so-called "employer branding") later in this chapter. However, internal recruitment shouldn't be ignored either. Successful projects still require a dedicated team for ongoing development, even in a SaaS world. Often, reallocating internal resources is the faster route to accelerate project growth. When the strategic relevance of the project is well-known within the company and potential career opportunities are clear, internal interest tends to follow quickly. In fact, some of the largest data streaming services among our clients started from internal demand and interest, allowing teams to scale rapidly.

Executive sponsorship is an absolute accelerator for internal team expansion, but it must be earned through a clear articulation of the platform's mission and vision, alongside demonstrated initial successes and a professional presentation backed by a consistent branding strategy.

4.3 Fundamental Tactics for Developing a Data Streaming Community

After kicking off this chapter with change management principles to successfully establish a Data Streaming Organization, we'll now combine them with the core aspects of marketing, such as personas, channels, and promotion.

Addressing Your Target Group

Anyone who has ever enthusiastically talked about their Kafka project at the (virtual) coffee machine will be familiar with the response "Oh, you're into literature?" Even though Apache Kafka has become the global de facto standard for data streaming, non-IT people usually associate Kafka with the author and data streaming with the latest Netflix series (which is not entirely wrong since Netflix runs its recommendation engine on Kafka). When talking about the advantages of data streaming technology, you can't rely on one common understanding for everybody.

That's why it's so crucial to employ different tonality, messaging, and forms of presentation for each persona, such as senior management, C-levels, architects, application owners, and developers. Each of those roles speaks a different language and focuses on different aspects of data streaming.

While developing these enablement and training plans, always keep in mind that everybody learns differently. This fact must be taken into account if you want to get everyone in the company to buy in. The larger the company, the longer it will take to gain a basic understanding of what a data streaming platform and terms like "event-driven" actually mean—and what opportunities they offer for IT and business units. This also means that we need patience combined with a continuous communication strategy on all channels and across all learning styles:

- **Visual learning**
- **Auditory learning**
- **Reading and writing**
- **Kinesthetic learning**

It's recommended to provide material for as many learning styles as possible, both stand-alone assets (videos, podcasts, blog articles) and perpetually accessible information in a single source of truth. Dedicated Kafka Days and Data Streaming Days are becoming increasingly popular at Confluent and offer the opportunity to convey the technology and added value via a wide

variety of channels within a single day. We'll cover some examples later in this chapter.

There might be dedicated personas or departments to target first, like solution architect groups or analytical teams. It's important to understand that even the low-hanging fruit in terms of use case recruitment will be at different stages of maturity themselves. Although the company may already be at Level 3 or 4 of the maturity curve, you'll always come across departments/business units/application teams that are still at Level 1 or 2. This requires a different approach compared to a team that has been working with the technology for a long time. This is another reason that a continuous and consistent communication strategy with one easily accessible single source of truth is indispensable.

After analyzing your target group and the corresponding personas, you can move on to the next step: **How and where do we reach them?**

Finding Your Communication Channels

One goal of a community within a data streaming organization is to proactively prevent reinventing the wheel. The same principle applies to creating the community itself: Many companies already have well-functioning communities (often role-based, such as architecture tribes and IT management roundtables, or topic-based, such as those focused on GenAI or microservices). It's worth exploring best practices internally:

How were successful communities started? Are existing groups willing to support the new one and offer a virtual handshake? What channels were used to establish the current communities, and who manages these? Where do these communities "live"? Familiarize yourself with the internal culture of collaboration. The structure and setup of these channels can be as diverse as the companies themselves, but a good starting point is to reach out to the internal communications department. You may even have a dedicated team for community management or employee experience. They're often eager to get involved early on, as this allows you to align the community's setup with their standards and leverage their existing resources and frameworks. Plus, having this department as an ally is always a strategic advantage.

Examples of common internal channels to promote your new community and its activities:

- Existing community channels where the target groups are already engaged
- Internal newsletters
- Dedicated mailings to target groups
- Existing educational formats (lunch & learns, town hall meetings)
- Promotional screens in the office (yes, the ones that also show you the daily menu of the café)
- Posters

Building Your Message

Once we've identified the "who" and "where," the next, more labor-intensive step is tackling the "how." As mentioned earlier, different individuals have various learning styles and respond to different communication approaches depending on their roles. The challenge now is to distill these differences into a common denominator, which is often less about technical details (sorry, tech enthusiasts) and more about delivering a strategic and emotional message. This message needs to be consistently and continuously communicated through the identified channels.

Aligning the core messages with your company's (technology) mission and vision ensures that the project's value is recognized within the company and positively associated with the corporate strategy. Using a familiar color palette and visual language can also enhance recognition.

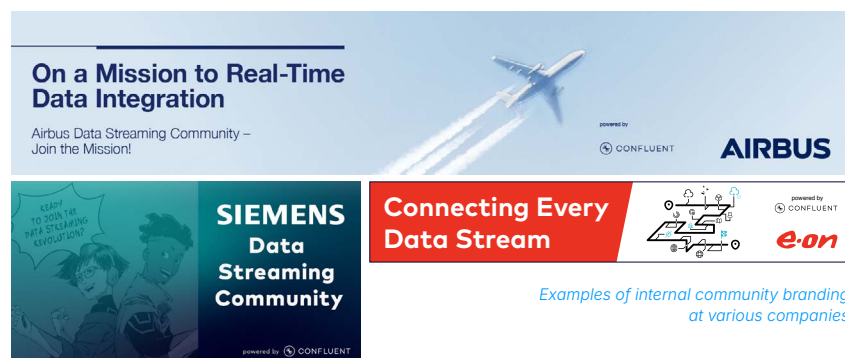
Based on our experience, we recommend a multi-phase launch of the internal community. Start by addressing the broad target audience, then refine the messaging to fit specific personas, conveying the added value in a way that resonates with each group.

Make sure to regularly position yourselves with new and growing projects, either within the company's existing internal formats or by establishing new ones. In other words, "Do good and talk about it."

Figuring out the "who," "where," and "how" is not an easy task. Confluent offers a dedicated workshop on the topic of positioning and branding.



Reach out to the Strategic Advisory team at Confluent to learn more about community building and the DSO.



Examples of internal community branding at various companies

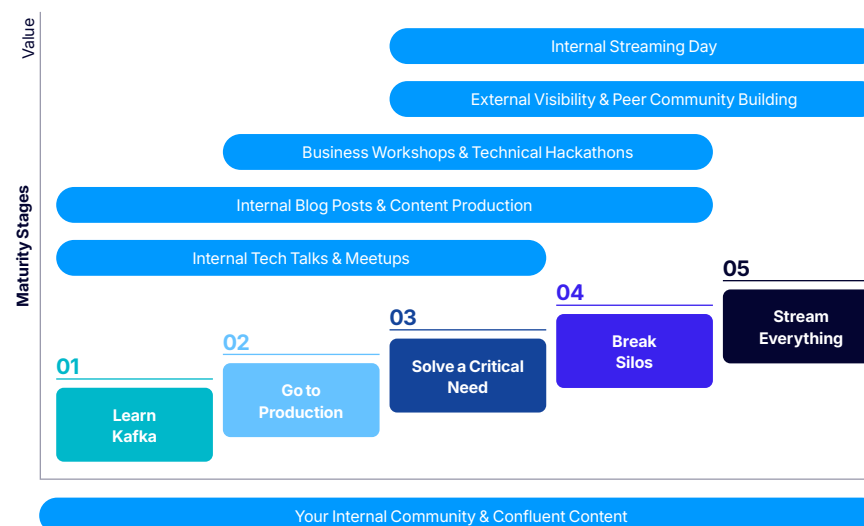
4.4 Strategies for Long-Term Engagement in a Data Streaming Community

From this point on, it's important to continuously meet the needs of all personas across the identified channels. If successful, this step will be repeated several times, especially in the first few months (or years, depending on the size of the company), as new teams continuously bring their use cases to the central data streaming platform. Different levels of knowledge and

expectations will be met along the journey, and it's recommended to meet them proactively instead of reactively.

A popular approach to kicking off and maintaining a positioning strategy is to host internal events. These can be designed on-site, hybrid or virtual, but we recommend to run not all events virtually to generate a full community experience.

Activities to Build a Community



Examples of community engagement programs, along the internal maturity curve

Kickoff Session and Series of Educational Talks

With one initial session, you want to make sure that everyone gets to the same level of understanding about what data streaming is. Hence, we recommend a 60-minute presentation titled "What is data streaming? Typical use cases in our industry."

It's important that the welcoming or opening notes to this presentation are given by the internal project team members who introduce themselves—not by the technology provider. At best, the mission and vision of the internal service can be communicated within the first 5-10 minutes. Also make sure to introduce the channels where interested teams can reach out to you (a simple email alias can do the trick) and where they can find more information.

The important part is to introduce not only informative sources and collateral but also future educational events, office hours, and/or interactive Q&A formats. The teams need to understand that what you're introducing really is a strategic initiative that is highly relevant for the company's digital future. Also make sure to include an opportunity for interested teams to share their first successes and use cases. Be as open and collaborative as possible and make sure all output is aligned with the internal branding that you as a project team/service owner team defined.

Get Your Teams Certified as Data Streaming Engineers

A core component of a sustainable data streaming strategy is an enabled technical workforce that is equipped with validated expertise in data streaming technologies and practices. Besides that, engineers love being certified. Hence, many companies, such as Citigroup and Virgin Australia, motivated their technical teams to become certified data streaming engineers. These certification exams are an easy route to including gamification and combining it with enablement offerings.



Full-Day Event: Data Streaming Day/Confluent Day

Depending on your company's culture and its position on the maturity curve, hosting a dedicated Data Streaming Day or Confluent Day might be a smarter, more impactful way to raise internal awareness. This approach is especially effective if several use cases are already live and ready to be showcased or if there's high interest in certain hot topics (e.g., GenAI, customer 360, digital twin). An additional option is to include the data streaming topic into a wider Data + AI Day.

The beauty of such an event is that you can leverage all types of event management strategies. Here are some ideas to make the most out of it:

- **Dedicated branding** for the event, ideally co-branded with the community and personalized either regionally or thematically (e.g., BMW's Kafka Wiesn with the slogan "O'gstreamt Is," or s.Oliver's Kafka Fashion Convention with the tagline "Fashion Is a Stream of Events").
- **Promotion materials** such as posters, flyers, digital slides in the office, and email newsletters to create buzz and anticipation.
- **Creative giveaways** for participants, such as community-branded T-shirts, stickers, or even customized cupcakes—get creative and make it fun and tasty!
- **Recordings and slide decks** from the event presentations to share in your community channels after the event; can also be repurposed for further content creation.



CONFLUENT

The Art of the Possible with Data Streaming in the Automotive Industry

Data Streaming with Apache Kafka with a discussion on use cases and real-life customer stories

April 21
01:00 pm CEST



Talk by Kai Waehner
Field CTO, Confluent

BMW Streaming Wiesn (Internal Oktoberfest)

The BMW Kafka Wiesn celebrated its five-year anniversary in 2024 and became the BMW Streaming Wiesn in 2025. This together the global BMW Group data streaming community in a hybrid event in the spirit of Oktoberfest and innovation. Every year, BMW Group showcases its most recent use cases, strategic initiatives, and the vision of its internal data streaming platform. One thing that cannot be missed: The Kafka pretzels!



Regardless of the format in which the data streaming platform and the associated community are introduced, everything has to happen on-brand and via the channels of the respective community. Continuity and permanence are essential, especially at the beginning, so that an internal brand can be built up. Another aspect to watch out for is to provide internal content and knowledge by the technology provider in equal

Michelin Kafka Day

At the Michelin Kafka Day in May 2024, Michelin's global data streaming community was committed to the topic and the vision. The agenda included a variety of formats, which were offered to more than 500 participants across three time zones. In addition to Michelin-branded macarons, the exclusive event shirts also sold like hotcakes on-site.



measures. Otherwise, the recipients might gain the impression of a one-sided advertising event, which could lead to a negative bias toward the technology vendor and, in turn, toward the product. It is better to act as a joint team right from the start, with the focus on internal use cases. If new topics are to be presented, it's better to directly relate them to internal projects, strategies, or visions.

External Engagement to Strengthen Brand and Community Recognition

When it comes to positioning and perception, the world beyond your company should not be overlooked. Once the first use cases have been successfully implemented on your data streaming platform and shared within the internal community, it's time to extend that communication externally. But what does that look like in practical terms?

1. Become part of the Confluent community and share your successes and experiences with other customers.

This fosters a valuable exchange with industry peers or others working on similar use cases. Why reinvent the wheel when others might have useful insights? Confluent offers various formats for this, including one-on-one phone exchanges, guest speaker spots at customers' internal Confluent Days, and industry-focused roundtables. Want to connect with multiple customers in your region at once? The Data in Motion Tour and the global data streaming conference *Current* provide excellent platforms to engage with customers across different industries and maturity levels.

2. Apache Kafka and Apache Flink®, as open source communities, also offer great opportunities to showcase technical expertise, share experiences, and receive feedback. More than 50,000 members meet regularly in more than 260 Meetups across 66 countries—and they're always looking for new hosts and speakers.

3. Your success stories deserve to be shared with the world—whether digitally or in print. Formats such as YouTube videos, written case studies, webinars, and press interviews provide ample opportunities to boost brand awareness and employer branding. After all, who wouldn't want to work for a company that's driving innovation with cutting-edge technology?

The beauty of these external formats is not only in the positive presentation of your project to the outside world but also in the new content they generate to further raise awareness internally for your community, service, and team.

Examples of external presence and positioning include:



- **Video of Michelin** showcasing its gold standard reputation for innovative mobility solutions



- **Speaking at Current**, the biggest event in data streaming



- **Participating in a webinar** alongside guest speakers and Confluent experts



- **Being featured on the Confluent Customer Hub**, surrounded by peers and industry leaders

Data Streaming Is a Journey: A Roadmap to Success

05

From Technology to Strategic Foundation

Data streaming is no longer just a technical enhancement. It has become a strategic capability at the heart of modern business and IT architecture. While data lakes, warehouses, and lakehouses still play important roles, they're no longer sufficient on their own. Success today depends on a new approach that puts data in motion at the center.

This shift requires more than adopting a new tool. It requires a mindset change. Moving from traditional request-response models and batch pipelines to a real-time, event-driven architecture enables organizations to unify operational and analytical workloads. This closes the long-standing gap between systems that run the business and those that analyze it. With a data streaming platform, the same real-time data that drives customer interactions and business transactions can also fuel dashboards, analytics, and AI systems.

Real Use Cases Across All Industries

This book has shared real examples from many industries. Companies in financial services, insurance, retail manufacturing, healthcare, energy, telecom, transportation, and gaming are moving away from siloed, batch-centric systems. They're placing real-time data at the core of their architectures. These are not experiments or limited use cases.

They're business-critical platforms for fraud detection, instant payments, connected vehicles, predictive maintenance, real-time personalization, and more. The specific challenges differ, but the pattern is the same: Organizations treat data streaming as a platform, not a point solution. They use it to build flexible, scalable foundations for innovation and resilience. This is what turns short-term wins into long-term strategic value. Confluent's data streaming platform, based on Apache Kafka and enhanced with Flink and Iceberg, delivers this foundation as a fully managed, cloud-native service. That means teams can focus on business impact rather than infrastructure management.

AI Needs Data Streaming to Work at Scale

Data streaming is also essential for enterprise AI. As generative and agentic AI applications emerge, they bring new demands. AI can't run on stale, fragmented, or delayed data. It needs context that is accurate, complete, and current. A data streaming platform acts as a context engine, continuously delivering fresh, governed data from operational systems into AI pipelines.

Not every intelligent agent will be built with Confluent. Some will rely on external AI platforms or SaaS services. All of them still will require trustworthy data, and this is what the data streaming platform provides. It acts as a context engine that delivers accurate, governed, and real-time information into any AI

application, whether it runs inside your own environment or in an external product. This ensures that every agent can operate safely, reliably, and with the right context at the right moment.

Community and Culture Drive Adoption

Beyond technology, people and culture make the difference. Successful organizations build strong internal data streaming communities. Architects, engineers, and business advocates lead by example. They share best practices, support teams, and promote data streaming as a shared capability across departments. This internal community transforms isolated use cases into reusable platforms. It connects IT and business toward common goals. It ensures that data streaming is recognized as a strategic foundation, not just another tool.

Organizations that embrace this community mindset evolve along a maturity curve. They start with small projects. Over time, they add data governance, self-service, and reusable components. Eventually, the data streaming platform becomes the digital nervous system of the enterprise. At that point, new use cases can be built faster, scaled globally, and integrated with analytics and AI in real time. This happens while maintaining security, ensuring compliance, and meeting critical SLAs.

This is not just about keeping up. It's about leading. Data streaming opens the door to use cases that were not possible before. Self-adjusting supply

chains. AI assistants that learn from live data. Hyper-personalized customer journeys. Resilient systems that detect and react to disruptions as they happen. These are no longer ideas for the future. They're happening now.

Your Next Step Starts Now

The need for real-time data will only increase. Data volumes are rising. AI is moving faster. Customer and regulatory expectations are growing. Businesses must respond with speed, precision, and confidence. Data streaming enables that response.

This book is just the beginning. The landscape will continue to evolve. New technologies, patterns, and customer stories will emerge. Future editions of this book will capture that progress. The goal is to provide a living resource that reflects the state of data streaming across the industry.

Whether you're starting your first project or scaling out across regions or lines of business, now is the time to act. Data streaming is not just about faster pipelines. It's about building a unified, intelligent, real-time foundation for your entire business.

Every journey begins with a first step, and you've seen where that journey leads. The future is real time. It's event-driven. It's AI-powered. And it's built on data streaming.

Take your next step today. Lead the way forward.

The Ultimate Data Streaming Team

Want to be part of the third edition of this book with your data streaming success story? Email us at use-case-book@confluent.io.



Bringing the second edition of The Ultimate Data Streaming Guide to life reminded us just how powerful teamwork really is! This book is the result of the collaboration, expertise, and encouragement of many individuals who contributed their time, knowledge, and support.

Thank you to our contributors, especially our customers, for sharing your stories and shaping this guide into a valuable resource. **To our reviewers and editors**, your commitment to precision has been invaluable. And to our readers and the data streaming community, your passion for innovation drives us. This book is for you and because of you.

Even though we won't be able to list everyone, we want to shout out some key team members:

Our managers **Kyle Gudgeon and Andrew Sellers**, for having our backs and making us stick to our commitments all the damn time. Our fearless project manager **Ronja Antesberger** for holding us all accountable to delivering by our deadlines. Again, Queen of Design **Elena Hasenbeck**, transforming vision into reality even when we just had sketches around painted boxes. Our linguistic hero **Florian Stocker** as well as special MVP **Susan Patten**—all on a mission to writing consistency!

Kate Sukhova and Emma Griffiths as well as the whole customer marketing team for collecting so many great real-life examples that are the heart and soul of this book. **Lyndon Hedderly, Jon McCullagh-Vines, and Nick Dearden** for all your valuable feedback, brainstorming time, and support on this project. **Cody van Kampen and the product marketing team** for support in building out a table of contents that resonates! **A ton of proofreaders, including family members and friends**, who did not hold back with feedback as well as motivation—equally.

We're already working on the next edition of the book, featuring new use cases, real-life examples, and more community success stories. In particular, we'll showcase how agentic AI is moving from prototypes into production across industries. These systems augment streaming platforms with autonomous agents that orchestrate workflows, anticipate risks, and act in real time. These stories will highlight how organizations are building not just reactive pipelines but intelligent, adaptive systems that continuously learn, collaborate, and deliver value at scale.

Want to be part of the third edition of this book with your data streaming success story? Email us at use-case-book@confluent.io.

Kai Waehner & Evi Schneider

Contact Us

Comments and thoughts on the book are encouraged. Or do you have a compelling data streaming example you'd like featured in the next version of the book? Please reach out at use-case-book@confluent.io

Stay connected with Kai Waehner on

 LinkedIn [kaiwaehner](#)

 X [kaiwaehner](#)

for ongoing discussions and updates.

For regular insights into data streaming trends, architectures, use cases, and success stories, follow Kai Waehner's blog:

 www.kai-waehner.de

You can also subscribe to his newsletter:

 www.kai-waehner.de/news

For those interested in building a data streaming community and hearing more about examples in this area, please reach out to Evi Schneider:

 evi@confluent.io

"This book is a fantastic resource for anyone looking to dive deep into the world of data streaming. It captures not only the technology but also the stories behind its adoption and the value it delivers."

Jay Kreps, Original Co-creator of Apache Kafka, Co-founder and CEO, Confluent

